

Java Server Reference Guide

ACI Java Infrastructure™



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Preface

ACI provides server management and administration facilities for Java servers built using ACI's Java Server Framework (AJSF) and Java Presentation Framework (AJPF). This user guide describes how to effectively use these facilities to configure and administer Java server processes in ACI products that use AJSF- and AJPF-based Java server processes. It includes overviews of the AJSF and AJPF and their data sources and describes the user interface windows (or browser pages) and properties used to configure and administer AJSF- and AJPF-based Java server processes.

This guide does not describe the ACI Java Management Framework (AJMF), which provides a monitoring and control facility based on Java Management Extensions (JMX) and a event processing environment. For detailed information on the AJMF and JMX, refer to the *ACI Java Infrastructure Application Management User Guide*.

Note: Because the look and feel of the user interface employed to configure and administer AJSF/AJPF applications can differ across ACI applications, no window or browser page screen shots are provided in this user guide. The fields displayed on these windows or pages are essentially the same across all AJSF- and AJPF-based applications. Refer to your application-specific documentation or ACI Online Help for guidance in accessing AJSF/AJPF configuration and administration windows or screens.

Audience

This user guide is intended for management and operations personnel responsible for configuring and administering the BASE24-eps User Interface (ESWEB), BASE24-eps Voice Authorization web pages, ATM (Java) Device Handler, IFX ATM Manager, Event Adapter, Application Management, and Remote Database Server applications, all of which are built on the AJSF. The BASE24-eps Voice Authorization web pages is also built on the AJPF. This guide contains both generic AJSF and AJPF information as well as AJSF and AJPF information that is specific to these applications.

Prerequisites

Readers of this user guide should be familiar with basic Java terminology and the user interfaces on which AJSF- and AJPF-based applications are deployed. Familiarity or experience with Java programming and Java virtual machines (JVMs) is recommended.

Additional Documentation

The ACI documentation set is arranged so that each ACI manual or guide presents a topic or group of related topics in detail. When one ACI manual or guide presents a topic that has already been covered in detail in another ACI manual or guide, the topic is summarized and the reader is directed to the other manual or guide for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information. This manual references the following ACI publications:

- The ***ACI Desktop User Interface Manual*** provides descriptions of task IDs associated with a specific ACI desktop user interface window or framework as well as information on configuring ACI desktop user functions.
- The ***ACI Application Management User Guide*** describes the AJMF as well as connector definition file configuration considerations for configuring multiple Application Management agents.
- The ***BASE24-eps ATM (Java) Configuration and Management Guide*** describes the least bills method dispense algorithm.
- The ***BASE24-eps Environment User Guide*** describes CCSID_VARIANTS and UI_AUDIT_LEVEL environment attributes. The CCSID_VARIANTS attribute is used in conjunction with the db.char.set.ebcdic property to define the IBM Coded Character Set Identifier (CCSID) variants supported by ATM Device Handler components. The UI_AUDIT_LEVEL environment attribute is used to control user auditing from the C++ User Interface process.
- The ***BASE24-eps Event Documentation*** reference guide describes all events that can be generated in a BASE24-eps system, including AJSF-based application events.
- The ***BASE24-eps Security Best Practices*** document describes cross-industry security best practices for the BASE24-eps product.
- The ***BASE24-eps Voice Authorization User Guide*** provides descriptions of task IDs associated with specific Voice Authorization web pages as well as information on configuring Voice Authorization web page functions.

This guide references Sun's *Java Secure Socket Extension (JSSE) 1.0.2 API User's Guide* for more information on AJSF properties used to initialize the Java Secure Socket Extension (JSSE) environment.

This guide references the following Public-Key Cryptography Standards specifications produced by RSA Laboratories:

- The Public-Key Cryptography Standards (PKCS) #12 specification, *Personal Information Exchange Syntax Standard* describes Version 1 and Version 3 X.509 certificates and files.
- The Public-Key Cryptography Standards (PKCS) #7 specification, *Cryptographic Message Syntax Standard* describes Secure MIME (S/MIME) and Cryptographic Message Syntax (CMS).

This guide references the book, *Concurrent Programming in Java*, by Doug Lea for the rationale and applications of several of the utility classes provided in the third-party JAR, `concurrent.jar`, which is used by the AJSF.

This guide contains references to the following International Organization for Standardization (ISO) publications:

- The ISO 3166 standard, *Codes for the Representation of Names of Countries*, describes three-character and two-character country codes used in the ISO standard.
- The ISO 639 standard, *Codes for the Representation of Names of Languages*, describes two-character language codes used in the ISO standard.

Software

This guide documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this guide. The customer is responsible for identifying and noting these changes.

Guide Summary

The following is a summary of the contents of this guide.

Section 1, “Introduction,” describes the ACI Java Server Framework and ACI Java Presentation Framework, and introduces you to the database, management, and administration facilities provided in these frameworks. It also introduces you to the properties used to configure Java server processes built on these frameworks.

Section 2, “Subsystems,” describes each of the subsystems provided in the framework layer of the Java Server Framework as well as the APIs and third-party JARs used by these subsystems.

Section 3, “Management,” shows you how to manage the configuration for Java server processes. This includes managing the configuration of alternate data sources, application-level properties, connections to and from external entities, and process relationships in clustered configurations for broadcasting messages.

Section 4, “Administration,” describes how to administer cache, TCP/IP sockets, configuration properties, and database usage for AJSF-based applications. It also describes how to monitor system usage, control event and trace message logging, implement field masking, use the optionally provided Secure Delete utility, and implement security for AJSF-based web application servers and ACI desktop user interface or browser clients.

Section 5, “Configuration Properties,” explains the three levels of Java properties in the AJSF—system, configuration, and application. It also describes properties common to the AJSF and an ACI utility you can use to encrypt passwords in the config.properties file.

Section 6, “Extending AJSF Security and Auditing,” describes how to extend User Security (USEC) authentication and AJSF user auditing functions to non-AJSF-based applications.

Section 7, “Event Adapter,” describes the AJSF-based Event Adapter process used to extract events from a Websphere MQ queue that APF products log to, reformat events if required, and write the events to configured output locations. The Event Adapter process is used when an APF product is installed on an IBM System z or Unix platform.

Section 8, “ACI Java Presentation Framework (AJPF), describes the ACI Java Presentation Framework used to develop browser-based user interfaces for ACI Enterprise Payment Solutions.

Appendix A, “BASE24-eps User Interface (ESWEB),” describes the AJSF properties specific to the BASE24-eps Java User Interface servlets or the Voice Authorization web page servlets in the ESWEB Server process for the ACI

desktop user interface or browser web pages. It also describes third-party JARs specific to the BASE24-eps Java User Interface servlets and provides a sample config.properties file used for the ESWEB process.

Appendix B, “BASE24-eps ATM (Java) Device Handler and IFX ATM Manager Processes,” describes the AJSF properties specific to BASE24-eps ATM (Java) Device Handler and IFX ATM Manager processes. It also provides a sample config.properties file used for a BASE24-eps ATM (Java) Device Handler process.

Appendix C, “Application Management,” describes the AJSF configuration for BASE24-eps Application Management agents.

Appendix D, “Remote Database Server,” is a placeholder for the AJSF-based Remote Database Server. The AJSF configuration for this server process is provided in the *BASE24-eps Extract and Reporting User Guide*.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this ACI publication. The publication number (for example, CF-DA000-02-ES for the *ACI Java Infrastructure Java Server Reference Guide* for BASE24-eps) appears on every page to assist readers in identifying the guide from which a page of information was printed. The publication date (for example, Sep-2009 for September, 2009) indicates the issue of the guide. The software release information (for example, R1.0v09.2 for release 1.0, version 09.2) specifies the software that the guide describes. This information matches the document information on the copyright page of the guide.

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Conventions Used in This Guide

This section explains how the security best practices symbol and IBM System z platform terminology is used in this guide.

Security Best Practices Symbol



This guide uses the symbol shown at the left to alert you to cross-industry best practices for protecting sensitive data that is stored, processed, or transmitted in a BASE24-eps system. Whenever you see this symbol, you should carefully review the information provided next to the symbol to ensure you are implementing cross-industry best practices for securing data. Refer to the ***BASE24-eps Security Best Practices*** document for more information about configuring the BASE24-eps product to comply with cross-industry best practices.

IBM System z Platform

Throughout this guide, the term “IBM System z platform” refers to IBM System z hardware running the z/OS operating system.

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1: Introduction

This section introduces you to the ACI Java Infrastructure (AJI) framework and the following frameworks from which AJI is composed:

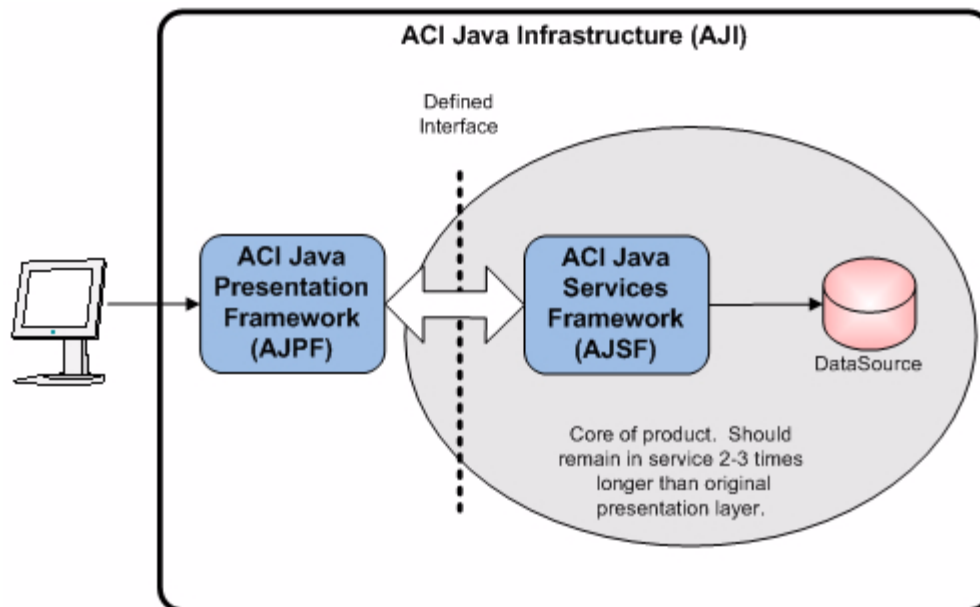
- ACI Java Server Framework (AJSF)
- ACI Java Presentation Framework (AJPF)
- ACI Java Management Framework (AJMF)

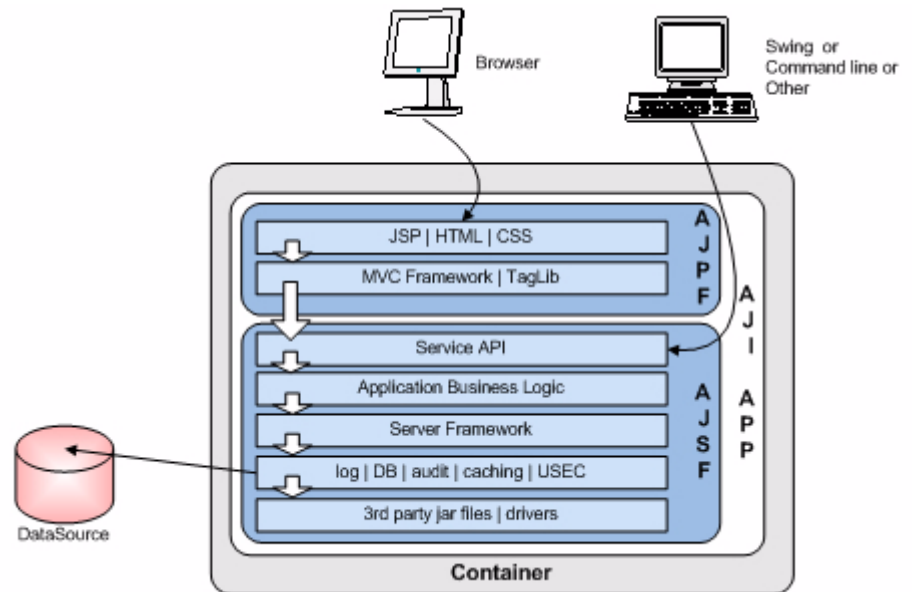
This section describes the AJSF and AJPF and introduces you to the database, management, and administration facilities provided in these frameworks. It also introduces you to the properties used to configure Java server processes built on these frameworks. The AJMF is described in the *ACI Java Infrastructure Application Management User Guide*.

ACI Java Infrastructure (AJI)

The ACI Java Infrastructure (AJI) is a framework of design patterns and services required for building integrated Java applications within the ACI Enterprise Payment Solution Payments Framework. AJI consists of two loosely coupled frameworks, the ACI Java Presentation Framework (AJPF) and the ACI Java Services Framework (AJSF), that are designed to allow for efficient, rapid presentation layer evolution while protecting and extending the life of core business logic at the server. The AJPF is built on top of the AJSF and adds web application presentation capabilities. All of the facilities of the AJSF are available to AJPF applications. AJI also includes the ACI Java Management Framework (AJMF), which uses Java Management Extension (JMX) technology for consolidating and centralizing the collection and publication of system management data and functionality across ACI products. All of the facilities of the AJSF are also available to AJMF applications.

The AJI is the next step in the architectural evolution of ACI's Java application environment. The AJI provides all of the core server and browser functionality previously provided by the ACI Java Server Framework (AJSF), while absorbing the User Security (USEC) and user auditing services previously provided by the ACI User Interface Framework (UIF).





This section introduces you to the AJSF and AJPF. For information on the AJMF, refer to the *ACI Java Infrastructure Application Management User Guide*.

ACI Java Server Framework

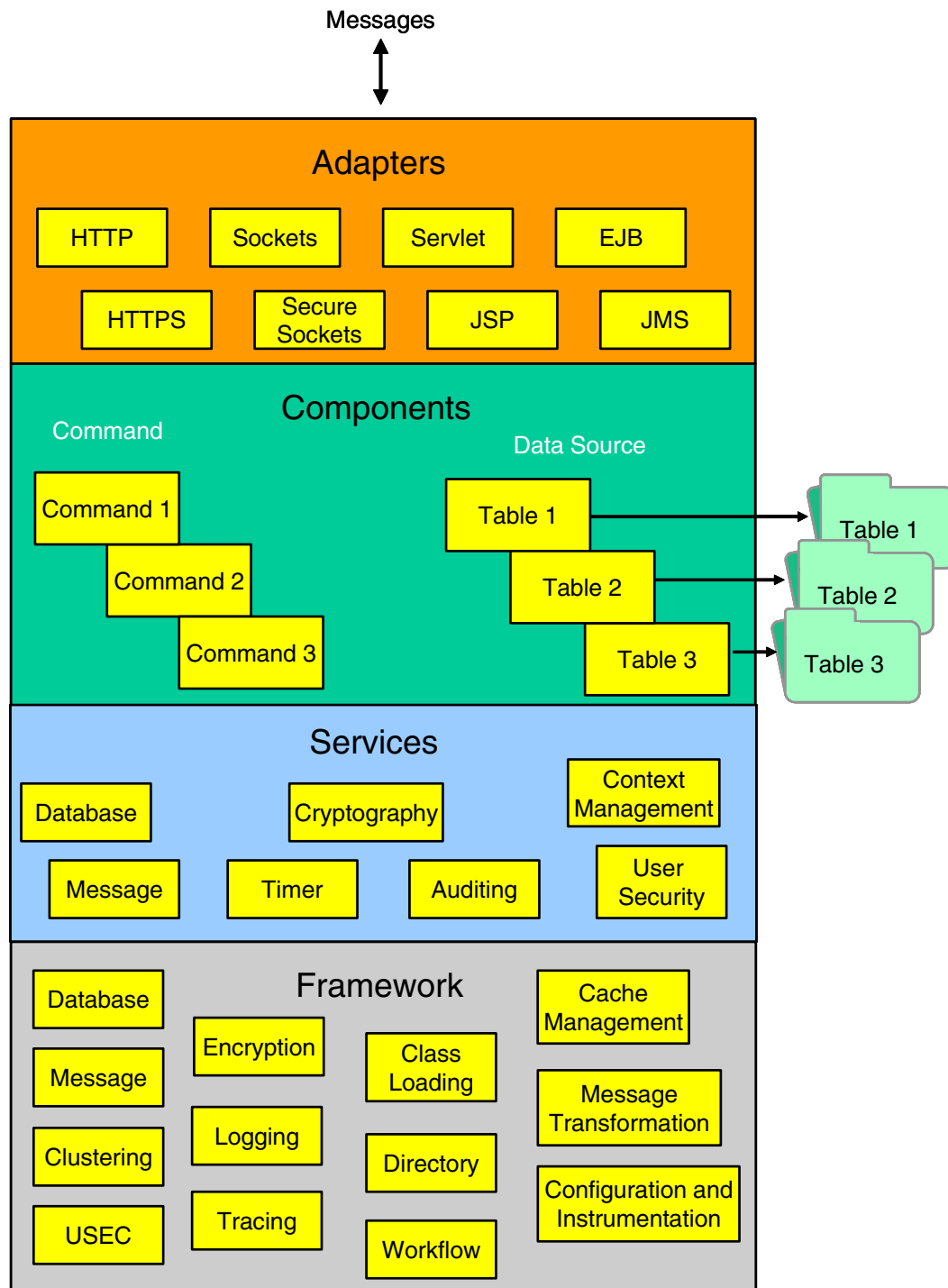
Java is an object-oriented programming language that uses a set of public application programming interfaces (APIs) to access operating system resources or other applications.

The ACI Java Server Framework allows you to manage and administer several ACI application products written in Java across multiple platforms using common management and administration functions built into ACI products. Java is an application-independent language that allows applications to be insulated from the different operating systems on various hardware platforms. ACI has deployed ACI Java Server Framework applications on the HP NonStop, IBM System z , Sun Solaris, IBM PSeries/AIX, HP-UX, and Windows platforms, however, not all platforms are supported for all AJSF-based applications. Check the application-specific system requirements to determine whether a platform is supported.

A benefit from writing Java-based applications is that operating system manufacturers and third-party organizations such as HP, IBM, and Sun Microsystems, build the OS-specific implementation free of charge to allow Java applications to run on their servers. By writing to the Java API, the ACI applications will run, without change and with only one version of the source code, on each platform that provides a Java Virtual Machine (JVM). The Java Server Framework uses the JVM to eliminate the need for platform-specific code.

Layers

The ACI Java Server Framework is comprised of four layers—adapters, components, services, and framework—as shown in the following illustration. Each of these layers is described on the following pages.



Adapters

Adapters provide a lightweight insulation layer for acquiring work from different mediums without impacting the underlying application. Currently, the AJSF provides adapters for Hypertext Transfer Protocol (HTTP), Secure Hypertext Transfer Protocol (S-HTTP), Sockets, Secure Sockets, Servlets, Java Server Pages (JSPs), Enterprise JavaBeans (EJBs), Java Message Service (JMS), and as a standalone Java application.

Components

Components are coarse-grained objects that make up the business logic of a particular application. Two types of components are supported—command components and data source components.

Command components are specific components that adhere to the command design pattern. These components are the business logic that makes up an application. Since these components all have the same interface, the framework has a layer that allows applications to be created by defining the order that the components are invoked through a Transaction Meta Map XML document. A Transaction Meta Map XML document is a work flow descriptor that defines the series of actions required to process a transaction. A number of ACI applications, such as the BASE24-eps ATM (Java) Device Handler processes and the BASE24-eps User Interface servlets, are built using this technique.

Data source components are encapsulation components that provide a single view to a data source. All transaction and user interface logic in command components use these components to access individual data sources. This layer performs routing to change the access to a data source from a database to potentially a message interface, an EJB, or a Lightweight Directory Access Protocol (LDAP) directory. Part of the framework consists of a code generator that builds a data source component for any relational database table through a metadata file. This same metadata file is also used to build database services described below.

Services

Services encapsulate specific entities of the application such as database and message services. They do not presently have a standard interface, but they do provide many commonly executed functions in a common reusable location. These are utility services that need to be present if the application requires them. For example, an online transaction processing application may require the following services:

- Auditing services
- Context management services
- Cryptography services
- Database services
- Message services
- Timer services
- User security services

Framework

The framework layer encapsulates all system resources including messaging, message routing, XML formatting, database access, logging, file system access, object caching, tracing, class loading and application configuration. The main function of the framework layer is to ensure that the applications written on top of it run on a variety of platforms and take advantage of the features of each platform from a scalability and performance standpoint. The framework layer can be broken down into the following subsystems, each of which is described in detail in [section 2](#).

- The Database subsystem provides support for database connectivity, database performance optimizing, and database auditing facilities.
- The Message subsystem provides the ability to message synchronously and asynchronously and includes callback functionality configured through a routing database to plug in message formatters at run time.
- The Clustering subsystem provides the ability to configure applications as multiple single-threaded applications on clustered servers such as the HP NonStop server, or as one or a few multithreaded applications on SMP servers such as Windows NT, Unix, and IBM System z.
- The Encryption subsystem performs a number of cryptographic functions such as DES encryption/decryption and adheres to the Java Cryptographic Extensions (JCE) API.
- The Logging subsystem provides for the logging of event messages generated by Java server processes using a third-party open source application called log4j. Log4j is an open source project, of the Apache group, which is based on the work of multiple authors.
- The Tracing subsystem provides for extensive tracing of all messaging, database access, cache management, and application logic using log4j.

- The Message Transformation subsystem provides a set of base classes for building and parsing external messages.
- The Directory subsystem contains an interface to naming and directory services.
- The Cache Management subsystem contains a Cache Manager component that can maintain cache information by Service type. Services make requests to place information in memory and retrieve information from memory. You can configure the database services for which data is cached and whether message context is stored in cache memory.
- The Configuration and Instrumentation subsystem provides mechanisms for configuring how Java server processes run, configuring how they connect to external entities, configuring the format and descriptions of event messages that can be generated by AJSF applications, and configuring sensitive data display masks used by the ACI desktop and browser presentation services. This subsystem also allows you to display and monitor database connections, socket connections, cache information, and threads in use.
- The class loading subsystem uses Java reflection to automatically load Java classes and instantiate instances of these classes at run time. This mechanism is used mainly in message, transaction map, and transaction meta map processing for dynamically creating message formatters for building and parsing messages. These classes are declared in the routing database (e.g., External Connection table and Transaction Map XML file) and invoked at run time.
- The user security (USEC) subsystem defines the users and access privileges for the ACI desktop and browser user interfaces. You can share the USEC subsystem among ACI Payment Framework (APF) products that use the ACI desktop or a browser user interface for access to multiple AJSF-based products. You can also *plug in* the USEC subsystem to third-party Java Authentication and Authorization Service (JAAS) authenticators for authentication with existing enterprise authenticators.
- The workflow subsystem uses transaction map and transaction meta map XML files to define the messages that can be processed and the transaction processing workflow to be used by an AJSF-based application for those messages. This subsystem follows the Java design pattern, command, in which objects are used to represent actions. It provides for loose coupling of business logic and allows for customer-specific modifications (CSMs) to be configured without affecting the standard application. Changes to the XML files are pluggable at runtime by simply warmbooting the application.

Application Programming Interfaces (APIs) and Third-Party JARs

The ACI Java Server Framework uses several industry-standard Java application programming interfaces (APIs), each of which is described below. Several of these APIs are provided by third-party JARs. ACI provides these third-party Java Archive (JAR) files with applications built on the AJSF.

Third-party JARs specific to AJSF-based applications are described in the application-specific appendices of this guide.

Note: Due to licensing requirements, the Java 2 Platform Enterprise Edition (J2EE) JAR is not provided by ACI. Visit the following website for more information on obtaining J2EE:

<http://java.sun.com/j2ee/index.jsp>

Axis

Axis is an Apache web services project that is essentially a SOAP engine — a framework for constructing SOAP processors such as clients, servers, gateways, etc.

Castor XML

Castor XML is an XML data binding framework. Unlike the two main XML APIs, DOM (Document Object Model) and SAX (Simple API for XML), which deal with the structure of an XML document, Castor enables you to work with the data defined in an XML document through an object model that represents that data.

Commons Codec

Commons Codec provides implementations of common encoders and decoders such as Base64, Hex, Phonetic and URLs.

Commons Discovery

The Discovery component is about discovering, or finding, implementations for pluggable interfaces. It provides facilities for instantiating classes in general, and for lifecycle management of singleton (factory) classes.

Commons-lang

The Lang Component provides a host of helper utilities for the java.lang API, notably string manipulation methods, basic numerical methods, object reflection, creation and serialization, and system properties. Additionally, it contains an inheritable enum type, an exception structure that supports multiple types of nested-Exceptions, basic enhancements to java.util.Date and a series of utilities dedicated to help with building methods, such as hashCode, toString and equals.

Commons-logging

The Logging package is an ultra-thin bridge between different logging libraries. Commons components can use the Logging API to remove compile-time and run-time dependencies on any particular logging package, and contributors can write Log implementations for the library of their choice.

Concurrent

This package provides standardized, efficient versions of utility classes commonly encountered in concurrent Java programming. This code consists of implementations of ideas that have been around for some time and is merely intended to save you the trouble of coding them. Discussions of the rationale and applications of several of these classes can be found in the second edition of the *Concurrent Programming in Java* book, by Doug Lea.

Java API for XML Binding (JAXB)

See the [“Message Transformation Subsystem”](#) topic in section 2 for information on how this API is used in the AJSF.

Java API for XML Processing (JAXP)

Provides a common interface for creating and using the standard Simple API for XML (SAX), XML Document Object Model (DOM), and Extensible Stylesheet Language Transformations (XSLT) APIs in Java, regardless of which vendor's implementation is actually being used. Third-party JARs used by this API are described below.

Xercesimpl. Xerces provides world-class XML parsing and generation. Fully-validating parsers implement the W3C XML and DOM (Level 1 and 2) standards, as well as the de facto SAX (version 2) standard. The parsers are highly modular and configurable. Initial support for XML Schema (draft W3C standard) is also provided.

XMLParserAPIs. The Java API for XML Processing (JAXP) provides a common interface for creating and using the standard SAX, DOM, and XSLT APIs in Java, regardless of which vendor's implementation is actually being used.

Java Authentication and Authorization Service (JAAS)

Java Authentication and Authorization Service (JAAS) is a set of APIs that can be used for the following two purposes:

- For authentication of users, to reliably and securely determine who is currently executing Java code, regardless of whether the code is running as an application, an applet, a bean, or a servlet.
- For authorization of users, to determine reliably and securely who is currently executing Java code, regardless of whether the code is running as an application, an applet, a bean, or a servlet.

JAAS authentication is performed in a pluggable fashion. This permits Java applications to remain independent from underlying authentication technologies. New or updated technologies can be plugged in without requiring modifications to the application itself. JAAS authorization extends the existing Java security architecture that uses a security policy to specify what access rights are granted to executing code. That architecture, as introduced in the Java 2 platform, was code-centric. That is, the permissions were granted based on code characteristics: where the code was coming from and whether it was digitally signed and, if so, by whom. With the integration of JAAS into the Java 2 Software Development Kit (SDK), the `java.security.Policy` API handles Principal-based queries, and the default policy implementation supports Principal-based grant entries. Thus, access control can now be based not just on what code is running, but also on who is running it.

Java Cryptography Extension (JCE)

The Java Cryptography Extension (JCE) provides a framework and implementations for encryption, key generation and key agreement, and message authentication code (MAC) algorithms. Support for encryption includes symmetric, asymmetric, block, and stream ciphers. The software also supports secure streams and sealed objects.

JCE is based on the same design principles found elsewhere in the Java Cryptography Architecture (JCA)—implementation independence and, whenever possible, algorithm independence. It uses the same *provider* architecture. Providers signed by a trusted entity can be plugged into the JCE framework, and new algorithms can be added seamlessly.

The JCE API covers the following:

- Symmetric bulk encryption, such as the Data Encryption Standard (DES), Rivest's Cipher 2 (RC2), and the International Data Encryption Algorithm (IDEA)
- Symmetric stream encryption, such as RC4
- Asymmetric encryption, such as Rivest, Shamir, and Adleman (RSA)
- Password-based encryption (PBE)
- Key agreement
- Message authentication codes (MACs)

See the [“Encryption Subsystem”](#) topic in section 2 for more information on how this API is used in the AJSF. Third-party JARs used by this API are described below.

Local_policy for Java Cryptography Extension (JCE). The local_policy for JCE provides the ability to do Full Function versus Limited Key Size Cryptography through the JCE API.

SunJCE provider for JCE. This is a standard cryptographic service provider named “SunJCE” (which implements a session key generator for DES). You must explicitly register the “SunJCE” provider with every JCE installation. The JCE and JSSE must be installed before an AJSF-based application can be installed. Note that with Java Virtual Machine 1.4 and above, the JCE and JSSE are already installed and registered.

US_export_policy for JCE. The US_export_policy for JCE provides the ability to do Full Function versus Limited Key Size Cryptography through the JCE API.

Bouncy Castle. The AJSF uses the Bouncy Castle cryptographic APIs JAR as an encryption library to provide a level of protection for sensitive data stored in the Application Configuration data source (APPLCNFG) for an AJSF-based application. The AJSF's ConfigPropertyBuilder password encryption utility also uses Bouncy Castle to encrypt passwords in the config.properties file.

BouncyCastle is a lightweight cryptography API in Java created by the Legion of the Bouncy Castle that provides the following:

- A provider for the Java Cryptography Extension (JCE) and Java Cryptography Architecture (JCA).
- A clean room implementation of the JCE.
- A library for reading and writing encoded Abstract Syntax Notation One (ASN.1) objects.
- Generators for Version 1 and Version 3 X.509 certificates and files compliant with the Public-Key Cryptography Standards (PKCS) #12 specification, *Personal Information Exchange Syntax Standard*.
- Generators/Processors for Secure MIME (S/MIME) and Cryptographic Message Syntax (CMS). CMS is a superset of the Public-Key Cryptography Standards (PKCS) #7 specification, *Cryptographic Message Syntax Standard*, from RSA Laboratories.
- Generators/Processors for Online Certificate Status Protocol (OCSP). Refer to the Internet Engineering Task Force (IETF) specification for OCSP (RFC 2560) for more information on OCSP.
- A signed JAR version suitable for the Java Development Kit (JDK) and the Sun JCE.

For more information on Bouncy Castle, refer to the following website:

<http://www.bouncycastle.org/>

Java DataBase Connectivity (JDBC)

See the “[Database Subsystem](#)” topic in section 2 for information on how this API is used in the AJSF.

Note: Because the JDBC driver is an API provided with the Java 2 Platform Enterprise Edition (J2EE), it is not provided by ACI.

Java Document Object Model (JDOM)

JDOM is an API for manipulating XML documents. JDOM is both Java-centric and Java-optimized. It behaves like Java, uses Java collections, is a completely natural API for current Java developers, and provides a low-cost entry point for using XML. Log4j uses JDOM.

While JDOM interoperates well with existing standards such as the Simple API for XML (SAX) and the Document Object Model (DOM), it is not an abstraction layer or enhancement to those APIs. Rather, it provides a robust, light-weight means of reading and writing XML data without the complex and memory-consumptive options that current API offerings provide.

Javadoc

All AJSF-based applications use Sun's Javadoc tool provided in the Java 2 SDK to generate API documentation (in HTML files) for all classes, methods, and constants used by the application. These files are delivered with all AJSF-based applications and are typically provided within a "docs" folder or an "apidocs" WinZip archive (apidocs.zip). Java programmers may find these documents useful for developing or customizing AJSF-based applications.

Java Message Service (JMS)

The Java Message Service (JMS) API is a messaging standard that allows application components based on the Java 2 Platform, Enterprise Edition (J2EE) to create, send, receive, and read messages. It enables distributed communication that is loosely coupled, reliable, and asynchronous.

See the "[Message Subsystem](#)" topic in section 2 for more information on how this API is used in the AJSF.

Java Servlets

Java Servlet technology provides a simple, consistent mechanism for extending the functionality of a Web server and for accessing existing business systems. The ACI desktop user interface runs as a servlet under the JVM.

JAX RPC

JAX RPC is used to develop, deploy, and invoke a Web Service using Java APIs for XML-based RPC (JAX-RPC).

Jaxen

The jaxen provides a Java XPath engine. Jaxen is a universal object model walker, capable of evaluating XPath expressions across multiple models. It currently supports dom4j, JDOM, and DOM.

Log4j

Log4j is an open source project of The Apache Software Foundation based on the work of many authors. It allows the developer to control which log statements are output with arbitrary granularity. It is fully configurable at runtime using external configuration files. Best of all, log4j has a gentle learning curve.

See the [“Logging Subsystem”](#) and [“Tracing Subsystem”](#) topics in section 2 for more information on how this API is used in the AJSF. Third-party JARs used by this API are described below.

Piccolo

Piccolo is an XML parser for Java that implements Simple API for XML (SAX) interfaces as a non-validating parser and attempts to detect all errors for XML well-formedness.

SOAP with Attachments API for Java (SAAJ)

SAAJ defines APIs for SOAP attachments.

Sun XML Datatypes Library

AJSF-based applications use the following third-party JARs from the Sun XML Datatypes Library package to support the data validation mechanisms. These JARs are also included in the Java Web Services Developer Pack (JWSDP).

- jax-qname
- namespace
- relaxngDatatype
- xsdlib

WSDL4J

The Web Services Description Language for Java Toolkit (WSDL4J) allows for the creation, representation, and manipulation of WSDL documents.

Xalan

Xalan-Java is an XSLT processor for transforming XML documents into HTML, text, or other XML document types.

XML Security

Provides security for XML documents.

ISO Country and Currency Codes

An `AJI_isocodes.jar` consolidates all ISO country codes and currency codes used by AJI-based applications. By consolidating these codes within the AJI, ISO country and currency mandate changes can be implemented in AJI-based applications (e.g., ACI desktop user interface server and clients) without any code changes.

The `AJI_isocodes.jar` must reside in the in the web application's `WEB-INF` directory and in the classpath or application path for each desktop client.

ACI Java Presentation Framework (AJPF)

The AJPF is based on JavaServer Faces (JSF) technology. JSF is a Java-based web application framework intended to simplify the development of browser-based user interfaces for Java Enterprise Edition (EE) applications. For more information on JSF, refer to the following URLs:

- <http://java.sun.com/javaee/jaserverfaces/>
- http://en.wikipedia.org/wiki/JavaServer_Faces

All of the general facilities of the AJSF are available to AJPF applications as well as AJSF applications. Like AJSF applications, AJPF applications use the user security (USEC) subsystem, user auditing services, event logging, password encryption, tracing, field masking, and database subsystem. In addition, AJPF uses the following tools to build web-based user interfaces:

- Faces Sun RI (Reference Implementation)
- ICE Faces AJAX and component library
- Facelets - templating and component creation

Although the AJPF can use many other tools based on the dependencies of the above, these are the main tools used to set up the AJPF framework and run a browser-based user interface.

Third-Party JARs

AJPF uses the following third-party JARs.

Backport-util-concurrent

Backport provides a concurrency library that works with uncompromised performance on all Java platforms currently in use, allowing development of fully portable concurrent applications.

Commons-beanutils

The BeanUtils component provides easy-to-use wrappers around Reflection and Introspection APIs to provide dynamic access to Java object properties.

Commons-collections

Commons-Collections builds upon the JDK Java Collections Framework classes by providing new interfaces, implementations and utilities.

Commons-digester

The Digester component provides a common implementation for reading XML configuration files to provide initialization of various Java objects within the system.

Commons-fileupload

The Commons FileUpload package adds robust, high-performance, file upload capabilities to servlets and web applications.

El Libraries

The EL libraries, (Unified) Expression Language, are part of the API used to reference the Java methods on JSF pages. With the unified expression language, the JSTL expressions and JSF expressions are handled in a similar fashion. These JARs are distributed with ICEfaces.

ICE Faces

ICEfaces is an open source J2EE Ajax framework for developing and deploying rich enterprise applications (REAs). For more information, visit www.icefaces.org.

Jsf-api

Provides the JavaServer Faces APIs.

Jsf-impl

Provides the JavaServer Faces implementation.

Krysalis jCharts

Krysalis jCharts is Java-based charting utility that outputs a variety of charts.

Java Server Properties

In Java, configuration often takes the form of properties. Java server processes running under the ACI Java Server Framework are configured using properties. Each property consists of a key name and a value. Both are string values. The AJSF uses three levels of properties—system, configuration, and application. The following paragraphs introduce each level of properties used for the AJSF. Each of the properties common to the AJSF are described in section 5. Application-specific properties are described in appendices.

System Properties

System properties are set by the operating system, by a user logged on to the operating system, or in a -D option JVM runtime argument or configuration properties file accessed when a Java server process is started. The AJSF requires a minimum of four system properties:

- Process ID
- Configuration filename (i.e., the location of the config.properties file used to bootstrap a Java server process)
- File encoding (specifies the character set used)
- File separator (specifies the default name-separator character, e.g., /).

The file encoding and file separator properties are set by the operating system and the process ID and configuration filename properties are set when a Java server process is started using the -D flag argument as shown in the example below.

```
-Dprocess.id=ESWEB  
-Dconfig.filename=/user/B24es_cur/ESWeb/ESConfig/uaf2/P6Q^GATE/config.  
properties
```

For some AJSF-based ACI applications, you can view the system properties currently defined for a Java server process on the System Properties window. For example, this window is accessed from **System Operations > Server Management > Server Administration > System Properties** on the ACI desktop user interface for BASE24-eps.

Configuration Properties

Configuration properties are used to bootstrap a Java server process when it is started. These properties are configured in a `config.properties` file. If the application is a Web application running in a Servlet container, the AJSF first tries to find the `config.properties` file in the web application's `WEB-INF` directory. If it is not found there, it then looks in the `WEB-INF/conf` directory. If it does not exist there, the AJSF looks in the `WEB-INF/config` directory. Lastly, the AJSF looks for the location of the `config.properties` in the `config.filename` system property. If the location is not specified in the `-Dconfig.filename` system property, it defaults to a file named “`config.properties`” located in the home directory of the user that started the process.

For any properties defined in the `config.properties` file that are also defined in the Application Configuration data source (APPLCNFG), the value in the `config.properties` file supersedes the value in the APPLCNFG, except for the `trace.enabled` property. For this property, the value in the APPLCNFG overrides the value in the `config.properties` file once the Java server process reads the APPLCNFG.

As stated above, the location of the `config.properties` file must be provided when a Java server is started. Only those properties necessary for a process to initialize should be defined in the `config.properties` file. These include the following:

- User ID and password for database access
- Database connection properties for accessing the database
- Location of the log4J configuration files for events and tracing
- Secure sockets layer (SSL) configuration properties

Initialization manager function properties for the Map Managers are configured in `config.properties` file or in the APPLCNFG. Beginning with AJSF 07.4, initialization function properties are only loaded from the APPLCNFG and not from the `config.properties` file. In either location, the entries in this list must be uniquely named (e.g., `initialization.function.DataValidatorManager`, `initialization.function.DataTypeValidationManager`, `initialization.function.MessageSchemaManager`, etc.). Also beginning with AJSF 07.4, initializers are either loaded from XML initialization map files or through specific configuration properties in the APPLCNFG table. They are mutually exclusive. For detailed information on XML initialization map files, see [section 9](#).

ACI provides a utility to encrypt the user password for database access that is stored in the `config.properties` file. This utility is described in [section 5](#).

For some AJSF-based ACI applications, you can view the configuration properties currently defined for a Java server process on the Configuration Properties window. For example, this window is accessed from **System Operations > Server Management > Server Administration > Configuration Properties** on the ACI desktop user interface for BASE24-eps.

Application Configuration Properties

The Application Configuration data source (APPLCNFG) defines application-level properties for Java server processes. For some AJSF-based ACI applications such as BASE24-eps, this file is maintained from the Application Configuration window (accessed from **System Operations > Server Management > Application** on the ACI desktop user interface). Common properties defined in this file include connection IDs, database properties, queue locations, whether data sources or user activities are audited, whether data is accessed from cache memory, and whether tracing is enabled. Properties for application-specific functions are also set in the APPLCNFG.

User Interface

AJSF applications use graphical user interface or web browser screens or windows to manage and administer the AJSF environment. These user interface screens or windows are introduced below and described in detail in sections 3 and 4.

Java Server Management

Java server management consists of maintaining the configuration for Java server processes. This includes managing the configuration of application-level properties, connections to and from external entities, process relationships in clustered configurations for broadcasting messages, and event messages.

ACI provides formatted screens or windows to maintain this information. Depending on the product in which the AJSF is employed, these screens or windows can be browser- or Java Swing-based. Although the look and feel may differ from one product to the next, the fields displayed and the meta data maintained are the same across all products. The following screens or windows are provided to maintain Java server configuration information:

- The Alternate Data Source data source (DATASRC) maintains alternate data source information. Currently, there is no user interface window for the DATASRC. The fields in the DATASRC are described in section 3.
- The Application Configuration window or screen maintains application-level properties in the Application Configuration data source (APPLCNFG). The fields on this window/screen are described in section 3. Application-level properties common to the AJSF are described in section 5. Product-specific application-level properties are described in product-specific appendices to this user guide.
- The External Connection Configuration window or screen maintains data in the External Connections data source (EXTRCNCT). The fields on this window/screen are described in section 3.
- The Listener Configuration window or screen maintains data in the Listener data source (LISTENER). The fields on this window/screen are described in section 3.
- The Process Configuration window or screen maintains data in the Process data source (PROCESS). The fields on this window/screen are described in section 3.

- The Event Configuration and Management window maintains data in the Event (Event) and Event Document (Event_Document) data sources. The fields on this window/screen are described in application-specific documentation.

Java Server Administration

The AJSF allows you to view and clear objects cached in the Cache Manager within the executing Java Virtual Machine (JVM) process and stop and start TCP/IP sockets. In addition, you can manage properties, view database usage, view system resource usage, and view system properties used by a Java server process.

ACI provides formatted screens or windows to perform these functions. Depending on the product in which the AJSF is employed, these screens or windows may be browser- or Java-based. Although the look and feel may differ from one product to the next, the fields and data displayed and the functions you can perform are the same across all products. The following six screens or windows are provided to administer Java server processes:

- The Cache Administration window or screen displays cached objects and allows you to clear objects from cache.
- The Configuration Properties window or screen displays the current properties for a Java server process from the config.properties, APPLCNFG, and inherited from the JVM. From this window or screen, you can also reload application-level properties after making changes to the APPLCNFG.
- The Database Administration window or screen displays database clients and connections used by Java server processes.
- The Socket Administration window or screen lets you stop and start TCP/IP socket connections used in the JVM.
- The System Monitor window or screen lets you view the dynamic use of system resources by a Java server process.
- The System Properties window or screen lets you view the system-level properties used by a Java server process.

The fields on the above windows/screens are described in section 4.

2: Subsystems

This section describes each of the subsystems provided in the ACI Java Server Framework (AJSF).

Database Subsystem

The database subsystem of the AJSF includes several features to enhance ACI applications. Each of these features is discussed below.

Note: Although the ACI Payments Manager browser application is configured in the AJSF database, it does not currently use JDAL as a database driver or any of the AJSF database generators or services described in this guide. The ACI Payments Manager browser application is strictly a client web application that uses the ACI Payments Manager user interface server for data access.

Code Generator

A code generator allows developers to build a database access service to any relational database table through a metadata (.gen or *Gen.xml) text file. This increases the speed at which applications can be developed. The interface to these services is modeled after the EJB entity bean interface. The database services convert all SQL statements into compiled statement objects and stores them for future use to avoid the overhead of recompiles inherent in most Open DataBase Connectivity (ODBC) and Java DataBase Connectivity (JDBC) applications. This greatly increases the performance of an application accessing a relational database.

The table names used in the `db.audited.table-name` and `db.cached.table-name` properties must match the physical table names to which they apply.

Transaction Integrity

Database services encapsulate the functionality to provide transaction protection across the different transaction protection monitors, such as the Transaction Monitoring Facility (TMF) on the HP NonStop platform. The database services also provide extensive tracing of all database access and failure recovery.

Auditing User Actions

The database auditing mechanism enables you to track all adds (inserts), updates (before and after images), deletes, and views of the application database by user ID and time of change. When a database service is audited, records are written in XML format to a User Audit Log data source (USRAULOG). You can turn auditing on or off using the `db.audited` property. The values for this property are `true` (auditing is turned on) or `false` (auditing is turned off). When auditing is

turned on, you can configure the database services that are audited when your ACI system is installed and the extent to which database services are audited in db. `audited.table-name` properties, where *table-name* is the physical name of data source to be audited. The values for these properties are none, medium, and full. For full auditing of a database table, header information as well as detail information about the before and after images of the affected record are logged to the USRAULOG. For medium auditing, only the header information is logged to the USRAULOG. For the none value, no auditing records are generated for the specified database table.

The names for database tables generic to the AJSF are listed in section 5 of this guide. Application-specific database table names are listed in the application-specific appendices of this guide.

Note: For the AJMF, `ajmf.audited.function` properties determine whether attribute save functions, attribute view functions, MBean Server functions, or operations functions are audited for AJI-based ACIJMX agents. For detailed descriptions of these properties, refer to the *ACI Java Infrastructure ACI Application Management User Guide*.

Additional AJSF properties (i.e., `cleanup.userauditlog` properties and a `timertask.function<text-string>` property) enable you to configure a User Audit Log Cleanup component to remove old data from the USRAULOG after a specified retention period expires. Application-specific properties can configure the directory location of an XML file for extracts as well as timed automatic extract of the USRAULOG by the User Audit Log XML Extract component.

Auditing cleanup and reporting timers automatically account for daylight savings time (summer time) on the server where the start time properties are configured.

The following table lists all of the properties required to implement user auditing in the user interface server process (e.g., ESWEB, PMWEB, or CRUXWEB) or ACIJMX agent, the property value(s) needed, and where you can find detailed descriptions of the property.

User Auditing Property	Value (Where Documented?)
<code>ajmf.audited.function</code>	true (<i>ACI Java Infrastructure ACI Application Management User Guide</i>)

User Auditing Property	Value (Where Documented?)
cleanup.userauditlog.retention.days	The number of days to leave audit records in the USRAULOG before they are removed. (section 5)
cleanup.userauditlog.cleanup.starttime	The time of day, in hh:mm format, the User Audit Log Cleanup component starts removing old rows from the USRAULOG. (section 5)
cleanup.userauditlog.timeperiod.hours	The number of hours between runs if the User Audit Log Cleanup component is to remove old audit rows from the USRAULOG more frequently than every twenty four hours. (section 5)
cleanup.userauditlog.batch.size	The number of rows to delete from the USRAULOG in a single batch when the User Audit Log Cleanup component removes old rows. (section 5)
db.audited	true (section 5)
db.audited. <i>table-name</i>	All applicable database tables (section 5, appendix A)
initialization.function< <i>text-string</i> >	com.aciworldwide.application.formatter.jaxb.JAXBContextManager (sections 5 and 6)
jsf.jaxb.context.< <i>text-string</i> >	com.aciworldwide.application.audit.formatter.xml.auditservice1_0_0 (sections 5 and 6)

User Auditing Property	Value (Where Documented?)
timertask.function<text-string>	com.aciworldwide.common. timertask.UserAuditLogCleanup
userauditlog.detail.column.size (Optional)	The size of the audit detail field (column) beyond the 1935 character limit of the JDAL driver if you are using a relational database for the User Audit Log data source (USRAULOG). (section 5)
xmltranmap.filename<text-string>	{root}/config/transmaps/ UserAuditTranMap.xml (sections 5 and 6)

With the above properties configured, you can generate auditing reports on an ad hoc basis from the User Audit wizard on the ACI desktop user interface. For more information, refer to the *ACI Desktop User Interface Manual*.

If you want to generate user auditing reports on an automated basis, the additional properties listed in the following table must be configured for the user interface server process (e.g., ESWEB, PMWEB, or CRUXWEB) or ACIJMX agent. For each property, the table specifies the value needed and where you can find a detailed description of the property.



User Auditing Property	Value (Where Documented?)
report.userauditlog.loc	The name of the directory where the user audit extract file is created. To meet cross-industry security best practices, any sensitive data in a user audit should be protected from unauthorized access. Therefore, the location where user audit data is extracted should be restricted. (section 5)

User Auditing Property	Value (Where Documented?)
report.userauditlog.name	The base name of the user audit extract file when automatically generating a user audit extract. (section 5)
report.userauditlog.offset.days	The number of days to subtract from the system date when the report timer fires. (section 5)
report.userauditlog.report.starttime	The time of day, in hh:mm format, at which the automatic extract is started. (section 5)
timertask.function<text-string>	com.aciworldwide.common.timertask.UserAuditReport (section 5)

Auditing records are written to the USRAULOG in XML message format and can be extracted for detailed reporting or analysis. Each record (row) contains a timestamp, a context description, ticket information, activity type, and data source name. Refer to the *ACI Desktop User Interface Manual* for a sample of the XML-formatted records written to the USRAULOG for database activity auditing. Refer to the *ACI Java Infrastructure ACI Application Management User Guide* for a description of the auditing records logged for the AJMF (ACIJMX agent).

Context descriptions provide context for the audited activity. For example, a typical context description for an application might be “CardInsertRq” or “VisaViewRequest”. You can check the source code to find all possible context descriptions for a AJSF-based application. Currently, the AJSF does not provide a context description for entries it writes to the USRAULOG.

Activity types used by the AJSF environment include:

- Delete
- Insert
- Update
- View

Each AJSF-based application may define additional activity types. For example, the USEC subsystem for the ACI desktop user interface includes the following activity types:

- AuditExtractReport
- ChangePassword
- Delete
- ExpirePassword
- Insert
- Logoff
- LogoffSelected
- Logon
- SessionExpired
- SessionTimeout
- Update
- VerifyPassword
- View

Refer to application-specific documentation for the context descriptions and activity types supported for a specific AJSF-based application.

Caching

The database caching mechanism coordinates cache synchronization across all executing Java server processes. You can configure the database tables cached when your ACI system is installed for individual database tables. Database caching improves performance for database rows or records accessed frequently and updated infrequently.

Database caching uses the Cache Manager to store database objects (see the [“Cache Management Subsystem”](#) topic for more information). The Cache Manager provides synchronization across all executing Java applications.

Drivers and APIs

Additional aspects of the database subsystem are the ACI-authored type 2 and type 3 (for the IBM System z) JDBC drivers written within a JDAL layer. These two drivers are only used by BASE24-eps. The JDBC API is the industry standard for database-independent connectivity between the Java programming language and a wide range of databases. The JDBC API provides a call-level API for SQL-based database access. JDBC technology allows ACI to use the Java programming language to exploit “Write Once, Run Anywhere” capabilities for applications that require access to enterprise data. These ACI-written drivers, Java Database Access Layer (JDAL) (type 2) and Java Database Access Layer 2 (JDAL2) (type 3), allow ACI’s Java applications to access Enscribe, c-tree, and DB2 data sources as a configurable JDBC driver at deployment time. They also provide an open industry standard API to the BASE24-eps applications data sources. The JDAL driver supports more of the ANSI SQL language than the JDAL2 driver does, while the JDAL2 driver is lighter weight and provides better performance for direct keyed access.

Alternate Data Sources

Through the use of the Alternate Data Source data source (DATASRC), you can configure specific data sources to be maintained in data formats that differ from the data format of the main application database. When attempting to access a data source for a particular Java database service, the database system will access the alternate data source if one is configured. If an alternate data source is not configured, it accesses the data source from the main application database. For more information on the DATASRC, refer to [section 3](#).

Pooling and Locking

Other features of the database subsystem include: connection pooling, statement pooling, *optimistic locking* and *pessimistic locking*. See the [glossary](#) for descriptions of optimistic and pessimistic locking.

Connections to the database are obtained from a static connections pool. Static connections allow all the expense of connecting to the database to be incurred only during process startup.

Statement pooling (available from JDBC 3.0) caches SQL queries that have been previously optimized and run so that, should they be needed again, they do not have to go through optimization preprocessing again (avoiding optimization steps, such as checking syntax, validating addresses, and optimizing access paths and execution plans). Statement pooling can be a significant performance booster.

Statement pooling and connection pooling in JDBC 3.0 can cooperate to share statement pools, so that connections can use a cached statement from another connection, thus incurring statement preparation overheads only once on the first execution of some SQL by any connection.

Message Subsystem

The framework has the ability to message synchronously and asynchronously and includes callback functionality configured through a routing database to plug in message formatters at run time. This allows you to write your own message formatters and dynamically configure them for easy implementations of custom interfaces to other applications. ACI has written message formatters for several ACI products including the following:

- Various XML formats
- ISO 8583:1993
- ANSI X9.15
- NCR NDC+
- Diebold 911/912
- IFX
- Payer authentication formats for Visa and MasterCard

Message interfaces and their endpoints are configured through a database and provide access to send Transmission Control Protocol/Internet Protocol (TCP/IP), HTTP, and JMS messages with different message parameters such as two-byte headers and end-of-text (EOT) characters, timeout settings and retry counts.

The Java Message Service (JMS) defines the standard for reliable Enterprise Messaging. Enterprise messaging, often also referred to as Messaging Oriented Middleware (MOM), is universally recognized as an essential tool for building enterprise applications. By combining Java technology with enterprise messaging, the JMS API provides a powerful tool for solving enterprise computing problems.

JMS parameters allow Java JMS applications to communicate with non-Java applications through WebSphere MQ or NET24-XPNET. The JMS implementation also has the capability to enable the JMS queue listeners to run multithreaded on symmetric multiprocessing (SMP) platforms (e.g., Sun Solaris). The framework performs connection pooling to TCP/IP sockets and JMS providers to eliminate the overhead of the connection operations during the transaction path. A JMS API has been implemented under NET24-XPNET for sending asynchronous messages, allowing applications written on the AJSF to look like any XPNET satellite process. On platforms such as Windows NT, Unix, and IBM System z, the application can message asynchronously through JMS providers such as IBM's WebSphere MQ and Sonic MQ.

Clustering Subsystem

The framework has the ability to configure applications as multiple single-threaded applications on clustered servers such as the HP NonStop server, or as one or a few multithreaded applications on SMP servers such as Windows NT, Unix, and IBM System z. The framework contains a database to define application processes and the mechanism for communicating among the processes using the message services defined above. Parts of the application that utilize the clustering services are the Database, Cache Management, and Configuration and Instrumentation subsystems.

For more information on configuring clusters of AII-based applications, refer to the [“Clustered Configuration”](#) topic in section 4.

Encryption Subsystem

The framework performs a number of cryptographic functions such as DES encryption/decryption. The encryption subsystem adheres to the Java Cryptographic Extensions (JCE) API. Currently, it is used with Thales e-Security, SafeNet (Eracom), and nCipher payShield hardware security modules in some ACI applications to verify and generate digital signatures, generate MAC keys, generate RSA keys, generate random numbers, verify and generate message digests, perform certificate chain validation, verify root certificates, verify self signed certificates, and generate and parse digital envelopes. The encryption subsystem can support both software and hardware encryption interfaces that adhere to the JCE.

Note: BASE24-eps ATM (Java) Device Handler processes do not use the JCE. Instead, they interface to the Transaction Security Services process through JMS to perform cryptographic functions. In this case, none of the *crypto* properties described in [section 4](#) need to be configured.

Logging Subsystem

You can configure the logging subsystem to log event messages generated by Java server processes to a variety of locations. The framework uses a third-party open source application called log4j. Log4j is an open source project, of the Apache group, which is based on the work of multiple authors. It allows the developer to control which log statements are output with arbitrary granularity and is fully configurable at runtime using external configuration files. You can configure log4j to send messages to the following destinations in the EventConfig.xml file:

- JVM console
- Daily rolling log files
- A listening JVM over a socket
- Windows NT Event Log
- Unix Syslog daemon
- JMS topic

ACI has also written appenders for the HP NonStop Event Management Subsystem (EMS), an SQL database, and WebSphere MQ for integrating with other ACI Payments Framework products and technologies. ACI also provides appenders for the Event Adapter process on an IBM System z or Unix platform to consolidate and format events generated from all system processes for viewing from the ACI desktop, system console, dated text files, or from Tivoli Enterprise Console (TEC) and/or Tivoli Netcool/OMNIBus product (IBM System z platform only).

These appenders and the Event Adapter process are discussed in sections 4 and 7 of this guide. The text of event messages can be configured to be extracted from a database or a properties file for internationalization. The event database also contains documentation about the probable cause, effect, and remedy for each event message. Refer to the following website for more information on log4j.

<http://logging.apache.org/log4j/docs/documentation.html>

Tracing Subsystem

The framework event environment provides extensive tracing of all messaging, database access, and cache management. The tracing subsystem can be configured to log trace messages to the same locations as the event logging subsystem because the tracing subsystem also uses log4j. The tracing subsystem is configured in the TraceConfig.xml file.

Message Transformation Subsystem

The message transformation subsystem is a set of base classes for building and parsing external messages. Currently, ISO 8583:1993, ANSI X9.15, IFX, and XML components exist for building messages. The framework also contains a tag-based extensible Data Object for sending messages between Java applications. This object can be serialized and deserialized using Java-specific transport mechanisms such as JMS and RMI.

The framework XML message subsystem is a part of the Message Transformation subsystem. The Java API for XML binding (JAXB) has been used for AJSF applications, and is preferred, for building XML formatters quickly that perform well. The framework XML message subsystem also automatically pools XML parsers for performance reasons.

Directory Subsystem

The directory services subsystem contains an interface to naming and directory services such as Lightweight Directory Access Protocol (LDAP), Network Information Service (NIS), Common Object Request Broker Architecture (CORBA) Common Object Services (COS) Naming, and files.

Cache Management Subsystem

Because of the need to build extended memory tables for database information that changes infrequently, the framework contains a Cache Manager component that can maintain cache information by Service type. Services make requests to place information in memory and retrieve information from memory. ACI application-specific user interfaces allow you to peruse the information in cache by service and object key. Cache management is used mostly by the database tables and the message services for maintaining message context. The Cache Manager has timers that you can configure using properties to automatically clean up cache pools on a regular basis such as nightly or even hourly. You can also manually clear cache through the application-specific user interface.

Configuration and Instrumentation Subsystem

The framework's configuration options are supplied in a Java properties file, XML documents, or a data source at runtime. Bootstrap properties such as the database location and logging subsystem configuration locations are defined in the config.properties file. ACI recommends that the rest of the application properties be configured in the Application Configuration data source (APPLCNFG). You can modify the APPLCNFG and refresh the in-memory copy of the APPLCNFG through the supplied user interface while the Java server application is running. Thus, you can keep the application running while reconfiguring it. You can also display all of the runtime properties (bootstrap, system, and application) for a Java server process through the user interface. Refer to section 4 of this guide for descriptions of these procedures.

XML documents control the logging of event and trace messages as well as the masking of sensitive cardholder data displayed on AJSF-based application windows or pages. In addition, some Java server processes use additional XML files for the scripting of work flow.

The Java Server Framework uses a number of data sources other than properties files and XML documents to configure how Java server processes connect to external entities and to format and describe event messages generated by AJSF applications.

The user interface for the AJSF also allows you to display and monitor database connections, socket connections, cache information and threads in use. Refer to [section 4](#) for more information on these user interface windows or screens.

Properties Files

The Java Server Framework uses two files to configure properties. The config.properties file configures properties used to bootstrap a Java server process. This properties file must be specified in an argument in the Java Virtual Machine (JVM) command used to start a Java server process. For more information on configuration properties, refer to the [“Configuration Properties”](#) topic provided in section 1. The Application Configuration data source (APPLCNFG) defines properties that configure how a Java server performs processing. The APPLCNFG is accessed through JDBC. For more information on application-level properties, refer to the [“Application Configuration Properties”](#) topic provided in section 1. If the same property is configured in both the config.properties file and the APPLCNFG, the value in the config.properties overrides the value in the

APPLCNFG, except for the trace.enabled property. For this property, the value in the APPLCNFG overrides the value in the config.properties file once the Java server process reads the APPLCNFG.

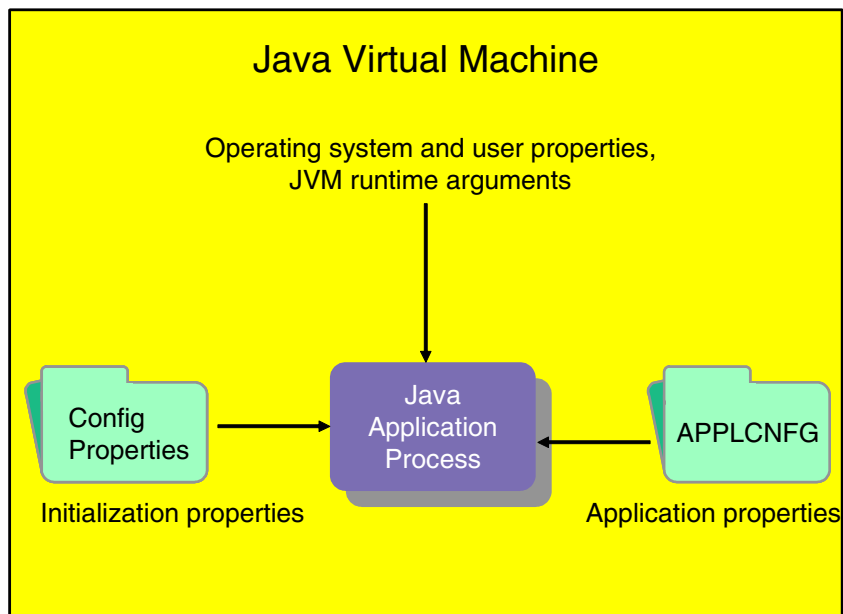
Note: ACI recommends that all configuration properties be defined in the APPLCNFG. Only those properties required to bootstrap the Java process need to be configured in the config.properties file.

During initialization, a Java server process loads properties from the config.properties file into an internal table. The process always attempts to find a property in this internal table before attempting to retrieve the property from the APPLCNFG. Therefore, if a property is configured in the config.properties file, the process does not even attempt to retrieve it from the APPLCNFG. The only exception is the trace.enabled property. The process uses the trace.enabled property from the config.properties file until database connections have been established. It then reads the APPLCNFG for the trace.enabled property. This allows for tracing of the logic up until the database connections have been established.

Note: Some AJSF applications can use additional properties files. For example, BASE24-eps ATM (Java) Device Handler processes use device type-specific properties files to define text for display on receipts and other configuration attributes for ATM terminals.

The Java Server Framework also uses properties defined at the system level. These properties are inherited from the operating system, set by the user logged on to the JVM, or set in JVM runtime arguments when a Java server process is started.

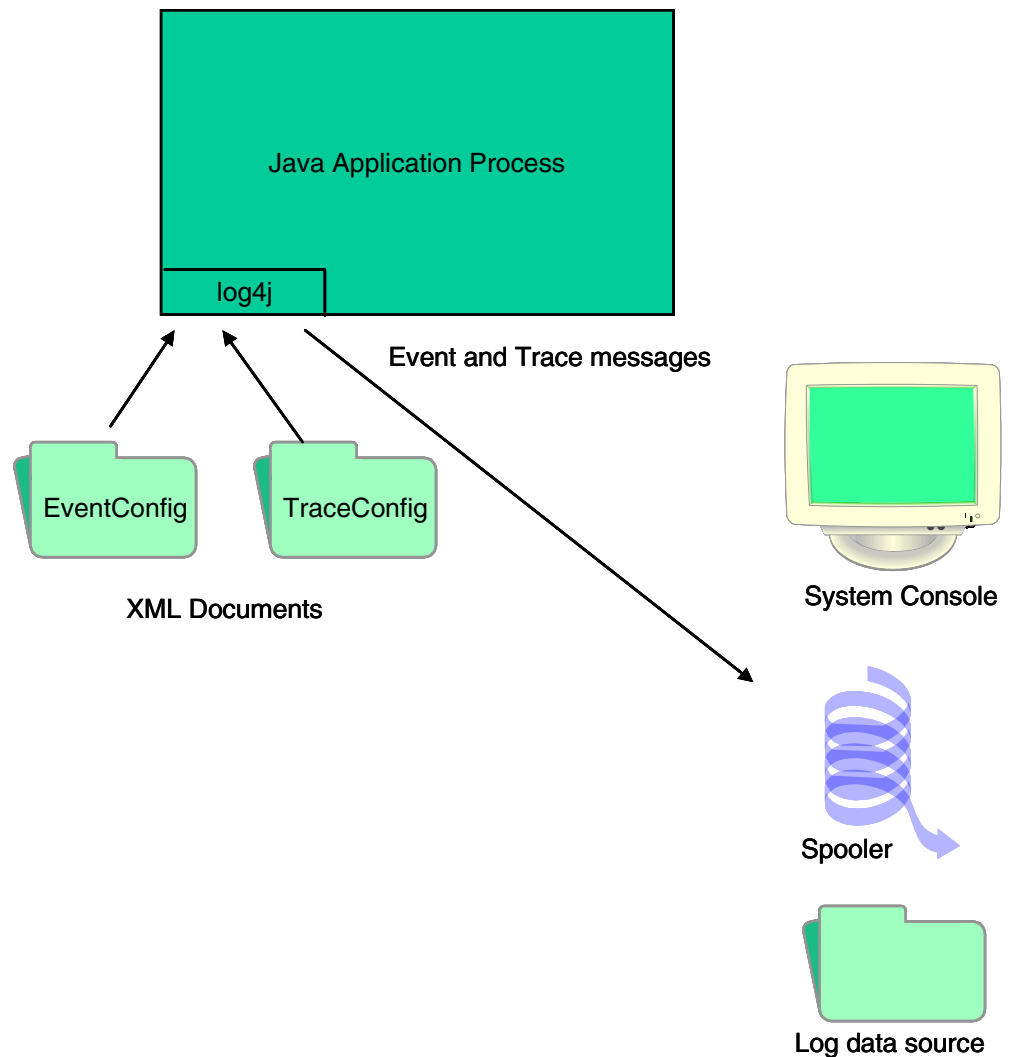
The relationships between properties and a Java server process are depicted in the following illustration.



XML Configuration Documents

All Java servers in the AJSF use two XML document files to configure how event and trace messages are generated by log4j. The EventConfig.xml file configures log4j for generating event messages. The TraceConfig.xml file configures log4j for generating trace messages.

Use of the EventConfig.xml and TraceConfig.xml files by log4j in a Java server process is depicted in the following illustration.



AJSF-based user interface window or browser page applications can use the FieldMasking.xml file to mask the display of sensitive cardholder data displayed on a user interface window or browser page.

Some Java server processes may use additional XML document files for the scripting of work flow. These XML document files are either described elsewhere in this guide or in an application-specific user guide or manual.

The EventConfig.xml, TraceConfig.xml, and FieldMasking.xml files are described in detail in section 4. Properties that specify the locations of these files are described in section 5.

Data Sources

The AJSF uses several data sources other than properties files and XML document files to configure how Java server processes connect to external entities and to configure and describe event messages that can be generated by Java server processes. Each of these data sources is introduced below and described in more detail in section 3.

Alternate Data Source

The Alternate Data Source data source (DATASRC) enables you to maintain certain data sources in a data format that differs from the main application database. This allows you to mix and match data formats as needed for your application. The DATASRC contains all the information needed to access an alternate data source, including the relative URL location, database driver, user ID and password, table prefix, and database connection parameters. The DATASRC does not currently have a user interface window. For detailed information on the [DATASRC](#), refer to section 3.

Event

Event messages generated by Java server processes are configured in the Event data source (Event). The Event also configures whether events are generated or suppressed. The Event is maintained from the Event Configuration and Management window on the ACI desktop user interface (accessed from **System Operations > Event > Event Management**).

Note: AJSF-based applications do not use tokens in event messages configured on the Event Configuration and Management window. Therefore, you cannot use token IDs (e.g., <1>, <2>, etc.) for text substitution in event messages generated by AJSF-based applications. AJSF-based applications use braces ({}) instead of tokens to denote text substitution in event messages. Brace-based text substitution is hard-coded in AJSF-based application code.

Event Documentation

The probable cause, effect, and remedy for an event are stored in the Event Documentation data source (Event_Document). Token descriptions for an event message are also stored in the Event_Document. The Event_Document is

maintained from the Event Configuration and Management window on the ACI desktop user interface (accessed from **System Operations > Event > Event Management**).

External Connection

The External Connections data source (EXTRCNCT) is used to configure connections from a Java server process to external entities. The Java server process uses the EXTRCNCT when it requests the external entity to perform some unit of work on its behalf. The protocol used to connect to the external entity is also configured in this data source. The EXTRCNCT is maintained from the External Connection Configuration window on the ACI desktop user interface (accessed from **System Operations > Server Management > External Connection**). For detailed information on this window, refer to section 3.

Hardware Security Module (HSM) Configuration

The Hardware Security Module (HSM) Configuration data source (HSMCNFG) is used to configure a physical connection to an Atalla AKB NSP or a Thales e-Security HSM for a Java server process. The Java server process uses the HSMCNFG when it needs a hardware security device to perform a cryptographic function. The HSMCNFG does not currently have a user interface window. For detailed information on the [HSMCNFG](#), refer to section 3.

Listener

The Listener data source (LISTENER) is used to configure connections from external entities to a Java server process. The Java server process uses the LISTENER when it needs to receive a unit of work, and it cannot be received through the standard Servlet mechanism. The protocol used by the external entity to connect to the Java server process is also configured in this data source. The LISTENER is maintained from the Listener Configuration window on the ACI desktop user interface (accessed from **System Operations > Server Management > Listener**). For detailed information on this window, refer to section 3.

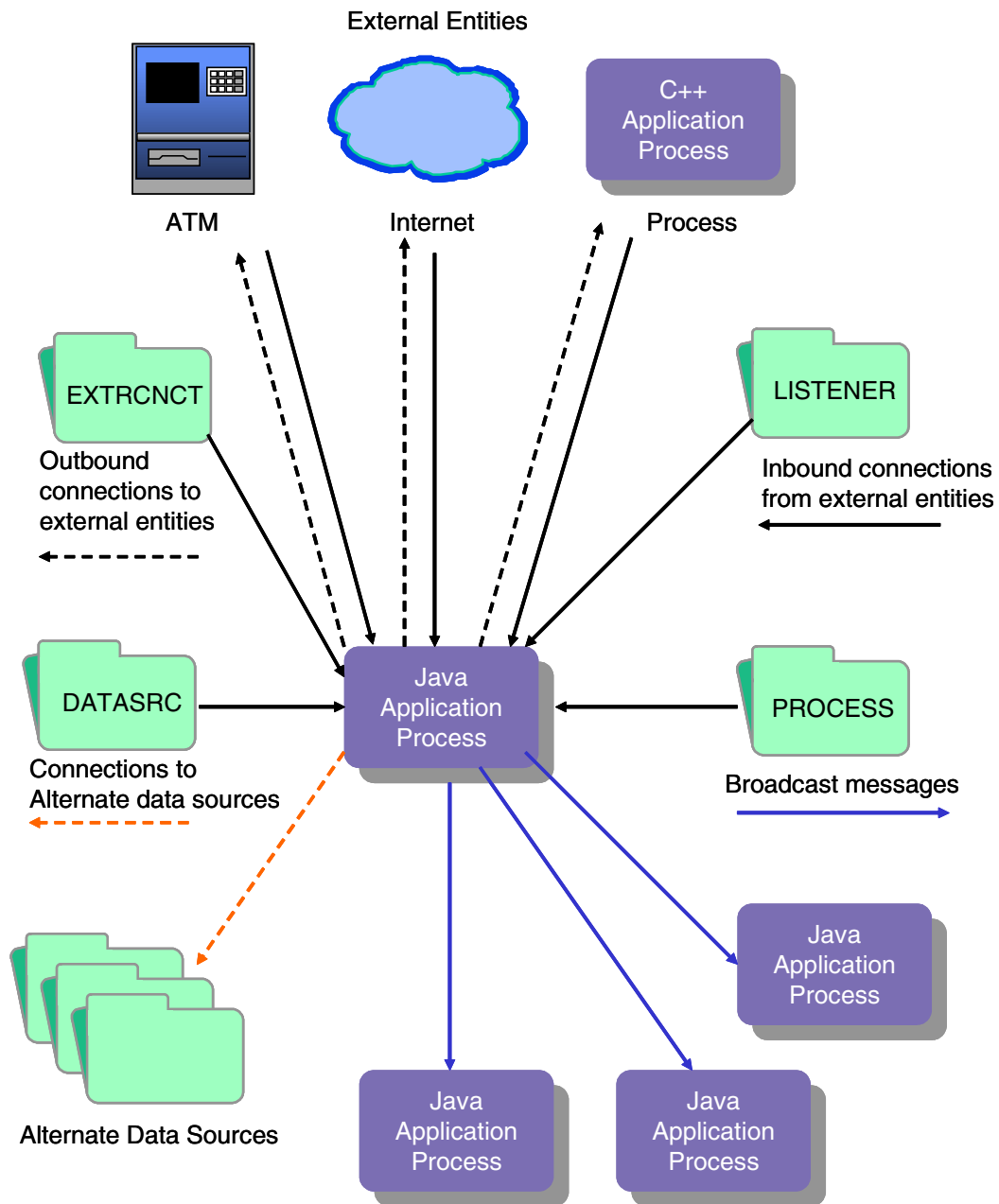
Process

The Process data source (PROCESS) is used to configure references to process entities running in the AJSF. A Java server process uses the PROCESS when it needs to broadcast messages to other processes running under the JVM, whether

on a clustered or SMP platform. In addition, some products (e.g., APF Admin, BASE24-eps) require an entry in the PROCESS for every process running in the JVM because drop down lists in the ACI desktop user interface are populated from it.

All processes configured in the PROCESS must be configured with a system property named “process.id”. This property is set in a JVM argument when the process is started (e.g., -Dprocess.id=P1A^JDH1). The PROCESS is maintained from the Process Configuration window on the ACI desktop user interface (accessed from **System Operations > Server Management > Process**). For detailed information on this window, refer to section 3.

The following graphic illustrates how the DATASRC, EXTRCNCT, LISTENER, and PROCESS are used in the AJSF.



USEC Subsystem

The User Security (USEC) application provides the authentication and authorization functions for the clients—both browser and ACI desktop user interface—in the ACI Payments Framework. USEC features a role based environment where users are granted permission to access specified application resources such as application screens. One feature of USEC that differentiates it from other user security applications is its ability to configure application data filtering so that user access rights can be granted based upon dynamic data. This feature is commonly used to create an easy to manage system where multiple institutions' data can be stored in one system data source, but users are only able to see data that they have permission to view. The USEC data structure lets a security administrator easily configure security groups to define access time frames and password groups that identify the characteristics of a user's password and how often it needs to be changed. The USEC subsystem also supports product integration with other third-party identity management systems such as IBM's Tivoli Access Manager.

Refer to the *ACI Desktop User Interface Manual* for details on the features of USEC, its data design, and the procedures for adding users.

Shared USEC Subsystem Message Context

You can share USEC subsystem message context among ACI Payment Framework (APF) products that use the ACI desktop or a browser user interface for access to multiple AJSF-based products. Different user interface web server processes (e.g., ESWEB, PMWEB, or CRUXWEB) provide ACI desktop or browser user interface access for different products and need to share message context when combined in a single system.

User Permissions and User Status Reporting

The USEC subsystem enables you to list summary user status information and provides hierarchical information for users, roles, permission groups, permissions, tasks, and field filters. You can generate this information on an ad hoc basis from the User Status and User Permissions windows, respectively, on the ACI desktop user interface. Refer to the *ACI Desktop User Interface Manual* for details on using these windows and for examples of the reports generated.

The following table lists all of the properties required to implement ad hoc user status and user permission reporting in the user interface server process (e.g., ESWEB, PMWEB, or CRUXWEB), the property value(s) needed, and where you can find detailed descriptions of the property.

User Auditing Property	Value (Where Documented?)
jsf.jaxb.context.<text-string>	com.aciworldwide.application. security.reporting.formatter.jaxb. userstatus com.aciworldwide.application. security.formatter.jaxb.usec1_0_0 (section 5)
timertask.function<text-string>	com.aciworldwide.application. security.timertask.UserDetailReport com.aciworldwide.application. security.timertask. UserPermissionsReport (section 5 and appendix A)
xmltranmap.filename<text-string>	xmltranmap.filename.3={root}/ config/transmaps/ UserStatusTranMap.xml xmltranmap.filename.2={root}/ config/transmaps/ UserSecurityTranMap.xml (section 5)

User security reporting timers automatically account for daylight savings time (summer time) on the server where the start time properties are configured.

You can also automatically generate reports for this information by configuring the following properties for the user interface server process (e.g., ESWEB, PMWEB, or CRUXWEB). The following table lists all of the properties required to

implement automatic user status and user permission reporting in the user interface server process (e.g., ESWEB, PMWEB, or CRUXWEB), the property value(s) needed, and where you can find detailed descriptions of the property.

User Auditing Property	Value (Where Documented?)
report.userdetail.loc	The name of the directory where the User Detail Report is created. (section 5)
report.userdetail.name	The base name to use for the file when automatically generating a User Detail Report. (section 5)
report.userdetail.starttime	The time of day, in hh:mm format, at which the User Detail Report is generated. (section 5)
report.userdetail.timeperiod.hours	The number of hours between generation of the User Detail Report if it is to be generated more frequently than every twenty four hours. (section 5)
report.userpermissions.loc	The name of the directory where the User Permissions Report is created. (section 5)
report.userpermissions.name	The base name to use for the file when automatically generating a User Permissions Report. (section 5)
report.userpermissions.starttime	The time of day, in hh:mm format, at which the User Permissions Report is generated. (section 5)

User Auditing Property	Value (Where Documented?)
report.userpermissions.timeperiod.hours	The number of hours between generation of the User Permissions Report if it is to be generated more frequently than every twenty four hours. (section 5)

Third-party JAAS Authentication

The USEC subsystem uses the Java Authentication and Authorization Service (JAAS). This architecture allows the USEC subsystem to plug into the authentication services of a enterprise access management system, such as the following:

- USEC 1.x
- Microsoft Active Directory
- Tivoli Access Manager for e-business
- Lightweight Directory Access Protocol (LDAP)

The USEC subsystem supports *pluggable* authenticators for each of the above enterprise authenticators. These authenticators are considered to be *pluggable* because they are interchangeable with other third parties. Thus, you can leverage a pluggable authenticator to grant “APF-related access” to your existing user base. This provides you with the flexibility of not having to re-add (duplicate) user IDs and manage them within multiple environments.

The USEC subsystem’s primary means of integrating with the enterprise environment is through the use of a pluggable authentication module. This industry standard means of coupling the enterprise's existing user definitions with product-specific permissions provides many benefits. The greatest of which is that users only need to remember one user ID and password.

You can configure your existing user base to be members of APF-related groups. These group names are known as the APF-configured roles, filters, and security groups. Any user requiring access to specific APF product resources must be assigned the specific product-related role, and in some cases, filter as well. APF roles, filters, and security groups are maintained within the USEC database.

You can write custom authenticators by extending the LoginModuleBase super class, which supports the following classes.

Class	Authenticator
JAASAuthenticator	USEC 1..x
ActiveDirectoryAuthenticator	Microsoft Active Directory
TivoliAuthenticator ¹	Tivoli Access Manager
LDAPAuthenticator	LDAP

¹ This authenticator is dependent upon the Tivoli “PD.jar” being present in the class path or web application deployment. This authenticator uses IBM’s pd.jar “PD API” to authenticator users and acquire user group names.

Each authenticator implements the same JAAS interface for performing authentication of a user (i.e., a user logon from the ACI desktop user interface or an ACI APF browser). Currently, these authenticators are used to perform the following authentication functions:

- Authenticate a user ID and password (encrypted/hashed)
- Perform any supported user validation
- Determine the groups with which a user is associated.

A group can be a role, filter, or security group. From the third-party authenticator’s perspective, these group names must be configured externally and associated to the existing third-party user base. This allows APF products to utilize the USEC authorization database and associate all externally defined users to specific ACI product-based roles and security constraints to control sessions within the APF. Likewise, filter groups are used to link external users to filterable data configured within the APF product base. Thus, you can limit external user accesss to specific data (e.g., specific institutions, business units, terminals, etc.).

USEC 1.x

The USEC 1.x is the default authenticator used when you define users within the USEC database. The USEC 1.x authenticator authenticates the user ID and password and derives the user’s roles and filters. It differs slightly from pluggable external authenticators because it performs specific validation relating to a user’s password. With the USEC 1.x authenticator, you can configure and manage

password-related validations such as the number of password failures allowed, how long passwords remain in effect before they must be changed, and how passwords are defined (i.e., minimum and maximum length, number of alpha characters required, number of numeric characters required).

Microsoft Active Directory

The Microsoft Active Directory authenticates the user ID and password against an Active Directory LDAP. User ID and password management is controlled outside the scope of the APF product base. You must configure groups within the Active Directory LDAP and each user requiring APF access as a member of the groups defined. Groups are USEC roles, filters, and security groups defined in the USEC database. You must prepend a group prefix to each of these groups defined in the Active Directory LDAP. These prefixes are defined by the following properties in the `ACTIVEDIRECTORY_jaas.config` file:

- `groupSuffixRole`
- `groupSuffixSecGrp`
- `groupSuffixFilter`

Tivoli Access Manager (TAM)

The Tivoli Access Manager JAAS authenticator (TAM) is used to authenticate users within an IBM Tivoli Directory Server (ITDS). User ID and password management is controlled within the TAM environment. You must configure groups within the ITDS and each user requiring APF access as a member of the groups defined. Groups are USEC roles, filters, and security groups defined in the USEC database. You must prepend a group prefix to each of these groups defined in the ITDS. These prefixes are defined by the following properties in the `Tivoli_jaas.config` file:

- `groupSuffixRole`
- `groupSuffixSecGrp`
- `groupSuffixFilter`

Tivoli Java SvrSslCfg Utility. The `com.tivoli.pd.jcfg.SvrSslCfg` class from the `PD.jar` is a required utility used to configure access from a Java application server (e.g., ESWEB, PMWEB, CRUXWEB, etc.) to a TAM environment using SSL. For more information on the `SvrSslCfg` utility, refer to the following web site:

http://publib.boulder.ibm.com/infocenter/tivihelp/v2r1/index.jsp?topic=/com.ibm.itame3.doc_5.1/am51_authJ_devref62.htm. The SvrSslCfg utility requires the following Jars:

- PD.jar
- ibmjsse.jar
- ibmjcefw.jar
- ibmjceprovider.jar
- local_policy.jar
- US_export_policy.jar
- ibmpkcs.jar
- jaas.jar

LDAP

The LDAP access authenticator is a generic authenticator that you can use with any LDAP (e.g., Open LDAP). This authenticator works much like the Microsoft Active Directory authenticator. The only difference is how it is configured and managed. User ID and password management is also controlled outside the scope of the APF product base and users must be assigned groups to acquire access to APF product resources. You must configure groups within the LDAP and each user requiring APF access as a member of the groups defined. Groups are USEC roles, filters, and security groups defined in the USEC database. You must prepend a group prefix to each of these groups defined in the LDAP. These prefixes are defined by the following properties in the LDAP_jaas.config file:

- groupSuffixRole
- groupSuffixSecGrp
- groupSuffixFilter

Group Prefixes

Although you have to prepend prefixes to USEC roles, filters, and security groups in third-party directories, you do not configure these prefixes as part of the groups in the USEC database.

Encryption in USEC

USEC employs encryption for the following items:

- User passwords stored in the USEC database are encrypted and stored using a one way SHA-1 hash, SHA-256 hash, or *salted* SHA-256 hash (SSHA-256).
- SSL can be configured for message encryption between USEC and other applications. This is performed using Java's Secure Socket extensions. For general information on setting up SSL, refer to the [“Security Considerations”](#) topic in section 4. For specific information on setting up SSL between the AJSF-based application server and Tivoli, refer to the [“Running the SvrSslCfg Utility”](#) topic in this section.
- Configuration files that hold database connection passwords can be encrypted using a proprietary algorithm by the [ConfigPropertyBuilder password encryption utility](#).

ACI Desktop User Interface and Browser Impacts

When you install and configure a JAAS authenticator other than USEC 1.x, ACI desktop user interface and browser functions are impacted as follows because they are essentially taken over by the third-party enterprise access management system:

- The Change Password window is disabled. A user cannot change their password on the ACI desktop user interface or browser.
- The following user security windows are not displayed, even if a user's permissions dictate otherwise:
 - ❑ User Security Profile
 - ❑ User Status

Because you can configure security groups regardless of the third-party JAAS authenticator you use, security constraints related to session management can still be adhered to. Even though a password group can be assigned to a security group containing valid user constraints, the constraints related to the password group are ignored when a third-party JAAS authenticator other than USEC 1.x is configured for the Java application server process.

Depending on the AJSF product, the user interface may incur additional impacts when using a JAAS authenticator other than USEC 1.x. For example, when using a JAAS authenticator other than USEC 1.x in BASE24-eps, the User ID field on the User Profile Relationship window does not display a selectable list of configured users. Instead, you must enter a valid user ID value in this field.

USEC Database Impacts

When using third-party JAAS authenticator other than USEC 1.x, the authentication tables in the USEC database are not accessed. This includes tables such as the following:

- Users (Users)
- User Status (UserStat)
- Password (Password)
- Password History (PswdHist)
- Password Group (PswdGrp)

Session management and session-related constraints are still supported regardless of the third-party authenticator used. The USEC 1.x authenticator continues to manage sessions based on security group assignment. However, password group constraints are not applied when using an external third-party authenticator.

The following is a list of security group constraints that are still enforced, regardless of the third-party authenticator used:

- Allowed start and end times
- Allowed days of week
- Bad logins allowed (will be temporarily disabled)
- Bad logins reset time (see above)
- Inactive days allowed
- Maximum sessions allowed
- Session timeout
- UI timeout

Third-Party Product Impacts

Minimal configuration is required within your third-party Active Directory, Tivoli, or LDAP to support pluggable USEC authentication.

Any user that requires access to the APF product base, must be added to or assigned to a *group*. For each role, security group, and filter defined in the USEC authorization database, a Tivoli, Active Directory, or LDAP group equivalent must be added to the third party. These groups can be assigned to your user base as required.

You can assign users to the following three types of groups:

- Roles are used to link a user to a set of permissions.
- Security groups are used to link a user to set of security constraints, mostly related to session control.
- Filters are used to link a user that has access to specific data values (e.g., institutions, business units, etc.).

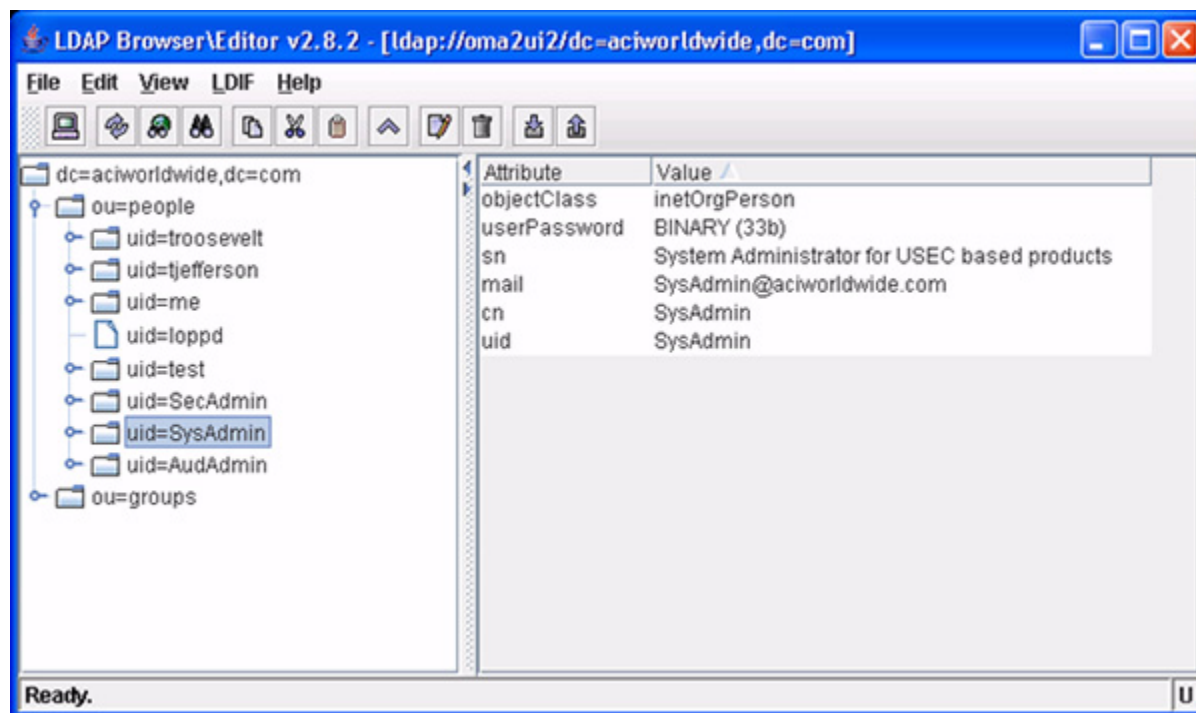
When adding group names, you must follow a naming convention of prefixing the actual group name with the following:

- For roles, the default convention is: `USEC.Role.roleID`, where *roleID* is the actual role defined in the USEC database.
- For security groups, the default convention is: `USEC.Group.securityGroupID`, where *securityGroupID* is the actual security group defined in the USEC database.
- For filters, the default convention is: `USEC.Filter.filterID`, where *filterID* is the actual filter defined in the USEC database.

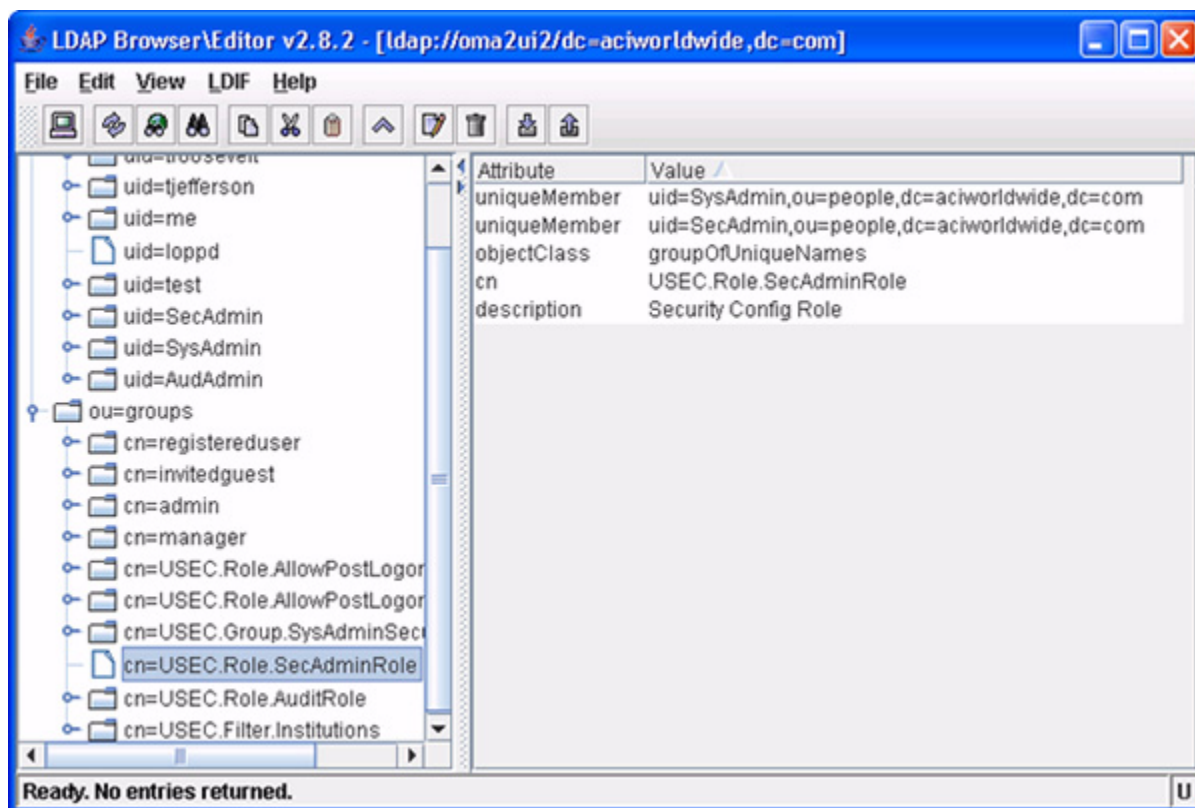
Note: The above prefix naming convention is not required or used for the standard USEC 1.x authenticator. Group prefix naming conventions are only needed for external authenticators such as Tivoli, ActiveDirectory, LDAP, or other custom authenticators.

Open LDAP Directory Example

The following example screen shots depict of a user and group in an open LDAP directory. The first screen shot depicts the attributes for the SysAdmin user.

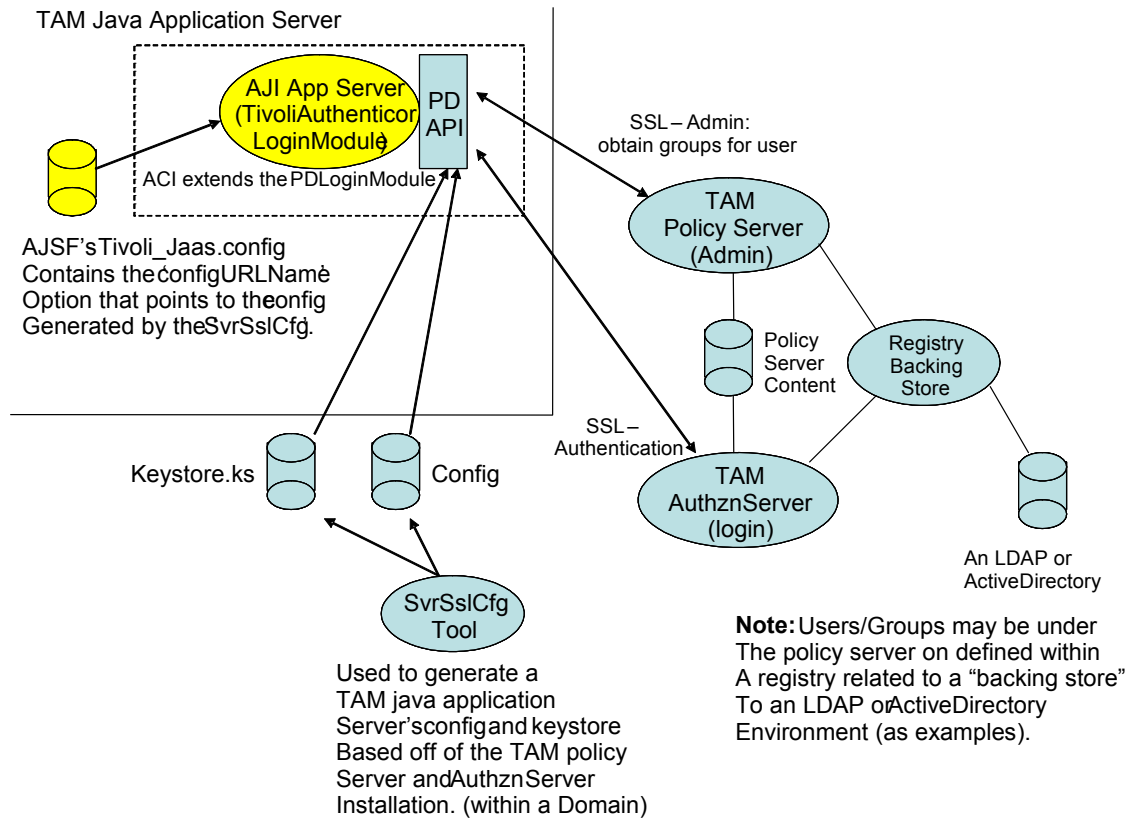


The next screen shot shows you the attributes for the SecAdminRole group, which also includes users assigned to the uniqueMember group.



Tivoli Access Manager Authenticator Configuration

The following illustration depicts the configuration and procedures for using the Tivoli Access Manager JAAS authenticator (TAM).



To implement the Tivoli Access Manager JAAS authenticator (TAM), perform the following:

1. Add the following Tivoli provided jars to the classpath for the Java application server process (e.g., ESWEB, PMWEB, CRUXWEB, etc.) that authenticates using the TivoliAuthenticator. These same jars are required by the SvrSslCfg utility.
 - PD.jar
 - ibmjsse.jar
 - ibmjcefw.jar
 - ibmjceprovider.jar
 - Local_policy.jar

- US_export_policy.jar
 - IbmPKCS.jar
 - Jaas.jar
2. Ensure that you have run the the SvrSslCfg Java Tivoli utility and generated the required Tivoli configuration and keystore files to be used by the PD API. You must configure the configURLName property in the AJSF Tivoli JAAS configuration file (Tivoli_jaas.config) to point to the output configuration file produced by this utility. To run the utility, the PD.jar must be in the utility's classpath. See the [“Running the SvrSslCfg Utility”](#) topic below for details on running the SvrSslCfg utility.

3. Define the AJSF system administrator user into the Tivoli environment.

Use the following information to load an initial system administrator user in the Tivoli environment:

Note: The group name prefix “USEC.Role.” must match the prefix defined in the groupSuffixRole property in the Tivoli_jaas.config file.

UserId:	SysAdmin
Password:	newuser01
Group Names:	USEC.Role.SysAdminRole USEC.Role.SecAdminRole USEC.Role.AuditRole USEC.Role.AllowPostLogonDebugRole USEC.Role.AllowPostLogonTraceRole

4. Create the Tivoli_jaas.config file (see the [“Tivoli_jaas.config”](#) topic for details). The configURLName property in this file must point to the Tivoli configuration file generated by the SvrSslCfg utility. You must specify the name of the Tivoli_jaas.config file in the java.security.auth.login.config system property when starting up the ACI Java user interface or browser process (e.g., ESWEB, PMWEB, CRUXWEB).
5. Create the [Tivoli_java.policy](#) file. Make sure the policy grants full access to the jaas.jar and pd.jar. You must specify the name of the Tivoli_java.policy file in the java.security.auth.policy system property when starting up the ACI Java user interface or browser process (e.g., ESWEB, PMWEB, CRUXWEB).

-Djava.security.auth.login.config=<directory-path>\Tivoli_jaas.config

-Djava.security.auth.policy=<directory-path>\Tivoli_java.policy

6. Add all of the preloaded USEC roles, filters, security groups for the SysAdmin user for your AJSF-based product into the Tivoli administrative environment. Once you have them all defined within the Tivoli directory, create a Tivoli-based user and password and assign the necessary groups (i.e., roles, filters, and security group) to this user.

Note: Make sure the group names are added with the corresponding suffix specified in your Tivoli JAAS configuration file:

- USEC.Role.<*role name(s)*> (Could be one or more).
 - USEC.Filter.* (May just be one initially, using an asterisk “*” as a wildcard for all filters).
 - USEC.SecGrp.<*security-group*> (Should just be one to define the user’s session constraints).
7. Make sure the TAM directory servers are active and start your AJSF-based web application server. You should be able to connect and log in to the APF product with a valid Tivoli user ID. If after logging in, the ACI desktop or browser displays access to product specific applications and possibly security administration, the Tivoli user ID has been successfully authenticated. The AJSF-based web application server has acquired the necessary group names during login to authorize the user’s access to APF product.

Running the SvrSslCfg Utility

You must provide the following parameter values when running the SvrSslCfg Tivoli java utility (located within the Tivoli provided PD.jar).

```
java.com.tivoli.pd.jcfg.SvrSslCfg -action value
                                   -cfg_action value
                                   .
                                   .
                                   .
                                   -key_file value
```

Parameter	Value
-action	config
-cfg_action	create
-admin_id	<i>admin_user_ID</i> (Tivoli directory’s Admin user)
-admin_pwd	<i>admin_password</i> (Admin user’s credential)

Parameter	Value
-appsvr_id	<i>application_server_name</i> (AJSF Java application server ID)
-appsvr_pwd	<i>application_server_password</i> (AJSF Java application server password) Note: This parameter is optional. If it is specified, the password must meet the current password rules in effect. If it is omitted, a default password is automatically generated.
-mode	{local remote} Indicates whether the Java application server processes requests remotely or locally.
-host	<i>host_name_of_application_server</i> (AJSF installed machine) Note: This parameter is optional. The default value is the local host.
-policysvr	<i>policy_server_name:port:rank</i> [,...]
-authzsvr	<i>authorization_server_name:port:rank</i> [,...]
-domain	Tivoli_Access_Manager_domain (if any)
-cfg_file	<i>fully_qualified_name_of_configuration_file</i> (output)
-key_file	<i>fully_qualified_name_of_keystore_file</i> (output)

For some of the required parameters related to the policy and authorization servers, you must obtain the following information from the Tivoli directory environment:

- A Tivoli administrative user ID and password.
- The Tivoli Directory domain for the policy and authorization servers.
- The qualified server name with port (and optional rank) of the Tivoli Policy server.
- The qualified server name with port (and optional rank) of the Tivoli Authorization server.

The SvrSslCfg utility generates two output files for which you must specify the file names:

- The configuration file, which is used and referenced in the Tivoli_jaas.config file from the AJSF environment.
- The keystore file, which is used and referenced from the configuration file.

The PD.jar contains the API that accesses all of the generated configuration in these output files. The API uses this output to build secure connections between the AJSF-based application server and the TAM environment (Policy and Authorization servers).

For detailed information on running the SvrSslCfg utility, refer to the following Tivoli Access Manager documentation:

http://publib.boulder.ibm.com/infocenter/tivihelp/v2r1/index.jsp?topic=/com.ibm.itame3.doc_5.1/am51_authJ_devref62.htm

Windows Batch Command for SrvSslCfg Utility

The following is a sample Windows batch command file for executing the SrvSslCfg utility.

```
set OUTPUTPATH=C:\abc\Tivoli\SSLConfig
set WASHOME="C:\Program Files\IBM\WebSphere\AppServer"
set JREPATH=%WASHOME%\java\jre\bin
set CLASSPATH=%WASHOME%\java\jre\lib\ext\PD.jar
set CONFIGS=-Dpd.cfg.home=%WASHOME%\java\jre
set PARAMS1=-action config -admin_id sec_master
set PARAMS2=-appsvr_id ajiserver -appsvr_pwd aji -polycysvr
oma2tivoli03.am.tsacorp.com:7135:1 -port 7135 -authzsvr
oma2tivoli03.am.tsacorp.com:7136:1
set PARAMS3=-mode remote -cfg_file "%OUTPUTPATH%\tivoli.config" -
key_file "%OUTPUTPATH%\tivoli.keystore" -cfg_action create

%JREPATH%\java -cp %CLASSPATH% %CONFIGS% com.tivoli.pd.jcfg.
SrvSslCfg %PARAMS1% -admin_pwd password %PARAMS2% %PARAMS3%

REM
REM Cleanup DOS Parameter values...
REM
set WASHOME=
set OUTPUTPATH=
set JREPATH=
set CLASSPATH=
set CONFIGS=
set PARAMS1=
set PARAMS2=
set PARAMS3=
```

JAAS Pluggable Authenticator System Properties

The following system properties are required for pluggable authentication with third-party Java Authentication and Authorization Service (JAAS) authenticators. These system properties must be configured for the application server container process that contains the “APF” Base-application.war file.

java.security.auth.policy — Specifies the relative URI path to the Java security policy file. This file consists of one or more grant entries, each of which consists of a number of permission entries.

Required: Yes

Example Entries -Djava.security.auth.policy=c:/lib/security/jsf_jaas.policy
-Djava.security.auth.policy=c:/lib/security/tivoli_java.policy

Examples of the jsf_jaas.policy and tivoli_java.policy files are shown below. In both policy files, Principal is a grouping of identities to which the permission is granted. In these examples, the asterisks are wildcards that grant the permission to all users. AllPermission is a permission that implies all other permissions.

jsf_jaas.policy file:

```
/** Java 2 Access Control Policy for the Enterprise Security */
/** Environment */

/* grant the JAAS core library AllPermission */
grant codebase "file:[path]jaas.jar", Principal * *
{
    permission java.security.AllPermission;
};
```

tivoli_java.policy file:

```
/** Java 2 Access Control Policy for the Enterprise Security */
/** Environment */

/* grant the JAAS and PD core library AllPermission */
grant codeBase "file:[path]PD.jar", Principal * *
{
    permission java.security.AllPermission;
};

grant codebase "file:[path]jaas.jar", Principal * *
{
    permission java.security.AllPermission;
```

```
};
```

java.security.auth.login.config — Specifies the relative URI path to a specific JAAS-based authenticator configuration file. Each file contains the default configuration for one of the supported authenticators—USEC 1.x (USECjsf_jaas.config or jsf_jaas.config), Microsoft Active Directory (ACTIVEDIRECTORY_jaas.config), Tivoli Access Manager (Tivoli_jaas.config), or LDAP (LDAP_jaas.config). See the [“Third-Party JAAS Authenticator Configuration Files”](#) topic for examples and property descriptions for each of these configuration files.

Required: Yes

Example Entries

- Djava.security.auth.login.config=c:/lib/security/jsf_jaas.config
- Djava.security.auth.login.config=c:/lib/security/ACTIVEDIRECTORY_jaas.config
- Djava.security.auth.login.config=c:/lib/security/Tivoli_jaas.config
- Djava.security.auth.login.config=c:/lib/security/LDAP_jaas.config

Third-Party JAAS Authenticator Configuration Files

Examples of the _jaas.config file for each of the supported authenticators are shown below. The properties listed in the ACTIVEDIRECTORY_jaas.config, Tivoli_jaas.config, and LDAP_jaas.config files are described following the sample configuration files. Each of these configuration files specifies the authenticator class contained in the security authentication package (com.aciworldwide.application.security.authentication). This package is located in the \JavaSF\lib\application.jar that is deployed with an AJSF-based application server.

These example configuration files are provided in the AJSF distribution's configuration folder (e.g., \JavaSF\config) and are provided as examples only. They are not to be viewed as supported code with thorough testing on each release, or with version updates as the third-party authenticators evolve.

Note: The login module name must be specified as “ESS” in each of these authenticator configuration files.

USECjsf_jaas.config (or jsf_jaas.config)

The jsf_jaas.config file contains the default configuration for the USEC 1.x authenticator used by the JAASAuthenticator class. This file must specify the JAASAuthenticator class contained in the security authentication package.

```
/** Login Configuration for the USEC Authenticator */  
  
ESS {    com.aciworldwide.application.security.authentication.  
JAASAuthenticator required debug=false;  
};
```

ACTIVEDIRECTORY_jaas.config

The ACTIVEDIRECTORY_jaas.config file contains the default configuration for the USEC authenticator used by the ActiveDirectoryAuthenticator class. This file must specify the ActiveDirectoryAuthenticator class contained in the security authentication package. Property descriptions are provided following the sample ACTIVEDIRECTORY_jaas.config file.

```
ESS {  
com.aciworldwide.application.security.authentication.  
ActiveDirectoryAuthenticator requisite  
    debug="true"  
    strongDebug="false"  
  
    java.naming.factory.initial="com.sun.jndi.ldap.LdapCtxFactory"  
    java.naming.security.authentication="simple"  
    java.naming.security.principal="{username}@amlab.tsalab.lab"  
  
    connectionURL="ldap://172.21.46.66:389"  
    secureSSL=false  
  
    scope=SUB  
    userBase="dc=amlab,dc=tsalab,dc=lab"  
  
    groupSearch="(&(objectclass=user)(userPrincipalName={userprincipal})) "  
    //groupSearch="(&(objectclass=user)(CN={username})) "  
    groupName="memberOf"  
  
    groupSuffixRole=USEC.Role.  
    groupSuffixSecGrp=USEC.Group.  
    groupSuffixFilter=USEC.Filter.  
  
    tryFirstPass=false
```

```
useFirstPass=false
storePass=fale
clearPass=false;
};
```

debug — A flag indicating whether to trace the steps through the authentication process. Valid values are as follows:

true = Yes, trace the authentication steps.
false = No, do not trace the authentication steps. Default.

strongDebug — A flag indicating whether more in-depth tracing of authentication processing is performed. This property should NOT be turned on in a production environment. Valid values are as follows:

true = Yes, perform in-depth tracing of the authentication steps.
false = No, do not perform in-depth tracing of the authentication steps. Default.

java.naming.factory.initial — Specifies the initial context factory used to produce LDAP connections.

java.naming.security.authentication — A string indicating the authentication mechanism to use. For the Sun LDAP service provider, this can be one of the following strings:

“none” = Use no authentication (anonymous)
“simple” = Use weak authentication (clear-text password)
“*sasl_mech*” = A space-separated list of SASL mechanism names. Use one of the SASL mechanisms listed by the Internet Assigned Numbers Authority (IANA) (<http://www.iana.org/assignments/sasl-mechanisms>). For example, “CRAM-MD5” means to use the CRAM-MD5 SASL mechanism described in RFC 2195 - IMAP/POP AUTHorize Extension for Simple Challenge/Response.

java.naming.security.principal — The name of the user/program that is doing the authentication. The value of this property depends on the security authentication scheme used. In the case of Active Directory, the principal is a fully qualified user’s domain name where the {username} replacement token is specified and later replaced with the userID of the actual user to be authenticated.

Example Entry: {username}@amlab.tsalab.lab

Using the above example value, when user “java.coder” logs on, the authentication request will be java.coder@amlab.tsalab.lab where {username} is replaced with “java.coder”.

connectionURL — The LDAP connection path specifying the server hosting the LDAP directory and the port used to connection to the LDAP server.

Example Entries: ldap://172.21.46.66:389
ldaps://172.21.24.66:636

secureSSL — A flag indicating whether SSL is used between the LDAP factory and the directory manager. Valid values are as follows:

true = Yes, SSL is used.
false = No, SSL is not used. Default.

If this property is set to true, the “java.naming.security.protocol” property is set to SSL, which tells the LDAP factory to communicate with the directory manager using ldaps instead of ldap. In addition, if this property is set to true, the process’s key and trust stores must contain the appropriate keys and imported certificates to properly establish an SSL connection with the Active Directory. If this property is set to false, then the basic http connection is used and it is assumed that the Active Directory accepts basic connections.

scope — A code indicating how a group search is performed from the user base directory level for the Active Directory. Valid values are as follows:

SUB = Sub tree level search. Search current level and all subsequent levels under that branch. Default.
ONE = Only search the current tree level.
OBJECT = Only search the current object within the current tree level.

userBase — Defines the directory *base* level for where user names reside. User names can reside at the base level or in a sub-tree level below the base. If not specified, the search base is the top-level context.

Each object in an Active Directory has a distinguished name (DN). The DN is unique from all other objects and contains the full information needed to retrieve the object. The DN contains the domain where the object resides and the path to the object. The DN is made up of the following attributes (or qualities):

- DomainComponentName (DC)
- OrganizationalUnitName (OU)
- CommonName (CN)

Example Entry: dc=amlab,dc=tsalab,dc=lab

groupSearch — Specifies the filter that is used to convert, via an LDAP search request, a user entry to a unique DN for a specific user. The unique DN contains group attributes (See the groupName directive below). This search criteria is used to obtain the given user's DN and attributes. The attributes contain the group names for which the user is a member.

There are 2 supported filter tokens: {username} and {userPrincipal}. The token used depends on how the LDAP directory is implemented to associate a given DN to a specific user. Some use just the user's ID/name while others may specify the full user domain principal. See usage below:

- Usage 1: “(&(objectclass=user)(userPrincipalName={userprincipal}))”
In this example, the userPrincipalName criteria is specified along with the {userPrincipal} filterable token. If this criteria is used, then the fully qualified user principal domain name is used (e.g., – Bill.Smith@am.acicorp.com) to acquire a unique DN and attributes.
- Usage 2: “(&(objectclass=user)(CN={username}))”
In this example, the Common Name criteria is specified along with the {username} filterable token. If this criteria is used, then the common name (e.g., authenticated userID = “Bill.Smith”) is used to acquire a unique DN and attributes.

groupName — A directive used to acquire the list of groups associated to the user DN returned from the group search LDAP search request.

- Usage 1 : groupName=“memberOf”
For the given user's DN, all “memberOf” attributes are obtained. Each attribute contains the group names associated with the authenticated user. These groups are the APF role(s), security group, and filter(s).

groupSuffixRole — The prefix used by the authenticator to acquire APF-related roles. This prefix must match the prefix used to define APF roles in the LDAP directory.

Example Entry: “USEC.ROLE.”

In this case, the directory must prefix all APF roles defined with “USEC.ROLE.”. For example, the APF fullInstAccRole role would be defined in the LDAP directory as USEC.ROLE.fullInstAccRole.

groupSuffixSecGrp — The prefix used by the authenticator to acquire APF-related security groups, which relate various session constraints to a given user. This prefix must match the prefix used to define APF security groups in the LDAP directory.

Example Entry: “USEC.SECGRP.”

In this case, the directory must prefix all APF security groups defined with “USEC.SECGRP.”. For example, the APF SecAdminSecGrp security group would be defined in the LDAP directory as USEC.SECGRP.SecAdminSecGrp.

groupSuffixFilter — The prefix used by the authenticator to acquire APF-related filters, which are identifiers that relate to a set of values. This prefix must match the prefix used to define APF filters in the LDAP directory.

Example Entry: “USEC.FILT.”

In this case, the directory must prefix all APF filters defined with “USEC.FILT.”. For example, the APF AllInstitution filter would be defined in the LDAP directory as USEC.FILT.AllInstitution.

tryFirstPass — A flag indicating whether the first JNDI LoginModule in the stack saves the password entered, and subsequent LoginModules also try to use it. If authentication fails, the LoginModules prompt for a new password and retry the authentication. Valid values are as follows:

true = Yes, use the tryFirstPass option.
false = No, do not use the tryFirstPass option. Default.

useFirstPass — A flag indicating whether the first LoginModule in the stack saves the password entered, and subsequent LoginModules also try to use it. LoginModules do not prompt for a new password if authentication fails (authentication simply fails). Valid values are as follows:

true = Yes, use the useFirstPass option.
false = No, do not use the useFirstPass option. Default.

storePass — A flag indicating whether the LoginModule saves the shared state if the login succeeds. The tryFirstPass and useFirstPass options do automatically enable this option. Valid values are as follows:

true = Yes, save the shared state if login is successful.
false = No, do not save the shared state if login is successful. Default.

clearPass — A flag indicating whether the LoginModule clears the shared state after committing. Valid values are as follows:

true = Yes, clear the shared state after committing.
false = No, do not clear the shared state after committing. Default.

Tivoli_jaas.config

The Tivoli_jaas.config file contains the default configuration for the Tivoli authenticator used by the TivoliAuthenticator class. Property descriptions are provided following the sample Tivoli_jaas.config file.

```
ESS {
com.aciworldwide.application.security.authentication.TivoliAuthenticator
required
    // Option.  Debugging the PDLoginModule super class of our Tivoli
Authenticator
    debug=false

    // Required: The java app server's config file built from SvrSslCfg that
    // contains all necessary connection information to the TAM policy and
    // authorization servers.
    //
    configURLName=file:///c:/am/tam_config_file.conf

    // Optional.  The TAM domain name if used.
    //accessMgrDomain=myDomain

    // Optional.  If creating a PDContext using a specified admin user and
    // password other than that related to the configURLName's config file.
    // java.naming.security.principal="Directory users admin. User's name"
    // java.naming.security.credentials="password"
    // credentialsEncrypted=false

    // Optional.  Group principal suffixes that define what's a USEC specific
    //
    //          group
    // broken into 3 categories:  Role, Group (constraints assoc w/ user),
```

```
//
//                                     Filter
// Defaults are:  ACI.ROLE.
//               ACI.GROUP.
//               ACI.FILTER.
//
groupSuffixRole="USEC.Role."
groupSuffixSecGrp="USEC.Group."
groupSuffixFilter="USEC.Filter."

//
// Optional.  Argument dictates the type of tivoli authentication,group
//            access to do based on the type of user.  Tivoli has the
//            directory of users and also support's registry users for
//            backing stores such as ActiveDirectory, LDAP, etc...
//            The default if not present is false (tivoli directory).
//
registryUsers=false
};
```

debug — A flag indicating whether to trace the steps through the authentication process. Valid values are as follows:

true = Yes, trace the authentication steps.
false = No, do not trace the authentication steps. Default.

configURLName — A file-based URL (e.g., file://) that points to the Java application server process's configuration file generated by the Tivoli Java SvrSslCfg utility. The contents of the generated configuration file define how the Java application server process communicates to the TAM environment over an SSL connection. This property is required.

Example: configURLName=file:///c:/data/jaas/Tivoli_gend_cfg/myappservr.cfg

accessMgrDomain — The domain name provided during login and group access through the PD API that the Tivoli Authenticator uses. This domain name must match what is specified in the Tivoli config file pointed to by the configURLName property. The accessMgrDomain property is optional and is not normally required.

Example: `accessMgrDomain=myDomain`

java.naming.security.principal — A specified administrative user name if you are creating a PDContext using a specified administrative user ID and password other than that related to the configuration file specified by the `configURLName` property. This administrative user ID and password override what is configured within the Tivoli configuration file produced by the `SvrSslCfg` utility (pointed to by the `configURLName` property). Normally, this information is self-contained in the generated Tivoli configuration file and thus is not needed. This property is optional.

Example Entry: `java.naming.security.principal=admin_user`

java.naming.security.credentials — The password for the administrative user if you are creating a PDContext using a specified administrative user ID and password other than that related to the configuration file specified by the `configURLName` property. This administrative user ID and password override what is configured within the Tivoli configuration file produced by the `SvrSslCfg` utility (pointed to by the `configURLName` property). Normally, this information is self-contained in the generated Tivoli configuration file and thus is not needed. This property is optional.

Example Entry: `java.naming.security.credential=mypassword`

The value of this property can be encrypted, and if so, the `credentialEncrypted` property must be set to true (`credentialsEncrypted=true`). Once obtained by the authenticator, the password is decrypted and passed accordingly in the PDContext.



To encrypt a value for this property, see the [“JAASConfigCredentialEncrypter Utility”](#) topic below.

credentialsEncrypted — A flag indicating whether the password value of the `java.naming.security.credentials` property is encrypted. Valid values are as follows:

`true` = Yes, the credentials password is encrypted.
`false` = No, the credentials password is not encrypted. Default.

groupSuffix[Role|Filter|SecGrp] — These optional properties specify the type of groups you can assign to Tivoli users within the Tivoli directory. In order to filter out USEC specific group assignments so that authorization data stored in USEC can be obtained for a given Tivoli configured user, these options specify the

prefix of the group names of interest for APF products. There are three prefix properties available—`groupSuffixRole`, `groupSuffixFilter`, and `groupSuffixSecGrp`. Groups from the TAM environment are acquired and set as group principals if they begin with the group prefix for any of these configured prefix strings. The defaults for these properties if not configured are “`ACI.ROLE.`”, “`ACI.SECGRP.`”, and “`ACI.FILTER.`”, respectively. Each prefix specifies a different kind of group. The role identifies what permissions are related to a user. The filter specifies what group of filterable values are associated to a user, and the security group specifies what specific constraints are to be processed for a user’s session.

Example Entries: `groupSuffixRole=“usec.role.”`
 `groupSuffixFilter=“usec.filter.”`
 `groupSuffixSecGrp=“usec.secgrp”.`

registryUsers — A flag indicating whether you are using a backing store through TAM, where users may be specified or maintained within a Tivoli backing store and not contained within the Tivoli directory itself. If this is the case, you must set this property to true. Most likely this is not needed and you can use the default value of false. Valid values are as follows:

true = Yes, a backing store is used.
false = No, a backing store is not used. Default.

LDAP_jaas.config

The `LDAP_jaas.config` file contains the default configuration for the USEC authenticator used by the `LDAPAuthenticator` class.

```
ESS {
com.aciworldwide.application.security.authentication.LDAPAuthenticator required
    debug="true"
    strongDebug="false"

    connectionURL="ldap://oma2ui2:389"
    secureSSL=false

    java.naming.factory.initial="com.sun.jndi.ldap.LdapCtxFactory"
    java.naming.security.authentication="simple"
    java.naming.security.principal="cn=Manager,dc=aciworldwide,dc=com"
    java.naming.security.credentials="secret"
    credentialsEncrypted=false
}
```

```
userCredentialAttr=userPassword
//userCredentialDecipherAlgo=SHA
scope=SUB
userBase="ou=people,dc=aciworldwide,dc=com"
userSearch="(uid={username})"

groupBase="ou=groups,dc=aciworldwide,dc=com"
//groupSearch="(uniqueMember={username})"
groupSearch="(uniqueMember={userdn})"
groupName="cn"

groupSuffixRole=USEC.Role.
groupSuffixSecGrp=USEC.Group.
groupSuffixFilter=USEC.Filter.

tryFirstPass=false
useFirstPass=false
storePass=false
clearPass=false;
};
```

The following properties in the LDAP_jaas.config file are essentially the same as those used for the ACTIVEDIRECTORY_jaas.config file. Refer to the definitions above for descriptions of these properties:

- debug
- strongDebug
- connectionURL
- secureSSL
- java.naming.factory.initial
- java.naming.security.authentication
- scope
- userBase
- groupSuffixRole
- groupSuffixSecGrp
- groupSuffixFilter
- tryFirstPass
- useFirstPass
- storePass
- clearPass

The remaining properties are either unique to the LDAP_jaas.config file or have a different usage from that used in the ACTIVEDIRECTORY_jaas.config file and are described below.

java.naming.security.principal — The name of the user/program that is doing the authentication. The value of this property depends on the security authentication scheme used.

For direct LDAP, this property value is a user name that is defined within the LDAP directory and has access to acquire credentials for users attempting to log on to the ACI desktop user interface application. The `java.naming.security.credentials` property defines the password associated with this administrative type of user.

Example Entry: `cn=Manager,dc=aciworldwide,dc=com`

java.naming.security.credentials — The password for the user binding to the LDAP directory. The value can be encrypted and if so, the `credentialsEncrypted` property must be set to true (`credentialsEncrypted=true`). Once obtained by the authenticator, the password is decrypted and passed accordingly to the LDAP directory in the user binding request.



To encrypt a value for this property, see the [“JAASConfigCredentialEncrypter Utility”](#) topic below.

credentialsEncrypted — A flag indicating whether the password value of the `java.naming.security.credentials` property is encrypted. Valid values are as follows:

`true` = Yes, the credentials password is encrypted.
`false` = No, the credentials password is not encrypted. Default.

userCredentialAttr — The name of the LDAP directory attribute that is requested on a directory lookup request. For the ACI desktop user interface application, this is the user’s password attribute. This value must match the attribute used within LDAP to hold the password/credential value of a given user.

Example Entry: `userCredentialAttr="userPassword"`

In the above example value, the “userPassword” attribute is requested to be retrieved from the LDAP directory.

The value contained with the attribute is the actual clear password or encrypted/hashed password. The value is compared against the password/credential provided by the authenticator for the given user.

If the password retrieved from the LDAP directory is encrypted or hashed, some LDAP implementations specify the one-way encryption algorithm with the password by prefixing the encrypted password/credential with the hash algorithm.

Example: “{SHA}aa3095599039gigghhed8”

In this case, the `userCredentialDecipherAlgo` property is not used as the LDAP implementation provides the decipher method used to compare credentials for authentication.

If no decipher algorithm is provided by the LDAP directory, then the password/credential is assumed to be in the clear unless the `userCredentialDecipherAlgo` property is defined. If the decipher method is specified in the `userCredentialDecipherAlgo` property, then that method is used to do the one-way hash of the password/credential provided to the authenticator for authentication. The resulting value is compared against the LDAP value returned.

userCredentialDecipherAlgo — A code that indicates how to encrypt (create a one-way hash value) of the credential to be authenticated against the LDAP directory for the given user. If this property is not configured and the LDAP directory does not prefix the password on a directory lookup request with the decipher algorithm, then the password is assumed to be in the clear. Valid values are as follows:

SHA or SHA-1	= SHA 1 algorithm.
SSHA or SSHA-1	= Salted SHA 1 algorithm.
SHA-2	= SHA 2 algorithm.
MD5	= Message-Digest algorithm 5.
MD2	= Message Digest Algorithm 2.
crypt	= Unix crypto.

userSearch — The filter that is used to convert, via an LDAP search request, a user entry to a unique DN for a specific user.

Example Entry: `userSearch="(uid={username})"`

For the above example filter, the LDAP directory searches the `userBase` directory location while obtaining all entries with the `uid` element equaling the replacement value (i.e. - `userID`) for the `{username}` replacement token. The token is replaced with the `userID` value being authenticated.

The user's credential attribute (specified by the `userCredentialAttr` property above) is obtained from the DN returned and is authenticated with the credential provided to the authenticator.

groupBase — Defines the directory *base* level for where group names reside. Group names can reside at the base level or in a sub-tree level below the base. If not specified, the search base is the top-level context.

Each object in an LDAP directory has a distinguished name (DN). The DN is unique from all other objects and contains the full information needed to retrieve the object. The DN contains the domain where the object resides and the path to the object. The DN is made up of the following attributes (or qualities):

- DomainComponentName (DC)
- OrganizationalUnitName (OU)
- CommonName (CN)

Example Entry: `ou=groups,dc=aciworldwide,dc=com`

groupSearch — Specifies the filter that is used to convert, via an LDAP search request, a group entry to a unique DN for a specific user. There are two supported filter tokens: `{username}` and `{userdn}`. The token used depends on how the LDAP directory is implemented to assign users to groups. Some use just the user's ID/name while others may specify the full user's directory distinguished name (fully qualified).

- Usage 1: “(uniqueMember={username})”

In this example, the `groupBase` directory is searched using the specified `groupSearch` filter where `{username}` is a replacement token. The token is replaced with the `userID` being authenticated.

- Usage 2: “(uniqueMember={userdn})”

In this example, the `groupBase` directory is searched using the specified `groupSearch` filter where the `{userdn}` is the replacement token. The token is the fully-qualified user distinguished name (not just `userID`). The user distinguished name is constructed using the `userID` and the LDAP directory to determine the actual distinguished name.

groupName — A directive used to acquire the list of groups associated to the user DN returned from the group search LDAP search request. The contents assigned to this property contain the actual groups (e.g., role, filter, security group) associated with the given user being authenticated.

JAASConfigCredentialEncrypter Utility



The JAASConfigCredentialEncrypter utility enables you to encrypt the credential used for creating a PDContext or LDAP directory binding that is stored in the java.naming.security.credentials property in the Tivoli_jaas.config or LDAP_jaas.config file, respectively.

JAASConfigCredentialEncrypter is a utility class that you can execute (via a batch/script process) to encrypt the credential specified in the Tivoli_jaas.config or LDAP_jaas.config file. This utility requires the standard installed JRE (1.4.2 or later). The utility encrypts the clear value specified for the java.naming.security.credentials property in the Tivoli_jaas.config or LDAP_jaas.config file and also sets the credentialsEncrypted property to a value of true if it is set to false.

The following is a sample run command for executing the JAASConfigCredentialEncrypter utility:

Example Command Syntax

```
Java -classpath application.jar;framework-core.jar  
-Djava.security.auth.login.config="jaas-config-filename" com.  
aciworldwide.application.security.authentication.  
JAASConfigCredentialEncrypter
```

where: *jaas-config-filename* is the name of the Tivoli_jaas.config or LDAP_jaas.config file.

Note: The class path jars are located within the lib section of the deployed APF application (i.e. – base-application).

Sharing USEC Message Context Among AJSF-based APF Products

If you are running multiple AJSF-based APF products on the same platform and need to integrate ACI desktop and browser user security functions among the products, you must modify the AJSF configuration for the appropriate ACI web access services user interface server processes (e.g., PMWEB, ESWEB, and APF Admin) to share USEC subsystem context.

For example, when an ACI Payments Manager user logs on to the ACI desktop user interface, the PMWEB process creates USEC session context and active session data. If the user accesses the TI Maintenance window, the PMWEB process requests a list of BASE24-eps limits via the LimitListRq request from the ESWEB process. If USEC subsystem context is not shared, the ESWEB process attempts to authorize the request based on the ACI Payments Manager user, the current USEC session context, and active session data. Because the USEC session context stored in the ESWEB process memory does not contain the session data of the ACI Payments Manager user since it was established only in the PMWEB process memory, a Session_Expired error is returned as a result.

To alleviate such errors, perform the following steps to share USEC subsystem context between the ACI applications:

1. Ensure that both of the ACI web access services user interface server processes are built using an AJSF code base of 07.2 or greater and that the JDBC driver is accessible on the server.
2. Set the context.type property in the APPLCNFG to a value of “DISK-WITH-CACHE” for both of the ACI web access services user interface server processes.
3. Set the user.security.version property in the APPLCNFG to a value of “usec1.1” for both of the ACI web access services user interface server processes to ensure that the USEC subsystem is running on-board in both processes.
4. Ensure that the Message Context data source (MSGCNTX) exists in the same database as the USEC subsystem tables for one of the ACI web access services user interface server processes. If not, create it. The following is a sample script for creating the MSGCNTX in an Oracle database.

```
CREATE TABLE MSGCNTX
(MSGID VARCHAR2(200) NOT NULL,
 SEQNUM NUMBER NOT NULL,
 CONTF LG VARCHAR2(1),
 EXPTM DATE,
 CONTEXT BLOB,
```

PRIMARY KEY (MSGID, SEQNUM))

5. Set the db.alt.datasource.name.msgctx property in the APPLCNFG to a value of “USEC” for the ACI web access services user interface server process that does not have the USEC tables in its database.
6. Add the following properties to the APPLCNFG with a value of “USEC” for the ACI web access services user interface server process that does not have the USEC tables in its database.

```
db.alt.datasource.name.actvsesn
db.alt.datasource.name.filtrole
db.alt.datasource.name.flds
db.alt.datasource.name.fldvals
db.alt.datasource.name.fltrfldvals
db.alt.datasource.name.fltrs
db.alt.datasource.name.funcfs
db.alt.datasource.name.permdetl
db.alt.datasource.name.permgrp
db.alt.datasource.name.pmgpxprm
db.alt.datasource.name.perms
db.alt.datasource.name.permtyp
db.alt.datasource.name.password
db.alt.datasource.name.pswdgrp
db.alt.datasource.name.pswdhist
db.alt.datasource.name.rolxpmgp
db.alt.datasource.name.roles
db.alt.datasource.name.secgrp
db.alt.datasource.name.taskflds
db.alt.datasource.name.taskfnfs
db.alt.datasource.name.tasks
db.alt.datasource.name.ticket
db.alt.datasource.name.usec
db.alt.datasource.name.usrxfilt
db.alt.datasource.name.userrole
db.alt.datasource.name.users
db.alt.datasource.name.usrstat
```

7. For the ACI web access services user interface server process that does not have the USEC tables in its database, add a row to the Alternate Datasource (DATASRC) table defining the “USEC” data source. The following is an example DATASRC configuration.

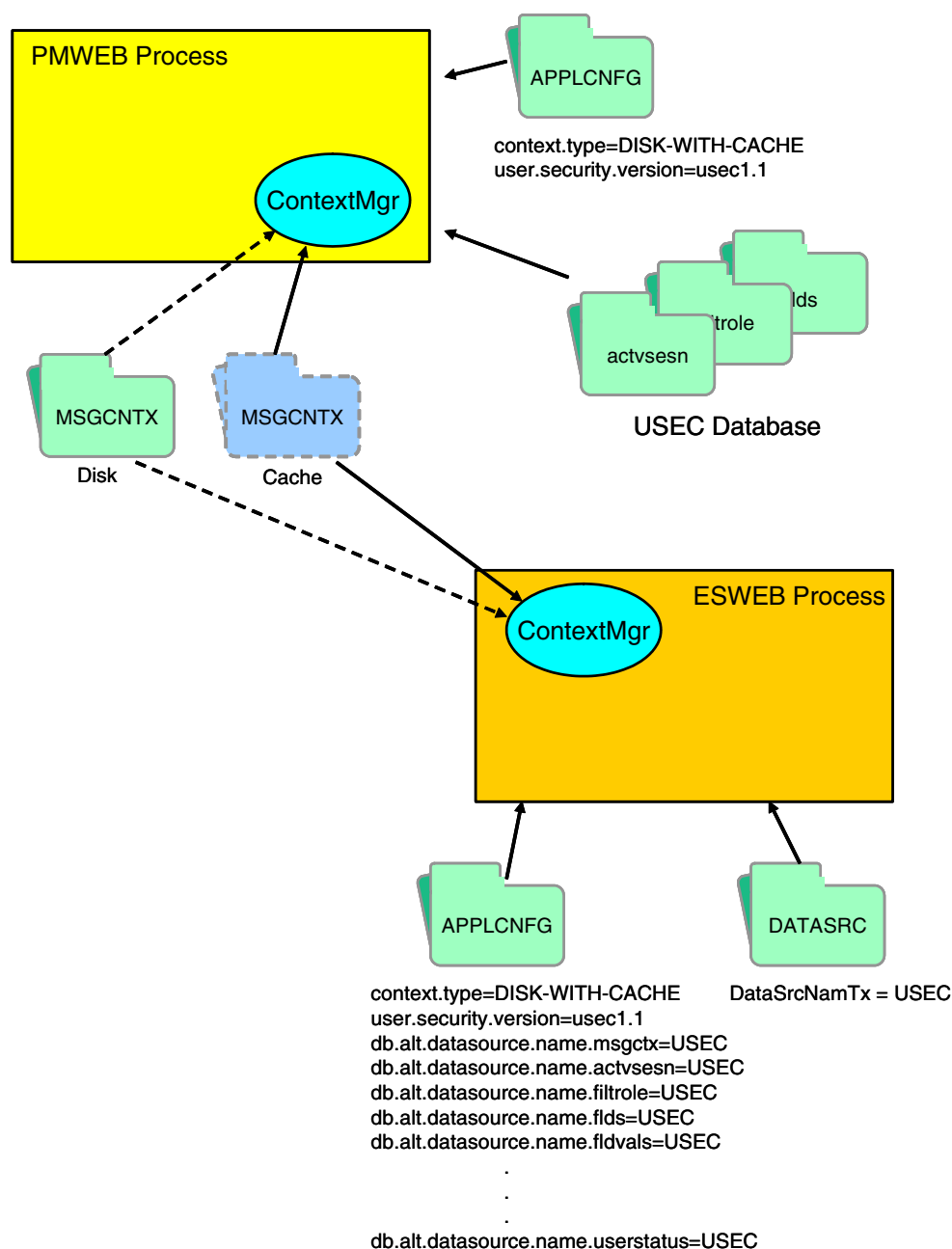
Field Name	Value
Data Source Name	USEC
Driver Name	com.microsoft.jdbc.sqlserver.SQLServerDriver

Field Name	Value
Data Source URL	jdbc:microsoft:sqlserver://oma2javadev1:1433;SelectMethod=cursor;DatabaseName=APFAdmin
Data Source Type	MSSQL
User ID	pserve
User Password	c67834a7b2e28c4e
Table Prefix	Not applicable
Maximum Connections	2
Start Connections	2
Connection Wait Time	10000

8. Stop and restart both of the ACI web access services user interface server processes.

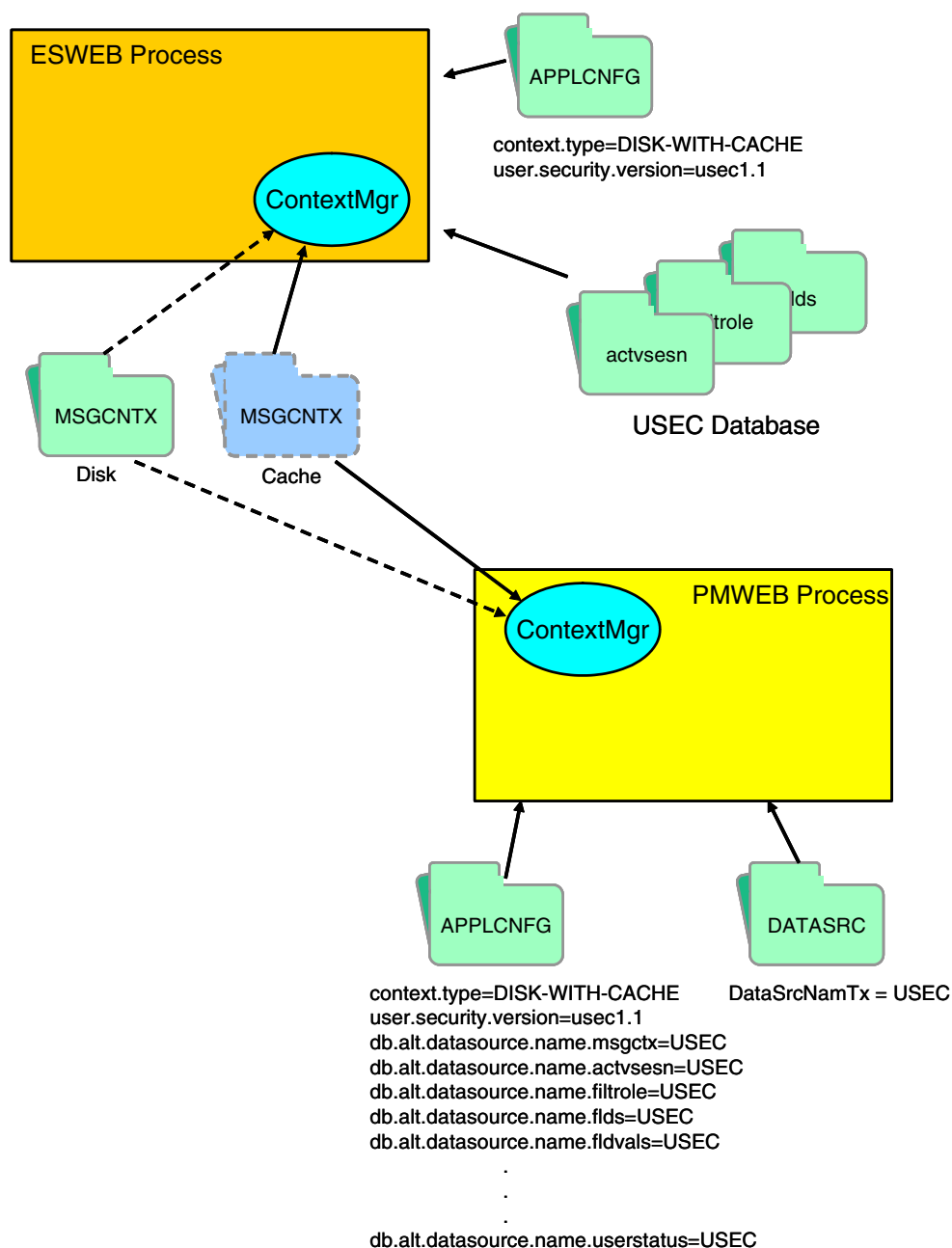
USEC Tables in ACI Payments Manager Database

The following illustration depicts the configuration requirements when USEC tables are maintained in the ACI Payments Manager web access services user interface server process (PMWEB) database and USEC context needs to be shared with the BASE24-eps ACI web access services user interface server process (ESWEB).



USEC Tables in BASE24-eps Database

The next illustration depicts the configuration requirements when USEC tables are maintained in the database for the BASE24-eps ACI web access services user interface server process (ESWEB) database and USEC context needs to be shared with the ACI Payments Manager web access services user interface server process (PMWEB).



Workflow Subsystem

The workflow subsystem uses transaction map and transaction meta map XML files to configure the messages that a AJSF-based application can process and the flow of work (transaction processing) to be performed on those messages.

In both configuration map files, classes are used to represent business logic. The transaction map file implements the Java command design pattern and the transaction meta map file implements the work flow. This provides for the loose coupling of business logic (i.e., actions are not hard-coded to proceed from one action to another) and allows for customer-specific modifications (CSMs) to be configured without affecting the standard application. Changes to the XML files are implemented at runtime by simply warmbooting the application.

Transaction map XML files implement a simple work flow by mapping symbolic keys (such as a message ID) into a formatter class to parse the message and a command class to process the message. A formatter class is used (if needed) to translate the message from an external (e.g., native format) into an internal format used by the application. The command class contains the business logic for processing the transaction. Transaction meta map XML files implement a complex work flow by defining the series of actions required to process a transaction and function as a work flow descriptor. The actions are performed in the bottom-up order in which they are arranged in the XML file. Actions can jump to other entries within the same transaction meta map file or to entries in other transaction meta map files. In addition, you can use conditions and if/then/else logic in transaction meta map files to direct the processing workflow. For example, you can redirect processing when errors occur.

AJSF applications use configuration properties (e.g., `xmltranmap.filename<text-string>` and `xmltransmetamap.filename.n`) to define the locations of transaction map and transaction meta map XML files. The Transaction Map Manager and Transaction Meta Map Manager read them at initialization and preload message handling details into memory. The data is then held in memory for as long as the application is running.

When the Message Service in a AJSF-based application receives or sends a message, it invokes a Formatter class to parse or format the message or a Formatter Controller class if multiple message formats are sent or received over the same connection. A Formatter Controller is only needed if multiple message formats are sent or received over the same connection. Once the type of message is determined, the Formatter Controller invokes a Formatter class to parse or build the message.

Formatter classes are implemented in the AJSF in the following three ways:

- In the Listener data source for inbound messages
- In the External Connection data source for outbound messages
- In the Transaction Map XML file for both inbound and outbound messages.

When a Formatter class parses a message, it determines the transaction map key. The transaction map key identifies a transaction map XML file entry, which specifies the specific Formatter class and Command Controller class for the message. The same tag used in the transaction map XML file can be used to access the actions to be performed for the message in corresponding transaction meta map XML files.

Example Transaction Map

The following is an example XML entry from the NCRDHTranMap.xml file used by the NCR NDC+ (Java) Device Handler process for BASE24-eps. In this entry, the message is an NDC deposit transaction from an ATM. The message is formatted by the NDCUnsolicitedRequestFormatter class and processing starts with the TransactionStateCommand Command Controller class.

```
<Entry>
  <tag>NDCRequestTranType21</tag>  <!-- NDC Deposit from ATM -->
  <fmtrClass>com.aciworldwide.base24.es.dh.ncr.formatter.
NDCUnsolicitedRequestFormatter</fmtrClass>
  <cmndClass>com.aciworldwide.common.command.TransactionStateCommand</
cmndClass>
</Entry>
```

Example Transaction Meta Map

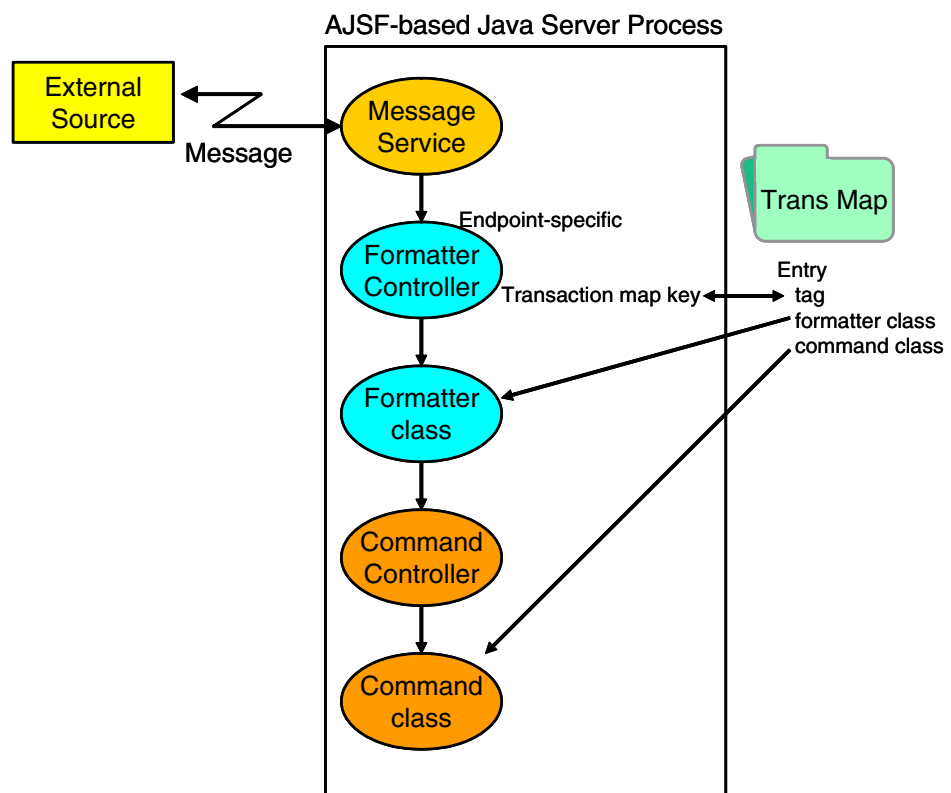
The next sample is a corresponding XML entry from the NCRDHTransMetaMap.xml file that specifies the actions to be performed for an NDC deposit transaction from an ATM. Notice that the <tag> value from the transaction map file is specified as the <currentTranMap> value in the transaction meta map file. In addition, this example specifies a condition that must be met for the following actions to be performed. The actions are performed in the reverse order (bottom up order) in which they are arranged in the meta map file.

```
<Transition>
  <currentTranMap>NDCRequestTranType21</currentTranMap> <!-- Deposit -->
  <condition>NOMAC</condition>
  <action>
    <setRespFormatterTranMap>NDCDeposit</setRespFormatterTranMap>
    <setRespErrorTranMap>DepositReversal</setRespErrorTranMap>
    <pushRqstTranMap>DetermineAuthorizerClass</pushRqstTranMap>
    <pushRqstTranMap>DepositRequest</pushRqstTranMap>
    <pushRqstTranMap>LastTransactionStatus</pushRqstTranMap>
    <pushRqstTranMap>ATDDepositRequestFetch</pushRqstTranMap>
    <pushRqstTranMap>InitialFinancialNativeMessage</pushRqstTranMap>
  </action>
</Transition>
```

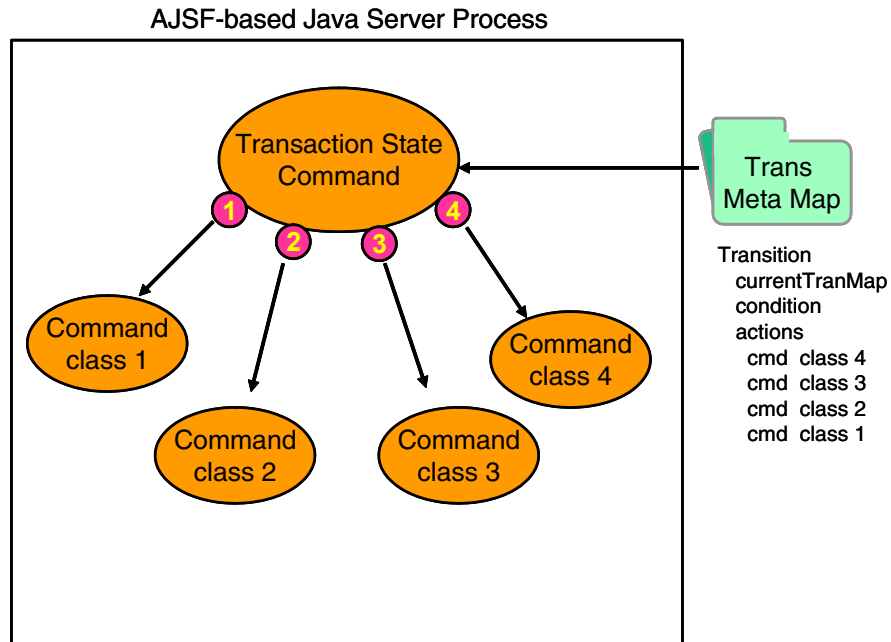
Workflow Summary

The following graphics illustrate how transaction map and transaction meta map XML files are used to direct the workflow in transaction processing.

The first diagram depicts transaction map processing. When the Message Service receives a message, it invokes an endpoint-specific Formatter Controller that knows enough about the message to determine the transaction map key, which is the key to entries in the transaction map file. The specified formatter class is invoked to format the message, and the Transaction State Command class is invoked to process the message.



The following diagram depicts transaction meta map processing. The Transaction State Command class invokes Command classes in the *bottom-up* order in which the classes are configured in the transaction meta map file.



3: Management

This section describes how to manage the configuration for Java server processes. This includes managing the configuration of alternate data sources, application-level properties, connections to and from external entities, connections to hardware security devices, and process relationships in clustered configurations for broadcasting messages. These configuration items are maintained in the following ACI Java Server Framework (AJSF) data sources:

- Alternate Data Source data source (DATASRC)
- Application Configuration data source (APPLCNFG)
- External Connections data source (EXTRCNCT)
- HSM Configuration data source (HSMCNFG)
- Listener data source (LISTENER)
- Message Context data source (MSGCNTX)
- Process data source (PROCESS)

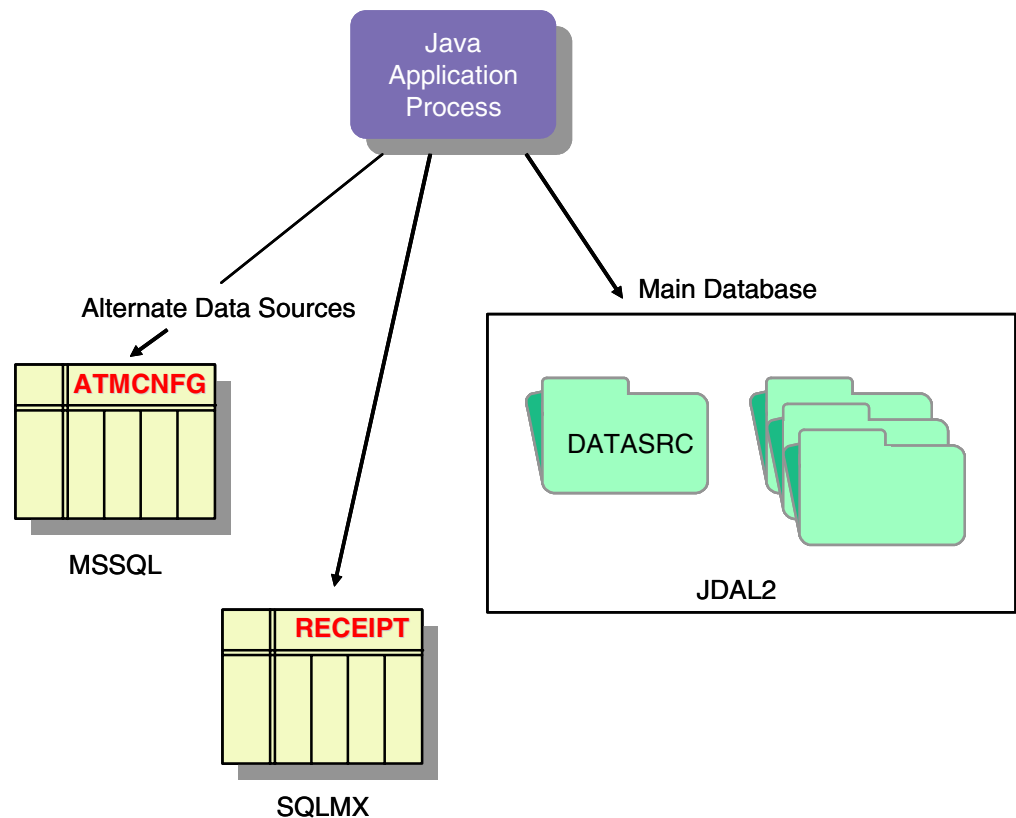
AJSF-based ACI applications provide formatted user interface screens or windows to maintain configuration data in these data sources. Although the look and feel of these screens and windows may differ from one product to the next, the fields displayed and the metadata maintained are the same across all products. This section describes how to configure these fields to meet the configuration requirements for your system.

Alternate Data Source Configuration

For various reasons, you may need to maintain certain data sources in databases other than the main application database. You can do this by configuring a `db.alt.datasource.name.db-service-name` property with the name of the data source and a corresponding Alternate Data Source data source (DATASRC) row that defines how to access the data source. This property and the DATASRC allow you to store data in different databases as needed for your application. The DATASRC contains all the information needed to access an alternate data source, including the relative URL location, database driver, user ID and password, table prefix, and database connection parameters.

Note: The AJSF does not currently support transactions that span multiple databases. Separate database transactions are required for each participating database.

When a Java server process needs to access the database, the database service first determines whether a `db.alt.datasource.name.db-service-name` property exists for the database service. If so, it reads the DATASRC from cache or disk and uses that information for the database access operation. If not, it accesses the database in the usual manner using the main database (i.e., “db.”) property values. The use of alternate data sources is depicted in the following illustration.



Field Descriptions

The following paragraphs describe each field in the DATASRC.

Data Source Name — The name of the alternate data source. This name must match the value of a `db.alt.datasource.name.db-service-name` property.

Driver Name — The class name of the JDBC driver to use for the alternate database connections. You should enter the entire package name as it is found in the JAR containing the driver. Refer to the vendor documentation for the JDBC driver for supporting information.

Data Source URL — The data source URI specification that allows the JDBC driver to access the correct alternate database. You can typically find the configuration requirements for this property in the vendor documentation for the JDBC driver. JDBC driver requirements may include information such as the host or IP address where alternate database resides, the port to connect to for the database, the name of database on the host, and other JDBC driver settings.

Data Source Type — the type of database to which the data source connects. Valid values are as follows:

DB2	= IBM DB2
JDAL	= ACI's Java Data Access Layer
JDAL2	= ACI's Java Data Access Layer 2
MSSQL	= Microsoft SQL Server
ORACLE	= Oracle
SQLMX	= HP NonStop SQL/MX

User ID — The JDBC user ID needed to authenticate a connection to the data source. This user ID is used to securely access the underlying data source and is typically established by a database or system administrator. ACI applications use this user ID to access the data source.

User Password — The JDBC password needed to authenticate a connection to the data source. This password is used to securely access the underlying data source and is typically established by a database or system administrator. The password stored in the DATASRC is encrypted.

Table Prefix — The prefix, if one exists, to prepend to this alternate data source table name. This allows you to use multiple databases with the same underlying data source names prepended with a unique identifier for each database, thereby making the full data source names in each database unique across all of the application's databases.

Maximum Connections — The maximum number of concurrent database connections that the Database Connection Manager (DBConnectionManager) can have open at one time to this alternate data source.

Start Connections — The number of connections the Database Connection Manager acquires to this data source when the Java server process is started. These connections remain active throughout the life of the process.

Connection Wait Time — The number of milliseconds the Database Connection Manager waits for another thread to free a connection to this data source. This value is used when the maximum number of connections has been reached as specified in the MaxConns field.

Example Configuration

The following is an example DATASRC configuration:

Field Name	Value
Data Source Name	ALTDATASRC1
Driver Name	com.ashna.jturbo.driver.Driver
Data Source URL	jdbc:JTurbo://172.16.1.53:1433/Contextv10/sp=false
Data Source Type	MSSQL
User ID	pserve
User Password	pserve
Table Prefix	Not applicable
Maximum Connections	50
Start Connections	1
Connection Wait Time	10

Application Configuration

The Application Configuration window or screen maintains configuration property values in the APPLCNFG. Using this window or screen, you can add, delete, or modify configuration property values and descriptions. You can enter the name of a specific Java server process for a configuration property or you can wildcard the process name so the configuration property applies to all Java server processes. You can also enter a user-defined description for a property or modify the description provided with preloaded configuration properties.

Note: ACI recommends that all configuration properties be defined in the APPLCNFG. Only those properties required to bootstrap the Java process need to be configured in the config.properties file.

The Property ID field contains the name of the configuration property and the Property Value field contains the value to which the property is set.

Note: Property names and values are case-sensitive.

The Process ID field specifies a logical name for an executing Java Virtual Machine process and is used to differentiate one process from another in the configuration of an application. The value you enter in this field must either match the value of the process.id property specified in a JVM runtime argument or configuration properties file when the process was started or be a wildcard character, an asterisk (*), representing all JVM processes. For the APF Admin application, the value in this field must also be defined in the Process Configuration data source (PROCESS) before you can enter it here. Using this field, you can associate more than one process with a property having the same name but having a different value. This enables you to configure a process so that it runs differently than other processes using the same property. When a Java process reads the APPLCNFG, it searches for its process-specific configuration properties first. If a process-specific property is not found, it then searches for the property with a wildcarded process ID. Thus, process-specific configuration properties take precedence over wildcard configuration properties.

Finally, the Property Description field enables you enter free-form user-defined text describing the property and its values. ACI provides descriptions for pre-loaded properties delivered from ACI.

External Connections Configuration

The External Connections Configuration window or screen maintains external connection configuration data in the EXTRCNCT. Using this window or screen, you can add, delete, or modify external connections to external entities (e.g., client website, ATM, C++ process, etc.). AJSF-based applications perform all data communications to external entities using entries in the EXTRCNCT. External entity-specific formatters are used to build and parse messages to each of the external entities.

Currently, the following protocols are supported to connect to an external entity. Additional protocols may be added in the future as required.

- Enterprise JavaBeans (EJB)
- Hypertext Transfer Protocol (HTTP)
- Java Message Service (JMS) - Asynchronous
- Java Message Service (JMS) - Synchronous
- Pathway (PATHSEND) protocol*
- Simple Mail Transfer Protocol (SMTP)
- Transmission Control Protocol/Internet Protocol (TCP/IP)
- Transmission Control Protocol/Internet Protocol (TCP/IP) - Client
- Transmission Control Protocol/Internet Protocol (TCP/IP) - Server

* The PATHSEND protocol enables AJSF-based applications to send Pathway messages to Pathway servers on the HP NonStop platform. AJSF-based applications that use the PATHSEND protocol must provide an adapter and message formatter that can handle Pathway messages. In addition, the following AJSF Java archive (.jar) files are required: framework-nsk.jar and common-nsk.jar.

The HP NonStop platform requires HP NonStop Server for Java Version 4.0 (NSJ4) or later and revision AAC (or later) of HP's JToolkit 2.0 for NonStop servers. JToolkit includes a .jar file (tdmext.jar) that must be included in the AJSF-based application's classpath. It also contains a library that must be bound into the JVM.

Several fields in the EXTRCNCT must be configured in accordance with the protocol used for the connection. These dependencies are described following a description of each of the fields in the EXTRCNCT under the [“Protocol Configuration Options”](#) topic below.

The External Connection ID field contains a user-defined logical name for an external connection. The Process ID field specifies the logical name for an executing Java Virtual Machine process that uses this connection. The value you enter in this field must match the value of the process.id property specified in a JVM runtime argument or in the configuration properties file when the process was started. For the APF Admin application, the value in this field must also be defined in the Process Configuration data source (PROCESS) before you can enter it here.

General Configuration Options

Several general configuration options in the EXTRCNCT describe the external connection, specify the formatter class used to build and parse messages for the connection, and configure the number of connections, timeout, and retry processing for the external connection.

The Connection Description field contains an optional user-defined description of the external connection. The Message Format Class field specifies the name of the invoked formatter class that builds request messages to and parses response messages from the external entity.

Note: The name of the invoked formatter class must be defined in the classpath of the ACI application or located in a JAR that is in the classpath.

The Timeout Limit field defines the maximum number of seconds that can elapse from the time a request is sent to an external entity to the time a response is received back from the external entity.

The Initial Connections and Maximum Connections fields are used only when one of the TCP/IP protocols is used. The Initial Connections field defines the initial number of socket connections to the defined external entity that the application creates at startup. If more connections are needed as transaction volume grows, the application uses the value in the Maximum Connections field to determine the maximum number of additional connections that can be created.

The Retry Attempts field specifies the maximum number of times the application process tries to establish a connection with the external entity if a message failure occurs. A message failure can occur at the system or application level.

Protocol Configuration Options

The protocol options that need to be configured for an external connection depend entirely on the type of protocol used for the connection. The Protocol Code field specifies the type of protocol to be used. The following paragraphs describe each of the options that can or must be configured for each possible protocol.

Note: All option values entered in the Protocol Options field must start, end, and be separated with a period (.). For example, if you enter the HDR2 option by itself in the Protocol Options field, you would type “.HDR2.”. If you enter both the HDR2 and NOREPLY option in the Protocol Options field, you would type “.HDR2.NOREPLY.”.

TCP/IP

When using the TCP/IP, TCP/IP - Client, or TCP/IP - Server protocols, you must configure one of the following options in the Protocol Options field: HDR2 or R2LF. You can also optionally enter any or all of the following options in the Protocol Options field: EOT, NOREPLY, and SSL. The options specified in this field define how TCP/IP data streams are tied to messages.

The HDR2 option indicates that the TCP/IP stream is prefaced with a two-byte binary message length. Note that this is a requirement for connecting to NET24-XPNET and BASE24-eps C++ servers using TCP/IP for pass-through message processing. The R2LF option indicates to read until a line feed character is encountered.

The optional EOT option indicates whether the ACI application removes the End-of-text (EOT) character from response messages received from the external entity. The optional NOREPLY option indicates that the ACI application does not expect to receive a response message in reply to a request message. Therefore, it does not store any context for the request message.



The optional SSL option indicates whether the ACI application uses Secure Sockets Layer (SSL) for the connection. Cross-industry security best practices recommend that you use technologies such as SSH, VPN, or SSL/TLS to encrypt all non-console administrative access for web-based management and other non-console administrative access.

The optional ASCII, EBCDIC, and UTF8 options specify the character encoding to be used for the message payload. If you configure more than one of these encoding options, the first option configured is used. If you do not configure an encoding option, the default character encoding defined for the Java Virtual Machine (JVM) is used.

Note: The JVM encoding property is platform-specific unless specified by the Java system property "file.encoding" as a command line argument (e.g., -Dfile.encoding=ISO8859-1).

In addition to the protocol options, you must specify the destination IP address and port number and the maximum number of connections allowed. The ACI application connects to the specified IP address and port number for sending requests. The Max. Connections Allowed field indicates the maximum number of connections that can be created as workload increases. A value of 0 or less in this field indicates that an unlimited number of connections can be created and they are acquired as they are needed. Any value greater than zero is initialized at process startup.

The following table depicts the valid protocol options and protocol-defined field values for the TCPIP, TCPIP-CLIENT, and TCPIP-SERVER protocols. The first column of the table lists the applicable fields on the External Connections Configuration window and the second column provides guidelines on how to set each applicable field.

Field	Description
Protocol Options	One of: HDR2 or R2LF. Optionally EOT to remove EOT from response. Optionally .NOREPLY. to indicate to send the message but don't wait for a response as none will be returned. Optionally .SSL. to enable a Secure Sockets Layer (SSL) connection.
Protocol Field #1	Destination IP address
Protocol Field #2	Port number
Protocol Field #3	Maximum Connections Allowed. Zero or less indicates unlimited and they will be acquired as they are needed. Anything greater than zero will be initialized at process startup.

HTTP



When using the HTTP protocol, you can optionally set the Protocol Options field to a value of “.SSL” to use the Secure Sockets Layer (SSL). Cross-industry security best practices recommend that you use technologies such as SSH, VPN, or SSL/TLS to encrypt all non-console administrative access for web-based management and other non-console administrative access.

You can also optionally set the NOREPLY option to indicate that the ACI application does not expect to receive responses from the external entity. The Visa 3-D Secure connection for the ACI Commerce Gateway product requires the SSL option.

In addition to the protocol options, you must specify the destination IP address and port number and the maximum number of connections allowed. The ACI application connects to the specified IP address and port number for sending requests.

The following table depicts the valid protocol options and protocol-defined field values for the HTTP protocol. The first column of the table lists the applicable fields on the External Connections Configuration window and the second column provides guidelines on how to set each applicable field.

Field	Description
Protocol Options	Optionally .SSL. to enable a Secure Sockets Layer (SSL) connection. Optionally .NOREPLY. to indicate to send the message but don't wait for a response as none will be returned.
Protocol Field #1	Destination URL
Protocol Field #2	HTTP Content type (to override default).
Protocol Field #3	Not applicable

JMS-ASYNC

When using the JMS - Asynchronous protocol, you can optionally enter the NOREPLY option to indicate that the ACI application does not expect to receive responses from the external entity.

When connecting to a BASE24-eps Integrated Server process, two additional options are required—MDSHDR and NONMQJMSTARGET. The MDSHDR option indicates to prepend a Message Delivery Subsystem (MDS) header to request messages. This option is required for all request messages sent to a BASE24-eps Integrated Server process. The NONMQJMSTARGET option is also required when connecting to a BASE24-eps Integrated Server process using WebSphere MQ. This option indicates to not include the MQRFH version 2 header (MQRFH2) in messages sent to a WebSphere MQ queue for a non-JMS application, such as the BASE24-eps Integrated Server process.

In addition to the above protocol options, you must specify whether the ACI application uses a JMS connection type of QUEUE or TOPIC. Use the QUEUE connection type if the ACI application connects to the external entity using a JMS or WebSphere MQ queue. Use the TOPIC connection type if the ACI application connects to the external entity using a JMS topic. A JMS topic identifies a *publish/subscribe* destination type for a JMS server. Currently, the TOPIC value is reserved for future use. If you are using a JMS or WebSphere MQ queue to connect, you must also specify the name of the destination queue in the Destination Queue Name field, and, optionally, the reply queue in the Reply Queue Name field. If you are using the NOREPLY option, you can leave the Reply Queue Name field empty.

You can have multiple AJSF application processes share the same external connection configurations by entering “{PROCESSID}” in the Reply Queue Name field. The “{PROCESSID}” syntax indicates that a response from the external entity is to be delivered to the specific process that sent the message. It indicates that the process ID of the process, as specified in the process.id JVM runtime argument or configuration properties file, is substituted as the reply queue name. For example, you can use the “{PROCESSID}” syntax for BASE24-eps ATM (Java) Device Handler process connections to the BASE24-eps Integrated Server process. All response messages from the BASE24-eps Integrated Server process must be sent to the originating BASE24-eps ATM (Java) Device Handler process for performance reasons because the transaction message context is maintained in memory instead of being written to disk.

The following table depicts the valid protocol options and protocol-defined field values for the JMS-ASYNC protocol. The first column of the table lists the applicable fields on the External Connections Configuration window and the second column provides guidelines on how to set each applicable field.

Field	Description
Protocol Options	Optionally .NOREPLY. to indicate not to store context as there will be no corresponding response messages.
Protocol Field #1	JMS Connection Type: QUEUE or TOPIC
Protocol Field #2	Destination queue name
Protocol Field #3	Reply to queue name

JMS-SYNC

When using the JMS - Synchronous protocol, there are no protocol options. You only need to specify the JMS connection type and destination queue name as described above for the JMS-Asynchronous protocol.

The following table depicts the valid protocol options and protocol-defined field values for the JMS-SYNC protocol. The first column of the table lists the applicable fields on the External Connections Configuration window and the second column provides guidelines on how to set each applicable field.

Field	Description
Protocol Options	Not applicable
Protocol Field #1	JMS Connection Type: QUEUE or TOPIC
Protocol Field #2	Destination queue name
Protocol Field #3	Not applicable

EJB

When using an Enterprise JavaBeans (EJB) protocol, you must configure the name of the bean to connect to. This protocol is currently reserved for future use.

The following table depicts the valid protocol options and protocol-defined field values for the EJB protocol. The first column of the table lists the applicable fields on the External Connections Configuration window and the second column provides guidelines on how to set each applicable field.

Field	Description
Protocol Options	Not applicable
Protocol Field #1	EJB name
Protocol Field #2	Not applicable
Protocol Field #3	Not applicable

SMTP

When using the SMTP protocol, no protocol configuration options are configured. You only need to select the SMTP option in the Protocol Code field on the Protocol Details tab of the External Connections Configuration window.

The following additional configuration must be in place in order to send email messages using SMTP.

- The JavaMail JAR file and its required JARs or the J2EE.jar file must exist in your class path.
- The `com.aciworldwide.framework.message.j2ee.mail.smtp` framework package contains the software used to connect to an SMTP host and send an email message to it. This package requires the following properties in the `config.properties` file for the AJSF application. Refer to section 5 for detailed descriptions of these properties.

Property	Description
<code>mail.smtp.host</code>	The name of the SMTP mail server.
<code>mail.smtp.from.addr</code>	The email address to be used as the “From” address.
<code>mail.smtp.auth.required</code>	A boolean flag indicating whether the SMTP mail server requires a user to be authenticated with it before accepting messages from the user.

Property	Description
mail.smtp.auth.user	The user name to use for authentication if the SMTP mail server requires authentication.
mail.smtp.auth.password	The password to use for authentication of the user if the SMTP mail server requires authentication.
mail.smtp.max.retries	The maximum number of times to retry establishing a connection to the SMTP host when sending a message.
initialization.function<text-string>	An initialization.function<text-string> property, where <text-string> is a unique text string for this initialization.function property name, must be configured for the following class: com.aciworldwide.framework.message.j2ee.mail.smtp.SMTPProcess

- The following must exist in the tranObj DataObject to send an email message:

Field	Description
MESSAGE_MAIL_RECIPIENT_ADDR_STR	The email address of the recipient. Note that according to RFC822, this value must be a US-ASCII string.
MESSAGE_MAIL_SUBJECT_STR	A string containing the Subject header field. This should be the Unicode-encoded subject line. The JavaMail classes perform the conversion between Unicode and the supplied subject character set encoding string value (<i>see</i> MESSAGE_MAIL_SUBJECT_ENCODING_STR).
MESSAGE_MAIL_SUBJECT_ENCODING_STR	An indicator of the character set of the supplied Subject header (e.g., iso-8859-1, us-ascii).

Field	Description
EXTERNAL_REQUEST_BYTE	A byte[] containing the email message (text) content. This should be a byte[] containing the Unicode encoded string. The JavaMail classes perform the conversion from Unicode to the character set identified by the supplied content character set encoding string value (<i>see</i> MESSAGE_MAIL_CONTENT_ENCODING_STR).
MESSAGE_MAIL_CONTENT_ENCODING_STR	An indicator of the character set of the supplied content (e.g., iso-8859-1, us-ascii).

The following is an example of the tranObj DataObject contents for sending an email message (contents shown in blue text):

```

InitializationManager.init();
DataObject tranObj = new DataObject();
tranObj.put(EXTERNAL_CONNECTION_ID_STR, "MLBMAIL");

tranObj.put(MESSAGE_MAIL_RECIPIENT_ADDR_STR, "bosakm@aciworldwide.com");
tranObj.put(MESSAGE_MAIL_SUBJECT_STR, "Página Imprimible Código de Moneda Nueva");
tranObj.put(MESSAGE_MAIL_SUBJECT_ENCODING_STR, "iso-8859-1");
tranObj.put(MESSAGE_MAIL_CONTENT_ENCODING_STR, "iso-8859-1");
tranObj.put(EXTERNAL_REQUEST_BYTE, "Página Imprimible Código de Moneda Nueva".getBytes());

MessageRouterService msgSrvc = new MessageRouterService();
msgSrvc.sendMessage(tranObj);

```

PATHSEND

When using the PATHSEND protocol, you must configure the PPD name of the Pathmon process (e.g., \$APMN), the server class of the Pathway server to connect to (e.g., SERVER-TSECIS), and the symbolic name of the destination server (e.g., TSECIS). The MDSHDR option indicates to prepend a Message Delivery Subsystem (MDS) header to request messages sent to the destination.

The configuration requirements for adding an external connection to the EXTRCNCT using the PATHSEND protocol are provided in the following table. The first column of the table lists the applicable fields on the External Connections Configuration window and the second column provides guidelines on how to set each applicable field for the PATHSEND protocol.

Field	Description
External Connection ID	User-defined connection ID.
Connection Description	User-defined text description of the connection (e.g., "PATHSEND - Send a message to an NSK Pathway server.").
Message Format Class	The message formatter class provided by the AJSF-based application.
Timeout Limit	The amount of time (in seconds) to wait before the request is timed out.
Initial Connections	Not applicable.
Retry Attempts	The number of times a failed request should be retried before failing the request.
Maximum Connections	Not applicable.
Protocol Code	PATHSEND
Protocol Options	The MDSHDR option indicates to prepend a Message Delivery Subsystem (MDS) header to request messages.
Pathmon PPD	The PPD name of the Pathmon process for the Pathway application (e.g., \$APMN).
Pathway Server Class	The name of the server class within the Pathway application. For example, this could be "SERVER-NCP" for sending requests to the Network Control Point (NCP) process.
Symbolic Destination	The symbolic name of the destination server (e.g., TSECIS).

HSM Configuration

For AJSF-based applications that require a hardware security module (HSM) to perform cryptographic functions, you must configure connections to the HSM in the HSM Configuration data source (HSMCNFG). In addition, you must configure the “HSMService” in an `initialization.function<text-string>` property in the APPLCNFG for the AJSF server process, as shown in the following example:

```
initialization.function.0.classname=com.aciworldwide.common.crypto.hsm.  
HSMService
```

Currently, the AJSF supports Atalla AKB Network Security Processors (NSPs) and Thales e-Security 8000 HSMs.

Note: The HSMService uses Java Native Interface (JNI) libraries while the Encryption subsystem uses Java Cryptographic Extensions (JCE) pluggable providers. Therefore, HSM vendor support in the HSMService is independent of the HSM vendor support in the Encryption subsystem, and vice versa.

Field Descriptions

The following paragraphs describe each field in the HSMCNFG.

HSM ID — A symbolic name assigned to the hardware security device or interface module (e.g., Boxcar or Racal Access Module) used to communicate with multiple hardware security devices. This field, along with the Process ID field, comprises the primary key to HSMCNFG records.

Process ID — A logical name for a Java server process. By configuring a process ID, a Java server process will only use HSMCNFG records with a matching process ID in the primary key. The value you enter in this field must match the value of the `process.id` property specified in a JVM runtime argument when the process was started.

Description Text — Free-form text to describe the hardware security device or interface module.

Vendor Number — A one-digit code used to identify the vendor for this hardware security device or interface module. Valid values are as follows:

- 0 = Atalla
- 1 = Thales e-Security

Model Number — A one-digit code used to identify the model of the hardware security device(s) accessed from this record. Valid values are as follows:

- 0 = Atalla AKB
- 1 = Thales e-Security HSM 8000

Address — The TCP/IP communication protocol address (e.g., 172.21.200.30) of the physical hardware security device or interface module used to communicate with the device.

Port Number — The TCP/IP communication protocol port number of the physical hardware security device or interface module.

Message Header Length — The length of message headers allowed for the hardware security device. This field is only used for Thales e-Security HSM 8000 devices and must match the value set when the device was installed. For Thales e-Security HSM 8000 devices, valid values are 1–255. Otherwise, this field should be set to 0.

Logical Connections — The maximum number of logical connections allowed to the hardware security device or interface module. Each logical connection is associated with a physical hardware security device or interface module. Multiple logical connections can be associated with the same physical device or interface module. The practical maximum value for this field is dictated by the maximum number of connections that the physical hardware security device or interface module supports.

PIN Pad Character — The hexadecimal character used by the hardware security device to pad PIN blocks. If you are using a Thales e-Security HSM 8000 device, you must set this field to F. Atalla AKB NSPs can support any hexadecimal value of 0–9 or A–F, depending on the type of PIN block used.

Max PIN Length — The maximum PIN length used for a Thales e-Security HSM 8000 device. The value of this field must match the installed value for the device. For a Thales e-Security HSM 8000 device, valid values are 4–12. Otherwise, you should set this field to 0.

Example Configuration

The following is an example HSMCNFG configuration:

Field Name	Value
HSM ID	HSMTHALES30
Process ID	AJSFAPP1
Description	THALES 8000 30
Vendor Number	1
Model Number	1
Address	172.21.200.30
Port Number	1500
Message Header Length	4
Logical Connections	5
PIN Pad Character	F
Maximum PIN Length	4

Listener Configuration

The Listener Configuration window or screen maintains Listener configuration data in the LISTENER. Using this window or screen, you can add, delete, or modify queues for which the AJSF application is to listen for messages received from an external entity (e.g., client website, ATM, C++ process, etc.) or another AJSF-based application. AJSF-based applications perform all data communications from external entities using entries in the LISTENER. External entity-specific formatters are used to translate messages from each of the external entities. JMS internal listener types are used for one AJSF-based application to communicate with another AJSF-based application. JMS internal listener types assume that the message is composed of a DataObject serialized in XML format.

Currently, the following protocols are supported to connect to an external entity. Additional protocols may be added in the future as required.

- Java Message Service (JMS) External Request
- JMS External Response
- JMS Internal Request
- JMS Internal Response
- Transmission Control Protocol/Internet Protocol (TCP/IP)
- Transmission Control Protocol/Internet Protocol (TCP/IP) - Client
- Transmission Control Protocol/Internet Protocol (TCP/IP) - Server

Note: The LISTENER is not used for the HTTP protocol. HTTP is the recommended protocol for most ACI Commerce Gateway installations due to its ability to easily traverse firewalls.

Several fields in the LISTENER must be configured in accordance with the protocol used for the queue connection. These dependencies are described following a description of each of the fields in the LISTENER under the [“Protocol Configuration Options”](#) topic below.

The Listener ID field contains the logical name of the Listener process for the connection. Listener processes use the value of the ListenerID field as the queue name they are to listen on. The Process ID field contains the logical name of the AJSF-based application process to receive messages from this Listener process. You can associate multiple AJSF-based application processes with a single Listener process. The value you enter in this field must match the value of the process.id property specified in a JVM runtime argument when the process was

started. For the APF Admin application, the value in this field must also be defined in the Process Configuration data source (PROCESS) before you can enter it here.

General Configuration Options

Several general configuration options in the LISTENER describe the external or internal connection, specify the formatter class, method, and adapter used to build and parse messages for the connection, and configure the number of connections for the external or internal connection.

The Maximum Connections field specifies the maximum number of socket connections that the application allows to be created by another entity that is attempting to communicate with it. This value is used when a TCP/IP protocol is configured in the Listener Type field.

The Message Format Class field specifies the name of the invoked formatter class that builds request messages to and parses response messages from the external entity. For connections from another AJSF-based application (i.e., the Listener Type field is set to a JMS-INTERNAL value), the com.aciworldwide.common.CommonXMLFormatter is used to translate messages; thus, you do not need to configure a formatter class in this field.

The Message Format Method field specifies the message format method to be invoked by the message format class specified in the Message Format Class field before sending or after receiving a message.

The Message Adapter Class field specifies the class name of the application adapter that is invoked by a request after the message formatter is invoked.

Note: The names entered in the Message Format Class and Message Adapter Class fields must be defined in the classpath of the AJSF-based application.

Protocol Configuration Options

The protocol options that need to be configured for a Listener process depend entirely on the type of listener used for the connection. The Listener Type field specifies the type of protocol to be used. The following paragraphs describe each of the options that can or must be configured for each possible protocol. Note that all option values entered in the Protocol Field Options field must start, end, and be separated with a period (.). For example, if you enter the HDR2 option by itself in

the Protocol Field Options field, you would type “.HDR2.”. If you enter both the HDR2 and NOREPLY option in the Protocol Field Options field, you would type “.HDR2.NOREPLY.”.

TCP/IP

When using the TCP/IP, TCP/IP - Client, or TCP/IP - Server protocols, you must configure one of the following options in the Protocol Field Options field: HDR2 or R2STR. You can also optionally enter one of the following options in the Protocol Field Options field: NOREPLY, SSL, or CLIENTAUTH. The options specified in this field define how TCP/IP data streams are tied to messages.

The HDR2 option indicates that the TCP/IP stream is prefaced with a two-byte binary message length. Note that this is a requirement for connecting to NET24-XPNET and BASE24-eps C++ servers using TCP/IP for pass-through message processing. The R2STR option indicates to read until a specified text string character is encountered in the message. You must enter this text string in the Termination String field.

The optional NOREPLY option indicates that the client does not expect to receive a response message from the AJSF-based application in reply to a request message. Therefore, it does not store any context for the request message.



The optional SSL option indicates to use the Secure Sockets Layer (SSL) in communications between the client and the AJSF-based application. With this option, the client authenticates the AJSF-based application before making a connection. Cross-industry security best practices recommend that you use technologies such as SSL/TLS to encrypt all non-console administrative access for web-based management and other non-console administrative access.

The optional CLIENTAUTH option indicates whether the AJSF-based application enforces SSL client authentication. With this option set, the AJSF-based application authenticates the client before accepting a connection. When using the CLIENTAUTH option, the AJSF-based application must have a database of valid client machines previously loaded.

The optional ASCII, EBCDIC, and UTF8 options specify the character encoding to be used for the message payload. If you configure more than one of these encoding options, the first option configured is used. If you do not configure an encoding option, the default character encoding defined for the Java Virtual Machine (JVM) is used.

Note: The JVM encoding property is platform-specific unless specified by the Java system property "file.encoding" as a command line argument (e.g., -Dfile.encoding=ISO8859-1).

The following table depicts the valid protocol options and protocol-defined field values for the TCPIP, TCPIP-CLIENT, and TCPIP-SERVER protocols. The first column of the table lists the applicable fields on the Listener Configuration window and the second column provides guidelines on how to set each applicable field.

Field	Description
Protocol Options	One of: HDR2 or R2STR; Optionally SSL and CLIENTAUTH (with SSL).
Protocol Field #1	Port to listen on.
Protocol Field #2	String to terminate read with if R2STR specified.
Protocol Field #3	Not applicable

JMS External

When using the JMS External Request or JMS External Response protocols, you can optionally enter MDSHDR and NONMQJMCLIENT in the Protocol Field Options field. No other protocol-specific information is configured.

The MDSHDR option indicates to prepend a Message Delivery Subsystem (MDS) header to request messages. This option is required for all request messages sent to a BASE24-eps Integrated Server process. The NONMQJMSTARGET option is also required when connecting to a BASE24-eps Integrated Server process using WebSphere MQ. This option indicates to not include the MQRFH version 2 header (MQRFH2) in messages sent to a WebSphere MQ queue for a non-JMS application, such as the BASE24-eps Integrated Server process.

The following table depicts the valid protocol options and protocol-defined field values for the JMS-EXTERNAL-REQUEST and JMS-EXTERNAL-RESPONSE protocols. The first column of the table lists the applicable fields on the Listener Configuration window and the second column provides guidelines on how to set each applicable field.

Field	Description
Protocol Options	Optionally MDSHDR and NONMQJMSCLIENT
Protocol Field #1	Not applicable
Protocol Field #2	Not applicable
Protocol Field #3	Not applicable

JMS Internal

You do not need to configure any options for the JMS Internal Request (JMS-INTERNAL-REQUEST) or JMS Internal Response (JMS-INTERNAL-RESPONSE) protocols.

Message Context

The ContextManager maintains message context for AJSF-based processes according to the setting of the [context.type](#) property. The context could be a message for an ACI desktop user interface session or an online transaction, depending on the AJSF-based application process. For example, the USEC subsystem uses both the Active Session table (ActSess) and session context in subsequent processing. Thus, this message data must be available in context for the entire session. Session data is added when a session is established (i.e., a user login) and removed when the session is removed (i.e., a user logout) or the session times out.

The AJSF ContextManager has several options to manage session context as described in the following table.

Contex.type Property Values	Description
MEM-IN-PROCESS (Default)	Store message context in the memory of the current process.
DISK	Store <i>shared</i> context on disk. More than one AJSF process can access the context managed by ContextManager.
DISK-WITH-CACHE	Store <i>shared</i> context on disk. More than one AJSF process can access the context managed by ContextManager. The AJSF process checks for context in a local cached copy prior to going to disk.
MEM-OUT-OF-PROCESS	Store <i>shared</i> context in the memory of another process (Context Manager process). More than one AJSF process can access the context managed by ContextManager.

Contex.type Property Values	Description
MEM-OUT-OF-PROCESS-WITH-CACHE	Store <i>shared</i> context in the memory of another process (Context Manager process). More than one AJSF process can access the context managed by ContextManager. The AJSF process checks for context in a local cached copy prior to going to another process.

When the context.type property is set to “MEM-OUT-OF-PROCESS” or “MEM-OUT-OF-PROCESS-WITH-CACHE,” the [context.manager.connection.id](#) property specifies the external connection ID of an entry in the External Connection data source (EXTRCNCT) that is used to send the context message to the Context Manager process. When the context.type property is set to “DISK”, the [context.retention.period](#) property indicates the number of days context is retained before the ContextManager deletes it.

The AJSF ContextManager uses the Message Context data source (MSGCNTX) to maintain message context. The following is an example of the layout of rows in the MSGCNTX.

```

(MSGID                VARCHAR(200) NOT NULL,
  SEQNUM              INTEGER NOT NULL,
  CONTFLG             CHAR(1) ,
  EXPTM              TIMESTAMP,
  CONTEXT             BLOB(1 M) ,
  CONTEXT_ROWID       ROWID NOT NULL
                      GENERATED ALWAYS,
  PRIMARY KEY (MSGID,
              SEQNUM) )

```

Process Configuration

The Process Configuration window or screen maintains process configuration data in the PROCESS. The PROCESS is used when a AJSF-based application needs to broadcast messages to other similar processes in a clustered configuration to keep cached application data in synchronization. If two or more AJSF-based application processes (e.g., the BASE24-eps Java User Interface process and BASE24-eps ATM (Java) Device Handler processes) access one or more of the same cached data sources, they must be configured in the PROCESS so that the shared cached data remains synchronized.

The Process ID field contains the logical name of the AJSF-based application and must match the value of the process.id property specified when the application was started.

The External Connection ID field specifies the key to an EXTRCNCT row that defines the internal protocol mechanism used to send system configuration messages to processes configured in a cluster for an application. The value in this field must match a value in the External Connection ID field in the EXTRCNCT.

The Process Description field is a user-defined text description of this AJSF-based application process.

Note: On the ACI desktop user interface, process IDs must be configured in the PROCESS before they are displayed in selectable lists on the following windows:

- Application Configuration
- Configuration Properties
- External Connection Configuration
- Listener Configuration
- System Monitor
- System Properties

For the ACI desktop user interface, you must enter the same value in the External Connection ID column that you enter in the Process ID column.

Clustered Configurations

AJI cluster functionality allows for configuring multiple AJI-based application processes on one server or across multiple servers. AJI cluster functionality uses the AJSF's message services to communicate from one AJI-based process to another. This functionality is useful for configuring applications for high availability and scalability. It is used the most for applications that deploy multiple processes that perform different types of work. For example, the BASE24-eps ATM (Java) Device Handler processes and the BASE24-eps Java User Interface servlets in the ESWEB server process access the same AJI data sources, for which the data is cached in each of the processes. Thus, these processes need to communicate to each other to keep their database cache in synchronization.

Supported Protocols

The configuration of clustering for an AJI application is dependent on the protocol selected. The current protocols supported for clustering are the following:

- HTTP
- TCP/IP
- JMS

The protocol to use for clustering depends on the application environment and the middleware available.

HTTP

One of the common configurations for messaging between processes is HTTP. HTTP is recommended if the AJI process receiving the message is running in a web application container such as WebSphere Application Server or Tomcat. HTTP is also the recommended protocol for most web application processes due to its ability to easily traverse firewalls. The InternalXMLServlet is used to transport the message. This must be configured in the web.xml file of the receiving AJI process. The following is an example of the settings that need to be added to an AJI web application's web.xml file for receiving admin messages:

```
<servlet>
  <servlet-name> InternalXMLServlet </servlet-name>
  <display-name> InternalXMLServlet </display-name>
  <servlet-class>com.aciworldwide.framework.
InternalXMLServlet </servlet-class>
```

```
</servlet>
```

TCP/IP

Another of the common protocols for messaging between AJI processes is TCP/IP. TCP/IP is recommended if the AJI process receiving the message is NOT running in a web application container such as WebSphere Application Server or Tomcat and the application does not utilize a Message Oriented Middleware (MOM) product such as Websphere MQ or XPNET. In this case, the protocol code column in the External Connection data source (EXTRCNCT) row for the sending AJI process would be set to TCPIP or TCPIP-CLIENT. The .HDR2. option should be used to aid the receiving process in determining when it has received a whole message. You can optionally configure the EOT and .NOREPLY. options.

The TCPIP-SERVER protocol must also be configured in a Listener data source (LISTENER) entry for the receiving AJI process.

JMS

The third option for messaging between processes is JMS. This is recommended for products that utilize a Message Oriented Middleware (MOM) product such as WebSphere MQ or XPNET. In this case the protocol code column in the EXTRCNCT row for the sending AJI process would be set to JMS-SYNC. The JMS connection type would be set to QUEUE or TOPIC and the destination queue name would be configured. For the receiving AJI process, a LISTENER entry would be configured with the JMS-EXTERNAL-REQUEST protocol and the optional MDSHDR and NONMQJMSCLIENT protocol options.

Configuration

The mechanism for keeping all the executing processes for an AJSF-based application in synchronization with each other in a clustered configuration involves the Process and External Connection data sources (PROCESS and EXTRCNCT, respectively) and optionally, the Listener data source (LISTENER).

The LISTENER is only used if the protocol for communicating between the processes is something other than HTTP. If the protocol used is a TCP/IP or JMS protocol, the LISTENER is required. HTTP is the recommended protocol for most installations due to its ability to easily traverse firewalls.

Process Data Source (PROCESS)

A PROCESS entry must be defined for every process in the clustered configuration. The process entries must point to different EXTRCNCT entries. The names can be anything desired.

The PROCESS is used when a AJI-based application needs to broadcast messages to other similar processes in a clustered configuration to keep cached application data in synchronization. It is also used for broadcasting general administrative messages. If two or more AJI-based application processes (e.g., the BASE24-eps Java User Interface (ESWEB) process and BASE24-eps ATM (Java) Device Handler processes) access one or more of the same cached data sources, they must be configured in the PROCESS so that the shared cached data remains synchronized.

External Connection Data Source (EXTRCNCT)

AJI-based applications perform all data communications to external entities using entries in the EXTRCNCT. External entity-specific formatters are used to build and parse messages to each of the external entities. The AJI reads the EXTRCNCT at start up and attempts to connect a process to all configured external entities. For administrative messages such as cache refreshes, the AJI uses an internal formatter and does not use the EXTRCNCT entry defined formatter even if one was configured.

Listener Data Source (LISTENER)

The LISTENER is used to configure listener entities such as JMS queues or TCP/IP listeners for which the AJI application is to listen for messages received from an external entity (e.g., client website, ATM, C++ process, etc.) or another AJI-based application process. The LISTENER is NOT used when a process needs to receive an HTTP message. It is assumed that the AJI process is running in a web application container (e.g., Tomcat or WebSphere Application Server) and the container is listening for the message. AJI-based applications perform all data communications from external entities using entries in the LISTENER. External entity-specific formatters are used to translate messages from each of the external entities. JMS internal listener types are used for one AJI-based application to communicate with another AJI-based application for administrative purposes. As with the External Connection table entries, for messages such as cache refreshes, the AJI uses an internal formatter and does not use the LISTENER entry defined formatter even if one was configured. JMS internal listener types assume that the

message is composed of a DataObject serialized in XML format. The AJI reads the LISTENER at start up and attempts to create listener entries for connectivity to all configured external entities.

Process ID

The mechanism for synchronizing AJI processes is dependent on the process.id system property that needs to be assigned to each executing AJI process. Process IDs must be unique among all processes accessing the same database catalog. The following is an example of how the process ID is set at runtime:

```
-Dprocess.id=AppProcess1
```

The PROCESS, EXTRCNCT and LISTENER all contain the process ID as part of their primary keys. These values must match the process.id system property value assigned to each process.

Synchronization Processing

The following is a step-by-step description of the process involved in broadcasting an administrative message:

1. An AJI process performs some function that requires it to notify all other processes such as updating a cached services database row.
2. The AJI administrative broadcast functionality, reads all rows in the process table. The process table defines the administrative external connection ID name to send to for each process.
3. For each PROCESS row, other than the processes own row, the AJI uses the PROCESS rows external connection ID and the executing AJI applications own process ID to look up the external connection to send to. As stated above a process knows its own process name by the process.id system property.
4. After finding the EXTRCNCT row, the AJI uses the information in that row to send the administrative message to the other process. If the message protocol is TCP/IP, each of the processes would have created a socket listener, at startup, as defined in the LISTENER. This socket listener is created to accept the administrative messages that are sent to the process.

Example HTTP Clustered Configuration

The example configuration below assumes that there are two processes configured on separate servers. The clustering mechanism can be used for processes on the same server also. This example depicts the configuration for utilizing the HTTP protocol for messaging between processes. Thus, a LISTENER configuration is not required. Each table indicates the value for each column in table row entry. For convenience, the name used for the column on an ACI desktop window or browser page is provided in parentheses.

PROCESS Entry 1:

PROCESS Column (UI Field Name)	Value
PrcsID (Process ID)	CMRCGWTY1
ExtrCnctID (External Connection ID)	CMRCGTWY1

PROCESS Entry 2:

PROCESS Column (UI Field Name)	Value
PrcsID (Process ID)	CMRCGWTY2
ExtrCnctID (External Connection ID)	CMRCGTWY2

EXTRCNCT Entry 1:

EXTRCNCT Column (UI Field Name)	Value
ExtrCnctID (External Connection ID)	CMRCGWTY1
MssgFrmtClasTx (Message Format Class)	<null>
PrtcCd (Protocol Code)	HTTP
PrtcOptnList (Protocol Options)	.SSL. Optionally .NOREPLY. to indicate to send the message, but not wait for a response as none will be returned.

EXTRCNCT Column (UI Field Name)	Value
PrtcDfnd1Tx ¹ (Destination IP Address)	<server-IP-address>/servlet/InternalXMLServlet
PrtcDfnd2Tx (Port Number)	application/xml
PrtcDfnd3Tx (Max. Connections Allowed)	<null>

¹ The value of the PrtcDfnd1Tx field must use an IP address or a domain name as shown in the following examples:

127.0.0.1/servlet/InternalXMLServlet
http://e24.portal.net/servlet/InternalXMLServlet

EXTRCNCT Entry 1:

EXTRCNCT Column (UI Field Name)	Value
ExtrCnctID (External Connection ID)	CMRCGWTY2
MssgFrmtClasTx (Message Format Class)	<null>
PrtcCd (Protocol Code)	HTTP
PrtcOptnList (Protocol Options)	.SSL. Optionally .NOREPLY. to indicate to send the message, but not wait for a response as none will be returned.
PrtcDfnd1Tx ¹ (Destination IP Address)	<server-IP-address>/servlet/InternalXMLServlet
PrtcDfnd2Tx (Port Number)	application/xml
PrtcDfnd3Tx (Max. Connections Allowed)	<null>

¹ The value of the PrtcDfnd1Tx field must use an IP address or a domain name as shown in the following examples:

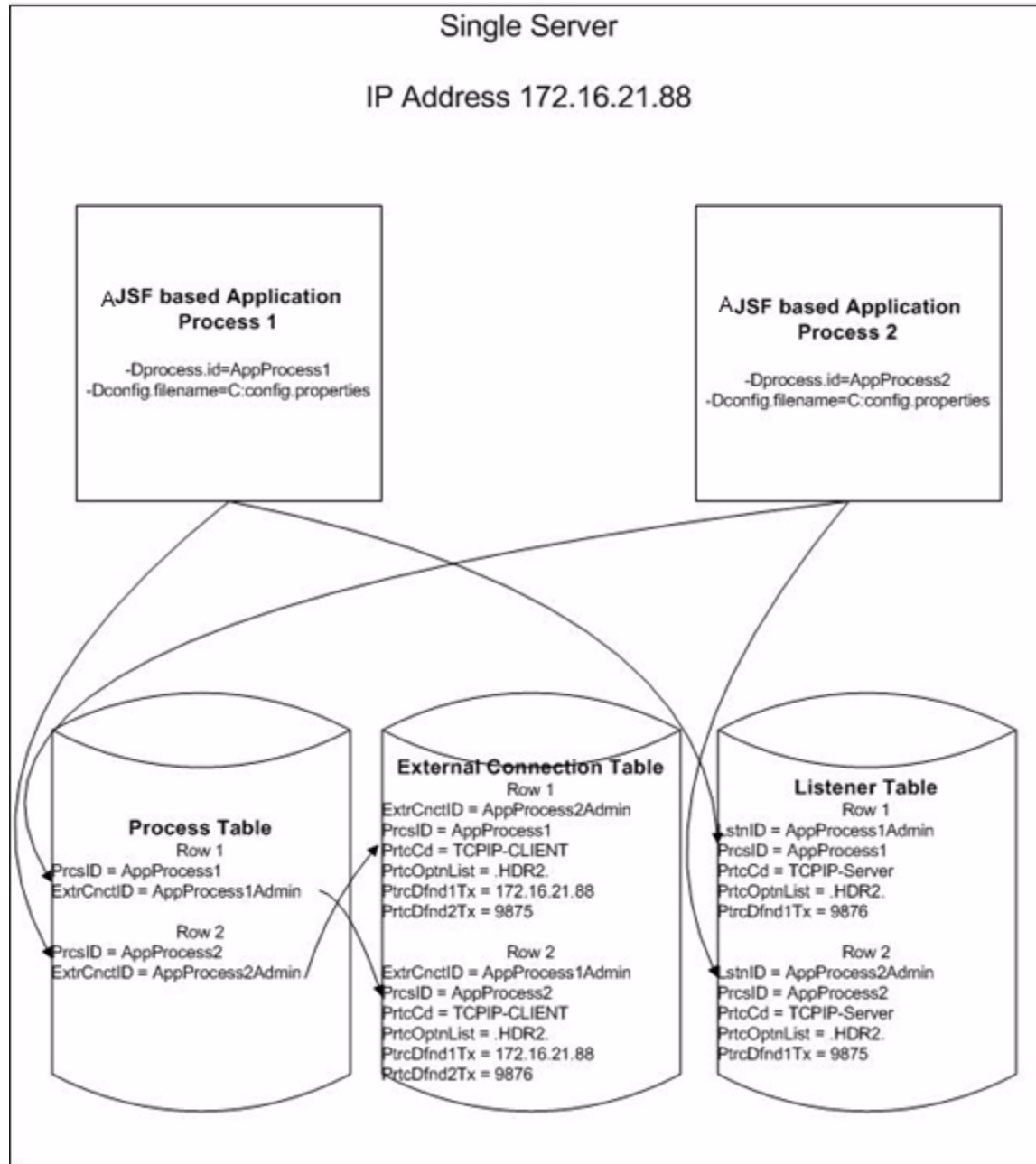
```
127.0.0.1/servlet/InternalXMLServlet  
http://e24.portal.net/servlet/InternalXMLServlet
```

Example TCP/IP Clustered Configurations

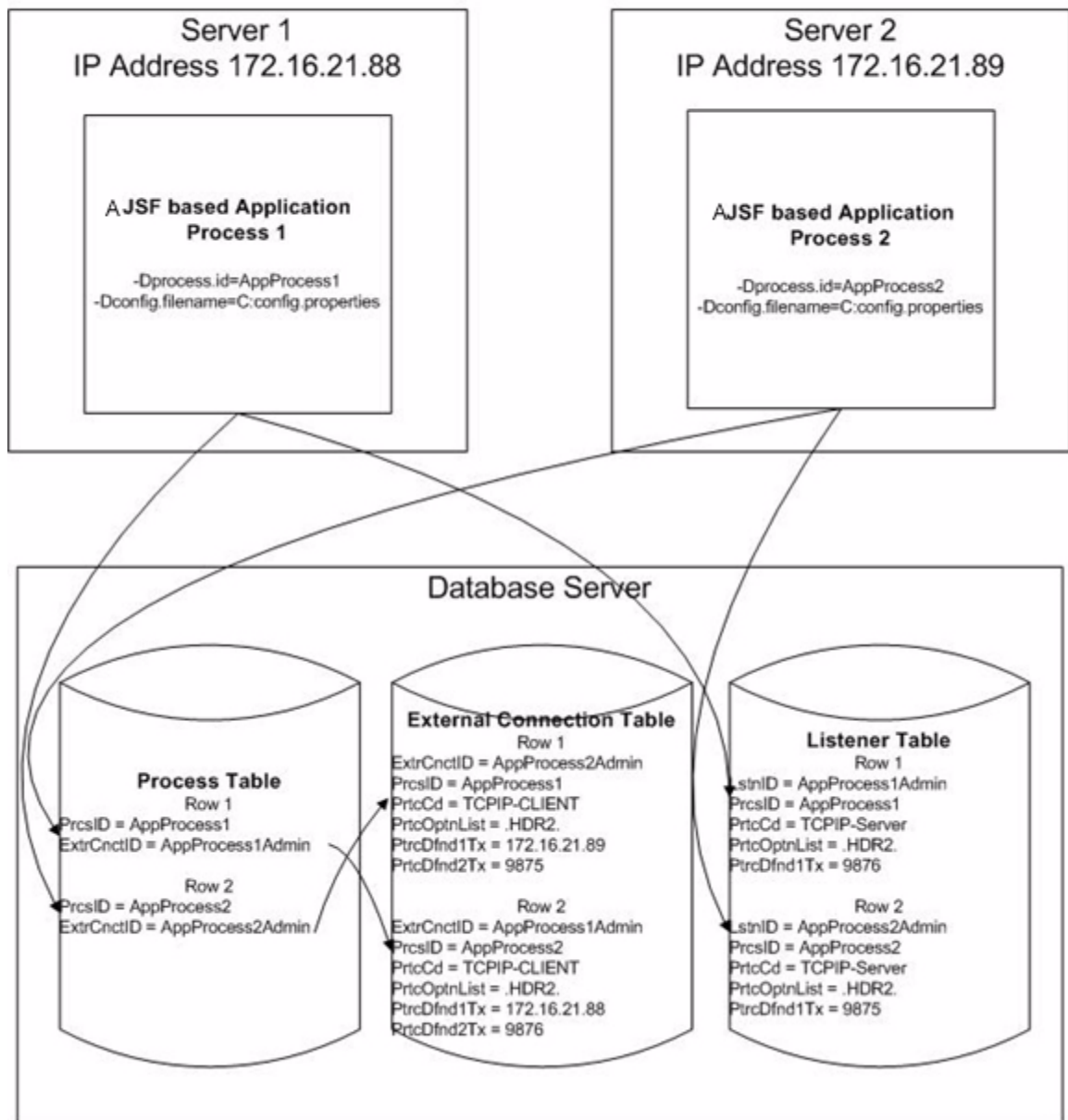
The following illustrations depict typical clustered TCP/IP configurations for single server and multiple server deployments, respectively, of AJI-based applications.

Note: The only difference in the configuration of for these two deployments is the IP address that identifies which server the processes resides on. For TCP/IP protocols, the IP address is stored in the PrctDfndlTx column in the EXTRCNCT row.

AJSF Single Server Application Configuration



AJSF Multiple Server Application Configuration



ACI Worldwide, Inc.

4: Administration

All ACI Java Server Framework (AJSF)-based applications provide facilities to perform the following administrative functions:

- View and clear objects cached in the Cache Manager within the executing Java Virtual Machine (JVM) process
- Stop and start TCP/IP sockets
- Reload application-level properties after making changes to the Application Configuration data source (APPLCNFG) and view system properties
- Monitor database usage
- Monitor system resource usage
- Log and monitor events
- Trace AJSF application processing
- Set up field masks for the masking of sensitive cardholder data displayed on AJSF-based user interface windows or browser pages.
- Securely delete files that contain sensitive cardholder data
- Set up secure communications using secure sockets layer (SSL) technology between clients and AJSF-based application servers and between the USEC subsystem and third-party authenticators.

This section provides guidelines and procedures for each of the above functions.

Cache Administration

The Cache Management feature of the AJSF allows fast access to objects that would otherwise take a long time to access. It involves keeping a copy of objects that are expensive to construct in cache memory after the immediate need for the object is over. The object can be expensive to construct for any number of reasons, such as requiring a lengthy computation or being fetched (retrieved) from a data source on disk.

When a data source is cached, the AJSF-based application first attempts to retrieve data from cache memory. If the row needed is not present in cache memory, the row is read from the data source on disk and is stored in cache memory. The row remains in the cache memory pool until the Cache Manager automatically cleans up cache pools on a regular basis such as nightly or even hourly (as configured in the `cleanup.database.cache.interval.hours`, `cleanup.dbcache.cleanup.starttime`, and `cleanup.dbcache.timeperiod.hours`) or you manually clear cache through the Cache Administration window or screen.

Note: A user interface for cache administration is not yet available for the ACI Payments Manager product.

The Service Name field on the Cache Administration window or screen identifies a Java class data entity (e.g., `EventManager`, `USECManager`) that interacts between the application and database or a database service that accesses a data source that is cached. Service identifiers are defined in metadata (`.gen` or `*Gen.xml`) text files

The Object Key field on the Cache Administration window or screen contains the key to each cached object for each database service. Key values separated by a colon indicate two sets of comma-delimited values:

- Values before the colon are the key column values used to read the cached row.
- Values after the colon are the key column values that were previously used to read the cached row.

Note: An object key of `FetchAllObj` indicates that an application performed a Fetch All function on a database service that was configured for caching in a `db.cached.table-name` property. This object contains all database table rows retrieved from the data source.

The Objects cached field displays the total number of objects cached by the Cache Manager within the executing Java Virtual Machine (JVM) process.

When you click the Clear Cache button on the Cache Administration window or screen, all objects currently cached are removed from the cache memory pool.

TCP/IP Socket Administration

Using the Socket Administration window or screen, you can stop and start inbound and outbound TCP/IP sockets for both clients and servers. Separate tabs are used to display inbound client, outbound client, inbound server, and outbound server sockets.

Note: A user interface for TCP/IP socket administration is not yet available for the ACI Payments Manager product.

Because client sockets are remote to the AJSF-based application, they are identified by both a host or IP address and a port number. Server sockets are local to the AJSF-based application and are identified with a port number only.

All outbound sockets are used for external TCP/IP connections configured in the EXTRCNCT. All inbound sockets are used for TCP/IP connections configured in the LISTENER.

Outbound connections configured with the TCP/IP or TCP/IP-CLIENT protocol in the EXTRCNCT are displayed on the Client Outbound tab of the Socket Administration window, while outbound connections configured with the TCP/IP-SERVER protocol in the EXTRCNCT are displayed on the Server Outbound tab.

Inbound connections configured with the TCP/IP or TCP/IP-CLIENT protocol in the LISTENER are displayed on the Client Inbound tab of the Socket Administration window, while inbound connections configured with the TCP/IP-SERVER protocol in the LISTENER are displayed on the Server Inbound tab.

Thus, the host or IP addresses and port numbers displayed on the Socket Administration window reflect all TCP/IP connections configured in the EXTRCNCT and LISTENER. If there are multiple connections in the EXTRCNCT or LISTENER for the same host or IP address and port number, you will see the same host or IP address and port number listed multiple times on the Socket Administration window, once for each configured connection.

For inbound sockets, the Socket Administration window or screen displays the number of sockets that were initiated remotely by a client or server for a connection. These are displayed in the Remote Connections column. In the server environment configured in the LISTENER, multiple client sockets can be established on a given TCP/IP port. The number of remote connections is a *snapshot*, at the time you accessed the Socket Administration window or refreshed the view, of the number of these sockets that have been accepted on the port.

For outbound sockets, the Socket Administration window or screen displays the number of initiated TCP/IP connections that are free. These are displayed in the Free Connections column. The maximum number of connections (sockets) to allow for a host/IP address and port combination is configured in the Max. Connections Allowed field. The number of free connections is a *snapshot*, at the time you accessed the Socket Administration window or refreshed the view, of the number of these sockets that are free (i.e., not currently in use by a thread of the AJSF application). In the AJSF, sockets (for each host/IP address and port combination) are pooled; allowing them to be shared among multiple threads.

For all socket connections, the Socket Administration window or screen displays the status of the socket, which can be Connected or Not Connected. You can stop a socket by changing the status from Connected to Not Connected and clicking Submit (). You can start a socket by changing the status from Not Connected to Connected and clicking Submit ().

The Total Connections field displays the total number of started TCP/IP connections. Note that if multiple connections are made to the same host or IP address and port, they are only counted once.

Property Administration

The AJSF allows you to view all properties currently in use by a Java server process. On the Configuration Properties window, you can view all of the properties defined to a Java server process in either the config.properties file or the Application Configuration data source (APPLCNFG). Any changes you make to properties in the APPLCNFG from the Application Configuration window do not take effect in processing until you click the Reload button on the Configuration Properties window. When you click the Reload button, the Java server process reads the APPLCNFG and reloads the property values into memory.

Note: A user interface for property administration is not yet available for the ACI Payments Manager product.

On the System Properties window, you can view all of the system-level properties currently defined to a Java server process.

To change properties in the config.properties file, you must use a text editor to change the property values in the file and then stop and restart all Java server processes that use the file.

To change system properties specified in the JVM runtime arguments, you must stop the Java server process and restart it with revised JVM runtime argument values.

To change system properties inherited from the operating system, refer to the vendor-specific operating system documentation.

Database Administration

The Database Administration window or screen enables you to view the Java DataBase Connectivity (JDBC) driver used to connect to the database, the number of clients currently using the driver, the user ID used to authenticate a connection to the database, the total number of connections, and the number of free connections. This window essentially provides a *snapshot* of the current database usage for any given Java server process. You can use this information to monitor database workload requirements and adjust database property values in the config.properties file as needed.

Note: A user interface for database administration is not yet available for the ACI Payments Manager product.

The JDBC data source driver is configured in the db.driver property. You can change the value of this property to access different types of data sources. The current value of the db.driver property in the config.properties file is displayed in the JDBC Driver Name field on the Database Administration window.

The Number of Clients field displays the current number of Java server process execution threads that are accessing the database through the JDBC driver. This number is dynamic as a thread removes a connection from the pool to perform a database access operation and then returns it to the pool when done. As work loads increases, this number will probably also increase.

The Data Source Name field displays the data source connection information required by a specific JDBC driver for a specific database. This is the data source URI specification that allows the JDBC driver to access the correct database and is configured in the db.datasource property in the config.properties file. Refer to the [db.datasource](#) property description in section 5 for more information.

The User ID field displays the user ID that Java server processes use to authenticate a connection to the database. This value is configured in the db.user.id property in the config.properties file.

The Total Connections field displays the current number of active connections from the Java server process to the database, including any static connections. Static connections are configured in the db.static.connections property in the config.properties file and remain active throughout the life of the Java server process. Static connections allow all the expense of connecting to the database to be incurred only during process startup.

Note: The number of current connections increases as work load increases and execution threads require connections that have not been established. The Java server process dynamically allocates more connections up to the maximum number configured in the `db.max.connections` property in the `config.properties` file.

Finally, the Free Connections field displays the current number of inactive connections available for the Java server process from the connection pool. This number is calculated by subtracting the number of current active connections from the maximum number of connections configured in the `db.max.connections` property in the `config.properties` file.

System Monitoring

The System Monitoring window enables you to view the current use of all AJSF system resources by a Java server process from a single window. From this window, you can view a summary of the following:

- Total number of active and free database connections and threads of execution (clients) currently accessing the database.
- Total number of TCP/IP socket connections and free TCP/IP socket connections as well as the total number of clients currently connected to the Java server process. The free (available) connections is calculated by subtracting the number of current connections from the maximum connections configured in the EXTRCNCT and LISTENER.
- Java config.properties file, log4j configuration file for generating event messages, log4j configuration file for tracing, and a flag indicating whether trace is currently enabled for the Java server process (i.e., the value of the trace.enabled property).
- The total number of objects currently cached by the Java server process.
- The total number of bytes in memory currently used by the Java server process and available in the operating system.
- The number of current active execution threads for the Java server process.

Note: A user interface for AJSF system resource monitoring is not yet available for the ACI Payments Manager product.

Because the System Monitoring window provides a summary of all AJSF system resource usage, it also contains hyperlinks to the following windows for a more detailed view of system usage by the Java server process:

- Database Administration window
- Socket Administration window
- Configuration Properties window
- Cache Administration window

Event Administration

Events generated by AJSF applications are logged according to the log4j configuration in the EventConfig.xml file specified by the event.log4j.config.url or event.config properties in the config.properties file. The event.config property allows you to use substitution variables when specifying the location of the EventConfig.xml file in the config.properties file. A substitution variable is identified by a starting value of “\${” and a closing value of “}”. The value between the braces is used as a property key to search for a value to substitute from the JVM system properties, and if not found, from the entries in the config.properties file or Application Configuration data source (APPLCNFG). This same functionality is available inside the EventConfig.xml file for specifying file locations. Thus, to specify a process-specific EventConfig.xml file located in a / config directory under the current directory, you could specify:

```
event.config=${user.dir}${file.separator}${process.id}_EventConfig.xml
```

Each of the substitution variables specifies a system, configuration (config.properties), or application (APPLCNFG) property. The AJSF substitutes the values of these properties to resolve the file location.

Also, within the EventConfig.xml file you could specify events to go to a process-specific file as shown in the following example.

```
<appender name="EVENT" class="com.aciworldwide.framework.log.
DailyRollingFileAppenderExt">
  <param name="File" value="${user.dir}${file.separator}${process.id}_Event.
log"/>
<!-- Character Encoding (z/OS Example): <param name="Encoding" value="IBM-1047"/
> -->
  <!-- Solaris Example: <param name="File" value="/logs/Events.log"/> -->
  <param name="DatePattern" value="'.'yyyy-MM-dd"/>
  <layout class="org.apache.log4j.PatternLayout">
    <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
  </layout>
</appender>
```

The EventConfig.xml file specifies the Appender and Layout components to be used for event messages generated by AJSF-based Java server processes. The Appender component specifies the output location for event messages while the Layout component specifies the format of event messages. The Filter component

allows you to determine whether an event should be raised based on multiple filter classes. The classes are consulted in the order configured. ACI uses filters to prevent an infinite loop for internal or system errors. The `org.apache.log4j.varia.DenyAllFilter` filter class is a generic filter class that denies the logging of all events.

The example `EventConfig.xml` file shown below contains different appenders for the platforms on which AJSF-based ACI applications are deployed. The appender configuration for an HP NonStop platform is named `NSK-EMS`. The appender configuration for an IBM System z or Unix platform is named by `MQJMS`.

Additional aggregates and variables may need to be uncommented or defined as required for your site.

A sample `EventConfig.xml` file is shown below.

Note: When the `EventConfig.xml` file is installed on an IBM System z platform, the encoding meta element in the XML declaration must be set to “IBM-1047” instead of “UTF-8” or the encoding meta element must be removed as shown in the following examples.

```
<?xml version="1.0" encoding="IBM-1047"?>
<?xml version="1.0" ?>
```

Note: All AJSF-based processes, except for the Event Adapter on IBM System z or Unix platforms, use an `EventConfig.xml` file similar to the example provided below. The Event Adapter uses its own `EvenConfig.xml` file, which is described in section 7.

```
<?xml version="1.0"?>
<!DOCTYPE log4j:configuration SYSTEM "log4j.dtd">
<log4j:configuration debug="false">

<!--
  Appenders define output destinations for event messages.
  The actual locations used is controlled by the <category>
  element toward the bottom of this file.

  This file comes configured to send events to files on the
  file system. When moving an application into production,
  this should be changed to a more robust location; depending
  upon platform the application is running on

  In this file are sample appenders for sending events to:
```

- 1) EMS for the HP NSK platform.
- 2) An MQ Queue for the Sun platform
- 3) A TCP/IP port when running with ES on the mainframe under CICS.

For more details on configuring the logging package, refer to

- 1) See the Logging project at the main Apache web site (www.apache.org)
- 2) For ACI supplied Appenders, refer to the JavaDocs delivered with the product.

-->

<!--

Appender for the system console (i.e., System.out)

-->

```
<appender name="SYSOUT" class="org.apache.log4j.ConsoleAppender">
  <param name="Target" value="System.out"/>
  <layout class="org.apache.log4j.PatternLayout">
    <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
  </layout>
</appender>
```

<!--

Appender for the system error console (i.e., System.err)

-->

```
<appender name="SYSERR" class="org.apache.log4j.ConsoleAppender">
  <param name="Target" value="System.err"/>
  <layout class="org.apache.log4j.PatternLayout">
    <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
  </layout>
</appender>
```

<!--

Appender for the sending events to an Error.log file on the filesystem.
This appender is enabled by default; mainly for development and installation purposes. It should be disabled (by commenting the following definition and reference in the category element below) when an application moves to production.

-->

```
<appender name="EVENT" class="com.aciworldwide.framework.log.
DailyRollingFileAppenderExt">
  <param name="File" value="C:/logs/Events.log"/>
  <!-- Character Encoding (z/OS Example): <param name="Encoding" value="IBM-
1047"/> -->
<!-- Solaris Example: <param name="File" value="/logs/Events.log"/> -->
  <param name="DatePattern" value="'. 'yyyy-MM-dd"/>
  <layout class="org.apache.log4j.PatternLayout">
    <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
  </layout>
```

```

</appender>

<!--
  Appender for the CRITICAL Events.log files.
  This is an example showing that events with different priorities can be
  logged to different destinations.  It is enabled by default and can
  be disabled if desired.

  Note that the Thrshold parameter specifies that only events with
  a Log4J priority of FATAL will be sent to this appender.
-->
<appender name="CRITICAL_EVENTS" class="com.aciworldwide.framework.log.
DailyRollingFileAppenderExt">
  <param name="Threshold" value="FATAL"/>
  <param name="File" value="C:/logs/CriticalEvents.log"/>
  <!-- Solaris Example: <param name="File" value="/logs/CriticalEvents.log"/>
-->
<!-- Character Encoding (z/OS Example): <param name="Encoding" value="IBM-1047"/
> -->
  <param name="DatePattern" value="'.'yyyy-MM-dd"/>
  <layout class="org.apache.log4j.PatternLayout">
    <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
  </layout>
</appender>

<!--
  The following appender would be used to send events to EMS on
  HP NSK servers.

  Note:
    Change the Collector value to if a different EMS collector
    is desired.
-->
<!--
<appender name="NSK-EMS" class="com.aciworldwide.framework.log.EMSAppender">
  <param name="Collector" value="$0"/>
  <layout class="org.apache.log4j.PatternLayout">
    <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
  </layout>
</appender>
-->

<!--
  The following appender would be used to send events to an MQ
  destination for ES; typically used on Unix servers (i.e., Sun, HP-UX,
  IBM AIX, etc).

  Notes:
    1) The ProviderUrl is the location of MQ JMS's ".bindings" file.
    2) Change the PREFIX portion of the QueueName parameter to match

```

the value specified when SETUPES was executed.

```
-->
<!--
  <appender name="MQJMS" class="com.aciworldwide.framework.log.JMSMQAppender">
    <param name="ProviderUrl" value="file:///db01/b24dev/jdh1/mq_jms_admin/JNDI-
Directory"/>
    <param name="QueueConnectionFactoryBindingName"
value="JmsQueueConnectionFactory"/>
    <param name="QueueName" value="PREFIX.EVTLOG"/>
    <param name="InitialContextFactory" value="com.sun.jndi.fscontext.
RefFSContextFactory"/>
    <layout class="org.apache.log4j.PatternLayout">
      <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
    </layout>
  </appender>
-->

<!--
  The following appender would be used by the Java DH on the mainframe
  to send events to the BASE24-es (CICS) logging environment via
  TCP/IP.

  Notes:
    1) The Host parameter is optional and defaults to "localhost".
    2) The Encoding parameter is optional. If not specified, the
       JVM's default character encoding is used. The values are
       those specified in the java.lang.* Package documentation.
-->
<!--
  <appender name="ES-CICS" class="com.aciworldwide.framework.log.
ACITCPAppender">
    <param name="Port" value="9211"/>
    <param name="Host" value="127.0.0.1"/>
    <param name="Encoding" value="ISO-8859-1"/>
    <layout class="com.aciworldwide.framework.log.ACISidistLayout">
      </layout>
    </appender>
-->

<!--
  The category used by Event Manager. All event messages are
  logged under this category. EventManager automatically prepends
  this category name to all requests. This example illustrates
  sending CRITICAL events to a different destination.
  In this case, it is to a different file; but it could be,
  for example, an SMTPAppender (to send an E-mail alert).
-->
  <category name="event" additivity="false">
    <priority value="info"/>
```

```

        <appender-ref ref="EVENT"/>
        <appender-ref ref="SYSOUT"/>
        <appender-ref ref="CRITICAL_EVENTS"/>
<!--      <appender-ref ref="NSK-EMS"/> -->
<!--      <appender-ref ref="MQJMS"/> -->
<!--      <appender-ref ref="ES-CICS"/> -->
    </category>

<!--
    The required root category.

    Note:
        Because the 'event' category above has it's additivity flag
        set to false, nothing should use this category.
-->
    <root>
        <priority value="debug"/>
        <appender-ref ref="SYSOUT"/>
    </root>
</log4j:configuration>

```

Event Appender

You must modify the EVENT appender in the EventConfig.xml file to reflect a valid location for event text log files (referred to in the “File” param). These would typically be placed in the same location as log files for the APF product. The “File” param definition in the following EVENT appender shows an example text file location definition:

```

<!-- Appender for the Error log files. -->
    <appender name="EVENT" class="com.aciworldwide.framework.log.
DailyRollingFileA
ppenderExt">
        <param name="File" value="#log_root#/EventAdapterEvents.log"/>
<!-- Character Encoding (z/OS Example): <param name="Encoding"
value="IBM-1047"/> -->
        <param name="DatePattern" value="'. 'yyyy-MM-dd"/>
        <layout class="org.apache.log4j.PatternLayout">
            <param name="ConversionPattern" value="%d{ISO8601} %-5p %c
[%t] %x - %m%n"
/>
        </layout>
    </appender>

```

JMSMQ Appender

All AJSF-based applications that write events to the WebSphere MQ event consolidation queue must use the JMSQAPPENDER definition in the EventConfig.xml file to reflect a valid queue name and MQJMS JNDI context. This enables any AJSF-based process, such as the ACI Application Management agent process, to log events to the same event queue for the APF product. The “ProviderUrl” and “InitialContextFactory” params need to refer to the location and class for the JNDI context that was established previously as part of the WebSphere MQ JMS installation and configuration. The “QueueName” param must be modified to point to the event MQ queue that is used by other APF product processes for event logging. The standalone Event Adapter process, described in section 7, reads from this queue and formats, filters, and writes the events to configured locations. The following is an example JMSMQAPPENDER appender.

```
<!--
```

```
The following appender would be used to send events to an MQ
destination for ES; typically used on Unix servers (i.e., Sun, HP-UX,
IBM AIX, etc) and the IBM System z.
```

```
Notes:
```

- 1) The ProviderUrl is the location of MQ JMS's ".bindings" file.
- 2) Change the PREFIX portion of the QueueName parameter to match the value specified when SETUPES was executed.
- 3) For the z/OS the following parameter may be necessary:

```
<param name="QueueEncoding" value="IBM-1047"/>
```
- 4) The Delivery Mode of the JMS message can be changed to either Non-Persistent or Persistent(DEFAULT):

```
<param name="QueueMessageDeliveryMode" value="Non-Persistent"/>
```

```
-->
```

```
<appender name="JMSMQAPPENDER"
class="com.aciworldwide.framework.log.JMSMQAppender">
  <param name="ProviderUrl" value="file:///user/mqjms/JNDI-Directory"/>
  <param name="QueueConnectionFactoryBindingName"
value="JmsQueueConnectionFactory"/>
  <param name="QueueName" value="EST.EVTLOG"/>
  <param name="InitialContextFactory" value="com.sun.jndi.fscontext.
RefFSContextFactory"/>
  <layout class="org.apache.log4j.PatternLayout">
    <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x - %m%n"
  />
  </layout>
</appender>
```

Event Message Documentation

Event messages that can be generated by AJSF applications, such as the APF Admin application, BASE24-eps ATM (Java) Device Handler and IFX ATM Manager processes, and the BASE24-eps Java User Interface servlets, are documented on the Event Message and Event Message Documentation tabs of the Event Configuration and Management window of the ACI desktop. AJSF-application messages are identified with subsystem IDs above 3000.

You can also find AJSF-application event documentation in application-specific event documentation reference guides. For example, the ***BASE24-eps Event Documentation*** guide describes all events that can be generated by BASE24-eps software in a BASE24-eps system, including AJSF-based application events.

Note: AJSF-based applications do not use tokens in event messages. Therefore, you cannot use token IDs (e.g., <1>, <2>, etc.) for text substitution in event messages generated by AJSF-based applications. AJSF-based applications use braces ({ }) instead of tokens to denote text substitution in event messages.

Additional References

You may find the following additional references useful in implementing log4j:

- Short introduction to log4j (<http://jakarta.apache.org/log4j/docs/manual.html>)
- Quick start to using log4j (<http://www.vipan.com/htdocs/log4jhelp.html>)
- Available appenders (<http://jakarta.apache.org/log4j/docs/api/index.html>)

Trace Administration

The AJSF provides two types of tracing—normal and JDAL.

Note: The Trace facility was designed as a diagnostic tool for development and test environments and not for general use in a production environment. ACI strongly recommends that it not be used in a production environment due to the overhead and impacts on performance.

Normal Tracing

Normal tracing allows you trace many different facets of AJSF processing (e.g., database connections, database content, performance, validator functions, etc.). Refer to the [“Selective Tracing”](#) topic for a list of all trace levels available.

Normal tracing is enabled with two application-level properties—`trace.enabled` and `trace.dataobject.enabled`. These properties are configured in the Application Configuration data source (APPLCNFG). Note that database tracing is included when performing a normal trace.

The `trace.enabled` property indicates whether normal tracing is performed. A value of `true` for this property enables normal tracing while a value of `false` disables it. The `trace.dataobject.enabled` property indicates whether trace messages are enabled for data object output from the Java server process when message tracing is enabled.

Examples of normal trace messages, including database trace messages, are shown below.

```
2004-08-30 12:25:44,012 trace.cache.put.com.aciworldwide.framework.CacheManager
com.aciworldwide.framework.CacheManager.put(Unknown Source)  main  - [04.1.0 -
Build 20040712-1016] Service: EXTRCNCT, category: DATABASE, key:
selextrcnctprocessid:1:JDH1 written to cache, a new service map entry was added
2004-08-30 12:25:44,013 trace.database.content.EXTRCNCT.com.aciworldwide.
common.DBService com.aciworldwide.common.DBService.traceFetch(Unknown Source)
main  - [04.1.0 - Build 20040712-1016] Fetch OK, Service Name: EXTRCNCT,
Statement ID: selextrcnctprocessid
Statement: SELECT ExtrCnctID, PrcsID, DscrTx, MssgFrmtClasTx, PrtcCd,
PrtcOptnList, PrtcDfnd1Tx, PrtcDfnd2Tx, PrtcDfnd3Tx, TimeOutNum, RtryAtmpNum,
InitConnNum, MaxConnNum, LastUpdtTm FROM extrcnct WHERE PrcsID = ?
Key Field 1: process_id_str Value: JDH1
```

```

2004-08-30 12:25:44,015 trace.database.content.EXTRCNCT.com.aciworldwide.
common.DBService com.aciworldwide.common.DBService.traceFetch(Unknown Source)
main - [04.1.0 - Build 20040712-1016] Fetch Result row #1:
  Field: external_connection_id_str Value: ATMSRPLY
  Field: process_id_str Value: JDH1
  Field: description_str Value: queue for sending out to atm
  Field: message_format_class_str Value: com.aciworldwide.base24.es.common.
formatter.ResponseFormatterController
  Field: protocol_code_str Value: JMS-ASYNC
  Field: protocol_option_list_str Value: .MDSHDR.NONMQJMSTARGET.
  Field: protocol_field_1_str Value: QUEUE
  Field: protocol_field_2_str Value: ATMSRPLY
  Field: protocol_field_3_str Value: {TERMINID}
  Field: timeout_int Value: 300
  Field: retry_attempts_int Value: 3
  Field: initial_connections_int Value: 1
  Field: maximum_connections_int Value: 1
  Field: last_update_time Value: 2004-04-15 13:30:11.438

```

JDAL Tracing

The ACI-written Java Database Access Layer (JDAL) (type 2) and Java Database Access Layer 2 (JDAL2) (type 3) database drivers allow you perform tracing of JDAL database functions. The `jdal.trace` system property indicates whether JDAL tracing is performed. A value of `true` for this property enables JDAL tracing while a value of `false` disables it. The `jdal.trace.method` system property indicates whether JDAL methods are traced when JDAL tracing is enabled. A value of `true` for this property enables JDAL method tracing while a value of `false` disables it.

Both of these properties are system properties specified in JVM runtime arguments. To ensure that JDAL tracing is disabled in a production environment, both of these properties should be set to `false` at runtime as shown below:

```

-Djdal.trace=false
-Djdal.trace.method=false

```

Examples of JDAL trace messages are shown below.

```

2004-08-30 12:25:43.993[main] JDAL (Trace) Sending SIDAL String: <GETNEXT>=<stmt
id="1718264953" typ="SELECT"> </stmt>
2004-08-30 12:25:43.998[main] JDAL (Trace) Receiving SIDAL String: Result =
ExtrCnctID='ESAUTH' ,PrctID='JDH1
',DscrTx='Connection to the BASE24-eps ISO93 Authorizer
',MssgFrmtClasTx='com.aciworldwide.base24.es.common.formatter.iso8583.es.
FormatterController

```

```
' ,PrtcCd='JMS-ASYNC'                                ' ,PrtcOptnList='.'
MDSHDR.NONMQJMSTARGET.
' ,PrtcDfnd1Tx='QUEUE
' ,PrtcDfnd2Tx='ATMISO
' ,PrtcDfnd3Tx='{PROCESSID}
' ,TimeOutNum='300' ,RtryAtmpNum='3' ,InitConnNum='1' ,MaxConnNum='1' ,LastUpdtTm='13
3013280529380000';
2004-08-30 12:25:43.999[main] JDAL (Trace) Statement: <stmt id="1718264953"
typ="SELECT"> </stmt> Error Code: jdal_ok Result More value: eof
2004-08-30 12:25:43.999[main] JDAL (Trace) Col (0) ExtrCnctID=ESAUTH
2004-08-30 12:25:43.999[main] JDAL (Trace) Col (1) PrcsID = JDH1
2004-08-30 12:25:44.000[main] JDAL (Trace) Col (2) DscrTx = Connection to the
BASE24-eps ISO93Authorizer
2004-08-30 12:25:44.000[main] JDAL (Trace) Col (3) MssgFrmtClasTx = com.
aciworldwide.base24.es.common.formatter.iso8583.es.FormatterController
2004-08-30 12:25:44.001[main] JDAL (Trace) Col (4) PrtcCd=JMS-ASYNC
2004-08-30 12:25:44.004[main] JDAL (Trace) Col (5) PrtcOptnList = .MDSHDR.
NONMQJMSTARGET.
2004-08-30 12:25:44.004[main] JDAL (Trace) Col (6) PrtcDfnd1Tx=QUEUE
2004-08-30 12:25:44.005[main] JDAL (Trace) Col (7) PrtcDfnd2Tx=ATMISO
2004-08-30 12:25:44.005[main] JDAL (Trace) Col (8) PrtcDfnd3Tx={PROCESSID}
2004-08-30 12:25:44.006[main] JDAL (Trace) Col (9) TimeOutNum = 300
2004-08-30 12:25:44.007[main] JDAL (Trace) Col (10) RtryAtmpNum = 3
2004-08-30 12:25:44.007[main] JDAL (Trace) Col (11) InitConnNum = 1
2004-08-30 12:25:44.008[main] JDAL (Trace) Col (12) MaxConnNum = 1
2004-08-30 12:25:44.009[main] JDAL (Trace) Col (13) LastUpdtTm =
133013280529380000
```

Trace Output Configuration

All trace messages are written to the logging location defined in the log4j configuration in the TraceConfig.xml file specified by the trace.log4j.config.url or trace.config properties. Because the trace log may contain sensitive data, you should secure this location by the file system so that only authorized users have access to the data. The trace.config property allows you to use substitution variables when specifying the location of the TraceConfig.xml file in the config.properties file. A substitution variable is identified by a starting value of “\${” and a closing value of “}”. The value between the braces is used as a property key to search for a value to substitute from the JVM system properties, and if not found, from the entries in the config.properties file. This same functionality is available inside the TraceConfig.xml file for specifying file locations. Thus, to specify a process-specific TraceConfig.xml file located in a /config directory under the current directory, you could specify:

```
trace.config=${user.dir}${file.separator}${process.id}_TraceConfig.xml
```

Each of the substitution variables specifies a system property. The AJSF substitutes the values of these properties to resolve the file location.

Also, within the TraceConfig.xml file you could specify trace data to go to a process-specific file as shown in the following example.

```
<appender name="TRACE" class="org.apache.log4j.FileAppender">
  <param name="File" value="${user.dir}${file.separator}${process.id}_
Trace.log"/>
  <param name="Append" value="false"/>
  <layout class="org.apache.log4j.PatternLayout">
    <param name="ConversionPattern" value="%d{ISO8601} %l [%t] %x -
%m%n"/>
  </layout>
```

The TraceConfig.xml file specifies the Appender and Layout components to be used for trace messages generated by AJSF-based Java server processes. The Appender component specifies the output location for trace messages while the Layout component specifies the format of trace messages. You can uncomment the Appender aggregate to be used for your system.

A sample TraceConfig.xml file is shown below. The File and DatePattern params specify the path and name of the trace output file. For this sample, the path and name would be: C:/CommerceFramework/Logs/trace.yyyy-MM-dd.log, where yyyy-MM-dd is the date the trace was started.

Note: When the TraceConfig.xml file is installed on an IBM System z platform, the encoding meta element in the XML declaration must be set to “IBM-1047” instead of “UTF-8” or the encoding meta element must be removed as shown in the following examples.

```
<?xml version="1.0" encoding="IBM-1047"?>
<?xml version="1.0" ?>
```

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE log4j:configuration SYSTEM "log4j.dtd">
<log4j:configuration debug="false">
  <!-- Appender for the system console for trace messages -->
  <appender name="SYSOUT" class="org.apache.log4j.ConsoleAppender">
    <param name="Target" value="System.out"/>
    <layout class="org.apache.log4j.PatternLayout">
      <param name="ConversionPattern" value="%d{ISO8601} %c %l [%t] %x -
%m%n"/>
    </layout>
```

```
</appender>

<!-- Appender for trace files for messages. -->
<appender name="TRACE" class="com.aciworldwide.framework.log.
DailyRollingFileAppenderExt">
  <param name="File" value="C:/CommerceFramework/Logs/trace.log"/>
<!-- Character Encoding (z/OS Example): <param name="Encoding" value="IBM-1047"/
> -->
  <!-- Solaris Example: <param name="File" value="/opt/CommerceFramework/Logs/
trace.log"/> -->
  <param name="DatePattern" value="'. 'yyyy-MM-dd"/>
  <layout class="org.apache.log4j.PatternLayout">
    <param name="ConversionPattern" value="%d{ISO8601}  %c  %l [%t] %x -
%m%n"/>
  </layout>
</appender>

<appender name="NULL" class="com.aciworldwide.framework.log.NullAppender">
  <layout class="org.apache.log4j.SimpleLayout"/>
</appender>

<!-- The category used by Trace Manager. All trace messages are logged under
this category. -->
<!-- TraceManager automatically prepends this category name to all requests.
For TraceManager, -->
<!-- the priority should be left at 'debug'. TraceManager always logs
messages with this priority. -->
<!-- In this example, trace messages are sent to both a trace file (identified
by an appender named -->
<!-- TRACE, and to the system error console. To only send message to the
trace file, simply remove -->
<!-- the line referring to SYSOUT -->
<category name="trace" additivity="false">
  <priority value="debug"/>
  <appender-ref ref="TRACE"/>
  <appender-ref ref="SYSOUT"/>
</category>

<!-- The following two categories illustrate one method to enable selective
tracing. Only categories whose -->
<!-- name starts with 'trace.test' will be logged to the trace file and the
system console. -->
<!-- This is accomplished by setting the priority of category named 'trace'
to 'fatal'. Note that this -->
<!-- works only for tracing because TraceManager always uses the 'debug'
priority. -->
<!-- <category name="trace.test" additivity="false">
  <priority value="debug"/>
  <appender-ref ref="TRACE"/>
  <appender-ref ref="SYSOUT"/>
</category>
<category name="trace" additivity="false">
```

```

    <priority value="fatal"/>
    <appender-ref ref="TRACE"/>
    <appender-ref ref="SYSOUT"/>
</category> -->
<!-- The required root category. Because the 'trace' category has it's-->
<!-- additivity flag set to false, nothing should use this category. -->
<root>
    <priority value="debug"/>
    <appender-ref ref="SYSOUT"/>
</root>
</log4j:configuration>

```

Application Management Integration with Tivoli

To integrate application management with Tivoli on an IBM System z platform, modify the TRACE appender definition in the TraceConfig.xml to reflect a valid location for text log files (referred to in the “File” param). Trace log files would typically be placed on \$ACIJMX_BASE/logs, but you could place them in the same location as log files for BASE24-eps processes. The “File” param defined below shows an example log file location.

```

<!-- Appender for the Error log files. -->
<appender name="TRACE" class="com.aciworldwide.framework.log.
DailyRollingFileAppenderExt">
    <param name="File" value="//user/acijmx/logs/Trace.log"/>
<!-- Character Encoding (z/OS Example): <param name="Encoding" value="IBM-1047"/
> -->
    <param name="DatePattern" value="'. 'yyyy-MM-dd"/>
    <layout class="org.apache.log4j.PatternLayout">
        <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x - %m%n"
/>
    </layout>
</appender>

```

Selective Tracing

Through the configuration of categories for an appender in the TraceConfig.xml file, you can select the type of data to be traced. Each category defines a certain type of data to be traced. By manipulating the priority value (i.e., fatal or debug) for a trace category, you can configure the TraceConfig.xml file to trace data from all categories except for those specified, or only from specified categories.

Because the AJSF only supports tracing when the priority value is set to debug, setting the priority to fatal effectively removes a category from being traced. AJSF-defined trace categories include the following:

Trace Category	Description
trace	Encompasses all possible trace categories. The names of all individual trace categories are affixed with “trace”.
trace.database.connection	Category for tracing database connections.
trace.database.transaction ¹	Category for tracing database transactions.
trace.database.content ¹	Category for tracing database access content.
trace.message.connection	Category for tracing message connections such as socket or Java Message Service (JMS) connections.
trace.message.content ¹	Category for tracing message content. Use this category for tracing message buffer contents.
trace.method	Category for tracing methods as they are invoked. Use this category for applications that use the <code>TraceManager traceMethod()</code> method.
trace.framework.performance	Category for tracing resource usage performance. This encompasses database access times and messaging response times.
trace.admin	Category for tracing administrative functionality.
trace.validator	Category for tracing validator functionality. Use this category for trace messages originating from validator classes.
trace.command	Category for tracing command functionality. Use this category for trace messages originating from command classes.

Trace Category	Description
trace.formatter	Category for tracing formatter functionality. Use this category for trace messages originating from formatter classes.
trace.datasource	Category for tracing data source functionality. Use this category for trace messages originating from data source classes.



¹ Enabling these trace functions may cause the tracing of sensitive data. Be sure that any resulting trace logs are secured according to cross-industry security best practices.

Note: Because the trace categories available may vary among AJSF-based applications, contact your ACI account representative to obtain a complete list of the trace categories available for your application.

Exclusive Category Configuration

The following configuration in the TraceConfig.xml file excludes specified categories from the trace output. In this case, you first set the priority value to debug for the trace category and then set the priority value to fatal for those categories you want to exclude from the output sent to the specified appenders. Trace data is then sent to the specified appenders for all categories, except for those with a priority value of fatal. In this example, trace output is excluded for the trace.database.connection, trace.framework.performance, trace.method, and trace.database.content categories.

```
<category name="trace" additivity="false">
  <priority value="debug"/>
  <appender-ref ref="TRACE"/>
  <appender-ref ref="SYSOUT"/>
</category>

<category name="trace.database.connection" additivity="false">
  <priority value="fatal"/>
  <appender-ref ref="TRACE"/>
  <appender-ref ref="SYSOUT"/>
</category>

<category name="trace.framework.performance" additivity="false">
  <priority value="fatal"/>
  <appender-ref ref="TRACE"/>
  <appender-ref ref="SYSOUT"/>
</category>
```

```
<appender-ref ref="SYSOUT"/>
</category>

<category name="trace.method" additivity="false">
  <priority value="fatal"/>
  <appender-ref ref="TRACE"/>
  <appender-ref ref="SYSOUT"/>
</category>

<category name="trace.database.content" additivity="false">
  <priority value="fatal"/>
  <appender-ref ref="TRACE"/>
  <appender-ref ref="SYSOUT"/>
</category>
```

Inclusive Category Configuration

The following configuration in the TraceConfig.xml file includes specified categories in the trace output. In this case, you first set the priority value to fatal for the trace category and then set the priority value to debug for those categories you want to include in the output sent to the specified appenders. Trace data is then sent to the specified appenders only for those categories with a priority value of debug. In this example, trace output is sent only for the trace.database.connection, trace.framework.performance, trace.method, and trace.database.content categories.

```
<category name="trace" additivity="false">
  <priority value="fatal"/>
  <appender-ref ref="TRACE"/>
  <appender-ref ref="SYSOUT"/>
</category>

<category name="trace.database.connection" additivity="false">
  <priority value="debug"/>
  <appender-ref ref="TRACE"/>
  <appender-ref ref="SYSOUT"/>
</category>

<category name="trace.framework.performance" additivity="false">
  <priority value="debug"/>
  <appender-ref ref="TRACE"/>
  <appender-ref ref="SYSOUT"/>
</category>

<category name="trace.method" additivity="false">
  <priority value="debug"/>
  <appender-ref ref="TRACE"/>
  <appender-ref ref="SYSOUT"/>
```



```
</category>

<category name="trace.database.content" additivity="false">
  <priority value="debug"/>
  <appender-ref ref="TRACE"/>
  <appender-ref ref="SYSOUT"/>
</category>
```

Field Masking



The AJSF provides a mask template storage and retrieval function. Client applications use this service to identify how sensitive data elements, such as card and account numbers should be presented. When a client application determines that it needs to present a sensitive data element like a card number, it requests the AJSF to provide the card number mask so that it can properly mask the leftmost or rightmost characters in the string. It also determines whether the user should be able to see the value in the clear.

Through the configuration of tags in the FieldMasking.xml file (the location of which is specified in the [xmlmaskmap.filename](#) property), you can control how sensitive data is masked for display in fields on AJSF-based user interface windows or browser pages. These controls enable you to comply with requirements specified in the *Payment Card Industry (PCI) Data Standards* for the masking of sensitive data. ACI provides three default field masks for amounts, account numbers, and primary account numbers (PANs) or card numbers and applies these masks to the appropriate fields on AJSF-based user interface windows or browser pages. Refer to the online help for a AJSF-based user interface or browser to determine whether ACI has applied one of these masks to a particular field on a user interface window or browser page.

Because some employees or other parties have specific needs to see full card numbers, usually for editing purposes, the field masking controls can allow values in masked fields to be displayed in the clear when a user has a masked field in focus or hovers the mouse over the masked field. Otherwise, the data in masked fields is masked as configured.

Mask definitions in the FieldMasking.xml control the character (e.g., #, *, etc.) used to mask data and the lefthand and righthand portions of the field value to remain unmasked. An additional control indicates whether the field value is displayed in the clear when the field is editable and is in focus or the mouse is hovering over the field. A sample of the FieldMasking.xml file with the default field masks provided by ACI is shown below. Detailed descriptions of its XML tags are provided following the sample.

```
<?xml version="1.0" encoding="UTF-8"?>
  <Mask id="AMOUNT">
    <MaskChar>#</MaskChar>
    <Description>Default PCI compliant Amount Mask</Description>
    <SeeInClear>true</SeeInClear>
    <UnMask>
      <Left>
```

```
        <Length>4</Length>
      </Left>
      <Right>
        <Length>4</Length>
      </Right>
    </UnMask>
  </Mask>

  <Mask id="ACCOUNT">
    <MaskChar>*</MaskChar>
    <Description>Default PCI compliant Account Mask</Description>
    <SeeInClear>true</SeeInClear>
    <UnMask>
      <Right>
        <Length>4</Length>
      </Right>
    </UnMask>
  </Mask>

  <Mask id="CARD">
    <MaskChar>*</MaskChar>
    <Description>Default PCI compliant Card Number Mask</Description>
    <SeeInClear>true</SeeInClear>
    <UnMask>
      <Left>
        <Length>6</Length>
      </Left>
      <Right>
        <Length>4</Length>
      </Right>
    </UnMask>
  </Mask>
```

<Mask> — The XML aggregate that contains a single field mask definition. The required id attribute specifies the name of the mask. AJSF-based applications reference this name when applying a mask to a particular field.

<MaskChar> — The character (e.g., *, #, etc.) used to substitute for masked field value characters. You can use any character in the defined character set (e.g., UTF-8) to mask data.

<Description> — An optional free-form description of the mask. This value is not used in processing and is for information only.

<SeelnClear> — A flag indicating whether the masked value is displayed in the clear when a masked field is editable and is in focus or the mouse is hovering over the masked field. Valid values are as follows:

- true = Yes, display the value in the clear when the field is editable and is in focus or the mouse is hovering over the field.
- false = No, do not display the value in the clear when the field is editable and is in focus or the mouse is hovering over the field. The field value is always masked.

Note: By using different masks for the same field (e.g., card number) on different windows, you can have a field value appear in the clear on one window, but be masked from view on another window.

<UnMask> — An XML aggregate that defines the left and right portions of the field value to remain unmasked (i.e., in the clear).

<Left> — The number of characters in the field value, starting from the left, to remain unmasked (i.e., in the clear). Valid values are from one character up to the maximum length of the field to be masked. This tag is optional. If you do not want to unmask any of the leftmost characters of a masked field value, you do not need to configure this tag.

<Right> — The number of characters in the field value, starting from the right, to remain unmasked (i.e., in the clear). Valid values are from one character up to the maximum length of the field to be masked. This tag is optional. If you do not want to unmask any of the rightmost characters of a masked field value, you do not need to configure this tag.

Masking Examples

If a mask is defined with four unmasked lefthand characters and four unmasked righthand characters and a masking character of *, then a clear value of 1234567890123456 would be displayed as the masked value 1234*****3456. If only the four leftmost characters are unmasked, the masked value would be displayed as 1234*****. If only the four rightmost characters are unmasked, the masked value would be displayed as *****3456. Finally, if no lefthand or righthand characters are defined as unmasked, the masked value would be displayed as *****.

Application Implementation

AJSF-based applications implement field masking by replacing the appropriate EComponent with its associated EMaskedComponent and setting the mask key for the component. The following table relates each of the available EMaskedComponents to its EComponent counterpart.

EMaskedComponent	EComponent
EMaskFilter (extension of MaskFilter)	MaskFilter (internal helper class)
EMaskedTextField	ETextField
EMaskedDataLabel	EDataLabel
EMaskedComboBox	EComboBox
EMaskedList	EList
EMaskedCurrencyField	ECurrencyField
EMaskedTableCellRenderer	ETableCellRenderer

EMaskedComponents display their value as masked (as defined by the mask template defined for their mask key), except for the following specific conditions, under which they display their value in the clear:

- The component is editable and is currently in focus.
- The mouse is currently hovering over the component and the <SeeInClear> tag for the mask is set to true.

Mask Retrieval

A Java system property named [GetAllMask](#) indicates whether a AJSF-based application retrieves masks from disk one at a time as they are needed or all at once the first time one needs to be used. Once a mask is retrieved, it is stored in cache memory so the application does not have to retrieve it from disk again. If you change the definition of a mask used by an application, you must warmboot the application by clearing the cache before the change becomes effective.

Secure Delete Utility



AJSF-based applications can optionally provide the SecureDelete command line utility to securely delete files that contain sensitive cardholder data. For example, you could use the SecureDelete utility to securely delete system dumps, trace files, and server to server database migration files once they are no longer needed for business purposes.

To help speed file remove disk operations, and to provide undelete functionality, some hardware vendors have implemented file delete functionality in their operating systems (e.g., Windows or IBM-AIX) by erasing pointers to files on the disk rather than actually erasing the files. This “feature” inadvertently allows unauthorized users (hackers) to scan the disk media for sensitive user data that may be left behind. Furthermore, advanced scanning tools are now on the market that allow unauthorized users to scan for residual magnetic images left behind even after a file has been overwritten. Thus, deleted files could be easily recovered if they are not erased (*wiped*), leaving sensitive information vulnerable to retrieval by unauthorized persons or companies. *File wiping* is the act of deleting a computer file securely, so that it cannot be restored by any means. By securely wiping files, you significantly reduce the chances that a file’s contents can be recovered. The SecureDelete utility not only removes the pointer to the file, but repeatedly overwrites the file multiple times with alternating byte arrays, which causes magnetic pattern changes on the disk itself, making recovery extremely difficult.

You can use the SecureDelete utility to wipe individual files, or all files in a specified directory. Directories themselves are not deleted. The utility displays the deletion progress with information including the names of files deleted, the operation outcome, and the last modification date of files.

Operation

The SecureDelete command line utility is delivered in a .jar file. You can display usage detail for operating the utility by running it without any parameters. To execute the utility without parameters, enter the following command:

```
java -jar secureddelete.jar
```

The following usage detail is displayed:

```
Securely deletes a file by overwriting the contents  
of the file numerous times before deleting the file.
```

The file is renamed before deletion occurs.

USAGE

SecureDelete [-d] [-f] [-p] [-r]

Parameter Definitions

- d List the fully qualified paths(including drive letter) of directories that need to have files deleted. Multiple directories should be separated with ';' and enclosed in quotes.
- f The fully qualified path of a specific file to be deleted. When this parameter is used, the Directory parameter is ignored.
- p Number of passes. Default is 3. Specifies the number of passes that write to each file. Range: 3-35 inclusive. Pass one writes all zeros followed by all ones, followed by a fixed random character. Successive passes follow this order.
- r Recursively delete directories. If this parameter is set, the directory specified in -d, will have all files and sub-directories securely deleted.

NOTE: Do not insert a space between the argument identifier and the value.

Example: -dC:\\ossi\\log\\history -r

File Testing

Before beginning its passes to overwrite the file, the utility tests the file to make sure it can open, write, and rename the file. The utility renames the file to “a” before it is wiped, and finally deleted. Thus, you need to make sure there are no files in the directory of a file to be deleted that are already named “a” before running the utility. If a file already exists in the running directory with a name of “a” when you run the utility, it terminates without renaming, wiping or deleting any files.

Secure Directory Deletion

There are two methods for deleting directories with the SecureDelete utility. You can either delete the contents only of a directory or you can delete the directory itself and all of its contents.

To delete the contents only of a directory, enter the directory name immediately following the -d switch.

For example, to securely delete the contents of a directory named “myFolder” at the root level of the C: drive, without removing the directory itself, you would enter the following command:

```
java -jar secureddelete.jar -dC:\myFolder
```

If the directory has spaces in its name (e.g., “my Folder”), you must enclose the directory name in double quotes as show below.

```
java -jar secureddelete.jar -d"C:\my Folder"
```

To securely delete a directory and its contents, add the -r switch to the command.

```
java -jar secureddelete.jar -dC:\myFolder -r
```

When deleting directories, you are prompted for confirmation.

```
Ok to Securely delete directory(s) 'directory-name'? 'y' to  
continue ->
```

Enter “y” or “Y” at the prompt to continue.

Command Line Output

The SecureDelete utility reports the number of files wiped and deleted and the number of directories removed. If a file in the specified directory tree cannot be removed (e.g., because it is a read only file or has a permission restriction), the file remains as is, and its parent directory is not removed.

Individual files that are wiped are listed and then deleted.

During a recursive directory delete operation, once all the files have been wiped and deleted, the utility removes all directories that no longer contain any files.

The following is example output from running the utility:

```
SecureDelete Beginning operation...
File afile.txt is a directory and will not be deleted.
File 'c:\aci\new\dirToDelete\aRealfile.txt' successfully
deleted(3 pass). Last modified date: Fri Sep 07 15:03:14 CDT 2007
DELETE SUMMARY - Files deleted: 1, Directories removed: 0,
Files skipped: 0
SecureDelete Completed.
```

Technical Information

The SecureDelete utility uses three 8-byte arrays to wipe files. Files are first overwritten with all zeros (0x00), followed by all ones (0xFF), and finally with all bytes set to AC (0xAC).

Specifying the “-p” parameter allows you to repeat these passes. If you specify a value for the “-p” parameter that is not a multiple of 3, the wipe process will complete with either all 0x00 or all 0xFF bytes instead of all 0xAC bytes written to the file just before deletion.

Security Considerations

This topic is provided for system administrators who want to know how the ACI desktop user interface or browser for APF products use today's communication security measures and what must be configured to implement these technologies within the ACI desktop user interface or browser server and clients. This topic provides both introductions to these technologies and step-by-step instructions for troubleshooting configuration problems related to these technologies.



Cross-industry best practices advocates using strong cryptography and security protocols such as secure sockets layer (SSL) / transport layer security (TLS) and Internet protocol security (IPSEC) to safeguard sensitive cardholder data during transmission over open, public networks. For example, the use of secure transfer protocols has been mandated by the Payment Card Industry (PCI) Data Security Standard for the transfer of sensitive and private data such as cardholder information, credit card numbers and passwords over the internet.

Secure Socket Layer (SSL) secures HTTP communications to and from ACI desktop user interface or browser clients and the AJSF-based web application server (e.g., ESWEB, PMWEB, CRUXWEB) for the ACI desktop user interface or browser servlets. SSL uses certificates, public, and private keys to authenticate the identity of both parties in the communication as well as encrypt the messages between both parties by creating a session key. The SSL protocol specifies how this session key gets created and how each party authenticates the other side of the communications. SSL over TCP/IP is used to secure communications between the USEC subsystem and other third-party applications, such as a third-party Java Authentication and Authorization Service (JAAS) authenticators.

SSL Overview

Because the AJSF-based web application server (e.g., ESWEB, PMWEB, CRUXWEB) must provide secure communication to clients, keys and certificates are needed in the configuration of the AJSF-based web application server. These keys and certificates are either stored within files called key stores (when using software encryption) or hardware security modules which act as protected key stores (for hardware based encryption). When using software encryption, these key store files are encrypted using a password.

SSL attempts to create a secure channel between a client and a server through which communications can be sent by providing message encryption/decryption as well as client and server authentication. SSL uses handshaking to communicate the public key (certificate) of each end to establish a session key. This session key

is then used to encrypt and decrypt the messages between the client and server. Each side of the communications can also be authenticated to make sure the other end is trusted. This is accomplished through the use of certificate verification during the handshaking. This topic provides an overview of the basic parts of SSL.

The AJSF-based web application server implements the Java Secure Socket Extension (JSSE) design to break up the certificate and key stores used during SSL as two different stores—keystores and truststores. Keystores (from JSSE's perspective) are databases of key pairs and certificates that are used to set up SSL authentication. Truststores are key stores that are used to verify the identities of other clients and servers. When a client or server is setting up an SSL session, it retrieves its certificates and keys from its key store. When it verifies the identities of other clients or servers, it retrieves trusted certification authority (CA) certificates from its truststores.

What is SSL?

Secure Socket Layer (SSL) is a protocol used to transmit sensitive data over the internet. SSL encrypts data using keys. SSL is called a *layer* because it is made up of three kinds of protocols that *layer* themselves inside packet communications that take place on TCP/IP communications. The current version supported by most SSL sites and devices today is specification level 3 and is implemented between a reliable connection-oriented network layer protocol (e.g. TCP/IP) and the application protocol layer (e.g., Hypertext Transfer Protocol (HTTP)).

One-Way and Two-Way SSL

SSL can be configured as one-way or two-way:

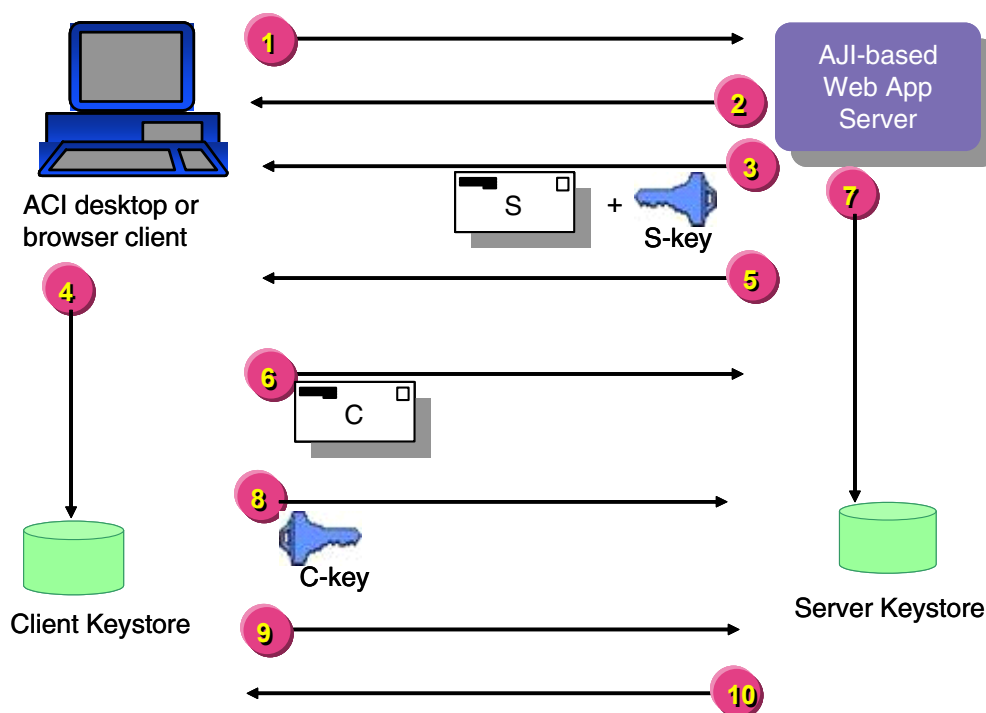
- With one-way SSL, the server is required to present a certificate to the client but the client is not required to present a certificate to the server. To successfully negotiate an SSL connection, the client must authenticate the server but the server will accept any client into the connection. One-way SSL is common on the Internet where customers want to create secure connections before they share personal data. Often, clients will also use SSL to log on so that the server can authenticate them.
- With two-way SSL, the server presents a certificate to the client and the client presents a certificate to the server. The server must be configured to require clients to submit valid and trusted certificates before completing the SSL connection. This is also referred to as *client authenticated* SSL because it requires a certificate from the client to also be verified at the server. In two-

way SSL the server requests the client's certificate and then authenticates it against its trust store. Then the client also sends its public key to the server for the session key to be created. Two-way SSL is less common than one-way SSL because it is expensive to manage in terms of expiry checks, creating certificates.

The following graphic depicts the steps involved in one-way and two-way SSL between ACI desktop user interface or browser clients and the AJSF-based web application server.

SSL authentication starts by a *handshake* between the server and the client. The server then sends a certificate and the public part of its key to the client. The client reads its trust store by reading a chain to authenticate the server's certificate. In one way SSL, only steps 1–4 and 9–10 are performed. In two way SSL, all steps are performed. Once steps 9 and 10 are performed, an SSL layer is established and a session key is created using the public key to create a secure communications channel. From then on, all application communications is encrypted using the session key.

Step descriptions are provided below the graphic.



1. The ACI desktop or browser client requests a connection to the AJSF-based web application server (says hello with a *handshake*).
2. The server says hello back.
3. The server sends its public server certificate and public key to the client.
4. The client validates the server certificate against its keystore.
5. The server requests the client's public certificate (for two-way SSL only).
6. The client sends its public client certificate to the server (for two-way SSL only).
7. The server validates the public client certificate against its keystore (for two-way SSL only).
8. The client sends its public key to the AJSF-based web application server (for two-way SSL only).
9. The client agrees on cipher specification.
10. The server agrees on cipher specification.

At this point, the SSL layer is established, a session key is created, and secure application communications can commence.

Key Types

There are various key types used in SSL connections defined mainly by the length of the key pair in bits. The main key types used in the AJSF-based web application server are:

- DES (Data Encryption Standard) keys are 64 bit keys (56 bits of which are used for security).
- 3DES or TripleDES (Data Encryption Standard) keys contain a 168 bit key (112 bits of which are used for security). These keys are also known by some providers as DESede keys.
- AES (Advanced Encryption Standard) keys are considerably larger than even DES or TripleDES keys because they contain 128, 192, or 256 bit keys (128 bits of which are used for security). This key type is sometimes called Rijndael although it is not technically exactly the same.

Certificates

Certificates consist of properties describing what they are used for, how they are constructed, and when they are valid. Common certificate properties include but are not limited to issuer, serial number, signature algorithm, public key, subject, valid from, valid to, and version properties.

The signature algorithm is not only used in signing but also with the public key in SSL communications to establish the session key.

The Valid To and Valid From properties are used to establish validity of the certificate.

The Subject and Issuer properties, which are used in establishing identity and a chain of trust, are comma-separated strings that contain (but are not limited to) the following values in the following order:

- Common Name (CN), also called machine name or *first and last name* by most devices. In end entity certificates, this contains the value that identifies the user of the certificate. This can be a machine name for LAN access, IP address, or web domain of the server that uses the certificate. When used in two-way SSL (see One-way and Two Way SSL Defined topic) to authenticate

the client, the common name is validated against the URL of the client. In intermediate and root certificates it can also be an additional identifier of the Organizational Unit.

- Organizational Unit (OU) is usually used as the division inside the company (or an identifier for the person) using the certificate.
- Organization (O) is usually used as the company or person using the certificate.
- Locality (L) is usually the city, village, town, etc. of the company (or person) using the certificate.
- State (S) is the state or province used to further define the locality.
- Country (C) is the country used to further define the State and Locality of the user of this certificate.

Root or certificate authority (CA) certificates can also contain a Certificate Revocation list (CRL), which is either an embedded list or the location (filename or URI) that points to a list of revoked certificates along with the reasons for revocation.

Certificate Types

Certificates are categorized according to the value of both the subject and the owner of the certificate as following:

- End Entity (EE). The owner is either an intermediate or root (CA) certificate. End entity certificates are defined as certificate subjects that use their private key or public key for purposes other than signing certificates.
- Intermediate Certificate. The owner is (signed by) a root (CA) or intermediate certificate.
- Root or Certificate Authority (CA) Certificate. A Certificate that is self-certified. The owner and the subject are the same value (i.e., signed by self).

Certificate Validation

There are three required criteria that must be satisfied for an end-entity certificate to be considered valid during an SSL session:

1. The certificate must not be expired. This means it must contain valid start and end dates and that the current date of use falls within this date range.

2. The certificate (client or server certificate) must be sent from the domain that it was issued to. Technically, this means that when it was created, the common name field (CN) of the certificate must contain the exact domain that it is being used from/to.
3. A valid chain of trust must be established with the certificate. This means that the certificate can be validated within the truststore by a root or certificate authority certificate. Intermediate certificates may be necessary to establish this chain of trust. Each certificate in the chain of trust must not be expired also.

For a certificate to be considered valid for use in SSL communications by the AJSF-based web application server or ACI desktop user interface client, the SSL protocol handler checks for what's called a valid chain of trust. The term chain of trust is also sometimes called a web of trust. A chain of trust is when the certificate being used can be resolved all the way up to its root or certificate authority (CA) certificate.

Resolution is done by comparing the Owner properties line of the certificate being verified to the Issuer properties line of all certificates in the keystore. The [“Establishing a Chain of Trust”](#) topic explains the individual properties that make up the Issuer and Owner lines. If the certificate resolved contains any Owner properties that are different from its Issuer properties (meaning it is not a root or CA certificate) then that certificate's Owner properties line is searched for inside the truststore and so on until a root or CA certificate is located thus completing the chain of trust. If no root or CA certificate can be located, a chain of trust can not be established and the certificate is considered to be invalid.

Only valid certificates can be used as client certificates through SSL within the AJSF-based web application server and ACI desktop user interface clients. This means that when you import a certificate into the SSL client keystore you need to import all certificates that serve to validate that client certificate's chain of trust. The entire chain of trust must be present or a runtime error will occur when the ACI desktop user interface client attempts to use that certificate. See the following topic for the procedures needed to establish a chain of trust.

Establishing a Chain of Trust

To verify the chain of trust for an end entity certificate, take the Issuer line of the end entity certificate and try to locate it in the Owner line of all the certificates (using your favorite keystore utility) within your truststore.

Note: If you use Sun's keytool, use the `-list use -v` command to list the certificate details.

If the located certificate is not a root or CA certificate, which is determined by comparing the Issuer line and Owner line to make sure they are identical, you may need to take that certificate's Owner line and locate a certificate within the truststore with a Subject line that matches it. Keep doing this procedure until you find the certificate that contains the same values for the Owner and Subject line and you have just verified the chain of trust.

Example. For example, let's say that you want to use a client certificate in communicating with the AJSF-based web application server (e.g., ESWEB) given the following certificate:

```
Alias name: servercert
Creation date: Apr 7, 2006
Entry type: keyEntry
Certificate chain length: 3
Certificate[1]:
Owner: CN=www.myserver.com, OU=IWT, O=ACI, L=Omaha, ST=Nebraska,
C=US
Issuer: CN=esweb-ca, OU=ESWEB, O=Development, C=US
Serial number: f9fdc7851d5e341a445b9b940b7b6afeaedfa230
Valid from: Fri Apr 07 10:06:28 CDT 2006 until: Sat Apr 07
10:06:28 CDT 2007
Certificate fingerprints:
    MD5: 39:D3:69:32:DD:A5:DF:8D:AF:25:42:79:F3:6C:89:A3
    SHA1:
4E:CF:B2:A4:EC:6B:48:D8:F4:49:6D:45:EB:D5:B9:F2:0B:2C:C8:64
```

The bolded line shows the Issuer line you need to locate in the Owner line of a certificate in your truststore. (Depending on how you inserted it, the full validation chain may follow the end-entity certificate and look a bit longer than the example above).

Searching through a properly loaded truststore, you would locate the following certificate:

```
Alias name: eswebca
Creation date: Apr 7, 2006
Entry type: trustedCertEntry

Owner: CN=esweb-ca, OU=ESWEB, O=Development, C=US
Issuer: CN=esweb-root, OU=ESWEB, O=Development, C=US
Serial number: f2048be98e1c117d1cdcd28107781873bc6c9f79
Valid from: Fri May 07 11:23:53 CDT 2004 until: Mon May 05
11:23:53 CDT 2014
Certificate fingerprints:
    MD5: F4:68:87:DC:C3:AC:D6:9E:E7:F9:48:A9:96:47:65:DB
    SHA1:
73:14:CB:45:88:E6:94:D1:5A:06:22:CC:70:D6:C1:BE:FA:1B:C3:E7
```

Now, because the Owner and Issuer line are not the same, this is an intermediate certificate. You must search through all intermediate certificates in the chain to locate the root or certificate authority (CA) certificate that belongs to the chain. In this case, only one intermediate certificate was used for the ESWEB server, so searching further in the truststore for the above certificates Issuer line you would locate the following certificate:

```
Alias name: pitroot
Creation date: Apr 7, 2006
Entry type: trustedCertEntry

Owner: CN=esweb-root, OU=ESWEB, O=Development, C=US
Issuer: CN=esweb-root, OU=ESWEB, O=Development, C=US
Serial number: 2cbb12b8d6f9c09d0027732412a69525ee1acab6
Valid from: Fri May 07 11:23:50 CDT 2004 until: Mon May 05
11:23:50 CDT 2014
Certificate fingerprints:
    MD5: 5C:E9:8F:83:24:7E:64:94:87:54:BE:F3:96:E8:38:FC
    SHA1:
28:E4:AC:FD:4E:FF:6A:27:26:5C:97:2B:88:A7:1B:0E:86:8A:C5:D3
```

Note that the Owner and Issuer are the same. This means you have finished your validation of the chain, and if you are using the end entity certificate with the alias of servercert, you can verify that it is trusted.

Cryptography Algorithms

Cryptography algorithms are mathematical operations that are performed on values (numbers or strings) in order to produce another value that represents the original value in an encrypted or hashed form.

The AJSF-based web application server uses symmetrical ciphers, certificates, and hashes for its cryptographic operations.

Symmetrical cipher algorithms are algorithms that can encrypt and decrypt data using the same key for both operations. Hash algorithms are algorithms that create a *fingerprint* of the data so that the data can be verified to be unmodified. It is virtually impossible to extract the original data from a hash *fingerprint*, although they are still vulnerable to attack.

The AJSF-based web application server uses symmetrical cipher algorithms during SSL and also when storing data containing private information to the database. SSL uses a symmetrical cipher to secure the data communications but first agrees

on the common key using the private and public key parts tied to the certificates of both parties in the communication. For data storage, the key used resides in a software or hardware key store (HSM).

The AJSF-based web application server uses hashes when private data (such as passwords) must be verified to be accurate.

Cipher Algorithms

Data Encryption Standard (DES). DES is a symmetric algorithm used for key encryption. It uses a 56-bit key along with a block cipher (encryption) method to break text to 64-bit blocks that are later encrypted.

TripleDES/DESe. Also known as Triple DES, 3DES is a form of DES with the difference of encrypting data three times. Unlike DES, 3DES uses three 64-bit keys to generate a 192-bit key that has been encrypted three times.

Hash Algorithms

SHA1. The SHA1 or Secure Hash Algorithm is used to hash values. Because this is a hash algorithm, anything that is encrypted is irreversibly encrypted. It is not a cipher like DES, but is one-way encryption algorithm. It is used to generate signatures and certificate fingerprints.

MD5. MD5 is a hash algorithm specifically used to generate certificate fingerprints and private keys as well as signatures.

SecureRandom. SecureRandom is a hash algorithm used to generate random numbers that are used as seeds for keys and unique IDs.

RSA. RSA is used for the storage form for SHA1 and MD5 hashes within the key stores used by the ACI desktop or browser application.

Hardware Security Modules

Hardware Security Modules (HSMs) are physical security measures used in the storing of and/or securing access to keys and certificates. While the AJSF-based web application server application does not mandate the use of hardware security modules due to the limit of data being stored, if volume reasons dictate, the AJSF-based web application server application is compatible with a number of hardware security modules that can aid in key and certificate storage security and scalability.

These hardware security modules can provide SSL handshaking hardware acceleration. See the [“Encryption Subsystem”](#) topic in section 2 for a current list of HSM vendors supported by the AJSF.

Troubleshooting SSL

General practice when experiencing SSL connectivity failures is to edit the config.properties or APPLCFG and set the SSL.debug property to all. After restarting the AJSF-based web application server, SSL keystore access and handshaking messages will be generated in the Tomcat or Websphere application server servlet container logs and the AJSF-based web application server logs. Tracing is not required to be turned on to get this output. Much of the Full Error examples shown in this section only show up in the web server logs if SSL.debug is on.

certificate_unknown (No trusted certificate found)

The cause of this error is that one or more certificates needed to establish the chain of trust of the end entity certificate were not found within the configured truststore.

This error can occur on outgoing and incoming SSL connections. In this situation, the AJSF-based web application server may be acting as a client to an external server or as the server to an authenticated SSL connection.

Full Error(s):

(In the AJSF-based web application server logs)

```
FATAL event.com.aciworldwide.framework.message.HTTPProcess
[jrpp-0] - An exception occurred in HTTPProcess sendMessage:
javax.net.ssl.SSLHandshakeException: sun.security.validator.
ValidatorException: No trusted certificate found
```

(In the Web Application Container Server stdout logs)

```
jrpp-0, SEND TLSv1 ALERT: fatal, description = certificate_unknown
jrpp-0, WRITE: TLSv1 Alert, length = 2
jrpp-0, called closeSocket()
jrpp-0, handling exception: javax.net.ssl.SSLHandshakeException: sun.
security.validator.ValidatorException: No trusted certificate found
jrpp-0, called close()
jrpp-0, called closeInternal(true)
```

Solution Checklist:

1. Check the truststore configuration and availability. Make sure that the SSL.client.truststore is properly set and verify the existence of the keystore at that location. Use keytool to list (or another keystore management program to open and display) the key store with the password that was encrypted with the ConfigPropertyBuilder password encryption utility or the Password Manager (PswdMgr) utility. For more details on how to use Sun's keytool utility to do this, refer to online keytool documentation (e.g., <http://java.sun.com/j2se/1.3/docs/tooldocs/win32/keytool.html>).
2. Verify that the chain of trust exists in the truststore as explained in the ["Establishing a Chain of Trust"](#) topic. The validation code in the system will fail the validation on the first certificate entry that was not located in the truststore so you can work backwards in validating the chain of trust. To do this:
 - a. First locate the error line (ALERT: fatal, description = certificate_unknown) in the Web Application Server stdout logs.
 - b. Next locate the first occurrence of the following text above the error line:
Subject: OU=
 - c. The line located contains the owner line, designated by OU=. Follow the procedures of establishing a chain of trust using the text following the OU=.

internal_error (handshake_failure)

This error can occur when the SSL debug output shows valid handshaking occurring, but suddenly the communication is terminated just after the server certificate is received. (This usually occurs shortly after ***** ServerHello, TLSv1** shows up in the Web Application Server logs).

Although there may be other causes to this error showing up, when this error happens shortly after the server sends the certificate (and two-way SSL is being requested by the server), the client certificate that is being read may be missing the key it was generated from. An end-entity certificate is used for the purpose of signing and a private or public key must exist for this to occur during SSL communications. In most cases, this keystore is the SSL client keystore, but this can happen in any keystore where the end-entity certificate was generated and the key used to generate it is somehow removed from the keystore.

Full Error(s):

(Web Application Server stdout logs, 1.4+ JVMs)

```
*** ServerHelloDone
[read] MD5 and SHA1 hashes:  len = 4
0000: 0E 00 00 00                      ....
*** Certificate chain
jrpp-1, handling exception: java.lang.NullPointerException
jrpp-1, SEND TLSv1 ALERT:  fatal, description = internal_error
jrpp-1, WRITE: TLSv1 Alert, length = 2
```

Or

(Web Application Server stdout logs, using earlier JVMs)

```
jrpp-1, WRITE:  SSL v3.1 Handshake, length = 36
jrpp-1, READ:  SSL v3.1 Alert, length = 2
jrpp-1, RECV SSLv3 ALERT:  fatal, handshake_failure
```

And

(AJSF-based web application server logs)

```
ERROR event.com.aciworldwide.commerce.common.https.HttpsProcess [jrpp-1] -
An exception occurred in HttpsProcess sendMessage:
javax.net.ssl.SSLHandshakeException: Received fatal alert: handshake_failure
    at com.sun.net.ssl.internal.ssl.SSLSocketImpl.b(DashoA6275)
    at com.sun.net.ssl.internal.ssl.SSLSocketImpl.b(DashoA6275)
    at com.sun.net.ssl.internal.ssl.SSLSocketImpl.a(DashoA6275)
    at com.sun.net.ssl.internal.ssl.SSLSocketImpl.a(DashoA6275)
    at com.sun.net.ssl.internal.ssl.AppOutputStream.write(DashoA6275)
```

Solution Checklist:

1. Check the truststore configuration and availability. Make sure that the SSL client.truststore is properly set and verify the existence of the keystore at that location. Use keytool (or another keystore management program) to open the keystore with the password that was encrypted using the ConfigPropertyBuilder password encryption utility or the Password Manager (PswdMgr) utility.
2. Verify that the chain of trust exists in the truststore as explained in the [“Establishing a Chain of Trust”](#) topic. The error will occur on the first certificate entry that was not located in the truststore so work backwards from the log to establish the chain of trust.

3. Validate that the entry in the keystore for the certificate contains an Entry type of **keyEntry**. If you cannot be certain of this, you may simply want to delete the certificate from the keystore and re-request it after generating a valid certificate key.

HTTPS hostname wrong: should be < ip or hostname >

This error is generated when the end entity certificate provided from the server or client does not match the URL of that server or client. The certificate may be valid in all other respects but the Common Name of the certificate (the string after the CN= line in the Subject property) must match the URL that uses it as the server certificate or client certificate.

This error can occur on outgoing and incoming SSL connections. In this situation, the AJSF-based web application server may be acting as a client to an external server or as the server to an authenticated SSL connection.

Full Error(s):

(In the AJSF-based web application server logs)

```
FATAL event.com.aciworldwide.framework.message.HTTPProcess [jrpp-0] - An
exception occurred in HTTPProcess sendMessage:
java.io.IOException: HTTPS hostname wrong:  should be <certificate URL/IP>
    at sun.net.www.protocol.https.HttpsClient.b(Unknown Source)
    at sun.net.www.protocol.https.HttpsClient.afterConnect(Unknown Source)
    at sun.net.www.protocol.https.AbstractDelegateHttpsURLConnection.
connect(Unknown Source)
    at sun.net.www.protocol.http.HttpURLConnection.getOutputStream(Unknown
Source)
    at com.sun.net.ssl.internal.www.protocol.https.HttpsURLConnectionOldImpl.
getOutputStream(Unknown Source)
    at com.aciworldwide.framework.message.HTTPProcess.sendMessage(Unknown
Source)
    at com.aciworldwide.framework.message.HTTPProcess.sendMessage(Unknown
Source)
```

Solution Checklist:

This error occurs when the URL for a response does not match the Common Name (CN) of the certificate that the client site is using as a Server certificate. The client site must either change its URLs to match the Common Name of the server certificates it is using or regenerate the certificates with a Common Name that matches the URL.

Default SSL Implementation

ACI provides a default Secure Sockets Layer (SSL)/Transport Layer Security (TLS) implementation with the ACI desktop user interface or browser application to enable you to quickly create a test SSL environment. To ensure security best practices, this test SSL environment should not be used in a production environment. ACI strongly recommends that security policy requirements and certificates be obtained from the relevant security department of your enterprise.

By default, the automated ACI desktop user interface or browser installation program configures SSL access to the AJSF-based web application server using HTTPS or SSL over TCP/IP where appropriate. SSL is an industry-standard way of protecting the privacy of data exchanged between client-server applications. Note that this default SSL installation uses self-signed certificates, one-way authentication where the client performs server authentication, and default keystore passwords.

The keytool utility from Sun Microsystems is used to manage ACI desktop user interface or browser client and server keystore contents. For the default installation, the automated install program uses the keytool utility to perform the following steps. Note a sample keytool command follows each step:

- Generate a server public/private key pair and a self-signed certificate.

```
keytool -genkey -v -keyalg RSA -keystore .\server.keystore.key -dname "CN=ComputerName, OU=AJSF, O=ACIW, L=OMAHA, ST=NE, C=US" -validity 365 -storepass <store password> -keypass <key password> -alias Web
```
- Export the self-signed trusted server public certificate.

```
keytool -export -rfc -keystore .\server.keystore.key -alias Web -storepass <store password> -file .\server.crt
```
- Import the self-signed trusted server certificate into the client trusted keystore.

```
keytool -import -alias Web -file .\server.crt -keypass <key password> -keystore .\client.truststore.key -storepass <store password> -noprompt
```

Refer to the following link for more information regarding the keytool utility from Sun Microsystems: <http://java.sun.com/j2se/1.4.2/docs/tooldocs/windows/keytool.html>

The following command files provided by the ACI desktop user interface or browser automated installation program are used to generate the keytool utility commands that perform the previously mentioned steps:

- Unix: {ACI desktop user interface or browser root}/Server/Web/config/security/GenKeyStore.sh
- Windows: {ACI desktop user interface or browser root}\Server\Web\config\security\GenKeyStore.bat

The server keystore as well as the client trusted keystore are distributed to the respective server and ACI desktop or browser client locations by the ACI desktop user interface or browser automated installation program. The process to distribute the client trusted keystore to a third-party client is outside the scope of the ACI desktop user interface or browser installation program. The following lists the {ACI desktop user interface or browser root} relative locations of the keystores and also lists the source location of the generated trusted keystore for third-party clients:

- ACI desktop or browser client trusted keystore: Client\run\UIDesktop\client.truststore.key
- Unix server keystore: Server/Web/config/security/server.keystore.key
- Unix third-party client trusted keystore: Server/Web/config/security/client.truststore.key
- Windows server keystore: Server\Web\config\security\server.keystore.key
- Windows third-party client trusted keystore: Server\Web\config\security\client.truststore.key

Using a Certificate Authority (CA)

In general, if you want to use a certificate authority (CA) or product designed to allow you to act as a CA, you need to perform the following steps:

1. Generate a server public/private key pair and certificate.
2. Export the server public key into a Certificate Signing Request (CSR).
3. Provide the CSR to a trusted certificate authority (CA) to obtain your certificate.
4. Import the *root* certificate (x.509) returned from the CA into the server keystore.

5. Import the *root* certificate (x.509) returned from the CA into the client trusted keystore.
6. Import the signed trusted server certificate (x.509) returned from the CA into the server keystore.
7. Validate the contents of the client and server keystore using the keytool utility.

Two-way SSL Authentication

Additional steps are required if you want to perform SSL with two-way authentication (i.e., both the client authenticates the server and the server authenticates the client). These steps regard the management of client-level certificates and are summarized as follows:

1. Generate client public/private key pair(s) and certificate(s).
2. Export the client public key into a certificate signing request (CSR).
3. Provide the CSR to a trusted certificate authority (CA) to obtain your certificate.
4. Import the *root* certificate (x.509) returned from the CA into the client keystore.
5. Import the *root* certificate (x.509) returned from the CA into the server trusted keystore.
6. Import the signed client certificate (x.509) returned from the CA into the client keystore.
7. Validate the contents of the client and server keystore using the keytool utility.

Keystore Passwords

In general, keystore passwords are required whenever a keystore is altered or restricted keystore data is used. A trusted keystore is a specialized form of a keystore used to store trusted certificates. The automated ACI desktop user interface or browser installation program sets the keystore passwords to default values. To change the keystore passwords from these default values, you must

perform the following steps. Note that all possible steps are listed below, but the actual steps required vary according to the SSL options used in your environment (e.g., two-way authentication, third-party client usage, etc.).

1. Change the keystore password for applicable ACI desktop user interface or browser client and server keystores and trusted keystores using the keytool utility.
2. Change the keystore or trusted keystore password value in the ACI desktop or browser client configuration and store the encrypted value using the Password Manager (PswdMgr) utility. This utility is described in the *ACI Desktop User Interface Manual*.
3. Change the keystore or trusted keystore password value stored in all applicable third-party client(s).
4. Change the keystore or trusted keystore value stored in the customer-provided web server (e.g., Websphere Application Server) or Tomcat servlet container configuration file.
5. Change the keystore or trusted keystore in the ACI desktop user interface or browser web application configuration file (e.g., config.properties) and store the encrypted value using the ConfigPropertyBuilder password encryption utility.

Apache Tomcat SSL Implementation

The following procedures discuss how to implement SSL within the Apache Tomcat servlet container for the BASE24-eps ACI desktop user interface or browser (AJSF-based web application server) and ACI desktop user interface or browser clients. SSL, known as Secure Sockets Layer, is a means for providing encryption of data between end points such as remote ACI desktop or browser clients, the AJSF-based web application server (within Tomcat), USEC, UAUD, and application servers. The UI Framework utilizes Sun-Microsystems global Java Secure Sockets Extension (JSSE) version 1.0.3_01 or later. The JSSE SSL implementation supports strong encryption. This version of JSSE supports SSL version 3.0

Sun's JSSE implementation is part of Sun's JRE 1.4 or later. The UIF relies on the following jars for its SSL functions:

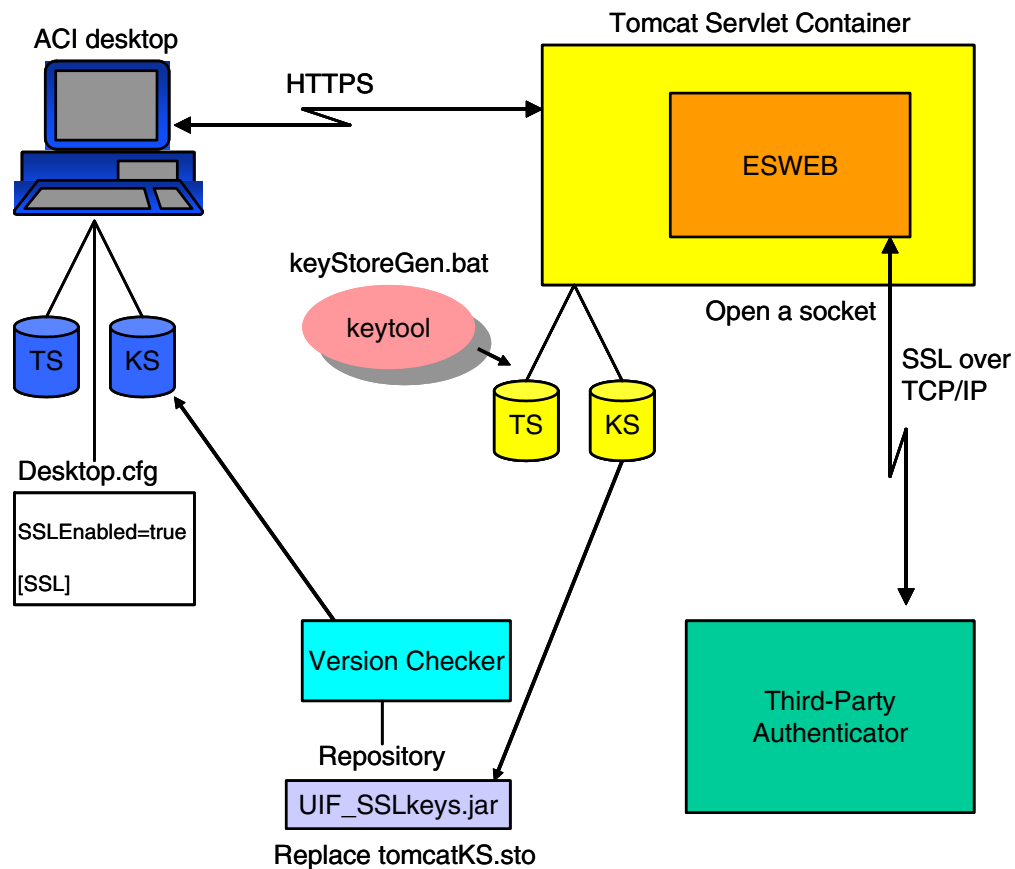
- jnet.jar
- jcert.jar
- jsse.jar.

SSL supports a flexible client-server authentication scheme and provides for algorithm-independent encrypted client-server communication. SSL runs as a layer between the Transport Control Protocol (TCP) and application layer protocols, such as RMI, HTTP, SMTP, and basic socket connections.

To implement SSL in Tomcat and ACI desktop user interface or browser clients, perform the following:

- Configure SSL for Tomcat.
- Generate certificates and keystores using keytool.
- Configure SSL for BASE24-eps ACI desktop user interface or browser clients.

This implementation is depicted in the following illustration and each of the above procedures is discussed in detail below.



Note: The configuration for SSL over TCP/IP to a third-party authenticator is not described here. See the “[Running the SvrSslCfg Utility](#)” topic in section 2 for detailed information on configuring SSL for such connections. In addition, the SSL configuration requirements for a Web access services server (such as the ACI Web Access Services server (Webgate)) deployed between ACI desktop or browser clients and the Tomcat server, is also not described here. See the “[Web Access Server SSL Implementation](#)” topic for more information.

Configure SSL for Tomcat

Use the following procedure to set up Tomcat to use SSL; or more specifically, to configure SSL within the Catalina web server that is bundled within Tomcat.

1. Create the certificate keystore using the Java keytool program.
2. On the machine where Tomcat is installed, uncomment the SSL HTTP/1.1 Connector entry in \$CATALINA_HOME/conf/server.xml file and modify it as necessary. The automated installation program defaults to a location such as _ACINESWeb\inssession-tomcat-6.0.20\conf\server.xml

Note: Tomcat version 5.5 is required for the HP NonStop s-series platform while Tomcat version 6.0 is required for the HP NonStop Integrity, IBM System z and Unix platforms.

For more information on Tomcat SSL security, refer to: <http://tomcat.apache.org/tomcat-6.0-doc/ssl-howto.html> or <http://tomcat.apache.org/tomcat-5.5-doc/ssl-howto.html>.

3. Save the server.xml file.

Generate Certificates and Keystores using Keytool

Use the following procedure to generate certificates and keystores for the Tomcat server and ACI desktop user interface or browser clients using Sun Microsystem’s keytool utility.

1. Install the JDK on your local machine, or if Java is already on your machine locate the \bin\keytool file. Refer to the following link for advanced details on the keytool utility: <http://java.sun.com/j2se/1.4.2/docs/tooldocs/windows/keytool.html>.

2. If you are using Java 1.4 or later, you can skip this step. If you are running a 1.3 JVM, download JSSE 1.0.3 (or later) from <http://java.sun.com/products/jsse> and either make it an installed extension on the system or else set an environment variable JSSE_HOME that points to the directory into which you installed JSSE.
3. An ACI provided SSL configuration package (i.e., SSL_config_instructions_and_work_area.ZIP) provides a batch script (keyStoreGen.bat) that you can run under Windows to generate your keystore and update the appropriate keystores for desktop or browser clients and the AJSF-based web application server. Unzip the contents of the SSL configuration package to a folder on your machine.
4. Configure and run the keyStoreGen.bat file to generate and update keystores. You can open the keyStoreGen.bat file with a text editor program such as Notepad or Wordpad to change its configuration before running it.

The keyStoreGen.bat file performs the following:

- Generates the Tomcat keystore using this command: %JAVA_HOME%\bin\keytool -genkey -alias Tomcat -keyalg RSA
Note: You can set the KEYALG parameter in the keyStoreGen.bat file to any key algorithm value supported by the keytool utility.
 - Generates the Tomcat certificate (tomcatKS.cer) to be used by the ACI desktop user interface or browser client. This is a self-signed certificate implementation. If you want to use your own certificate, see the [“Deploying a Customer SSL Certificate”](#) topic below.
 - Generates the desktop keystore (dtKS.stor).
 - Generate the desktop certificate (dtKS.cer).
 - Imports the Tomcat certificate into the desktop keystore. (Used for server authentication. This is similar to the browser action when a user is given the option to view and accept a server certificate).
 - Imports the desktop certificate into the Tomcat server keystore (Used for client authentication).
- a. Change the OUTPUT param to point to the directory where you want to store your keystores and certificates

```
set OUTPUT=C:\SSL_work_area\httpServerKS\output
```

- b. Modify the following param values (except for the %OUTPUT%, %WSNAME%, and %2% variables) as needed for the Tomcat server:

```
set WSNAME=Tomcat
set WSKEYSTORE=%OUTPUT%\tomcatKS.sto
set WSCERTKEY=%OUTPUT%\tomcatKS.cer
```

```
set WSCERTINFO="CN=omaweelb, OU=UIF, O=ACIW, L=OMAHA,  
ST=NE, C=US"
```

- WSNAME – The name of the AppServer (e.g., Tomcat)
- WSKEYSTORE – The file name of the Tomcat keystore (a reference to this keystore must be add to the Tomcat's server.xml file as described later).
- WSCERTKEY – The file name of the Tomcat certificate.
- WSCERTINFO – The owner information for the Tomcat certificate.

Note: You must change CN param to the machine name that the Tomcat server resides on. This value is used to match the HTTP URL used by the desktop or browser to connect to Tomcat.

You can also change the OU, O, L, ST, and C parameters to match your organization.

- WSALIAS – The alias used for key generation.
- WSKSPASS – The Keystore password.

You can also modify the KEYALG and VALIDDAYS parameters to meet your security requirements.

- c. Although not necessary, you can modify the certificate parameters for the desktop or browser client as desired. The naming conventions for these parameters are the same as for the AJSF-based web application server parameters, except they all begin with “DT” instead of “WS”.
 - d. Run the keyStoreGen.bat script. The following files are created in your specified output folder (e.g., C:\SSL_work_area\httpServerKS\output):
 - dtKS.cer
 - dtKS.sto
 - tomcatKS.cer
 - tomcatKS.sto
5. Access the uif_sslkeys.jar from the desktop or browser installation (in the UIF\lib\java_1.5 or java_1.3 folder). Un-jar this jar file, replace the dtKS.sto keystore with the dtKS.sto keystore created in the above step, and re-jar this file. Place this file on your desktop or browser client, or in the Version Checker repository so that all of your desktop or browser clients will be updated. You have now updated the desktop or browser client to include the server (Tomcat) certificate.

Note: If you replace the dtKS.sto file in the uif_sslkeys.jar in the Version Checker repository, the file is automatically uploaded to ACI desktop user interface or browser clients when a new session is started.

6. Copy the tomcatKS.sto keystore created above to the appropriate directory for the Tomcat server. Note the path to this directory. A sample would be _ACI\ESWeb\insession-tomcat-6.0.20\config\security
7. Update the Tomcat server.xml file located in the Tomcat configuration directory (e.g., _ACI\ESWeb\insession-tomcat-6.0.20\conf) as follows:

Uncomment or edit the keystorePass and keystore parameters in the connector section of the server.xml file.

```
<Connector port="8443"
maxThreads="150" minSpareThreads="25" maxSpareThreads="75"
enableLookups="false" disableUploadTimeout="true"
acceptCount="100" debug="0" scheme="https" secure="true"
clientAuth="false" sslProtocol="TLS" keystorePass="defaultWS"
keystore="<Path from step 6.>\tomcatKS.sto" />
```

Set the keystorePass parameter to the value of the WSKSPASS parameter used in the keyStoreGen.bat file to generate the AJSF-based web application server certificate in step 4.

Set the keystore parameter to the location of the AJSF-based web application server keystore from step 6 (e.g., _ACI\ESWeb\insession-tomcat-6.0.20\config\security\tomcatKS.sto).

8. Restart Tomcat (e.g., double-click the startesweb.bat file). The AJSF-based web application server is now configured to run either using HTTP or HTTPS (i.e. SSL).

Configure SSL for an ACI Desktop User Interface or Browser Client

Now that the AJSF-based web application server is configured to support SSL, you can now configure each ACI desktop client or browser to support SSL as well. For each deployed desktop or browser client, perform the following:

1. Modify the desktop.cfg (local.cfg) file as follows:
 - a. Add or set the SSLEnabled property to “true” to the Desktop section.

```
[Desktop]
...
SSLEnabled=true
Host=TomcatServerName
...
```


- b. Add the https port entry in the Hosts section, setting the port to the value configured in the Tomcat server.xml file above. (Note that the servlet name may vary by product release; match the one used in your vanilla deployment).

```
[Hosts]
TomcatServerName=<tomcat_machine_name>:8443/ESServer/
ESGenericServlet
```

- c. Add the following SSL section if not already present:

```
[SSL]
#
# The provider and protocol handler package need to be
# specified unless otherwise defined in the java.security
# file. They should be set to sun's implementation(s).
#
Provider=com.sun.net.ssl.internal.ssl.Provider
ProtocolHandlerPkgs=com.sun.net.ssl.internal.www.protocol

#
# If auto update is true, the specified key store file
# below will automatically be extracted by locating it in
# any jar in the classpath. This typically would come
# from the # UIFSSLKeys.jar. If the keystore should not
# be extracted upon startup, then set this property to
# false.
#
KeyStoreAutoUpdate=true
KeyStorePassword=defaultDT
KeyStore=dtKS.sto
KeyStoreType=JKS
```

2. Restart the desktop or browser client. Upload the new uif_sslkeys.jar if it appears in the Version Checker window. At this point, no URL connections errors should appear in the desktop.err file. This means that an HTTP SSL connection was established between the desktop or browser client and the AJSF-based web application server. You should also see a lock icon (🔒) on the right-hand bottom corner of the desktop or browser application frame. This confirms that the desktop or browser is started with the SSLEnabled property set to true.
3. Log on to the desktop or browser and test.

Deploying a Customer SSL Certificate

If required for your security protocols, you can modify the keyStoreGen.bat file to load an existing certificate rather than creating and loading a self-signed certificate.

If you have a production certificate that is located in another master keystore, you must export the key certificate to a .cer file and then import it into the AJSF-based web application server's keystore using keytool. Use the following procedure to perform these tasks:

1. Edit the keyStoreGen.bat file and add the following section to the end of the file.

```
set CUSTCERTKEY=%OUTPUT%\customerKS.cer    where customerKS.cer
      is the name of the certificate
set MASTERKEYSTORE=%OUTPUT%\masterKS.sto
set MASTERALIAS=ALIAS
set MASTERKSPASS=%3%

echo Production certificate export: %CUSTCERTKEY% ...
%KEYTOOL% -export -rfc -keystore %MASTERKEYSTORE% -alias
%MASTERALIAS% -storepass %MASTERKSPASS% -file %CUSTCERTKEY%
```

2. Execute the keyStoreGen.bat file to export the certificate.
3. Edit the keyStoreGen.bat file again. Delete the lines added in step 1 and add the following section to the end of the file. Make sure that the CUSTCERTKEY parameter points to the certificate exported in step 2.

```
set CUSTCERTKEY=%OUTPUT%\customerKS.cer    where customerKS.cer
      is the name of the production certificate

echo Desktop import of Customer provided AppServer certificate
(For server authentication) %KEYTOOL% -import -alias %WSALIAS%
-file %CUSTCERTKEY% -keypass %WSKSPASS% -keystore
%DTKEYSTORE% -storepass %DTKSPASS% -noprompt
```

4. Execute the keyStoreGen.bat file to import the certificate into the keystore for the AJSF-based web application (Tomcat) server.

Web Access Server SSL Implementation

If you are using ACI Web Access Services (WebGate) on the HP NonStop platform, refer to the *Web Access Services Administrator Guide* for detailed information on configuring SSL over your IP connections. Enabling SSL in this environment requires the purchase of one of the following ACI Enterprise Security Services.

- SG215 - SafeTGate:UI Support Module (SSL for UI). The product module to be licensed if you only want to secure ACI desktop or browser client connections to ACI Web Access Services (WebGate). This module is less expensive than SSL support for all IP connectivity.

- SG210 - SafeTgate: SSL. The product module to be licensed if you want to secure all XPNET IP connections, ACI Web Access Services connections, and even run a stand-alone copy of a SafeTGate process as an SSL proxy server. This module allows you to provide SSL over all IP connections on the HP NonStop platform.

If you are using a third-party web server on an IBM System z or Unix platform, refer to the configuration instructions in the web server vendor's documentation.

Implementing SSL Between ESWEB and XMLS

Perform the following steps to enable SSL between the ESWEB server process and the XML Server process (XMLS) in BASE24-eps.

1. Ensure that the following new properties are configured in the config.properties file for the ESWEB process.
SSL.client.truststore.password
SSL.client.truststore
2. On the External Connection Configuration window, select the External connection record for the BASE24-eps C++ XMLServer process (e.g., External Connection ID=BASE24-ES/C++XMLSERVER, Process ID=BASE24-ESUI).
3. In the Protocol Options field on the Protocol Details tab, append the "SSL." option and save the record.
4. For the IBM System z and Unix platforms, configure SSL in the ICE-XS process for the XML UI Server (XMLS) process. Refer to the *ACI Communication Services for Advanced Networking (ICE-XS) Administrator Guide* for the steps to configure SSL.

For HP NonStop platforms, configure SSL for the TCPSRV process in the NET24-XPNET TCPIP subsystem using the SafeTGate SSL Library for HP NonStop platforms. Refer to the *NET24-XPNET TCP/IP Protocol Implementation Guide* and the *ACI Enterprise Security Services (SafeTGate) SSL Library Guide* for the steps to configure SSL. This is the TCPSRV process used to *pass through* requests from the ESWEB process through XPNET to the XMLS process.

5. Extract the certificate from the ICE-XS keystore file or the certificate file and add it to the ESWEB truststore.
6. Restart the ESWEB process.

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5: Configuration Properties

Java servers built using the ACI Java Server Framework (AJSF) are configured using properties configured at the system level, in a configuration properties file read at startup, or in the Application Configuration data source read at startup. This section describes each of the properties common to ACI applications built on the AJSF. Properties specific to an ACI application built on the AJSF are described in an appendix to this user guide or in an application-specific manual. Properties specific to the AJMF are described in the *ACI Java Infrastructure ACI Application Management User Guide*.

Java server properties are defined for ACI applications when the application is installed. While most properties can remain set to their installed values, some may require adjustment after installation to meet changing transaction processing requirements.

System-Level Properties

This subsection describes the minimum system properties required for Java server processes built on the AJSF, as well as optional system properties. Application-specific system properties are described in the application-specific appendices of the user guide. Some system properties are set at runtime, while others are propagated from the JVM on which the Java server process is run or from the user logged onto the JVM. You can view all the system-level properties for a Java server process on the System Properties window or screen.

Required System Properties

The following system properties are required for AJSF-based applications.

config.filename — The fully-qualified location of the config.properties file (e.g., /user/B24es_cur/ESWeb/ESConfig/uaf2/P6Q^GATE/config.properties). This is a runtime system property that is either set with the -D flag argument when a Java server process is started as shown in the following example or defaults to a file named “config.properties”.

```
JAVA -Dconfig.filename=/user/B24es_cur/ESWeb/ESConfig/uaf2/P6Q^GATE/config.properties
```

file.encoding — The character encoding set used for the default locale (e.g., ISO8859-1).

Note: To support multibyte encoding, this property must be set to match the value (e.g., UTF-8) of the [MULTIBYTE_ENCODING](#) property.

file.separator — The platform-dependent name or file separator character (e.g., / on UNIX). ACI recommends that you set this property to / since it is valid on a variety of hardware platforms (e.g., Unix, HP NonStop, Windows, etc.).

process.id — The symbolic name assigned to a Java server process (e.g., P6Q^ESWEB01). This is a runtime system property that is either set with the -D flag argument when a Java server process is started or is set in the config.properties file. If present in both places, the value of the -D flag argument takes precedence. This property is used by several components and services in the AJSF. The value of this property must match the symbolic name configured for the Java server process on the Process Configuration window.

Required:	Yes
Example Entry:	JAVA -Dprocess.id=JDH2
Usage:	CommonXMLFormatter DBConnectionManager ExternalConnectionComponent FrameworkAdmin JMSMessageRouterService MessageRouterService MessageService EventLogAppender LogManager TimerTaskManager JMSUtils

System Property Configuration

The file.encoding and file.separator system properties are set by the JVM. Several options are available for configuring the process ID and configuration filename system properties as discussed below.

For a AJSF application that is not running as a Web server, the process ID and configuration filename properties are set as follows. The process ID can be set in a -D option JVM runtime argument or it can be set inside the configuration properties file. If it is set in both locations, the -D argument value takes precedence. The configuration filename property must either be set when a Java server process is started using the -D option JVM runtime argument (-Dconfig.filename), or it defaults to a file named “config.properties”.

For Web-based AJSF applications, the process ID and configuration filename servlet initialization parameters may be retrieved from the web.xml file instead.

Non-Web Application

When the configuration filename property is specified by the -Dconfig.filename argument, the following rules apply for resolving the path to the configuration file. You can specify an absolute path (e.g., c:/config/myconfig.properties) or a relative path (e.g., ../config/myconfig.properties) to the configuration file. If you specify a relative path, the path is resolved based on the current working directory of the JVM. You can also use the following shorthand notations in the argument value to specify the JVM’s classpath to search for the configuration file or the home directory of the user who started the JVM.

- For the classpath, you can use “{classpath}/” or “\${classpath}/” to represent the classpath in the argument value (e.g., “{classpath}/myconfig.properties” or “\${classpath}/myconfig.properties”). Note that the argument value must start with “{classpath}/” or “\${classpath}/” for this shorthand substitution notation.
- For the home directory, you can use “{user.home}/” or “\${user.home}/” to represent the home directory of the user who started the JVM (e.g., “{user.home}/myconfig.properties” or “\${user.home}/myconfig.properties”). In a Windows environment, both of these examples are expanded to “C:/Documents and Settings//myconfig.properties”. Note that the argument value must start with “{user.home}/” or “\${user.home}/” for this shorthand substitution notation.

When the configuration filename property is not specified by the -Dconfig.filename argument, the AJSF uses a default file name of “config.properties” and searches for this file in the following order:

1. The current working directory.
2. The classpath.
3. The user’s home directory.

Web Application

For AJSF applications running as Web servers, the process ID and configuration properties filename can be retrieved from the process.id and config.filename servlet initialization parameters, respectively, in the web.xml file. In addition, the restart.seconds servlet initialization parameter can also be retrieved from the web.xml file.

The restart.seconds parameter specifies the minimum amount of time the web container waits before attempting to restart the servlet after it throws an `UnavailableException` during initialization. If this parameter is not defined in the web.xml file, it defaults to five seconds. The container attempts to re-initialize the servlet with a newly instantiated servlet interface when it receives a request for the servlet. If the minimum wait time has not yet expired, the request is blocked and queued until the application successfully initializes. ACI web applications, such as the ACI desktop user interface, can throw an `UnavailableException`, for example, if the servlet fails to open a JDBC connection (i.e., database unavailable). Use of this parameter allows the web container to make the servlet temporarily unavailable without impacting the availability of other web applications running in

the container. How the web container actually handles an `UnavailableException` may vary among web container providers (e.g., ACI Web Application Services - Java Services, WebSphere, Weblogic, etc.).

Note: Any requests received by the web container for an unavailable servlet are blocked and queued until the servlet application is successfully initialized.

The following is an example of the servlet initialization parameters for the `InitializationHTTPServlet`.

```
<servlet>
  <servlet-name>InitializationHTTPServlet</servlet-name>
  <display-name>Initialization</display-name>
  <servlet-class>com.aciworldwide.framework.InitializationHTTPServlet</
servlet-class>
  <init-param>
    <param-name>process.id</param-name>
    <param-value>MyProcessId</param-value>
  </init-param>
  <init-param>
    <param-name>config.filename</param-name>
    <param-value>/WEB-INF/config/myconfig.properties</param-value>
  </init-param>
  <init-param>
    <param-name>restart.seconds</param-name>
    <param-value>8</param-value>
  </init-param>
</load-on-startup>100</load-on-startup>
```

If the value of the `config.filename` parameter begins with a slash character (“/”), the file location is assumed to be absolute (relative to the Web application’s context root). If the value of the `config.filename` parameter does not begin with a slash character, the AJSF searches for the file in the following locations in the order specified.

1. `/WEB-INF/conf`
2. `/WEB-INF/config`
3. `/WEB-INF`
4. Use the servlet’s classloader

If the `config.filename` parameter is not specified, the AJSF searches for a file named “`config.properties`” in the locations and order specified above. If the AJSF still cannot find the configuration file, it searches for the file using the “[Non-Web Application](#)” method described above.

Optional System Properties

The following system properties are optional for AJSF-based applications.

GetAllMask — A flag indicating whether a AJSF-based application retrieves field masks from disk one at a time as they are needed or all at once the first time one needs to be used. Once a field mask is retrieved, it is stored in cache memory so the application does not have to retrieve it from disk again. Valid values are as follows:

true = Yes, retrieve all field masks the first time one needs to be used.
false = No, retrieve field masks one at a time as they are needed. Default.

The GetAllMask property is set with the `-D` flag argument when a Java server process is started (i.e., `-DGetAllMask=true` or `-DGetAllMask=false`). Refer to the “[Field Masking](#)” topic in section 4 for more information on field masks.

jdal.trace — A flag indicating whether the ACI-written JDBC driver, Java Database Access Layer (JDAL) performs database tracing. Valid values are as follows:

true = Yes, JDAL performs database tracing.
false = No, JDAL does not perform database tracing. Default.

Note: The Trace facility was designed as a diagnostic tool for development and test environments and not for general use in a production environment. ACI strongly recommends that it not be used in a production environment due to the overhead and impacts on performance. Therefore, you should only enter this property in a `-D` JVM runtime argument when instructed by ACI for problem resolution.

jdal.trace.method — A flag indicating whether the JDAL driver performs method tracing when JDAL tracing is enabled. When this property is set to true, trace method signatures through the AJSF JDAL driver are included in the trace output for problem resolution. Valid values are as follows:

true = Yes, JDAL performs method tracing.

false = No, JDAL does not perform method tracing. Default.

Note: The Trace facility was designed as a diagnostic tool for development and test environments and not for general use in a production environment. ACI strongly recommends that it not be used in a production environment due to the overhead and impacts on performance. Therefore, you should only enter this property in a -D JVM runtime argument when instructed by ACI for problem resolution.

Configuration Properties

Properties defined in the `config.properties` file are read by a Java server process when it initializes. A minimum of four configuration properties are required to bootstrap a Java server process built on the AJSF—`db.user.id`, `db.user.password`, `event.log4j.config.url`, and `trace.log4j.config.url`. Several database properties (i.e., properties that start with “db.”) used to access the application database must also be defined in the `config.properties` file. These configuration properties, along with any other properties required to initialize an application process, must be defined in the `config.properties` file specified at runtime in the `config.filename` property for a Java server process. You can view the properties configured in the `config.properties` file, along with the properties configured in the Application Configuration data source, on the Configuration Properties window. You must edit the `config.properties` file to change the values of configuration properties in this file. Modified property values in the `config.properties` file do not take effect until you stop and restart the Java server process.

Password Encryption Utility



The `ConfigPropertyBuilder.class` command window utility enables you to encrypt the password value for the following properties:

- `acijmx.oncf.xpnet.system.xxxx.user.password`
- `db.user.password`
- `keystore.password`
- `SSL.client.keystore.password`
- `SSL.client.truststore.password`
- `SSL.server.keystore.password`
- `SSL.server.truststore.password`

Cross-industry best practices recommends that all passwords for database access be encrypted. Note that the database password must be encrypted before the ACI application can access the database. After you have entered the password and associated user ID properties in the `config.properties` file, enter the following command at the Unix or DOS prompt to encrypt passwords. Once the passwords are encrypted in the `config.properties` file, you must stop and restart the AJI application that uses the `config.properties` file.

Note: If you are changing the value of the `db.user.password` property for BASE24-eps AJI applications, you must also change the password in the `db-user-id_UID` parameter in the `config.csv` for the IBM System z and Unix platforms. Refer to the *BASE24-eps Environment Management User Guide* for information on updating the `db-user-id_UID` parameter.

Syntax

```
java -cp CLASSPATH ConfigPropertyBuilder directory/config.properties
      -UserPropertiesOnly
```

CLASSPATH is a classpath that includes the `framework-tools.jar`, `framework-core.jar`, and `bouncycastle.jar`. The `ConfigPropertyBuilder.class` is distributed in the `framework-tools.jar`.

directory is the qualified name of the directory in which the `config.properties` file is located.

`-UserPropertiesOnly` is a parameter indicating to prompt for user properties only. If you do not provide this parameter when you run the `ConfigPropertyBuilder` utility, the utility prompts for all AJSF and user password properties if they are present in the specified `config.properties` file.

If you need to encrypt user properties, they must be defined in the `config.properties` file as follows:

```
#####
#
# The following section of the file is a place holder where user
# properties to be encrypted may be placed.
#
# It is critical that the section begin with "Begin User Encrypted
# Values" and end with "End User Encrypted Values".
#
#####

##### Begin User Encrypted Values #####
#acijmx.oncf.xpnet.system.xxxx.user.password=placeholder
##### End User Encrypted Values #####
```

When you run the utility, it prompts you for the db.user.id property value and each of the AJSF and user password property values, or just the user password property values, as shown below. If you are not using one of the password properties, simply press **Enter** to skip it. You must enter clear values correctly for the correct value to be generated.

Prompting AJI Encrypted Values...

DB UserID or Password Change requires both db.user.id and db.user.password...

Enter db.userid:

db-user-value

Enter db.user.password:

db-password-value

Enter keystore.password:

keystore-password-value

Enter SSL.client.keystore.password:

ssl-client-keystore-password-value

Enter SSL.client.truststore.password:

ssl-client-truststore-password-value

Enter SSL.server.keystore.password:

ssl-server-keystore-password-value

Enter SSL.server.truststore.password:

ssl-server-truststore-password-value

Processing User Encrypted Values.

Enter oncf.xpnet.system.xxxx.user.password:

user-password-value

Enter <KEY> to use for encrypting oncf.xpnet.system.xxxx.user.password:

key-value

oncf.xpnet.system.xxxx.user.password Encrypted value is: *encrypted-value*

The utility creates a new config.properties file with encrypted values for the password properties entered. It also renames the original config.properties file to config.properties.xx where xx starts at 01 and is increased by one with each execution of the utility.

The utility also displays the encrypted and decrypted values of each entered password property value as shown below.

password-property-name Encrypted value is: *encrypted-value*
password-property-name Decrypted value is: *unencrypted-value*

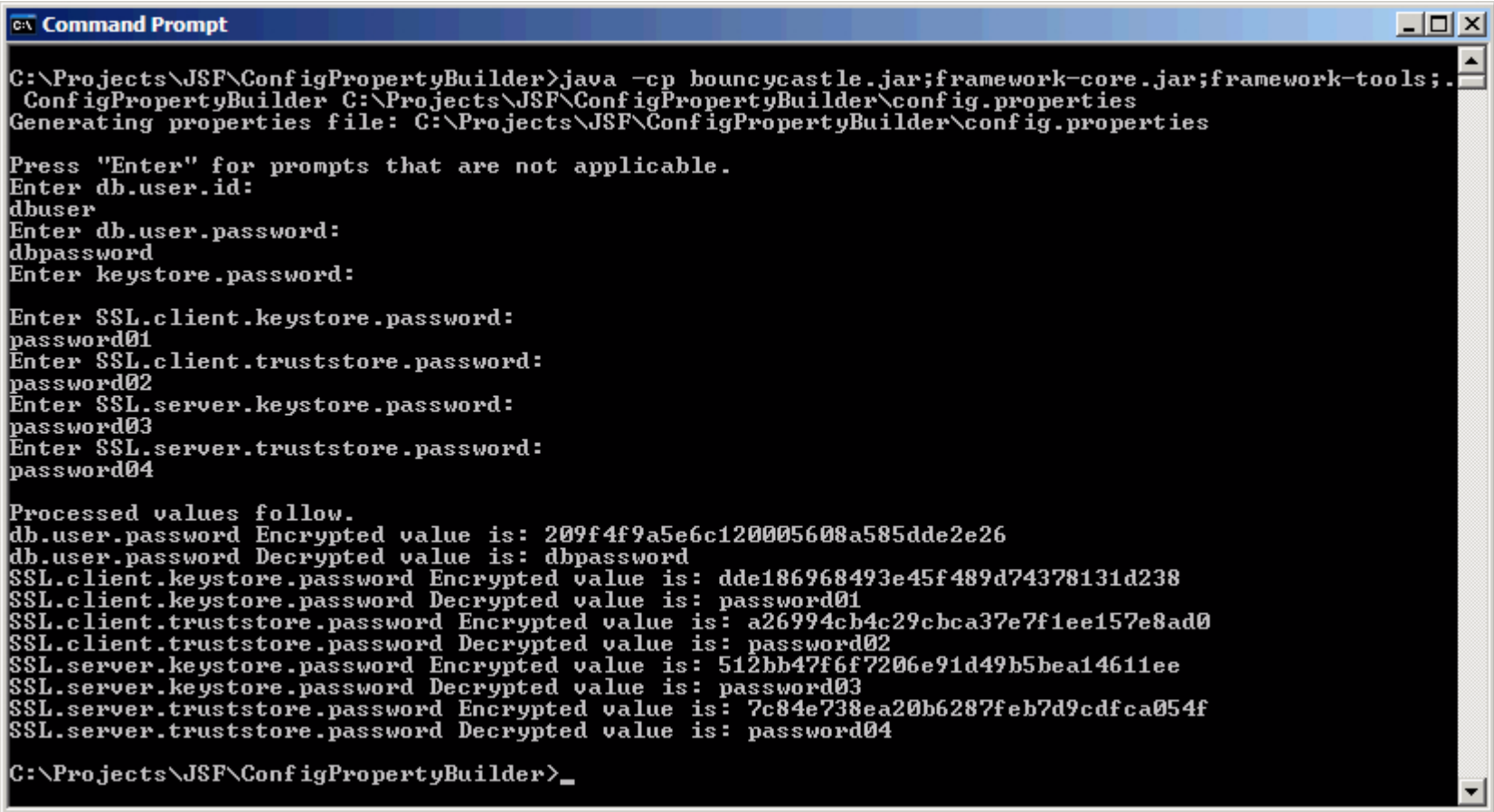
If you do not enter a value for a prompted password property, it is not processed.

Note: After a password property value is encrypted, it will be much longer than the original value and will contain both alphabetic and numeric characters.

Example Command. The following page depicts an example command for running the ConfigPropertyBuilder.class command window utility from a DOS prompt. The command was run from a specific directory to simplify the class path needed and the config.properties file location was fully-qualified. The -UserPropertiesOnly parameter was not specified.

Example Command Syntax

```
C:\Projects\JSF\ConfigPropertyBuilder>java -cp bouncycastle.  
jar;framework-core.jar;framework-tools.jar; ConfigPropertyBuilder  
C:\Projects\JSF\ConfigPropertyBuilder\config.properties
```



```
C:\> Command Prompt

C:\Projects\JSF\ConfigPropertyBuilder>java -cp bouncycastle.jar;framework-core.jar;framework-tools;.
ConfigPropertyBuilder C:\Projects\JSF\ConfigPropertyBuilder\config.properties
Generating properties file: C:\Projects\JSF\ConfigPropertyBuilder\config.properties

Press "Enter" for prompts that are not applicable.
Enter db.user.id:
dbuser
Enter db.user.password:
dbpassword
Enter keystore.password:

Enter SSL.client.keystore.password:
password01
Enter SSL.client.truststore.password:
password02
Enter SSL.server.keystore.password:
password03
Enter SSL.server.truststore.password:
password04

Processed values follow.
db.user.password Encrypted value is: 209f4f9a5e6c120005608a585dde2e26
db.user.password Decrypted value is: dbpassword
SSL.client.keystore.password Encrypted value is: dde186968493e45f489d74378131d238
SSL.client.keystore.password Decrypted value is: password01
SSL.client.truststore.password Encrypted value is: a26994cb4c29cbca37e7f1ee157e8ad0
SSL.client.truststore.password Decrypted value is: password02
SSL.server.keystore.password Encrypted value is: 512bb47f6f7206e91d49b5bea14611ee
SSL.server.keystore.password Decrypted value is: password03
SSL.server.truststore.password Encrypted value is: 7c84e738ea20b6287feb7d9cdfca054f
SSL.server.truststore.password Decrypted value is: password04

C:\Projects\JSF\ConfigPropertyBuilder>_
```


Application Configuration Properties

Most properties, other than those required to bootstrap a Java server process, are configured in the Application Configuration data source (APPLCNFG). The advantage of placing these properties in the APPLCNFG is that you can change property values in the APPLCNFG without stopping and restarting the Java server process. Properties in the APPLCNFG are maintained from the Application Configuration window. When you add, modify, or delete properties in the APPLCNFG, the changes do not take effect until after you reload the APPLCNFG into cache memory by selecting a process ID on the Configuration Properties window and click the Reload button. The Configuration Properties window displays all APPLCNFG and config.properties file properties for a Java server process.

Note: Some AJSF-based applications, such as BASE24-eps ATM (Java) Device Handler processes, use additional properties files. Some of the properties used by BASE24-eps ATM (Java) Device Handler processes are defined in ATM Bundle and device property files.

AJSF Properties

The following properties are generic to the AJSF. Any application-specific properties are either described in an application-specific appendix to this user guide or in an application-specific manual.

baos.manager.init.count — The number of `ByteArrayOutputStream` objects initially created by the Byte Array Output Stream Manager. Byte array output streams are pooled to avoid the cost of initializing the internal `byte[]` each time one is created. If this property is not configured, it defaults to 4.

Required: No
Example Entry: `baos.manager.init.count=2`
Usage: Byte Array Output Stream Manager

baos.manager.init.stream.size — The initial size of the `ByteArrayOutputStream` objects to create. If this property is not configured, it defaults to approximately 4K.

Required: No
Example Entry: `baos.manager.init.stream.size=2024`
Usage: Byte Array Output Stream Manager

baos.manager.max.pool.size — The maximum number of `ByteArrayOutputStream` objects that can be pooled. If this property is not configured, it defaults to 10.

Required: No
Example Entry: `baos.manager.max.pool.size=12`
Usage: Byte Array Output Stream Manager

cleanup.userauditlog.retention.days — The number of days to leave audit rows in the User Audit Log data source (USRAULOG) before they are removed. The default value is 30 days.

The Payment Card Industry (PCI) Data Security Standard, which is used as the framework for Visa's Cardholder Information Security Program (CISP), states that you should retain audit trail history "for a period that is consistent with its effective use, as well as legal regulations." An audit history usually covers a period of at

least one year, with a minimum of three months available online. The PCI Data Security Standard also states that you should “promptly back-up audit trail files to a centralized log server or media that is difficult to alter.”

Required: No
Example Entry: `cleanup.userauditlog.retention.days=15`
Usage: User Audit Log Cleanup component

cleanup.userauditlog.cleanup.starttime — The time of day, in hh:mm format, the User Audit Log Cleanup component starts removing old rows from the User Audit Log data source (USRAULOG). The default value is 00:15 (12:15 am). This time is in the time of the executing server where the application is running and is automatically adjusted for daylight savings time (summer time).

Required: No
Example Entry: `cleanup.userauditlog.cleanup.starttime=00:15`
Usage: User Audit Log Cleanup component

cleanup.userauditlog.timeperiod.hours — The number of hours between runs if the User Audit Log Cleanup component is to remove old audit rows from the User Audit Log data source (USRAULOG) more frequently than every twenty four hours. The default value is 24 hours.

Required: No
Example Entry: `cleanup.userauditlog.timeperiod.hours=12`
Usage: User Audit Log Cleanup component

cleanup.userauditlog.batch.size — The number of rows to delete from the User Audit Log data source (USRAULOG) in a single batch when the User Audit Log Cleanup component removes old audit rows. If this property is not configured, it defaults to 500. If this property is set to 0, the User Audit Log Cleanup component attempts to remove all old audit rows in a single batch. If the USRAULOG has not been cleaned for an extended period of time, this could cause the maximum number of row locks for the platform to be exceeded, thereby causing user audit log cleanup to fail.

Required: No
Example Entry: `cleanup.userauditlog.timeperiod.hours=500`
Usage: User Audit Log Cleanup component

cleanup.database.cache.interval.hours — The number of hours to leave data rows in cache memory before they are removed. The default value is 24 hours.

Required: No
 Example Entry: cleanup.database.cache.interval.hours=48
 Usage: Database Cache Cleanup component

cleanup.dbcache.cleanup.starttime — The time of day, in hh:mm format, the Database Cache Cleanup component starts removing old data rows from cache memory. The default value is 23:59 (midnight). This time is in the time of the executing server where the application is running and is automatically adjusted for daylight savings time (summer time).

Required: No
 Example Entry: cleanup.dbcache.cleanup.starttime=23:15
 Usage: Database Cache Cleanup component

cleanup.dbcache.timeperiod.hours — The number of hours between runs if the Database Cache Cleanup component is to remove old data rows from cache memory more frequently than every twenty four hours. The default value is 24 hours.

Required: No
 Example Entry: cleanup.dbcache.timeperiod.hours=12
 Usage: Database Cache Cleanup component

context.manager.connection.id — When the context.type property is set to MEM-OUT-OF-PROCESS or MEM-OUT-OF-PROCESS-WITH-CACHE, this property specifies the external connection ID of an entry in the External Connection data source (EXTRCNCT) that is used to send the context message to the Context Manager process. The value of this property must match a value configured in the EXTRCNCT.

Required: Yes, if the context.type property is set to MEM-OUT-OF-PROCESS
 Example Entry: context.manager.connection.id=JSRCTXMGR
 Usage: ContextManager

context.retention.period — The number of days, from the start of the current day, message context is retained in the Message Context data source (MSGCNTX) before it is deleted by the Context Manager process. This property is used only if the context.type property is set to a value of DISK.

Required: Yes, if the context.type property is set to DISK

Example Entry: context.retention.period=3

Usage: Message Context Component

context.type — Indicates the type of context management the Context Manager performs. Valid values are as follows:

- | | |
|--------------------|---|
| DISK | = The transaction (or session) context is maintained in the Message Context data source (MSGCNTX) on disk. This allows for response messages to be returned to any process that is configured to listen on the response queue. |
| DISK-WITH-CACHE | = A copy of the Message Context data source (MSGCNTX) is maintained in cache by the local process. If the local process cannot find the transaction (or session) context in cache, it attempts to retrieve the context from another process or from disk. |
| MEM-IN-PROCESS | = The transaction (or session) context is maintained in the memory of the process generating the context. This means that response messages must be returned to the process that originated the message. |
| MEM-OUT-OF-PROCESS | = The transaction (or session) context is maintained in the memory of a separate process configured as a Context Manager. This allows for response messages to be returned to any process that is configured to listen on the response queue. |

MEM-OUT-OF-PROCESS-WITH-CACHE = The transaction (or session) context is maintained in cache by a separate process configured as a Context Manager. If the Context Manager process cannot find the transaction context in cache, it attempts to retrieve the context from another process or from disk. This allows for response messages to be returned to any process that is configured to listen on the response queue.

Note: For performance reasons, ACI recommends that this property be set to a value of MEM-IN-PROCESS for BASE24-eps ATM (Java) Device Handler processes.

Required: Yes
 Example Entry: context.type=MEM-IN-PROCESS
 Usage: ContextManager

crypto.jce.prng.algname — The name of the SHA-1 Pseudo-Random Number Generation (SHA1PRNG) algorithm to use if the provider specified by the crypto.jce.prng.provider property uses a name other than the default value of SHA1PRNG.

Required: No
 Example Entry: crypto.jce.prng.algname=CRYPTOKI
 Usage: JCE Provider

crypto.jce.prng.provider — The name of the SHA-1 Pseudo-Random Number Generation (SHA1PRNG) provider.

Required: No
 Example Entry: crypto.jce.prng.provider=au.com.eracom.crypto.provider.ERACOMProvider
 Usage: JCE Provider

crypto.jce.aescipher.algname — The name of the Advanced Encryption Standard (AES) cipher algorithm to use if the provider specified by the crypto.jce.aescipher.provider property uses a name other than the default value of AES.

Required: No
Example Entry: `crypto.jce.aescipher.algname=Rijndael`
Usage: JCE Provider

crypto.jce.aescipher.provider — The name of the Advanced Encryption Standard (AES) cipher algorithm provider.

Required: No
Example Entry: `crypto.jce.aescipher.provider=sun.security.provider.Sun`
Usage: JCE Provider

crypto.jce.descipher.algname — The name of the Data Encryption Standard (DES) algorithm to use if the provider specified by the `crypto.jce.descipher.provider` property uses a name other than the default value of DES/CBC/PKCS5Padding.

Required: No
Example Entry: `crypto.jce.descipher.algname=DES`
Usage: JCE Provider

crypto.jce.descipher.provider — The name of the Data Encryption Standard (DES) algorithm provider.

Required: No
Example Entry: `crypto.jce.descipher.provider=com.sun.crypto.provider.SunJCE`
Usage: JCE Provider

crypto.jce.desedecipher.algname — The name of the Triple Data Encryption Standard (3DES) algorithm to use if the provider specified by the `crypto.jce.desedecipher.provider` property uses a name other than the default value of DESede.

Required: No
Example Entry: `crypto.jce.desedecipher.algname=DESede`
Usage: JCE Provider

crypto.jce.desedecipher.provider — The name of the Triple Data Encryption Standard (3DES) algorithm provider.

Required: No
Example Entry: `crypto.jce.desedecipher.provider=au.com.eracom.crypto.provider.ERACOMProvider`
Usage: JCE Provider

crypto.jce.cast5cipher.algname — The name of the block cipher developed by Carlisle Adams and Stafford Tavares (CAST5) to use if the provider specified by the `crypto.jce.cast5cipher.provider` property uses a name other than the default value of CAST5.

Required: No
Example Entry: `crypto.jce.cast5cipher.algname=DESede`
Usage: JCE Provider

crypto.jce.cast5cipher.provider — The name of the CAST5 block cipher algorithm provider.

Required: No
Example Entry: `crypto.jce.cast5cipher.provider=au.com.eracom.crypto.provider.ERACOMProvider`
Usage: JCE Provider

crypto.jce.sha1.algname — The name of the Secure Hash Algorithm (SHA-1) to use if the provider specified by the `crypto.jce.sha1.provider` property uses a name other than the default value of SHA.

Required: No
Example Entry: `crypto.jce.sha1.algname=SHA`
Usage: JCE Provider

crypto.jce.sha1.provider — The name of the SHA-1 hash algorithm provider.

Required: No
Example Entry: `crypto.jce.sha1.provider=au.com.eracom.crypto.provider.ERACOMProvider`
Usage: JCE Provider

crypto.jce.md5.algname — The name of the MD5 algorithm to use if the provider specified by the crypto.jce.md5.provider property uses a name other than the default value of MD5. The MD5 algorithm is used to create a message digest of input data and is used to verify data integrity.

Required: No
Example Entry: crypto.jce.md5.algname=MD5
Usage: JCE Provider

crypto.jce.md5.provider — The name of the MD5 algorithm provider.

Required: No
Example Entry: crypto.jce.md5.provider=au.com.eracom.crypto.provider.ERACOMProvider
Usage: JCE Provider

crypto.jce.hmacsha1.algname — The name of the Hash-based Message Authentication Code (HMAC) Secure Hash Algorithm (SHA-1) to use if the provider specified by the crypto.jce.hmacsha1.provider property uses a name other than the default value of HMACSHA1.

Required: No
Example Entry: crypto.jce.hmacsha1.algname=HMACSHA1
Usage: JCE Provider

crypto.jce.hmacsha1.provider — The name of the HMAC SHA-1 hash algorithm provider.

Required: No
Example Entry: crypto.jce.hmacsha1.provider=au.com.eracom.crypto.provider.ERACOMProvider
Usage: JCE Provider

crypto.jce.rsa.algname — The name of the Rivest, Shamir, and Adleman (RSA) cipher algorithm to use if the provider specified by the crypto.jce.rsa.provider property uses a name other than the default value of RSA.

Required: No
Example Entry: crypto.jce.rsa.algname=RSA
Usage: JCE Provider

crypto.jce.rsa.provider — The name of the RSA cipher algorithm provider.

Required: No
Example Entry: `crypto.jce.rsa.provider=au.com.eracom.crypto.provider.ERACOMProvider`
Usage: JCE Provider

crypto.jce.sha1rsasig.algname — The name of the SHA-1 with RSA digital signature algorithm to use if the provider specified by the `crypto.jce.sha1rsasig.provider` property uses a name other than the default value of `SHA1withRSA`.

Required: No
Example Entry: `crypto.jce.sha1rsasig.algname=SHA1withRSA`
Usage: JCE Provider

crypto.jce.sha1rsasig.provider — The name of the SHA-1 with RSA digital signature algorithm provider.

Required: No
Example Entry: `crypto.jce.sha1rsasig.provider=au.com.eracom.crypto.provider.ERACOMProvider`
Usage: JCE Provider

crypto.jce.dbcipher.algname — The name of the Triple Data Encryption Standard (3DES)/Electronic Code Book (ECB)/Public Key Cryptographic Standards (PKCS) 5 Padding algorithm to use if the provider specified by the `crypto.jce.dbcipher.provider` property uses a name other than the default value of `DESede/ECB/PKCS5Padding`.

Required: No
Example Entry: `crypto.jce.dbcipher.algname=DESede/ECB/PKCS5Padding`
Usage: JCE Provider

crypto.jce.dbcipher.provider — The name of the `DESede/ECB/PKCS5Padding` algorithm provider.

Required: No
Example Entry: `crypto.jce.dbcipher.provider=sun.security.provider.Sun`
Usage: JCE Provider

crypto.pool.ceiling — The maximum number of objects that can be pooled for a particular combination of cryptographic object and key. This indicates the maximum number of concurrent accesses needed for a cryptographic object and ensures that no more than the specified number of keys per key type, algorithm, or alias are stored in the pool. The default value for this property is 5.

If you are using an ACI application that accesses a hardware security device from a vendor other than SafeNet (Eracom) using the JCE, you should leave this property set to its default value or set it to a value less than the default.

If you are using an ACI application that accesses a SafeNet (Eracom) hardware security device using the JCE, you can use the following calculation to determine the appropriate value for this property.

KeyCount	= (number of DES keys + number of 3DES keys + number of HMAC keys) $\times$ 2 (modes of operation (encrypt/decrypt))
NumConnections	= 65534 $\times$ ~90% = 59000 (Allow approximately 10% padding for overages)
crypto.pool.ceiling value	= NumConnections $\div$ KeyCount (Round down for the final result)

In the above calculation, 65534 is the maximum number of *sessions* or connections available to a SafeNet (Eracom) security device. When some ACI applications, such as the ACI Access Control Server (ACS), use a large number of keys in a concurrent manner, the ACS may use all of these connections. The ceiling specified by this property prevents this from occurring.

Required:	No
Example Entry:	crypto.pool.ceiling=5
Usage:	JCECryptoPool

crypto.provider — The name of the class used to provide cryptographic services (i.e., key and certificate management).

Required:	Yes, if using the Certificate Manager or Cryptography class
Example Entry:	crypto.provider=com.aciworldwide.framework.crypto.JCEProvider
Usage:	Certificate Manager Cryptography

datatype.jar***datatype.jar.aci.***

datatype.jar.csm.* — Identifies the relative Universal Resource Identifier (URI) location of application data types JAR files. Each *.jar file can contain an XML schema or likely multiple schemas where each schema expresses a data type constraint. The AJSF loads these constraints for use by the applications during data validation. The value of the config.home property is used to resolve the relative URI to an absolute location.

Required: No

Example Entries: datatype.jar.aci.ESXMLDataTypes=lib/es-datatypes.jar

db.alt.datasource.name.db-service-name — The name of an alternate data source for the database service indicated by *db-service-name*. The value of this property must match a user-defined data source name configured in the Alternate Data Source data source (DATASRC). This property enables you to maintain specified data sources in a data format that is different from the main application database. For example, if your BASE24-eps database is an SQL database, you could maintain specified data sources in flat JDAL files by configuring those alternate JDAL data sources in the DATASRC and specifying them in this property. When the database service attempts to access a data source, it first looks to see whether a *db.alt.datasource.name.db-service-name* property is configured for the database service that accesses that data source. If so, it accesses the alternate data source specified by the property. If not, it accesses the data source in the usual manner.

Required: No

Example Entry: db.alt.datasource.name.inst=Institution

Usage: DBService

db.audited — A flag indicating whether all database table activity is fully audited, except for those *db.audited.table-name* properties set to a value of medium or none. This includes all ACI desktop user interface or browser web page activity, database activity initiated by a timer task, or any other database activity that uses the database table. Cross-industry best practices recommends



that you maintain and monitor an audit trail of all user activity. Changes made directly to the database without using a user interface (e.g., DALCI or GoldenGate replication changes) are not audited. Valid values are as follows:

- `true` = Yes, all database table activity is fully audited, except for those `db.audited.table-name` properties set to a value of medium or none.
- `false` = No, no database table activity is audited, except for those `db.audited.table-name` properties set to a value of full or medium.



db.audited.table-name — Indicates the type of user auditing to be performed for a particular physical database table, represented by *table-name*. These properties provide more granularity than the `db.auditing` property for the auditing of specific database tables. If the `db.audited` property is set to a value of `true`, `db.audited.table-name` properties set to a value of medium or none limit the auditing performed for those database tables. If the `db.audited` property is set to a value of `false`, only database tables for those `db.audited.table-name` properties set to a value of full or medium are audited. Cross-industry best practices recommends that you maintain and monitor an audit trail of all user activity. The lowercase *table-name* should be set to a physical database table name. Auditing includes all ACI desktop user interface activity or browser web page activity, database activity initiated by a timer task, or any other database activity that uses the database table. Changes made directly to the database without using a user interface (e.g., DALCI or GoldenGate replication changes) are not audited. You can configure multiple occurrences of this property, with one for each database table requiring auditing. Valid values are as follows:

- `none` = No auditing is performed for this database table.
- `medium` = Medium auditing is performed for this database table. For this value, only header information is logged in auditing records.
- `full` = Full auditing is performed for this database table. For this value, header information as well as detail information about the before and after images of the affected record are logged.

If this property is not defined for a AJSF database table, it defaults to a value of none unless BASE24-eps is installed. When BASE24-eps is installed, all of these properties default to a value of full.

table-name values generic to the AJSF are listed in the following table.

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
applcnfg	Application Configuration data source (APPLCNFG)	Application Configuration window
datasrc	Alternate Data Source data source (DATASRC)	Not Applicable
event	Event data source (Event)	Event Message and Event Suppression Information tabs of the Event Configuration and Management window
event_document	Event Document data source (Event_Document)	Event Configuration and Management window
extrcnct	External Connections data source (EXTRCNCT)	External Connection Configuration window
fltxrole ¹	Filters X Roles data source (FLTXROLE)	User Security Profile Configuration window (Filters tab) User Security Role Configuration window (Filters tab)
fields ¹	Fields data source (Fields)	
fldvals ¹	Field Values data source (FldVals)	
flxfldvl ¹	Filter Field Values data source (FLXFldVl)	
filters ¹	Filters data source (Filters)	
functs ¹	Functions data source (Functs)	User Security Role Configuration window (Permissions tab)

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
hsmcnfg	Hardware Security Module (HSM) Configuration data source (HSMCNFG)	Not applicable
languag	Language data source (LANGUAG)	Not applicable
listener	Listener data source (LISTENER)	Listener Configuration window
password ¹	Password data source (Password)	User Security Profile Configuration window (Password tab)
pswdgrp ¹	Password Group data source (PswdGrp)	User Security Group Configuration window (Password Group tab)
pswdhist ¹	Password History data source (PswdHist)	User Security Profile Configuration window (Password tab)
permdetl ¹	Permission Detail data source (PermDetl)	User Security Role Configuration window (Permissions tab)
permgrp ¹	Permission Group data source (PermGrp)	User Security Role Configuration window (Permission Groups tab)
perms ¹	Permissions data source (Perms)	User Security Role Configuration window (Permissions tab)
permtype ¹	Permissions Type Code data source (PermType)	User Security Role Configuration window (Permissions tab)
pmgpxprm	Permission Group Permissions data source (PMGPXPRM)	Not applicable

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
process	Process data source (PROCESS)	Process Configuration window
roles ¹	Roles data source (Roles)	User Security Role Configuration window (Roles tab)
rolxpmgp ¹	Role Permission Groups data source (RoIXPmGp)	User Security Role Configuration window (Roles tab)
secgrp ¹	Security Group data source (SecGrp)	User Security Group Configuration window (Security Group tab)
taskflds ¹	Task Fields data source (TaskFlds)	User Security Role Configuration window (Permissions tab)
taskfncs ¹	Task Functions data source (TaskFnCs)	
tasks ¹	Tasks data source (Tasks)	
usraulog	User Audit Log data source (USRAULOG)	Not applicable
versions	Versions data source (VERSIONS)	Not applicable
webappopts	Web Application Options data source (WEBAPPOPTS)	Not applicable
webapps	Web Applications data source (WEBAPPS)	Not applicable
webusropts	Web Application User Options data source (WEBUSROPTS)	Not applicable

Refer to the application-specific appendices of this user guide or an application-specific manual for a complete list of database tables and the data sources accessed by an application.

Required: No
Example Entry: db.audited.applcnfg=full
Usage: Database Component

db.broadcast — A flag indicating whether to send broadcast messages to other JVMs when performing one of the following functions: clearing the cached database entries for all services, clearing the cached database entries for a specific service, or reloading the application configuration properties for the process into cache memory. Valid values are as follows:

true = Yes, send broadcast messages for certain database caching operations.
(Default)
false = No, do not send broadcast messages for certain database caching operations.

Required: No
Example Entry: db.broadcast=true
Usage: Framework Administration

db.cached.*table-name* — Indicates whether the data source indicated by the database table name, represented by *table-name*, is accessed from cache memory or from disk. Valid values are as follows:

true = Yes, perform caching for this database table.
false = No, do not perform caching for this database table.

This property indicates whether the database access performed on the specified service uses the database table caching mechanism provided by the CacheManager. For all SQL queries, except the `fetchbyDataObject` method, the database table cache is searched first for the data. If the data is not found, the cache is updated with the data retrieved from the data source. Thereafter, it is retrieved from cache until the timer for the `cleanup.database.cache.interval.hours` property expires or the database service adds, updates or deletes a row of the data source or the Java server process is stopped and restarted, at which time, cache memory is flushed. This functionality should be used for tables that are updated infrequently, or read frequently and contain a small number of rows. Database table names for tables must match the physical table name. You can configure multiple occurrences of this property, with one for each database table requiring caching support.

Refer to the [db.audited.table-name](#) property description for a listing of database tables generic to the AJSF. Refer to the application-specific appendices of this user guide or an application-specific manual for a complete list of database tables and the data sources accessed by an application.

Required: Yes
Example Entry: db.cached.applcnfg=true
Usage: Cache Manager

db.connection.wait.time — The number of milliseconds the Database Connection Manager waits for another thread to free a database connection. This property value is used when the maximum number of connections has been reached as specified in the db.max.connections property.

Required: Yes
Example Entry: db.connection.wait.time=10
Usage: DBConnectionManager

db.datasource — The data source URI specification that allows the JDBC driver to access the correct database. You can typically find the configuration requirements for this property in the vendor documentation for the JDBC driver. JDBC driver requirements may include information such as the host or IP address where database resides, the port to connect to for the database, the name of database on the host, and other JDBC driver settings.

Required: Yes
Example Entries: BASE24-eps ATM (Java) Device Handler:
db.datasource=jdbc:JTurbo://172.16.1.53:1433/
DeviceHandlerv32/sp=false

BASE24-eps Java User Interface Servlets:
db.datasource=jdbc:JDAL:-MDBV=CSV:-AI=ES:-STRT=/G/
moat/ui6sdata

ACI Commerce Gateway:
db.datasource=jdbc:JTurbo://127.0.0.1:1433/ACIDB/sp=false
(MS SQL Server):
db.datasource=jdbc:JTurbo://127.0.0.1:1433/ACIDB/sp=false
(Oracle 8i):
db.datasource=jdbc:oracle:thin:@127.0.0.1:1521:ACIDB

ACI Access Control Server:

db.datasource=jdbc:JTurbo://127.0.0.1:1433/ACISC/sp=false

(MS SQL Server):

db.datasource=jdbc:JTurbo://127.0.0.1:1433/ACISC/sp=false

(Oracle 8i):

db.datasource=jdbc:oracle:thin:@127.0.0.1:1521:ACISC

ACI Payments Manager browser:

(MS SQL Server):

db.datasource=jdbc:JTurbo://172.16.150.31/CERTEGYPM/
sp=false

ACI Payments Manager desktop:

(DB2):

db.datasource=jdbc:db2:DB2A

APF Admin:

db.datasource=jdbc:oracle:thin:@oma3server:1521:ORCL

Usage:

DBConnectionManager

db.driver — The class name of the JDBC driver to use for all database connections. Enter the entire package name as it is found in the JAR containing the driver. Refer to the vendor documentation for the JDBC driver for supporting information.

Required: Yes

Example Entries: MS SQL Server (JTurbo):

db.driver=com.ashna.jturbo.driver.Driver

Oracle 8i:

db.driver=oracle.jdbc.OracleDriver

JDAL:

db.driver=com.aciworldwide.jdbcdriver.jdal.JDALDriver

DB2:

db.driver=COM.ibm.db2.jdbc.app.DB2Driver

Usage:

DBConnectionManager

db.max.connections — The maximum number of concurrent database connections that the Database Connection Manager (DBConnectionManager) can have open at one time.

Required: Yes
Example Entry: db.max.connections=50
Usage: DBConnectionManager

db.static.connections — The number of connections the Database Connection Manager acquires when the Java server process is started. These connections remain active throughout the life of the process.

Required: Yes
Example Entry: db.static.connections=2
Usage: DBConnectionManager

db.table.prefix — The prefix, if one exists, to prepend to all data source table names in the application database. This allows you to use multiple databases with the same underlying data source names prepended with a unique identifier for each database, thereby making the full data source names in each database unique across all of the application's databases.

Required: No
Example Entries: db.table.prefix=Prod_
db.table.prefix=PM074.
Usage: Database

db.type — The type of database being used in the current JVM executing environment. Valid values are as follows:

DB2 = IBM relational database
JDAL = ACI's Java Data Access Layer database
JDAL2 = ACI's Java Data Access Layer 2 database
MSSQL = Microsoft SQL Server database
Oracle = Oracle database
SQLMX = HP NonStop SQL/MX database

Required: Yes
Example Entry: db.type=MSSQL
Usage: Database Service

db.user.id — The JDBC user ID needed to authenticate a connection to the database. This user ID is used to securely access the underlying database and is typically established by a database or system administrator. ACI applications use this user ID to access the database.

Required: Yes
 Example Entry: db.user.id=myuser
 Usage: DBConnectionManager



db.user.password — The JDBC password needed to authenticate a connection to the database. This user ID is used to securely access the underlying database and is typically established by a database or system administrator. For security, the password value stored in the config.properties file must be encrypted using the [password encryption utility](#) provided by ACI. Cross-industry best practices recommends that you do not use vendor-supplied default passwords, that you use passwords containing both numeric and alphabetic characters with a minimum password length of at least seven characters, that you change passwords at least every 90 days, and that a new password is not the same as any of the last four passwords used.

Note: For AJI applications in BASE24-eps, the value of the db.user.password property must match the value of the *db-user-id_UID* parameter in the CONFCSV (or config.csv), where *db-user-id* is the database connection user ID value. Thus, whenever you change the password value of the db.user.password property, you must also change the password value of the *db-user-id_UID* parameter. For more information on the *db-user-id_UID* parameter and changing its value, refer to the *BASE24-eps Environment Management User Guide*.

Required: Yes
 Example Entry: db.user.password=657e1f6fecea9ea8
 Usage: DBConnectionManager

DigestCredAlgorithm — A code indicating the type of secure hash algorithm (SHA) used for encoding (encrypting) passwords stored in the USEC database (i.e., Password and Password History data sources). Hash algorithms compute a fixed-length digital representation (known as a *message digest*) of an input data sequence (the *message*) of any length.. The SHA-1 algorithm generates a message digest that is 160 bits long, while the SHA-256 and SSHA-256 generate a message digest of 256 bits. Generally speaking, the longer the message digest, the harder it is to determine the original message. The SSHA-256 algorithm uses a 64 bit (8 byte) random key to *salt* the message before it is hashed (encoded) thereby creating an even more secure message digest.

This property has no impacts to existing user passwords that are currently using the SHA-1 encoding. When verifying a password, the USEC subsystem still supports password verifications (authentication) based on the existing encoded password in the Password data source.

To use the SHA-256 or SSHA-256 algorithms, the Password element in the Password data source (PASSWORD) and Password History data source (PSWDHIST) must be changed from char(28) to varchar(64).

Note: If you currently have users with passwords encrypted by the SHA-1 algorithm and do not configure this property, requests to change a password results in the new password being stored as a salted SHA-256 hash value (encoded). This enables you to gradually migrate to salted SHA-256 support.

Valid values are as follows:

SHA-1 = Use the SHA-1 algorithm to encrypt stored passwords.
SHA-256 = Use the SHA-256 algorithm to encrypt stored
 passwords.
SSHA-256 = Use the salted SHA-256 algorithm (SSHA-256) to
 encrypt stored passwords. Default.

Required: No
Example Entry: DigestCredAlgorithm=SHA-1
Usage: USEC subsystem

event.config

event.log4j.config.url — The fully-qualified URI location of the log4j configuration file (EventConfig.xml) for logging event messages.

Although both the event.config and event.log4j.config.url properties are valid, the event.config property is newer and allows you to use substitution variables when specifying the location of the EventConfig.xml file in the config.properties file. For more information on using substitution variables in this property value, refer to the [“Event Administration”](#) topic in section 4.

The event.log4j.config.url property has been replaced by the event.config property.

Required: Yes

Example Entries: `event.config=C:/ACI/APFAdmin_07.2/Server/Web/config/EventConfig.xml`
`event.log4j.config.url=file:///C:/src/jdh/config/BASE24-Common/EventConfig.xml`
`event.log4j.config.url=file:///C:/SecureCommerce/CommerceGateway/Config/EventConfig.xml`
`event.log4j.config.url=http://localhost:8080/config/EventConfig.xml`
`event.log4j.config.url=file:///C:/Data/pm_source/current_uis/pm/pmserver/Config/EventConfig.xml`

Usage: EventManager

event.sockets.encoding — The character encoding set (e.g., ISO8859-1) used for event logging with the ACI Transmission Control Protocol (TCP) Appender. If this property is not set, the default system encoding is assumed.

Required: No

Example Entry: `event.sockets.encoding=ISO8859-1`

Usage: ACI TCP Appender

event.sockets.host — The name of the host to send event messages to when using the ACI TCP Appender to log event messages. If this property is not configured, it defaults to localhost.

Required: No

Example Entry: `event.sockets.host=remotehost`

Usage: ACI TCP Appender

event.sockets.port — The TCP port number to connect to, acting as a client, when using the ACI TCP Appender to log event messages.

Required: Yes, if using the ACI TCP Appender

Example Entry: `event.sockets.port=8004`

Usage: ACI TCP Appender

filter.field.task-id.n — Indicates field filter support for specific task IDs, where *task-id* is the task ID associated with a specific ACI desktop user interface window, browser page, or framework and *n* is a sequential number starting with 0 and increased by 1 for multiple filters associated with the same task ID. Task IDs are described in the *ACI Desktop User Interface Manual*.

Note: Filter properties are created and used for internal ACI development. Do not alter these properties at all after your system is installed.

This property guides the Java process to map incoming data values from logical elements in a message to a field ID understood by the filtering implementation configured for the USEC subsystem. USEC filtering is configured on the Filters tab of the User Security Role Configuration window. Values for this property are entered in the following three-part format, with each part separated by a semicolon:

Identifier;field ID;logical message element

where:

Identifier is a code indicating whether the property value is to be used only for a filter, only for USEC maintenance, or for both a filter and for USEC maintenance. Valid values are:

F = Filter only

M = USEC maintenance only

B = Both filter and USEC maintenance

Field ID is the name of the field as known by the USEC subsystem. Currently, filtering is supported for the following fields: InstitutionID, Issuer, and Profile.

Logical message element is the logical element in the message that contains the data.

Filtering support also requires a filter class specification in the appropriate transaction map XML file.

Required: No

Example Entries:

filter.field.InstitutionConfiguration.0=B;InstitutionID;instdInstitutionID

filter.field.InstitutionConfiguration.1=M;Issuer;instdInstitutionID

hc.bad_text — A list of characters for which hack character validation is performed.

Required: No

Example Entry: hc.bad_text=~`!@#\$\$%^_+=+|\:'.',./;

Usage: HackCharacterValidation

initialization.function<text-string> — Specifies the fully-qualified Java package and class name to be started at initialization by the Initialization Manager, where <text-string> is a unique text string for this initialization.function property name.

Beginning with AJSF 07.4, initialization function properties are only loaded from the APPLCNFG and not from the config.properties file. Also beginning with AJSF 07.4, initializers are either loaded from XML initialization map files or through specific configuration properties in the APPLCNFG table. They are mutually exclusive. For detailed information on XML initialization map files, see [section 9](#).

Required: No

Example Entries:

User Security Management:

```
initialization.function.0.classname=com.aciworldwide.  
framework.TransactionMapManager  
initialization.function.1.classname=com.aciworldwide.  
common.MessageRouterService  
initialization.function.2.classname=com.aciworldwide.  
framework.pools.SAX2ParserManager  
initialization.function.3.classname=com.aciworldwide.  
application.formatter.jaxb.JAXBContextManager  
initialization.function.4.classname=com.aciworldwide.  
application.security.USECManager  
initialization.function.5.classname=com.aciworldwide.  
application.pci.masking.formatter.jaxb.JAXBContextManager
```

ACI Payments Manager:

```
initialization.function.6.classname=com.aciworldwide.pm.  
common.server.PMAuditService  
initialization.function.7.classname=com.aciworldwide.pm.  
common.server.PMServer  
initialization.function.8.classname=com.aciworldwide.pm.  
core.batch.BatchController
```

HSM Configuration:

```
initialization.function.9.classname=com.aciworldwide.  
common.crypto.hsm.HSMService
```

BASE24-eps User Interface:

```
initialization.function.DataValidatorManager=es.server.  
common.DataValidatorManager  
initialization.function.DataTypeValidationManager=es.server.  
common.DataTypeValidationManager  
initialization.function.JAXBContextManager=com.  
aciworldwide.application.formatter.jaxb.  
JAXBContextManager  
initialization.function.MessageSchemaManager=es.server.  
formatter.xml.MessageSchemaManager  
initialization.function.OLTPAssignManager=es.server.  
common.OLTPAssignManager  
initialization.function.SAX2ParserManager=com.  
aciworldwide.framework.pools.SAX2ParserManager  
initialization.function.TransactionMapManager=es.server.  
framework.TransactionMapManager
```

Usage: InitializationManager

isDisableUserOnFailedLogonAttempts — A flag indicating whether a user's status is automatically set to Disabled/System when the user exceeds the allowed number of failed logon attempts (configured in the Bad Login Tries field on the User Security Profile Configuration window). If this property is set to true, the user cannot log on again until the security administrator resets the user's status. If this property is set to false, the user can attempt to log on after the period configured in the Reset Bad Logins field on the User Security Profile Configuration window expires. Valid values are as follows:

true	= Yes, reset the user's status to Disabled/System when the user exceeds the allowed number of failed logon attempts. The user cannot log on again until the security administrator resets the user's status.
false	= No, do not reset the user's status to Disabled/System when the user exceeds the allowed number of failed logon attempts. The user can attempt to log on after the period configured in the Reset Bad Logins field on the User Security Profile Configuration window expires. Default.

Required: No

Example Entry: isDisableUserOnFailedLogonAttempts=true

Usage: USEC subsystem

jsf.jaxb.context.<text-string> — The Java API for XML binding (JAXB) context path for manual user audit or user security reporting from the ACI desktop user interface or for user authentication with non-AJSF-based applications, where <text-string> is a unique text string for this jsf.jaxb.context property name.

Required: Yes, for user audit or user security reporting.

Example Entry: jsf.jaxb.context.0=com.aciworldwide.application.audit.
formatter.jaxb.useraudit
jsf.jaxb.context.1=com.aciworldwide.application.security.
reporting.formatter.jaxb.userstatus
jsf.jaxb.context.2=com.aciworldwide.application.security.
formatter.jaxb.usec1_0_0

Usage: User audit or user security reporting
User authentication for non-AJSF-based applications

keymanager.provider — The pathname of the provider for key management services.

Required: Yes
Example Entry: `keymanager.provider = com.aciworldwide.framework.crypto.
KeyStoreService`
Usage: KeyManager

keystore.filename — The path name of the data source containing the Java keystore. Public and private keys as well as certificates are stored in the Java keystore. For keystores maintained in a hardware security device, this property must be set to null. See the [keystore.type](#) property for more details.

Required: Yes
Example Entry: `keystore.filename=C://SecureCommerce//ACS//Config//
Keystore//keystore.bin`
Usage: Key Store Service

keystore.hardware — A flag indicating whether the keystore is maintained within a hardware security device or in software. Valid values are as follows:

true = Yes, the keystore is maintained within a hardware security device.
false = No, the keystore is not maintained within a hardware security device. It is maintained in software.

The value of this property must be set in accordance with the value of the [keystore.type](#) property.

Required: Yes
Example Entry: `keystore.filename=false`
Usage: Key Store Service



keystore.password — The password used to access the Java keystore. Public and private keys as well as certificates are stored in the Java keystore. Therefore, the value of this property should be treated with the upmost confidentiality. For security, the password value stored in the config.properties file must be encrypted using the [password encryption utility](#) provided by ACI. Cross-industry best practices recommends that you do not use vendor-supplied default passwords, that you use passwords containing both numeric and alphabetic characters with a minimum password length of at least seven characters, that you change passwords at least every 90 days, and that a new password is not the same as any of the last four passwords used. This property is currently used only for SafeNet (Eracom) JCE functionality.

Required: Yes, for SafeNet (Eracom) JCE functionality.
 Example Entry: `keystore.password=512bb47f6f7206e90d0d440da775982b`
 Usage: Eracom JCE Provider
 Key Store Service

keystore.provider — The name of the Java Keystore provider used by the Key Store Service. The AJSF allows you to use any hardware or software Java keystore provider. The value of this property must be set in accordance with the value of the [keystore.type](#) property. The following are currently supported values for the `keystore.provider` property:

SUNJCE = Software-based Java Cryptography Extension (JCE) keystore provided by Sun.
 IBMJCE = Software-based JCE keystore provided by IBM.
 nCipher.KM = nCipher keystore.
 ERACOM = Hardware keystore provided by SafeNet (Eracom).

Required: Yes
 Example Entry: `keystore.provider=SunJCE`
 Usage: Key Store Service

keystore.type — Indicates the type of keystore provided. The value depends on the keystore provider and the configuration used. Valid values are as follows:

CRYPTOKI = Keystore provided by nCipherKM.
 JCEKS = Software-based Java Cryptography Extension (JCE) keystore provided by Sun or IBM.
 nCipher.sworld = Hardware keystore provided by SafeNet (Eracom).

The `keystore.provider`, `keystore.hardware`, and `keystore.filename` properties must be set in accordance with the value in the `keystore.type` property as shown in the following table.

keystore.type	keystore.provider	keystore.hardware	keystore.filename
CRYPTOKI	nCipherKM	false	<i>file-pathname</i>
JCEKS	SUNJCE, IBMJCE	false	<i>file-pathname</i>
nCipher.sworld	ERACOM	true	null

Required: Yes
Example Entry: `keystore.type = JCEKS`
Usage: Key Store Service

mail.smtp.host — The name of the Simple Mail Transfer Protocol (SMTP) mail server. This property, along with all the other mail.smtp properties, must be configured in the config.properties file to support the sending of SMTP e-mail messages to an SMTP host system.

Required: Yes, if using SMTP
Example Entry: `mail.smtp.host=lng001.tsa.com`
Usage: SMTP Process

mail.smtp.from.addr — The e-mail address to be used as the from address in e-mails sent to an SMTP host system. This property, along with all the other mail.smtp properties, must be configured in the config.properties file to support the sending of SMTP e-mail messages to an SMTP host system.

Required: Yes, if using SMTP
Example Entry: `mail.smtp.from.addr=SMTPTester@aciworldwide.com`
Usage: SMTP Process

mail.smtp.auth.required — A flag indicating whether the receiving SMTP mail server requires the user specified in the mail.smtp.auth.user property to be authenticated before accepting messages from the user. Valid values are as follows:

true = Yes, authenticate the user.
false = No, do not authenticate the user. Default.

This property, along with all the other mail.smtp properties, must be configured in the config.properties file to support the sending of SMTP e-mail messages to an SMTP host system.

Required: Yes, if using SMTP
Example Entry: `mail.smtp.auth.required=false`
Usage: SMTP Process

mail.smtp.auth.user — The user name to use for authentication if the SMTP mail server requires authentication and the mail.smtp.auth.required property is set to true. This property, along with all the other mail.smtp properties, must be configured in the config.properties file to support the sending of SMTP e-mail messages to an SMTP host system.

Required: Yes, if using SMTP
Example Entry: mail.smtp.auth.user=SMTPTester
Usage: SMTP Process



mail.smtp.auth.password — The password to use for authentication of the user specified in the mail.smtp.auth.user property if the SMTP mail server requires authentication and the mail.smtp.auth.required property is set to true. This property, along with all the other mail.smtp properties, must be configured in the config.properties file to support the sending of SMTP e-mail messages to an SMTP host system. For security, the password value stored in the config.properties file should be encrypted. Cross-industry best practices recommends that you do not use vendor-supplied default passwords, that you use passwords containing both numeric and alphabetic characters with a minimum password length of at least seven characters, that you change passwords at least every 90 days, and that a new password is not the same as any of the last four passwords used.

Required: Yes, if using SMTP
Example Entry: mail.smtp.auth.password=Mypassword1
Usage: SMTP Process

mail.smtp.max.retries — The maximum number of times to retry establishing a connection to the SMTP host when sending a message. This property, along with all the other mail.smtp properties, must be configured in the config.properties file to support the sending of SMTP e-mail messages to an SMTP host system.

Required: Yes, if using SMTP
Example Entry: mail.smtp.max.retries=3
Usage: SMTP Process

message.environment — Indicates the type of messaging environment used for the AJSF-based application. Valid values are as follows:

DIRECT = Direct messaging between processes.
JMS = Messaging through the Java Message Service (JMS)
EJB = Messaging through Enterprise JavaBeans (EJB). Reserved for future use.

Required: Yes
Example Entries: message.environment=JMS
message.environment=DIRECT
Usage: CommonHTTPServlet
CommonXMLServlet
CommonJSPPage

message.jms.queue.externalconnectionid — The external connection ID configured in the External Connection data source (EXTRCNCT) for JMS. This property is required if the message.environment property is set to JMS. The value of this property must match an external connection ID value in the EXTRCNCT for the JMS service.

Required: Yes, if the message.environment property is set to JMS
Example Entries: message.jms.queue.externalconnectionid=NCR-JMS
message.jms.queue.externalconnectionid=DIEBOLD-JMS
message.jms.queue.externalconnectionid=COMMERCEGATEWAY
Usage: CommonHTTPServlet
CommonXMLServlet
CommonJSPPage

message.jms.Connection.DisableOnStartup — A flag indicating whether the JMS Message Router Service's capability to start JMS connections at startup is disabled. Disabling JMS connections at startup enables the application to determine when to start accepting messages from the middleware component they connect to. Valid values are as follows:

true = Yes, disable the capability to start JMS connections at startup.
false = No, do not disable the capability to start JMS connections at startup.

Required: No
Example Entry: message.jms.Connection.DisableOnStartup=true
Usage: JMS Message Router Service

message.jms.queuesFromJNDI — A flag indicating whether the ACI application defines the queue name or the queue name is retrieved from a Java Naming and Directory Interface (JNDI) server. If this property is set to true, it allows the JNDI server to manage the queue to queue name mapping. If this property is set to false, the ACI application defines the actual queue name. This property is only used when using JMS for messaging. A setting of true is preferred. Valid values are as follows:

true = Yes, the queue name is retrieved from a JNDI server.
false = No, the queue name is not retrieved from a JNDI server. The application must define the queue name.

Required: No
Example Entry: message.jms.queuesFromJNDI=true
Usage: JMS Listener Manager
JMS Connection Manager
JMS Session Manager

message.jms.QueueConnectionFactory — The name of the queue connection factory used to automatically create queues for point-to-point messaging.

Required: Yes, if using a JMS messaging provider such as WebSphere MQ or NET24-XPNET.
Example Entries: message.jms.QueueConnectionFactory=JmsQueueConnectionFactory
message.jms.QueueConnectionFactory=QUEUE.CONNECTION.FACTORY
Usage: JMS Connection Manager

message.jms.jndi.initialcontextfactory — The name of the initial context factory for the JNDI service provider specified by the message.jms.jndi.providerurl property. The context factory creates an initial context instance for reaching all other parts of the namespace of the underlying naming/directory service.

Required: Yes, if using a JMS messaging provider such as WebSphere MQ or NET24-XPNET.

Example Entries: `message.jms.jndi.initialcontextfactory=com.sun.enterprise.naming.SerialInitContextFactory`
`message.jms.jndi.initialcontextfactory=com.sun.jndi.ldap.LdapCtxFactory`
`message.jms.jndi.initialcontextfactory=com.sun.jndi.fscontext.RefFSContextFactory`
 (WebSphere MQ):
`message.jms.jndi.initialcontextfactory=com.sun.jndi.fscontext.RefFSContextFactory`
 (Open JMS):
`message.jms.jndi.initialcontextfactory=org.exolab.jms.jndi.rmi.RmiJndiInitialContextFactory`
`message.jms.jndi.initialcontextfactory=progress.message.jclient.QueueConnectionFactory`

Usage: JMS Listener Manager
 JMS Connection Manager
 JMS Queue Manager

message.jms.jndi.providerurl — The URL location of the Java Naming and Directory Interface (JNDI) processing provider (e.g., LDAP, DNS, CORBA, etc.).

Required: Yes, if using a JMS messaging provider such as WebSphere MQ or NET24-XPNET.

Example Entries: `message.jms.jndi.providerurl=t3://localhost:7001`
 (Open JMS):
`message.jms.jndi.providerurl=rmi://localhost:1099/JndiServer`
 (WebSphere MQ):
`message.jms.jndi.providerurl=file:C:/Program files/ibm/MQSeries/Java/JNDI-Directory`
`message.jms.jndi.providerurl=file:C:/Data/Program Files/IBM/WebSphere JNDI`

Usage: JMS Listener Manager
 JMS Connection Manager
 JMS Queue Manager

message.jms.use.nsk.threads — A flag indicating whether to use HP NonStop message threading. This property must be set to a value of true if the AJSF application is installed on an HP NonStop platform with the JDAL JDBC driver. This property is not used when the AJSF is installed on an SMP platform. Valid values are as follows:

true = Yes, use HP NonStop message threading.
false = No, do not use HP NonStop message threading. Default.

Required: Yes, if the AJSF is installed on an HP NonStop platform
Example Entry: message.jms.use.nsk.threads=true
Usage: Message Listener Implementation

message.property.handling — A code indicating how the application handles messages to and from other applications. Valid values are as follows:

JMS = The application communicates through Java Message Service (JMS) to other applications using JMS. JMS send/receive properties are used.
MDS = The application communicates through JMS to a BASE24-eps application that requires the BASE24-eps proprietary message header. Message Delivery Subsystem (MDS) header properties are used.
Default.

Required: Yes
Example Entry: message.property.handling=JMS
Usage: Message Properties

msgcntx.context.column.size — The default the size, in bytes, of the binary large object (BLOB) data written to the context column in the Message Context data source (MSGCNTX) table for message context that can span multiple database rows.

Required: Yes
Example Entry: msgcntx.context.column.size=3800
Usage: Message Context

MULTIBYTE_ENCODING — A code defining the type of encoding to be applied for the transfer of multiple-byte data between the ACI desktop or browser client and Java servers in the BASE24-eps system. Because this encoding value must be synchronized with the MULTIBYTE_ENCODING Environment attribute

value in the Environment data source (Environment) for C++ servers, you cannot enter it on the Application Configuration window. Instead, both the `MULTIBYTE_ENCODING` property and the `MULTIBYTE_ENCODING` Environment attribute are set from the Content Encoding field on the Locale Content Configuration window. Whenever you change the value of this field, you must either restart the ESWeb Server process or reload the ESWeb Server process from the Configuration Properties window (accessed from **System Operations > Server Management > Server Administration > Configuration Properties**).

Warning: The value of the Content Encoding field on the Locale Content Configuration window should not be changed after the database has been populated with multibyte data. Changing this value after populating multibyte fields may have unpredictable results.

Currently, UTF-8 is the only supported encoding value. This property is only used if you have licensed the Multibyte Character Set Service (EE-SE250) module. The value of this property is used to set the `multibyte.encoding` system property. This system property is then used to encode multibyte messages in both the application client and in the UIF for messages containing multibyte fields.

The value of this property must be the same as the [file.encoding](#) system property for the JVM.

Required:	No
Usage:	User Interface Framework

report.userauditlog.loc — The name of the directory where the user audit extract file is created. For manually started extracts, the user audit extract file name is specified in the Extract file name field on the Audit Report Wizard window on the ACI desktop user interface. If not specified in this field, the name defaults to `AudExt_YYMMDD_hhmmss.xml`, where `YYMMDD_hhmmss` is the year, month, day, hour, minutes, and seconds when the file was created. For automatic extracts, the user audit extract file name is specified by the `report.userauditlog.name` property. The `report.userauditlog.loc` property defaults to a value of “{report location},” which must be changed to a valid location after installation.

Note that if the specified directory does not already exist, the audit extract file is not created.



To meet cross-industry security best practices, any sensitive data in a user audit should be protected from unauthorized access. Therefore, the location where user audit data is extracted should be restricted.

If a `timertask.function<text-string>` property is not configured for “com.aciworldwide.common.timertask.UserAuditReport,” automatic extracts of the user audit data are not performed.

Required: Yes, if using user auditing
Example Entries: /user/B24es_054/ESWeb/ESConfig/uaf2/P6Q^GATE/reports/
Usage: User Audit Log XML Extract

report.userauditlog.name — The base name of the user audit extract file when automatically generating a user audit extract. This base name is suffixed with “_YYMMDD_hhmmss.xml” to create the file name of the user audit extract file, where *YYMMDD_hhmmss* is the year, month, day, hour, minutes, and seconds when the file was created. If this property is not configured, it defaults to a base name of “AudExtAuto”.

A `timertask.function<text-string>` property must be configured for “com.aciworldwide.common.timertask.UserAuditReport” before this property is used in processing. If a `timertask.function<text-string>` property is not configured for “com.aciworldwide.common.timertask.UserAuditReport,” automatic extracts of the user audit data are not performed.

Required: No
Example Entry: report.userauditlog.name=Auto_User_Extract
Usage: User Audit Log XML Extract

report.userauditlog.offset.days — The number of days to subtract from the system date when the report timer fires. It is used to calculate the date of the user audit activity to be reported on for an automatic user audit extract. For example, if this property is set to 2, the report includes one day’s data from two days prior to the report execution. If this property is set to 7, the report includes one day’s data from seven days prior to the report execution. If this property is not configured, it defaults to 1, which means user audit activity is extracted for yesterday. Automated extracts always extract all of the specified day’s user audit data to the User Audit Report extract XML file.

A `timertask.function<text-string>` property must be configured for “com.aciworldwide.common.timertask.UserAuditReport” before this property is used in processing. If a `timertask.function<text-string>` property is not configured for “com.aciworldwide.common.timertask.UserAuditReport,” automatic extracts of the user audit data are not performed.

Required: No
Example Entry: 7
Usage: User Audit Log XML Extract

report.userauditlog.report.starttime — The time of day, in hh:mm format, at which automatic extracts of a previous day’s audit data from the User Audit Log data source (USRAULOG) to an XML file are started. If this property is not configured, it defaults to 01:00, which indicates that the automatic extract starts at 1:00 am. The `report.userauditlog.offset.days` property indicates the number of days to schedule between automatic extracts.

A `timertask.function<text-string>` property must be configured for “com.aciworldwide.common.timertask.UserAuditReport” before this property is used in processing. If a `timertask.function<text-string>` property is not configured for “com.aciworldwide.common.timertask.UserAuditReport,” automatic extracts of the user audit data are not performed.

This start time automatically accounts for daylight savings time (summer time) on the server where the property is configured. This time is in the time of the executing server where the application is running.

Required: No
Example Entry: 01:00 (1:00 am)
Usage: User Audit Log XML Extract

report.userdetail.loc — The name of the directory where the User Detail Report is created. This property defaults to a value of “{report location}”, which must be changed to a valid location after installation.

Note that if the specified directory does not exist, the User Detail Report is not created.

A `timertask.function<text-string>` property must be configured for “com.aciworldwide.common.timertask.UserDetailReport” before this property is used in processing. If a `timertask.function<text-string>` property is not configured for “com.aciworldwide.common.timertask.UserDetailReport,” the User Detail Report is not automatically generated.

Required: Yes, if using user detail reporting
Example Entries: /user/B24es_064/ESWeb/ESConfig/uaf2/P6Q^GATE/reports/
Usage: User Detail Report

report.userdetail.name — The base name to use for the file when automatically generating a User Detail Report. This base name is suffixed with “_YYMMDD_hhmmss.xml” to create the file name of the User Detail Report file, where *YYMMDD_hhmmss* is the year, month, day, hour, minutes, and seconds when the file is created. If this property is not configured, it defaults to a base name of “UserDetailReport”.

A `timertask.function<text-string>` property must be configured for “com.aciworldwide.common.timertask.UserDetailReport” before this property is used in processing. If a `timertask.function<text-string>` property is not configured for “com.aciworldwide.common.timertask.UserDetailReport,” the User Detail Report is not automatically generated.

Required:	No
Example Entry:	report.userdetail.name=User_Detail_Extract
Usage:	User Detail Report

report.userdetail.starttime — The time of day, in hh:mm format, at which the User Detail Report is generated. If this property is not configured, it defaults to 04:00, which indicates that the automatic report generation starts at 4:00 a.m. The `report.userdetail.timeperiod.hours` property indicates the number of hours between generation of the User Detail Report.

A `timertask.function<text-string>` property must be configured for “com.aciworldwide.common.timertask.UserDetailReport” before this property is used in processing. If a `timertask.function<text-string>` property is not configured for “com.aciworldwide.common.timertask.UserDetailReport,” the User Detail Report is not automatically generated.

This start time automatically accounts for daylight savings time (summer time) on the server where the property is configured. This time is in the time of the executing server where the application is running.

Required:	No
Example Entry:	report.userdetail.starttime=04:00 (4:00 am)
Usage:	User Detail Report

report.userdetail.timeperiod.hours — The number of hours between generation of the User Detail Report if it is to be generated more frequently than every twenty four hours. The default value is 24 hours, which means the User Detail Report is generated daily.

A `timertask.function<text-string>` property must be configured for “com.aciworldwide.common.timertask.UserDetailReport” before this property is used in processing. If a `timertask.function<text-string>` property is not configured for “com.aciworldwide.common.timertask.UserDetailReport,” the User Detail Report is not automatically generated.

Required: No
Example Entry: report.userdetail.timeperiod.hours=12
Usage: User Detail Report

report.userpermissions.loc — The name of the directory where the User Permissions Report is created. This property defaults to a value of “{report location}”, which must be changed to a valid location after installation.

Note that if the specified directory does not exist, the User Permissions Report is not created.

A `timertask.function<text-string>` property must be configured for “com.aciworldwide.common.timertask.UserPermissionsReport” before this property is used in processing. If a `timertask.function<text-string>` property is not configured for “com.aciworldwide.common.timertask.UserPermissionsReport,” the User Permissions Report is not automatically generated.

Required: Yes, if using user permissions reporting
Example Entries: /user/B24es_064/ESWeb/ESConfig/uaf2/P6Q^GATE/reports/
Usage: User Permissions Report

report.userpermissions.name — The base name to use for the file when automatically generating a User Permissions Report. This base name is suffixed with “_YYMMDD_hhmmss.xml” to create the file name of the User Permissions Report file, where *YYMMDD_hhmmss* is the year, month, day, hour, minutes, and seconds when the file is created. If this property is not configured, it defaults to a base name of “UserPermissionsReport”.

A `timertask.function<text-string>` property must be configured for “com.aciworldwide.common.timertask.UserPermissionsReport” before this property is used in processing. If a `timertask.function<text-string>` property is not configured for “com.aciworldwide.common.timertask.UserPermissionsReport,” the User Permissions Report is not automatically generated.

Required: No
Example Entry: `report.userpermissions.name=User_Permissions_Extract`
Usage: User Permissions Report

report.userpermissions.starttime — The time of day, in hh:mm format, at which the User Permissions Report is generated. If this property is not configured, it defaults to 03:00, which indicates that the automatic report generation starts at 3:00 am. The `report.userpermissions.timeperiod.hours` property indicates the number of hours between generation of the User Permissions Report.

A `timertask.function<text-string>` property must be configured for “com.aciworldwide.common.timertask.UserPermissionsReport” before this property is used in processing. If a `timertask.function<text-string>` property is not configured for “com.aciworldwide.common.timertask.UserPermissionsReport,” the User Permissions Report is not automatically generated.

This start time automatically accounts for daylight savings time (summer time) on the server where the property is configured. This time is in the time of the executing server where the application is running.

Required: No
Example Entry: `report.userpermissions.starttime=03:00 (3:00 am)`
Usage: User Permissions Report

report.userpermissions.timeperiod.hours — The number of hours between generation of the User Permissions Report if it is to be generated more frequently than every twenty four hours. The default value is 24 hours, which means the User Permissions Report is generated daily.

A `timertask.function<text-string>` property must be configured for “com.aciworldwide.common.timertask.UserPermissionsReport” before this property is used in processing. If a `timertask.function<text-string>` property is not configured for “com.aciworldwide.common.timertask.UserPermissionsReport,” the User Permissions Report is not automatically generated.

Required: No
Example Entry: `report.userpermissions.timeperiod.hours=168 (weekly)`
Usage: User Permissions Report

sax2.parser.init.count — The number of Simple API for XML (SAX2)-compliant XML parsers to be initially created by the InitializationManager at process startup. It is more efficient to have a pool of parser instances available for clients rather than having a client create a new one as needed. If this property is not configured, it defaults to 4.

Required: No
 Example Entry: sax2.parser.init.count=5
 Usage: SAX2 Parser Manager

SSL.client.keystore — The pathname of the data source containing the keystore for Secure Sockets Layer (SSL) client security. The value in this property is used to initialize the JSSE environment by loading the value of the property in the config.properties file to the javax.net.ssl.keyStore system property for the JVM. Refer to the *Java Secure Socket Extension (JSSE) 1.0.2 API User's Guide* (available at java.sun.com) for more information.

Required: Yes, if an external connection configured in the External Connections data source (EXTRCNCT) requires HTTP/S.
 Example Entry: SSL.client.keystore = SSLKeyStore
 Usage: HTTPSConnectionFactorySunJSSE

SSL.client.keystore.password — The password required to access the SSL keystore for client security. The value in this property is used to initialize the JSSE environment by loading the value of the property in the config.properties file to the javax.net.ssl.keyStorePassword system property for the JVM. Refer to the *Java Secure Socket Extension (JSSE) 1.0.2 API User's Guide* (available at java.sun.com) for more information. For security, the password value stored in the config.properties file must be encrypted using the [password encryption utility](#) provided by ACI. Cross-industry best practices recommends that you do not use vendor-supplied default passwords, that you use passwords containing both numeric and alphabetic characters with a minimum password length of at least seven characters, that you change passwords at least every 90 days, and that a new password is not the same as any of the last four passwords used.



Required: Yes, if an external connection configured in the External Connections data source (EXTRCNCT) requires HTTP/S.
 Example Entry: SSL.client.keystore.password =
 512bb47f6f7206e90d0d440da775982b
 Usage: HTTPSConnectionFactorySunJSSE

SSL.client.truststore — The pathname of the data source containing the truststore for Secure Sockets Layer (SSL) client security. The value in this property is used to initialize the JSSE environment by loading the value of the property in the config.properties file to the javax.net.ssl.trustStore system property for the JVM. Refer to the *Java Secure Socket Extension (JSSE) 1.0.2 API User's Guide* (available at java.sun.com) for more information.

A *truststore* is a keystore that is used when making decisions about who to trust. If you receive some data from an entity that you already trust, and if you can verify that the entity is the one it claims to be, then you can assume that the data really came from that entity.

Required: Yes, if an external connection configured in the External Connections data source (EXTRCNCT) requires HTTP/S.
 Example Entry: SSL.client.truststore = SSLTrustStore
 Usage: HTTPSConnectionFactorySunJSSE



SSL.client.truststore.password — The password required to access the SSL truststore for client security. The value in this property is used to initialize the JSSE environment by loading the value of the property in the config.properties file to the javax.net.ssl.trustStorePassword system property for the JVM. Refer to the *Java Secure Socket Extension (JSSE) 1.0.2 API User's Guide* (available at java.sun.com) for more information. For security, the password value stored in the config.properties file must be encrypted using the [password encryption utility](#) provided by ACI. Cross-industry best practices recommends that you do not use vendor-supplied default passwords, that you use passwords containing both numeric and alphabetic characters with a minimum password length of at least seven characters, that you change passwords at least every 90 days, and that a new password is not the same as any of the last four passwords used.

Required: Yes, if an external connection configured in the External Connections data source (EXTRCNCT) requires HTTP/S.
 Example Entry: SSL.client.truststore.password = 209f4r9ahe6c12ooo4608a5
 Usage: JSSEManager

SSL.debug — A flag indicating whether to log trace messages for Secure Sockets Layer (SSL) processing to the standard out location (STDOUT). Valid values are as follows:

all = Yes, log trace messages for Sun SSL processing.
 true = Yes, log trace messages for IBM SSL processing.
 false = No, do not log trace messages for SSL processing. Default

Note: The Trace facility was designed as a diagnostic tool for development and test environments and not for general use in a production environment. ACI strongly recommends that it not be used in a production environment due to the overhead and impacts on performance.

Required: No
 Example Entry: SSL.debug= false
 Usage: JSSEManager
 HTTPSConnectionFactorySunJSSE

SSL.protocol.handler.pkgs — The pathname of the handler's implementation package name to be added to the list of packages searched by the Java URL class. This is a system property.

Required: Yes, if an external connection configured in the External Connections data source (EXTRCNCT) requires HTTP/S.
 Example Entry: java.protocol.handler.pkgs=com.sun.net.ssl.internal.www.protocol myApp
 Usage: HTTPSConnectionFactorySunJSSE

SSL.server.keystore — The pathname of the data source containing the keystore for Secure Sockets Layer (SSL) server security. The value in this property is used to initialize the JSSE environment by loading the value of the property in the config.properties file to the javax.net.ssl.keyStore system property for the JVM. Refer to the *Java Secure Socket Extension (JSSE) 1.0.2 API User's Guide* (available at java.sun.com) for more information.

Required: Yes, if an external connection configured in the External Connections data source (EXTRCNCT) requires HTTP/S.
 Example Entry: SSL.server.keystore = SSLKeyStore
 SSL.server.keystore=../../config/security/server.keystore.key
 Usage: JSSEManager

SSL.server.keystore.password — The password required to access the SSL keystore for server security. The value in this property is used to initialize the JSSE environment by loading the value of the property in the config.properties file to the javax.net.ssl.keyStorePassword system property for the JVM. Refer to the *Java Secure Socket Extension (JSSE) 1.0.2 API User's Guide* (available at java.sun.com) for more information. For security, the password value stored in the config.properties file must be encrypted using the [password encryption utility](#) provided by ACI. Cross-industry best practices recommends that you do not use vendor-supplied default passwords, that you use passwords containing both



numeric and alphabetic characters with a minimum password length of at least seven characters, that you change passwords at least every 90 days, and that a new password is not the same as any of the last four passwords used.

Required: Yes, if an external connection configured in the External Connections data source (EXTRCNCT) requires HTTP/S.

Example Entry: `SSL.server.keystore.
password=512bb47f6f7206e90d0d440da775982b`

Usage: JSSEManager

SSL.server.truststore — The pathname of the data source containing the truststore for Secure Sockets Layer (SSL) server security. The value in this property is used to initialize the JSSE environment by loading the value of the property in the config.properties file to the javax.net.ssl.trustStore system property for the JVM. Refer to the *Java Secure Socket Extension (JSSE) 1.0.2 API User's Guide* (available at java.sun.com) for more information.

A *truststore* is a keystore that is used when making decisions about who to trust. If you receive some data from an entity that you already trust, and if you can verify that the entity is the one it claims to be, then you can assume that the data really came from that entity.

Required: Yes, if an external connection configured in the External Connections data source (EXTRCNCT) requires HTTP/S.

Example Entries: `SSL.server.truststore = SSLTrustStore
SSL.server.truststore=../../config/security/server.truststore.key`

Usage: JSSEManager

SSL.server.truststore.password — The password required to access the SSL truststore for server security. The value in this property is used to initialize the JSSE environment by loading the value of the property in the config.properties file to the javax.net.ssl.trustStorePassword system property for the JVM. Refer to the *Java Secure Socket Extension (JSSE) 1.0.2 API User's Guide* (available at java.sun.com) for more information. For security, the password value stored in the config.properties file must be encrypted using the [password encryption utility](#) provided by ACI. Cross-industry best practices recommends that you do not use vendor-supplied default passwords, that you use passwords containing both numeric and alphabetic characters with a minimum password length of at least seven characters, that you change passwords at least every 90 days, and that a new password is not the same as any of the last four passwords used.



Required: Yes, if an external connection configured in the External Connections data source (EXTRCNCT) requires HTTP/S.

Example Entry: SSL.server.truststore.
password=7c84e738ea20b628fd407625f148bfb2

Usage: JSSEManager

timertask.manager.processid — The process ID of the Timer process that is to run all cleanup and reporting tasks for the executing JVM. The functions to be executed are specified in `timertask.function<text-string>` properties.

Required: Yes, if cleanup tasks are required.

Example Entries: `timertask.manager.processid=P6S^ESWEB01`
`timertask.manager.processid=TMRTASKS`

Usage: Timer Task Manager

timertask.function<text-string> — The name of a class to be executed by the Timer process, where `<text-string>` is a unique text string for this `timertask` function property name. Each class designates cleanup tasks to be performed for the AJSF application. Additional properties must be configured in conjunction with most timed tasks to be performed. These properties are identified in parentheses following each example timer task class entry.

Required: Yes, if cleanup or reporting tasks are required.

Example Entries: (APF Admin):
`timertask.function.0.classname=com.aciworldwide.common.timertask.UserAuditLogCleanup` (see also the following property descriptions: `cleanup.userauditlog.retention.days`, `cleanup.userauditlog.cleanup.starttime`, `cleanup.userauditlog.timeperiod.hours`, `cleanup.userauditlog.batch.size`)
`timertask.function.1.classname=com.aciworldwide.common.timertask.UserAuditReport` (see also the following property descriptions: `jsf.jaxb.context.<text-string>`, `jsf.jaxb.formatter.formatted.output`, `report.userauditlog.offset.days`, `report.userauditlog.report.starttime`, `report.userauditlog.loc`, `report.userauditlog.name`, and `xml.parser.validate.document`)
`timertask.function.2.classname=com.aciworldwide.application.security.timertask.UserPermissionsReport` (see also the following property descriptions: `jsf.jaxb.context`.

`<text-string>`, `report.userpermissions.loc`, `report.userpermissions.name`, `report.userpermissions.starttime`, `report.userpermissions.timeperiod.hours`)
`timertask.function.3.classname=com.aciworldwide.application.security.timertask.UserDetailReport` (see also the following property descriptions: `jsf.jaxb.context.<text-string>`, `report.userdetail.name`, `report.userdetail.timeperiod.hours`, `report.userdetail.starttime`, `report.userdetail.loc`)
 (BASE24-eps ATM (Java) Device Handler process):
`timertask.function.0.classname=com.aciworldwide.base24.es.dh.common.timer.timertask.ReversalSAFTimerTask` (see also the following property descriptions: `dh.reversal.saf.intra.batch.delay`, `dh.reversal.saf.max.rowcount`, `dh.reversal.saf.run.delay.minutes`, `dh.reversal.saf.start.delay.minutes`, `dh.reversal.saf.msg.retries`, `dh.reversal.saf.send.interval`)
`timertask.function.1.classname=com.aciworldwide.base24.es.dh.common.timer.timertask.StatementDataCleanupTimerTask` (see also the following property descriptions: `dh.statementdata.cleanup.starttime`, `dh.statementdata.cleanup.timeperiod.hours`, `dh.statementdata.cleanup.retention.days`)

Usage: Timer Task Manager

trace.enabled — A flag indicating whether trace messages are enabled for components or services executing in the Java server process. Message tracing is configured in the configuration file specified by the `trace.log4j.config.url` property. Valid values are as follows:

`true` = Yes, message tracing is enabled.
`false` = No, message tracing is not enabled.

Note: The Trace facility was designed as a diagnostic tool for development and test environments and not for general use in a production environment. ACI strongly recommends that it not be used in a production environment due to the overhead and impacts on performance.



In addition, the enabling of message tracing may cause the logging of sensitive data. Be sure that any resulting trace logs are secured according to cross-industry security best practices.

Required: Yes
 Example Entry: `trace.enabled=false`
 Usage: TraceManager

trace.config

trace.log4j.config.url — The fully-qualified URI location of the log4j configuration file for logging trace messages.

Although both the trace.config and trace.log4j.config.url properties are valid, the trace.config property is newer and allows you to use substitution variables when specifying the location of the TraceConfig.xml file in the config.properties file. For more information on using these substitution variables in the trace.config property, refer to the [“Trace Output Configuration”](#) topic in section 4.

Required: Yes

Example Entries: trace.log4j.config.url=file:///C:/src/jdh/config/BASE24-Common/TraceConfig.xml
 trace.log4j.config.url=http://localhost:8080/config/TraceConfig.xml
 trace.log4j.config.url=file:///C:/Data/pm_source/current_uis/pm/pmserver/Config/TraceConfig.xml
 trace.config=C:/ACI/APFAdmin_07.2/Server/Web/config/TraceConfig.xml

Usage: TraceManager

usec.host — The name of the remote host (name or IP address) where the USEC subsystem is executing. The USEC Interface (USECIF) component uses this property value to communicate with the USEC subsystem. This property is only required if USEC is running offboard the ESWEB server process.

Required: No

Example Entries: usec.host=usecprod1
 usec.host=1.160.10.240

Usage: USEC Interface component

usec.port — The remote port number through which the USEC Interface (USECIF) component communicates with the USEC subsystem. This property is only required if USEC is running offboard the ESWEB server process.

Required: No

Example Entry: usec.port=4672

Usage: USEC Interface component

userauditlog.detail.column.size — If you are using a relational database for the User Audit Log data source (USRAULOG), this property allows you increase the size of the audit detail column beyond the 1935 character limit of the JDAL driver. By doing so, you can improve audit performance by avoiding record splits of detail entries over 1935 characters in length.

Required: No
Usage: User Auditing

user.security.AllowFilterMigration — A flag indicating whether field filters that currently exist for a permission ID are automatically migrated to each user associated with the permission ID when the ESWEB process is started and the user.security.version property is set to usec1.2. Valid values are as follows:

true = Yes, migrate field filters to users when the user.security.version property is set to usec1.2.
false = No, do not migrate field filters to users when the user.security.version property is set to usec1.2.

Required: No
Usage: User Security

user.security.version — A code indicating what version of the USEC subsystem is installed. The version dictates whether the USEC subsystem is to run onboard/in-process in the ESWEB server or as an outboard process and whether some user security functionality is available. For example, USEC version 1.1 or later is required to run the application onboard/in-process and to support JAAS

pluggable third-party authentication. USEC version 1.2 or later is required to assign filters directly to a user through the User Security Profile Configuration window. Valid values are as follows:

- usec1.1 = USEC version 1.1 is installed. The USEC subsystem runs onboard/in-process (i.e., within the servlet container for the ESWEB server). JAAS pluggable third-party authentication is supported. Field filters are not automatically migrated from permission IDs to users when the ESWEB process is started. Default.
- usec1.2 = USEC version 1.2 is installed. The USEC subsystem runs onboard/in-process (i.e., within the servlet container for the ESWEB server). JAAS pluggable third-party authentication is supported as well as the ability to assign filters directly to a user through the User Security Profile Configuration window. If field filters currently exist for a permission ID, they are automatically migrated to each user associated with the permission ID when the ESWEB process is started and the user.security.AllowFilterMigration property is set to true.

Required: Yes
Usage: User Security

website.accountformat — The format of account numbers entered at the web site. All characters, other than the number sign (#) character, are for formatting the entered account number. The # character represents a character of the entered account number. For example, if the characters of the entered account number are 1234567890123456 and this property is set to a value of #####-#####-#####-#####, the account number would be formatted as 1234-5678-9012-3456. If this property is not configured, it defaults to a value of #####-#####-#####-#####.

Required: No
Example Entry: website.accountformat=#####-#####-#####-#####
Usage: WebsiteConfig

website.phoneformat — The format of phone numbers entered at the web site. All characters, other than the number sign (#) character, are for formatting the entered phone number. The # character represents a character of the entered phone number. For example, if the characters of the entered phone number are 4023907600 and this property is set to a value of (###) ###-#####, the phone number would be formatted as (402) 390-7600. If this property is not configured, it defaults to a value of (###) ###-#####.

Required: No
Example Entry: `website.phoneformat=(###) ###-####`
Usage: WebsiteConfig

website.character.encoding — The internal character encoding set (e.g., ISO8859-1) used for all data entered from JSP screens on the website. The default value is ISO8859-1. The value of this property should match the value in the `i18n.jsf` file located in the website's `web-inf` directory.

Required: No
Example Entry: `website.character.encoding=ISO8859-1`
Usage: CommonJSPPage

website.datestyle — A code indicating the format in which dates are displayed on a browser user interface pages. The value of this property is used by the Java `DateFormat` class and is set for the process so all browser pages display dates in the same way. The `DateFormat` class provides many class methods for obtaining default date/time formatters based on the default or a given locale and a number of formatting styles. The formatting styles for the class include `FULL`, `LONG`, `MEDIUM`, and `SHORT`. Valid values of this property are as follows:

- 0 = `FULL` displays the full date, including the day of the week, month, day, year, and calendar as in `Wednesday, April 14, 2004 AD`
- 1 = `LONG` displays the month, day, and year as in `April 14, 2004`
- 2 = `MEDIUM` displays the month (abbreviated), day, and year as in `Apr 14, 2004`
- 3 = `SHORT` displays the month, day, and year numerically as in `04.14.04`

Required: No
Example Entry: `website.datestyle=1`
Usage: AdminPage
AppPage
WebsiteConfig

website.defaultlanguage — A user-defined language code used to access a row from the Language data source (`LANGUAG`) to obtain a locale value (see the `website.defaultlocale` property description below). The locale value specifies the default language to be used for the AJSF-based application. If this property is not configured, it defaults to a value of `USA`.

Required: No
Example Entry: website.defaultlanguage=USA
Usage: WebsiteConfig

website.defaultlocale — The default locale to use for localizing the AJSF-based application. For the AJSF, the locale is a string value comprised of the ISO language code and ISO country code, where the locale string value is formatted as follows:

ll_CC

where:

ll is the two-byte alpha language indicator from ISO-639 in lower-case and *CC* is the two-byte alpha country code from ISO-3166 in upper-case. Refer to the following website for a list of all current two-byte alpha country codes from ISO-3166: http://www.iso.org/iso/country_codes/updates_on_iso_3166.htm

Note: The language indicator and country code are separated by an underbar (_).

The combination of language indicator and country code allows you to differentiate between different usages of a particular language in different countries. For example, a language code of en_US specifies the English language as used in the United States while a code of en_GB specifies the English language as used in Great Britain; a language code of es_ES specifies the Spanish language as used in Spain while a code of es_MX specifies the Spanish language as used in Mexico.

Required: Yes, if used with a Web browser-based user interface.
Example Entry: website.defaultlocale=en_US
Usage: AdminPage
AdminUser
CommonJSPPages
CommonXMLFormatter
HTTPServletBase
Language
WebsiteConfig
XMLServletBase

website.domain — The default domain name for a website application.

Required: No
Example Entry: website.domain=aciworldwide.com
Usage: WebsiteConfig

website.helpURL — The URL for web page help.

Required: Yes
Example Entry: http://{host}:{port}/ESWeb/help (where {host} and {port} are replaced with user-defined values during installation)
Usage: Any AJPF application, such as Voice Authorization

website.httpsport — If the website.secureSSL property is set to true, this property defines the port used for SSL security.

Required: No
Example Entry: website.httpsport=443
Usage: WebsiteConfig

website.inactivetimeout — The number of seconds that can elapse without activity before a web connection is considered to have timed out. For example, this property could be used to identify ATMs that are not active because of communication problems. This property should be set to a value greater than the longest anticipated disconnect period (e.g., for a dial-up ATM). This property is only used with a Web browser-based user interface. If this property is not configured, it defaults to 20 seconds.

Required: No
Example Entry: website.inactivetimeout=15
Usage: WebsiteConfig

website.secureSSL — A flag indicating whether the application uses Secure Sockets Layer (SSL) security for web connections. Valid values are as follows:

true = Yes, SSL security is used.
false = No, SSL security is not used.

Required: No
Example Entry: website.secureSSL=false
Usage: WebsiteConfig

website.timestyle — A code indicating the format in which times are displayed on a browser user interface pages. The value of this property is used by the Java DateFormat class and is set for the process so all browser pages display times in the same way. The DateFormat class provides many class methods for obtaining default date/time formatters based on the default or a given locale and a number of formatting styles. The formatting styles for the class include FULL, LONG, MEDIUM, and SHORT. Valid values of this property are as follows:

- 0 = FULL displays the hour, minute, second, am/pm indicator, and time zone as in 3:30:42pm CST
- 1 = LONG displays the hour, minute, second, and am/pm indicator as in 3:30:32pm
- 2 = MEDIUM displays the hour, minute, and am/pm indicator as in 3:30pm
- 3 = SHORT displays just the hour and minute as in 3:30

Required: No

Example Entry: website.timestyle=1

Usage: AdminPage
AppPage
WebsiteConfig

website.xml.menumap.filename<text-string> — The name of a Transaction Menu Map XML file, where *<text-string>* is a unique text string for this website.xml.menumap.filename property name. Transaction Menu Map XML files specify schemas used to manage browser user interface menu items. These schemas provide input for all possible menu items that can be presented on the browser user interface. Some menu items may not be displayed for a specific user due to the User Security (USEC) application permission definitions for the user. These schema files also allow you to customize the menu layout. For example, you can customize the schema to change the menu layout, such as the order in which menu items are presented, and to change the tasks associated with individual menu items. This property should be defined in the initialization section of the config.properties file because the Menu Map Manager reads it at initialization to preload menu handling details into memory. The data is held in memory for as long as the application is running.

Note: The JSFMenuMap.xml file contains general information that is distributed with all AJSF-based applications using a browser user interface.

Required: Yes, for Web browser-based applications.
Example Entries: website.xml.menu.map.filename.0=JSFMenuMap.xml
website.xml.menu.map.filename.1=ESMenuMap.xml
website.xml.menu.map.filename.1=PMMenuMap.xml

ACI Payments Manager browser:

website.xml.menu.map.filename0=C:/pm/current_uis/pm/
browserUI/Config/PMMenuMap.xml
website.xml.menu.map.filename1=C:/pm/current_uis/pm/
browserUI/Config/JSFMenuMap.xml

Usage: MenuMapManager
WebsiteConfig

xml.builder.use.indent — A flag indicating whether the Builder class builds XML messages with indentation and new line characters. These formatted XML messages are easier to read and are useful in a testing or development environment. ACI recommends that you do not use indented XML messages in a production system. Valid values are as follows:

true = Yes, use indentation when building XML documents.

false = No, do not use indentation when building XML documents. Default.

Required: No
Example Entry: xml.builder.use.indent=false
Usage: Builder

xmlmaskmap.filename — The name of the Field Masking XML file that contains definitions for PCI masking of data displayed on user interface windows or browser pages.

Required: Yes, for PCI field masking
Example Entries: xmlmaskmap.filename={root}/ESConfig/FieldMasking.xml
where {root} is where the FieldMasking.xml file is installed.
xmlmaskmap.filename=C:/Data/pm_source/current_uis/pm/
pmserver/Config/FieldMasking.xml
Usage: PCI Masking

xmltranmap.filename<text-string> — The name of a Transaction Map XML file, where <text-string> is a unique text string for this xmltranmap.filename property name. Transaction Map XML files map symbolic keys (such as a message ID) into formatter classes to parse the message and command classes to

process the transaction. This property should be defined in the initialization section of the config.properties file because the Transaction Map Manager reads it at initialization to preload message handling details into memory. The data is held in memory for as long as the application is running.

Required: Application-dependent

Example Entries:

APF Admin:

```
xmltranmap.filename.0={root}/config/transmaps/
FrameworkTranMap.xml
xmltranmap.filename.1={root}/config/transmaps/
UserAuditTranMap.xml
xmltranmap.filename.2={root}/config/transmaps/
UserSecurityTranMap.xml
xmltranmap.filename.3={root}/config/transmaps/
UserStatusTranMap.xml
xmltranmap.filename.4={root}/config/transmaps/
PCIMaskingTranMap.xml
```

where {root} is where the application is installed.

BASE24-eps Java User Interface servlets:

```
xmltranmap.filename.0={root}/ESConfig/
UserAuditTranMap.xml
xmltranmap.filename.1={root}/ESConfig/
UserStatusTranMap.xml
xmltranmap.filename.2={root}/ESConfig/
PCIMaskingTranMap.xml
xmltranmap.filename.3={root}/ESConfig/
UserSecurityTranMap.xml
```

where {root} is where the application is installed.

BASE24-eps Voice Authorization servlets:

```
xmltranmap.filename.0={root}/VAConfig/VATranMap.xml
```

where {root} is where the application is installed.

BASE24-eps ATM Diebold 911/912 and NCR NDC+ (Java) Device Handler processes:

```
xmltranmap.filename.0=C:/ECommerce/Version041/config/
DHCommonTranMap.xml
xmltranmap.filename.1=C:/ECommerce/Version041/config/
DieboldDHTranMap.xml
xmltranmap.filename.2=C:/ECommerce/Version041/config/
BaseDHSurchargeTranMap.xml
```


xmltranmap.filename.3=C:/ECommerce/Version041/config/
NCRDHTranMap.xml

BASE24-eps ATM IFX ATM Manager process:

xmltranmap.filename.0=/user/ESDH041/config/DHBase/
DHCommonTranMap.xml

xmltranmap.filename.1=/user/ESDH041/config/IFX/
IFXFrameworkTranMap.xml

xmltranmap.filename.2=/user/ESDH041/config/IFX/
IFXATMTranMap.xml

xmltranmap.filename.3=/user/ESDH041/config/DHBase/
ClassicDHTranMap.xml

ACI Payments Manager desktop (PMWEB) server:

xmltranmap.filename.0=C:/Data/pm_source/current_uis/pm/
pmserver/Config/FrameworkTranMap.xml

xmltranmap.filename.1=C:/Data/pm_source/current_uis/pm/
pmserver/Config/UserSecurityTranMap.xml

xmltranmap.filename.2=C:/Data/pm_source/current_uis/pm/
pmserver/Config/UserStatusTranMap.xml

xmltranmap.filename.3=C:/Data/pm_source/current_uis/pm/
pmserver/Config/UserAuditTranMap.xml

xmltranmap.filename.4=C:/Data/pm_source/current_uis/pm/
pmserver/Config/PCIMaskingTranMap.xml

xmltranmap.filename.5=C:/Data/pm_source/current_uis/pm/
pmserver/Config/PMTransMap.xml

ACI Payments Manager browser:

xmltranmap.filename.0=C:/pm/current_uis/pm/browserUI/
Config/PMTransMap.xml

xmltranmap.filename.1=C:/pm/current_uis/pm/browserUI/
Config/PCIMaskingTranMap.xml

Usage: TransactionMapManager

xmltransmetamap.filename.*n* — The fully-qualified name of a Transaction Meta Map XML file, where *n* is a sequential number starting at 0 and increased by 1 for each filename specified. Transaction Meta Map XML files define a series of actions required to process a transaction. You can think of them as work flow descriptors. This property should be defined in the initialization section of the config.properties file because the Transaction Meta Map Manager reads it at initialization to preload message handling details into memory. The data is held in memory for as long as the application is running.

Required: Application-dependent

Example Entries:

BASE24-eps ATM Diebold 911/912 and NCR NDC+ (Java) Device Handler processes:

xmltransmetamap.filename.0=C:/ECommerce/Version041/config/DHCommonTransMetaMap.xml

xmltransmetamap.filename.1=C:/ECommerce/Version041/config/DieboldDHTransMetaMap.xml

xmltransmetamap.filename.2=C:/ECommerce/Version041/config/BaseDHSurchargeTransMetaMap.xml

xmltransmetamap.filename.3=C:/ECommerce/Version041/config/DieboldDHSurchargeTransMetaMap.xml

xmltransmetamap.filename.4=C:/ECommerce/Version041/config/NCRDHTransMetaMap.xml

xmltransmetamap.filename.5=C:/ECommerce/Version041/config/NCRDHSurchargeTransMetaMap.xml

BASE24-eps ATM IFX ATM Manager process:

xmltransmetamap.filename.0=/user/ESDH041/config/DHBase/DHCommonTransMetaMap.xml

xmltransmetamap.filename.1=/user/ESDH041/config/IFX/IFXFrameworkTransMetaMap.xml

xmltransmetamap.filename.2=/user/ESDH041/config/IFX/IFXATMTransMetaMap.xml

xmltransmetamap.filename.3=/user/ESDH041/config/DHBase/ClassicDHTransMetaMap.xml

Usage: TransMetaMapManager

xmltransmetamap.trace.paths — A flag indicating whether the Transaction Meta Map Manager traces path states and writes trace messages to the file specified by the xmltransmetamap.trace.paths.filename property. Valid values are as follows:

true = Yes, trace path state information.

false = No, do not trace path state information.

Required: No

Example Entry: xmltransmetamap.trace.paths=false

Usage: TransMetaMapManager

xmltransmetamap.trace.paths.filename — The name of a file to which path states traced by the Transaction Meta Map Manager are written. You can then analyze the file output to ensure that transactions are being processed as expected.

Required: No

Example Entry: `xmltransmetamap.trace.paths.filename=C://src//jdh//config//TraceState`

Usage: `TransMetaMapManager`

xmltransmetamap.trace.paths.errorconditions — A flag indicating whether the Transaction Meta Map Manager includes error conditions when tracing transaction path states. If so, error conditions are written to the file specified by the `xmltransmetamap.trace.paths.filename` property. Valid values are as follows:

`true` = Yes, include error conditions when tracing path state information.

`false` = No, do not include error conditions when tracing path state information.

Required: No

Example Entry: `xmltransmetamap.trace.paths.errorconditions=false`

Usage: `TransMetaMapManager`

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6: Extending AJSF Security and Auditing

You can extend the ACI Java Server Framework (AJSF) User Security (USEC) authentication and user auditing functions to non-AJSF-based applications.

This section describes the USEC and user auditing APIs and how to configure them for non-AJSF-based applications.

Overview

User authentication and auditing functionality is provided using XML messages. The XML schemas provide version capabilities such that new functionality can be introduced while previous versions of functionality are still supported.

The AJICoreLib1_0_0.xsd XML schema defines commonly used AJI core XML elements and attributes, while other XML schemas provide user session management (i.e., login and logout), identity management (i.e., change password), and user audit (i.e., audit insert) functionality through a web service or XML interface. These XML schemas are described below. Corresponding Web Services Description Language (WSDL) files are also provided for each of these XML schemas to support web services for each of these functions.

User Security Authentication

For user security, the functionality available to non-AJSF-based applications is currently limited to the following optional user authentication functions: login, logout, and change password. These functions are provided in the following XML schemas.

XML Schema	Description
SessionServiceMessage1_0_0.xsd	Provides XML definitions for version 1.0.0 LoginRq/Rs and LogoutRq/Rs XML messages.
IdentityManagementMessage1_0_0.xsd	Provides XML definitions for version 1.0.0 ChangePasswordRq/Rs XML messages.

Messages

The User Security Transaction Map file (i.e., UserSecurityTranMap.xml) provides support for the following messages:

- AJI*1.0.0*ChangePasswordRq. AJI version 1.0.0 change password request messages.
- AJI*1.0.0*LoginRq. AJI version 1.0.0 login request messages.
- AJI*1.0.0*LogoutRq. AJI version 1.0.0 logout request messages.

The supporting code for the user security authentication functions listed above is located within the application.jar and is included in the Base-Application.war deployable web application archive file.

Note: Refer to the IdentityManagementMessage1_0_0.xsd and SessionServiceMessage1_0_0.xsd files for additional information regarding message contents.

User Auditing Functions

For user auditing, the functionality available to non-AJSF-based applications is the ability to insert auditing entries into the User Audit Log data source (USRAULOG) through a web service or XML interface.

A user auditing XML schema that contains additional information regarding message contents is provided for version 1.0.0 messages (i.e., AuditServiceMessage1_0_0.xsd).

Messages

The User Audit Transaction Map file (i.e., UserAuditTranMap.xml) provides support for the following messages:

- **AJI*1.0.0***AuditInsertRq. AJI version 1.0.0 audit insert request message.

The supporting code for the user auditing functions listed above is located within the application.jar and is included in the Base-Application.war deployable web application archive file.

Note: Refer to the AuditServiceMessage1_0_0.xsd file for additional information regarding message contents.

Implementation

To extend optional USEC authentication and user auditing for non-AJSF-based applications, you must configure several properties for the Java Web Server process (e.g., ESWEB, PMWEB, APF Admin) in the config.properties file or the APPLCNFG and add a LISTENER record for the remote application.

Properties Configuration

The following properties must be configured in the config.properties or APPLCNFG for the Java Web Server process:

xmltranmap.filename<text-string> — The name of a Transaction Map XML file, where <text-string> is a unique text string for this xmltranmap.filename property name. An xmltranmap.filename<text-string> property must exist that refers to the UserSecurityTranMap.xml file if you are supporting the user security functions and another xmltranmap.filename<text-string> property must exist that refers to the UserAuditTranMap.xml file if you are supporting the user auditing functions.

Example Entries: xmltranmap.filename.0=UserAuditTranMap.xml file
 xmltranmap.filename.1=UserSecurityTranMap.xml file

jsf.jaxb.context.<text-string> — The Java API for XML binding (JAXB) context path for user authentication and auditing with non-AJSF-based applications, where <text-string> is a unique text string for this jsf.jaxb.context property name. A jsf.jaxb.context.<text-string> property must be added to refer to the following packages:

- com.aciworldwide.application.audit.formatter.xml.auditservice1_0_0
- com.aciworldwide.application.security.formatter.xml.identitymanagement1_0_0.jaxb
- com.aciworldwide.application.security.formatter.xml.sessionservice1_0_0.jaxb

Example Entries: jsf.jaxb.context.0=com.aciworldwide.application.audit.
formatter.xml.auditservice1_0_0
jsf.jaxb.context.1=com.aciworldwide.application.security.
formatter.xml.identitymanagement1_0_0.jaxb
jsf.jaxb.context.2=com.aciworldwide.application.security.
formatter.xml.sessionservice1_0_0.jaxb

initialization.function<text-string> — Specifies the fully-qualified Java package and class name to be started at initialization by the Initialization Manager, where <text-string> is a unique text string for this initialization.function property name. An initialization.function<text-string> property must be configured for each of the following classes when a LISTENER entry is configured (see below) and JAXB is utilized:

- com.aciworldwide.common.MessageRouterService
- com.aciworldwide.application.formatter.jaxb.JAXBContextManager
- com.aciworldwide.application.pci.masking.formatter.jaxb.
JAXBContextManager

Beginning with AJSF 07.4, initialization function properties are only loaded from the APPLCNFG and not from the config.properties file.

XML Transaction Map Files

The UserAuditTranMap.xml file must contain the following transaction entry:

```
<Entry>
  <tag>AJI*AuditInsertRq*1.0.0</tag>
  <fmtrClass>com.aciworldwide.application.audit.formatter.xml.
auditservice1_0_0.AuditInsertFormatter</fmtrClass>
  <cmndClass>com.aciworldwide.common.command.AuditCommand</
cmndClass>
</Entry>
```

The UserSecurityTranMap.xml file must contain the following transaction entries:

```
<Entry>
  <!-- AJI 1.0.0 Change Password Request -->
  <tag>AJI*ChangePasswordRq*1.0.0</tag>
  <fmtrClass>com.aciworldwide.application.security.formatter.
xml.identitymanagement1_0_0.ChangePasswordFormatter</fmtrClass>
  <cmndClass>com.aciworldwide.application.security.
authentication.command.ChangePasswordCommand</cmndClass>
```

```
</Entry>
<Entry>
  <!-- AJI 1.0.0 Login Request -->
  <tag>AJI*LoginRq*1.0.0</tag>
  <fmtrClass>com.aciworldwide.application.security.formatter.
xml.sessionservice1_0_0.LoginFormatter</fmtrClass>
  <cmdClass>com.aciworldwide.application.security.
authentication.command.LoginWithContextCommand</cmdClass>
</Entry>
<Entry>
  <!-- AJI 1.0.0Logout Request -->
  <tag>AJI*LogoutRq*1.0.0</tag>
  <fmtrClass>com.aciworldwide.application.security.formatter.
xml.sessionservice1_0_0.LogoutFormatter</fmtrClass>
  <cmdClass>com.aciworldwide.application.security.
authentication.command.LogoutWithContextCommand</cmdClass>
</Entry>
```

Listener Configuration

In addition to the above property and transaction map configuration, you must configure an entry in the Listener data source (LISTENER) to connect incoming authentication requests from the non-AJSF-based application to the USEC subsystem.

The following table defines an example LISTENER entry to provide user authentication for a remote application using TCP/IP.

Field Name	Value/Description
Listener ID	User-defined
Process ID	Java Web Server (e.g., P1A^ESWEB)
Maximum Connections	1
Listener Description	Free-format text
Message Format Class	com.aciworldwide.application.formatter. CommonXMLFormatterController
Message Format Method	Not applicable.
Message Adapter Class	com.aciworldwide.common.adapter. GenericSyncListenerAdapter

Field Name	Value/Description
Listener Type	TCPIP-SERVER
Port Number	TCP/IP port number for the non-AJSF-based application.
Termination String	Not applicable.
Protocol Field Options	.HDR2.

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7: Event Adapter

The *ACI Event Adapter* is a Java server process that can be used by any ACI Payments Framework application running on an IBM System z or Unix platform to consolidate and standardize the collection, formatting, and distribution of event messages. The Event Adapter is a standalone application that runs in the AJSF environment and is responsible for filtering, formatting, and distributing a stream of events to various endpoints for consumption.

Overview

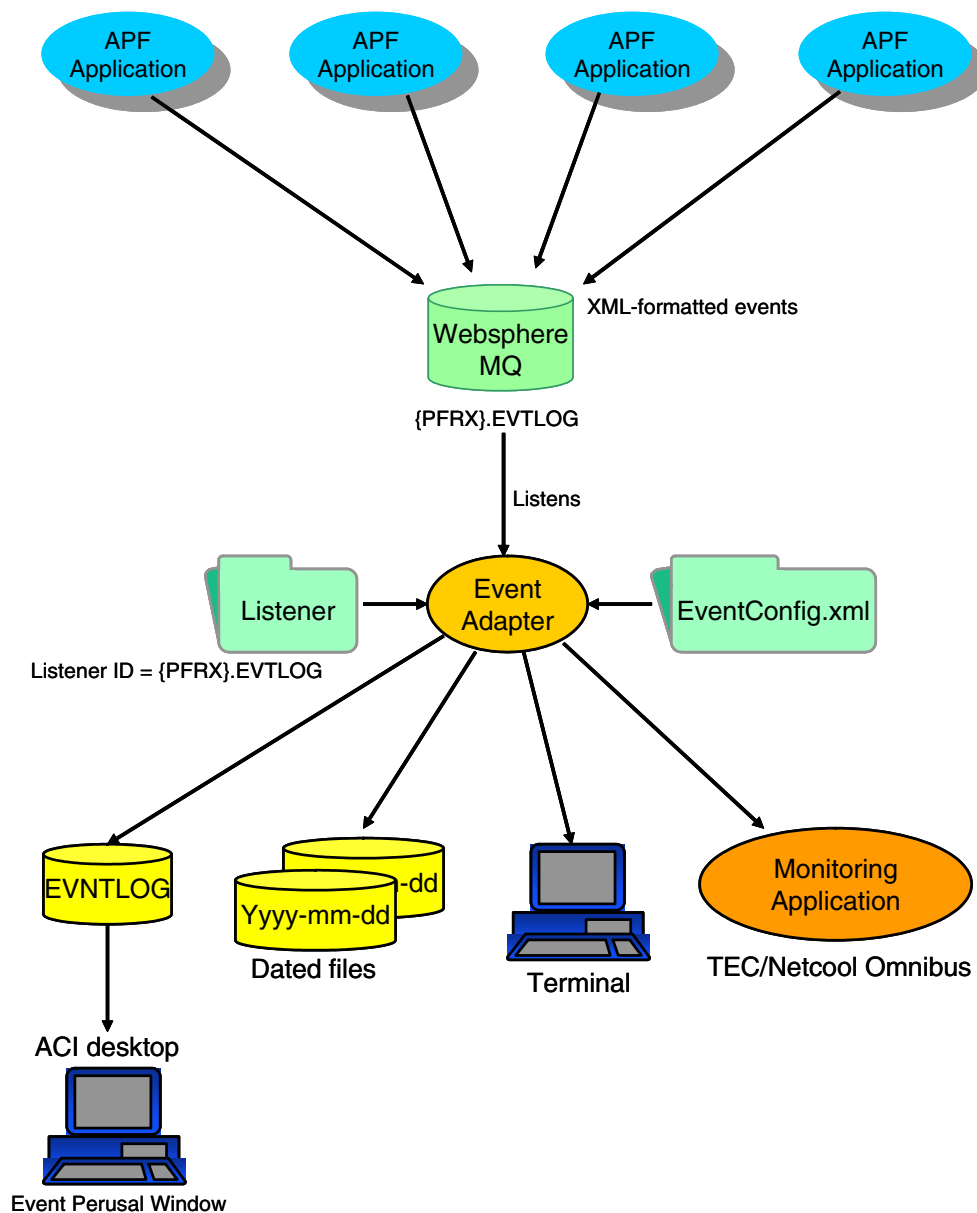
The purpose of the ACI Event Adapter is to facilitate standard, centralized event logging and management for the ACI Payments Framework (APF). The ACI Event Adapter performs the following:

- Acts as a common collection point for all events, to serialize the events in order of time received.
- Specifies a common content/format (or at minimum, common information) for all events logged.
- Examines and prepares the event set for automation.
- Distributes events to a wide variety of endpoints, using a number of delivery mechanisms.
- Provides message filtering and reformatting services.

When using the ACI Event Adapter, APF event messages are logged as XML messages that can be manipulated for a variety of viewing and reporting purposes or for integration with other user applications. The Event Adapter receives events in an ACI-defined EventSchema XML format from a WebSphere MQ queue. The adapter parses the events and in some cases normalizes them to an internal Java data object. It then uses standard Java-based LOG4J processing to *broadcast* the events to the various defined locations, in varying formats, using various mechanisms. The Event Adapter process writes events to each location configured by an appender in the EventConfig.xml file. Each appender can also reformat or filter events before writing them to the appender output location.

The LOG4J module (Apache project) uses a standard methodology for development of modules that are responsible for distribution of log/event messages. The basic approach for development of unique delivery mechanisms is to develop a LOG4J appender that performs the specific delivery required. In addition, filters may be supplied to route a specific class of event (e.g., events of a critical nature for system operation) to a specific appender.

The following illustration depicts Event Adapter processing.



The Event Adapter uses its own EventConfig.xml file to determine how events are filtered, formatted, and where they are output. The adapter can format and write events to any or all of the following output locations:

- The Event Log data source (EVNTLOG) for event perusal from the Event Perusal window on the ACI desktop user interface.
- Dated log files.

- System console, SYSERR, or SYSOUT.
- Enterprise monitoring application, such as IBM's Tivoli Enterprise Console/Netcool Omnibus.
- Any other location for which a valid appender is configured in the EventConfig.xml file for the Event Adapter process.

EventConfig.xml File

ACI Event Adapter event processing is configured using its own EventConfig.xml file, which is read at startup.

Among other things, the EventConfig.xml file specifies the following:

- The locations to which event messages are to be sent.
- The appenders and formatters used for each location.
- The filtering to use for each location.

Note: When the ACI Event Adapter is installed on an IBM System z platform, the encoding meta element in the XML declaration must be set to “IBM-1047” instead of “UTF-8” or the encoding meta element must be removed as shown in the following examples.

```
<?xml version="1.0" encoding="IBM-1047"?>
<?xml version="1.0" ?>
```

A sample EventConfig.xml file for the Event Adapter is shown below.

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE log4j:configuration SYSTEM "log4j.dtd">
<log4j:configuration debug="false">

    <!--
        This EventConfig.xml file is specific to the test environment for
        the EventPerusal adapter. Specific comments have been added to this
        file to denote entries that'd need to be changed appropriately for
        different customer environments. These specific comments start with
        "EXAMPLE ENTRY BELOW".
    -->

    <!-- Appender for the system console (i.e., System.out) -->
    <appender name="SYSOUT" class="org.apache.log4j.ConsoleAppender">
        <param name="Target" value="System.out"/>
        <layout class="org.apache.log4j.PatternLayout">
            <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
        </layout>
        <filter class="com.aciworldwide.event.filter.InternalEventFilter"/>
        <filter class="org.apache.log4j.varia.DenyAllFilter"/>
    </appender>
```

```
<!-- Appender for the system error console (i.e., System.err) -->
<appender name="SYSERR" class="org.apache.log4j.ConsoleAppender">
  <param name="Target" value="System.err"/>
  <layout class="org.apache.log4j.PatternLayout">
    <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
  </layout>
  <filter class="com.aciworldwide.event.filter.InternalEventFilter"/>
  <filter class="org.apache.log4j.varia.DenyAllFilter"/>
</appender>

<!-- Appender for the Error log files. -->
<appender name="EVENT" class="com.aciworldwide.framework.log.
DailyRollingFileAppenderExt">
  <!-- EXAMPLE ENTRY BELOW -->
  <param name="File" value="//db01/b24dev/users/event_user/logs/
EventAdapterEvents.log"/>
<!-- Character Encoding (z/OS Example): <param name="Encoding" value="IBM-1047"/
> -->
  <param name="DatePattern" value="'. 'yyyy-MM-dd"/>
  <layout class="org.apache.log4j.PatternLayout">
    <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
  </layout>
  <filter class="com.aciworldwide.event.filter.InternalEventFilter"/>
  <filter class="org.apache.log4j.varia.DenyAllFilter"/>
</appender>

<!-- Appender for the Error log when running in a clustered environment.
      Use this appender to write events to the EVENTLOG SQL table. Note
that
      you must make sure each JVM has a different process id system property
      specified
-->
<!--
<appender name="EVENT"
  class="com.aciworldwide.common.log.EventLogAppender">
  <layout class="org.apache.log4j.PatternLayout">
    <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
  </layout>
  <filter class="com.aciworldwide.event.filter.InternalEventFilter"/>
  <filter class="org.apache.log4j.varia.DenyAllFilter"/>
</appender>
-->

<!-- Appender for the CRITICAL Events files. -->
<appender name="CRITICAL_EVENTS"
  class="com.aciworldwide.framework.log.DailyRollingFileAppenderExt">
  <param name="Threshold" value="FATAL"/>
  <!-- EXAMPLE ENTRY BELOW -->
```

```

    <param name="File" value="//db01/b24dev/users/event_user/logs/
EventAdapterCriticalEvents.log"/>
<!-- Character Encoding (z/OS Example): <param name="Encoding" value="IBM-1047"/
> -->
    <param name="DatePattern" value="'.'yyyy-MM-dd"/>
    <layout class="org.apache.log4j.PatternLayout">
        <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
    </layout>
    <filter class="com.aciworldwide.event.filter.InternalEventFilter"/>
    <filter class="org.apache.log4j.varia.DenyAllFilter"/>
</appender>

<appender name="EVENTTEXTAPPENDER"
    class="com.aciworldwide.framework.log.DailyRollingFileAppenderExt">
    <param name="File" value="/db01/b24dev/users/event_user/logs/Events.log"/>
<!-- Character Encoding (z/OS Example): <param name="Encoding" value="IBM-1047"/
> -->
    <param name="DatePattern" value="'.'yyyy-MM-dd"/>
    <layout class="org.apache.log4j.PatternLayout">
        <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
    </layout>
    <filter class="com.aciworldwide.event.filter.SystemEventFilter"/>
    <filter class="com.aciworldwide.event.filter.InternalEventFilter"/>
</appender>

<!-- Appender for the DATABASE -->
<!-- comment out this appender element if not needed and remove DATABASE log
    from category element below -->
<appender name="EVENTDATABASEAPPENDER"
    class="com.aciworldwide.event.appender.EventDatabaseAppender">
    <layout class="org.apache.log4j.PatternLayout">
        <param name="ConversionPattern" value="%d{ISO8601} %-5p %c [%t] %x -
%m%n"/>
    </layout>
    <filter class="com.aciworldwide.event.filter.SystemEventFilter"/>
    <filter class="org.apache.log4j.varia.DenyAllFilter"/>
</appender>

<!-- Appender for the output of events to Tivoli Enterprise Console-->
<appender name="EVENTTIVOLIAPPENDER"
    class="com.aciworldwide.event.appender.EventTivoliAppender">
    <param name="ConfigFile" value="/db01/b24dev/users/event_user/EvtAdapter/
config/TivoliConfigFile"/>
    <filter class="com.aciworldwide.event.filter.SystemEventFilter"/>
    <filter class="com.aciworldwide.event.filter.InternalEventFilter"/>
</appender>

<appender name="NULL" class="com.aciworldwide.framework.log.NullAppender">
    <layout class="org.apache.log4j.SimpleLayout"/>
</appender>

```

```
<!-- The category used by Error Manager. All event messages are logged under
this category. -->
<!-- ErrorManager automatically prepends this category name to all requests. -
->
<!-- This example illustrates sending CRITICAL events to a different
destination. In this case, -->
<!-- it is to a different file; but it could be, for example, an SMTPAppender
(to send an E-mail alert). -->

<category name="event" additivity="false">
  <priority value="info"/>
  <appender-ref ref="EVENT"/>
  <!-- <appender-ref ref="SYSOUT"/> -->
  <appender-ref ref="CRITICAL_EVENTS"/>
  <appender-ref ref="EVENTTEXTAPPENDER"/>
  <!-- <appender-ref ref="EVENTDATABASEAPPENDER"/> REMOVE COMMENTS TO ENABLE
-->
  <!-- <appender-ref ref="EVENTTIVOLIAPPENDER"/> REMOVE COMMENTS TO ENABLE --
>
</category>

<!-- The required root category. Because the 'event' category has it's-->
<!-- additivity flag set to false, nothing should use this category. -->
<root>
  <priority value="debug"/>
  <appender-ref ref="SYSOUT"/>
</root>
</log4j:configuration>
```

Event Adapter Appenders

The Event Adapter appenders provided in the EventConfig.xml file at installation are described below.

SYSOUT and SYSERR Appenders

The SYSOUT and SYSERR appenders write events to the standard system output and system error locations, respectively, for the Event Adapter process. Typically, these output streams are defined to the system console and are commonly used to display informational and error messages, respectively.

Note: ACI recommends that the SYSOUT appender not be used to avoid the unnecessary duplication of events from the ACIJMX agent in the Event Log data source (EVNTLOG).

EVENT Appenders

The EVENT appenders filter, format, and write events to the error log file. Two EVENT appenders are provided, one for clustered WebSphere MQ environments and one for nonclustered WebSphere MQ environments.

CRITICAL_EVENTS Appender

The CRITICAL_EVENTS appender writes only critical events generated by the Event Adapter to a specified log file.

Event Text Appender

The EVENTTEXTAPPENDER filters, formats, and writes events to a rolling daily log file. The underlying file is rolled over at a user-chosen frequency that is specified by the DatePattern option. It is possible to specify monthly, weekly, half-daily, daily, hourly, or minutely rollover schedules. This pattern should follow the java.text.SimpleDateFormat conventions. For detailed information on the DatePattern, please see the log4j JavaDoc information on the logging.apache.org website.

Event Database Appender

The EVENTDATABASEAPPENDER filters, formats, and writes events to the Event Log data source (EVNTLOG), from which the contents can be displayed on the Event Perusal window of the ACI desktop user interface.

ACI TEC Appender (EVENTTIVOLIAPPENDER)

The *ACI TEC appender* enables the ACI Event Adapter to integrate with Tivoli Enterprise Console (TEC) 3.9. The appender uses IBM-supplied Java EIF (Event Integration Facility) classes to generate an event stream to the TEC and OMNIBus products on the TEC server.

Event streams are sent to Netcool OMNIBus by way of an EIF *probe* that allows EIF-enabled applications to route events to the Netcool platform.

The ACI TEC appender receives an event as a Java data object and produces an EIF Event object that contains a number of name/value pairs that correspond to a supplied .BAROC file. It performs a straightforward mapping of ACI-defined event severities (Trace/Debug, Informational, Warning, Error, Critical) to TEC-defined severity levels.

Use of the ACI TEC appender requires a TEC configuration file (TivoliConfigFile script file), which is supplied by ACI. This script file provides environment information for both the EIF classes as well as contact/address information for the (usually remote) TEC server.

If there is a problem contacting the TEC server via the EIF classes, the EIF creates a cache file of events on the local file system to be logged when the communication to the TEC server is reestablished.

Platform Implementation

The following topics describe how the Event Adapter can be implemented on a Unix or IBM System z platform.

Unix Platform

For a Unix platform, the Event Adapter process typically extracts events from the queue it is configured to listen to, reformats the events, and writes them to the Event Log data source (EVNTLOG). You can then view events from the Event Perusal window on the ACI desktop user interface. You can also direct events to the SYSOUT, SYSERR, EVENTTEXTAPPENDER, and EVENTDATABASEAPPENDER locations.

IBM System z Platform

For an IBM System z platform, the Event Adapter can perform the same functions that it does on a UNIX platform, as well as distributing events to the Tivoli Enterprise Console (TEC) and/or Tivoli Netcool/OMNIBus product. The Event Adapter uses IBM's Event Integration Facility (EIF) library as the API for TEC integration.

Both IBM Tivoli Enterprise Console (TEC) and IBM Tivoli Netcool/OMNIBus are enterprise-level event collection and management systems. They collect and consolidate event and alarm information in real-time from networks, hardware, and software throughout an environment, and provide for sophisticated, centralized, and comprehensive event management and handling (e.g., filtered views, automated problem diagnosis and resolution, intelligent rules-based processing).

The Tivoli Enterprise Console (TEC) is in wide use throughout the Tivoli install base. However, IBM's current direction is to migrate TEC customers to the newer, more robust Netcool products, particularly Netcool OMNIBus.

Netcool/OMNIBus offers an EIF *probe* that allows an EIF-enabled application to route its events to the Netcool platform.

Tivoli monitoring components such as TEC typically reside on a Tivoli Enterprise Monitoring Server (TEMS) or Tivoli Enterprise Portal Server (TEPS) hardware and software environment dedicated to the monitoring platform. The TEMS/TEPS need to be modified to import the ACI event format. This is done using an ACI-supplied Tivoli BAROC file containing information defining ACI event content.

ACI event messages do not contain sensitive cardholder information that impact PABP or PCI compliance. For general security considerations, the TEMS/TEPS hardware is expected to reside in a trusted domain with the payments processing systems.

The Event Adapter that generates the event stream to TEC needs to be modified to incorporate the Tivoli Event generation classes. This is done by copying the Tivoli EIFSDK directory to the ACI Event Adapter server directory and adding the JAVA classpath to the Event Adapter startup shell script. The Tivoli Configuration script file specifies the TEC server IP address and port that the Event Adapter uses to send events for the ACI TEC appender.

The topics below describe the above configuration requirements for integrating the Event Adapter with Tivoli. For detailed instructions on installing and configuring the Event Adapter for Tivoli, see the [“Integrating the ACI Event Adapter with the IBM Tivoli Enterprise Console \(TEC\) or Tivoli Netcool/OMNIBus”](#) topic later in this section.

TivoliConfigFile Script File

The Tivoli Configuration script file specifies the server IP address and port. The location of this file (e.g., /db01/b24dev/users/event_user/EvtAdapter/config) must match the value of the ConfigFile param for the Tivoli Enterprise Console event adapter appender in the EventConfig.xml file. A sample of the TivoliConfigFile script file is shown below.

```
ServerLocation=172.21.33.117
ServerPort=5529
BufEvtPath=./TivoliEventAdapter.cachefile
```

The ServerLocation and ServerPort parameters should be set to the IP address and listening port of the Event Server on the Tivoli Enterprise Console server. The BufEvtPath parameter defines the cache file that is used if the Event Server is temporarily unavailable. Using ./ in front of the filename tells the event adapter to place the cache file in the same directory where the ACI Event Adapter process was started. You can change this parameter to place the cache file in any desired directory that is accessible to the Event Adapter process.

Tivoli Enterprise Console Server Configuration

The ACI-provided Basic Recorder of Objects in C (BAROC) file, `aci_event.baroc`, must be imported into a rulebase on the Tivoli Enterprise Console. The `aci_event.baroc` file must be located in a directory accessible to the Enterprise Console Server (e.g., `c:\IBM\Tivoli\aci\aci_event.baroc`). The `aci_event.baroc` file describes the classes of events that ACI supports.

APF products use only a single XML schema for all events issued by APF modules, so only a single `.BAROC` file is required for input.

ACI provides a Tivoli `.BAROC` file that defines the expected event contents for ACI events to the Tivoli management server. This file must be installed onto the Tivoli server and integrated into the event base resident on the server via straightforward manual steps.

Downloading EIF Libraries

EIF libraries are supplied by IBM and must be downloaded and licensed by customers that intend to use TEC or OMNIBus. See the [“Integrating the ACI Event Adapter with the IBM Tivoli Enterprise Console \(TEC\) or Tivoli Netcool/OMNIBus”](#) topic for detailed instructions.

Updates to the IBM-supplied EIF library can be issued periodically. These classes need to be updated on IBM product directories.

Listener Configuration

The following table describes the Listener configuration needed for the Event Adapter. This record is preloaded in the Listener data source (LISTENER) during installation. No value is needed for any fields in the LISTENER that are not listed (e.g., PrtcDfndlTx, RspnMsgFrmtMthdTx, etc.).

Field Name	Value
LstnID	\${NET}.EVTLOG
PrclsID	EVENT_ADAPTER
MaxConnNum	10
DscrTx	Event listener that listens to the Event Queue.
RspnMsgFrmtClasTx	com.aciworldwide.event.formatter. EventMessageFormatter
RqstAdptClasTx	com.aciworldwide.event.adapter. EventMessageAdapter
Type	JMS-EXTERNAL-REQUEST

Application Properties

The following application properties configured in the `config.properties` file or the `APPLCNFG` are specific to the Event Adapter process.

eventlog.messagetext.column.size — The maximum length, in bytes, allowed for an event message. If this property is not configured, it defaults to 3800 bytes.

Required: No
Example Entry: `eventlog.messagetext.column.size=8000`
Usage: EventMessageFormatter Component

eventlog.retention.days — The number of days' worth of events the Event Adapter retains in the Event Log data source (EVENTLOG). The default value is 7.

Required: No
Example Entry: `eventlog.retention.days=7`
Usage: EventAdapterCleanup Timer Task

eventlog.cleanup.starttime — The time of day, in hh:mm format, the Event Adapter starts removing old events from the Event Log data source (EVENTLOG). The default value is 00:30 (12:30 am).

This start time automatically accounts for daylight savings time (summer time) on the server where the property is configured. This time is in the time of the executing server where the application is running.

Required: No
Example Entry: `eventlog.cleanup.starttime=00:30`
Usage: EventAdapterCleanup Timer Task

eventlog.timeperiod.hours — The number of hours between runs if the Event Adapter is to remove old events from the Event Log data source (EVENTLOG) more frequently or less frequently than every twenty-four hours. The default value is 24 hours.

Required: No
Example Entry: `eventlog.timeperiod.hours=25`
Usage: EventAdapterCleanup Timer Task

initialization.function<text-string> — Specifies the fully-qualified Java package and class name to be started at initialization by the Initialization Manager, where <text-string> is a unique text string for this initialization.function property name.

Beginning with AJSF 07.4, initialization function properties are only loaded from the APPLCNFG and not from the config.properties file.

Required: No

Example Entry: initialization.function.0.classname=com.aciworldwide.
common.MessageRouterService

Sample Config.properties File

The following is an example of the config.properties file used for the Event Adapter process. This file may need to be edited during installation for site-specific information.

```
# This config.properties file is specific to the test environment for the
EventPerusal adapter.
# Specific comments have been added to this file to denote entries that'd need
to be changed
# appropriately for different customer environments. These specific comments
start with
# "EXAMPLE ENTRY BELOW" at the beginning of the line.
#
# There are also variables inserted used to automate installation in an ES
environment.
# Variables used are:
#   #home_root# - the base path for installation of EventAdapter (eg: /db01/
b24dev/ccb/C62B)
#
#   Data Source properties
#
db.user.id=ADMIN
db.user.password=

# EXAMPLE ENTRY BELOW
#   JDAL-specific configuration; please modify appropriately.
#
db.datasource=jdbc:JDAL:-MDBV=CSV:-AI=ES:-STRT=#home_root#/config:-numthreads=50

#
# EXAMPLE ENTRY BELOW
#   db.max.connections defines the maximum number of connections allowed to
connect to
#   the database.
#
db.max.connections=5

#
# EXAMPLE ENTRY BELOW
#   db.static.connections defines the number of connections to acquire to the
database
#   when the jvm starts. These connections will stay active through the life of
the
#   executing JVM.
#
db.static.connections=5
```

```
#
# db.connection.wait.time defines the amount of time, in milliseconds, to wait
# for
# another thread to free a Database connection in the Database connection
# manager.
# This is used when the maximum number of connections have been acquired as
# specified
# in the db.max.connections property.
#
db.connection.wait.time=10

#
# db.driver specifies the class name of the JDBC driver to use for all
# database connections.
# The whole package name as it is found in the jar file containing the driver
# should be specified.
#
db.driver=com.aciworldwide.jdbcdriver.jdal2.JDAL2Driver
#
#
# db.type specifies the type of database the application is installed on.
# Valid values are:
#     DB2
#     ORACLE
#     MSSQL
#     SQLMX
#
db.type=JDAL2

#
# LogManager properties.
# The Log4J config file location for trace and event
#
# EXAMPLE ENTRY BELOW
# Event & trace configuration configuration; please modify appropriately.
#
event.log4j.config.url=file://#home_root#/EvtAdapter/config/EventConfig.xml
trace.log4j.config.url=file://#home_root#/EvtAdapter/config/TraceConfig.xml

#
# Messaging properties.
#
# Valid values for message.environment are:
#     DIRECT
#     JMS
#     EJB (Not yet fully implemented)
#
message.environment=JMS

#
# OPENJMS
#
```

```
#message.jms.jndi.initialcontextfactory=org.exolab.jms.jndi.rmi.  
RmiJndiInitialContextFactory  
#message.jms.jndi.providerurl=rmi://localhost:1099/JndiServer  
#message.jms.QueueConnectionFactory=JmsQueueConnectionFactory  
  
#  
# MQ Series  
#  
# EXAMPLE ENTRY BELOW  
# JNDI-specific configuration configuration; please modify appropriately.  
#  
message.property.handling=MDS  
message.jms.jndi.initialcontextfactory=com.sun.jndi.fscontext.  
RefFSContextFactory  
message.jms.jndi.providerurl=file://#home_root#/queuing/mq_jms_admin/JNDI-  
Directory  
message.jms.QueueConnectionFactory=JmsQueueConnectionFactory  
  
# Optional Property variable used by the EventMessageFormatter to allow the  
installer  
# to adjust the maximum Event log message length.  
# Default value is 3800 bytes.  
#  
eventlog.messagetext.column.size=8000  
  
#  
# EXAMPLE ENTRY BELOW  
# Properties to define the cleanup tasks that will be performed  
# Properties to set:  
# eventlog.retention.days - number of days to retain rows in the log  
# eventlog.cleanup.starttime - a 24hr clock time specifies when to start  
# eventlog.timeperiod.hours - number of hours between runs  
#  
eventlog.retention.days=7  
eventlog.cleanup.starttime=00:30  
eventlog.timeperiod.hours=25  
  
#  
# EXAMPLE ENTRY BELOW  
# Setup TimerTask entries to tell the framework that EVENT_ADAPTER is to  
initialize  
# the EventAdapterCleanup TimerTask  
#  
timertask.manager.processid=EVENT_ADAPTER  
timertask.function.0.classname=com.aciworldwide.event.timertask.  
EventAdapterCleanup
```

Event Message XML Schema

The following is the XML schema used by APF processes to log event messages. It is shown here to provide a basic understanding of the XML structure and data used in APF event messages.

Note: When the ACI Event Adapter is installed on an IBM System z platform, the encoding meta element in the XML declaration must be set to “IBM-1047” instead of "UTF-8" as shown in the following example.

```
<?xml version="1.0" encoding="IBM-1047"?>

<?xml version="1.0" encoding="UTF-8" ?>
- <!--edited with XML Spy v4.4 U (http://www.xmlspy.com) by ACI Worldwide (ACI Worldwide)-->
<xsd:schema targetNamespace="http://www.aciworldwide.com" xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:evt="http://www.aciworldwide.com">
  <xsd:element name="Event">
    <xsd:complexType>
      <xsd:sequence>
        <xsd:element name="DateTime" type="xsd:dateTime" />
        <xsd:element name="TaskID" type="xsd:string" />
        <xsd:element name="ThreadID" type="xsd:string" minOccurs="0" />
        <xsd:element name="UniqueEventID" type="xsd:string" />
        <xsd:element name="RelatedEventID" type="xsd:string" minOccurs="0" />
        <xsd:element name="Generator" type="xsd:string" />
        <xsd:element name="GeneratorBuildVsn" type="xsd:string" minOccurs="0" />
        <xsd:element name="Severity" type="evt:SeverityType" />
        <xsd:element name="EventText" type="xsd:string" minOccurs="0" />
        <xsd:element name="Subject" type="evt:SubjType" minOccurs="0" />
        <xsd:element name="SubsystemID" type="evt:SSIDType" />
        <xsd:element name="EventNumber" type="xsd:int" />
        <xsd:element name="SisErrorList" minOccurs="0">
          <xsd:complexType>
            <xsd:sequence>
              <xsd:element name="SisError" type="evt:SisErrQualifierType"
maxOccurs="unbounded" />
            </xsd:sequence>
          </xsd:complexType>
        </xsd:element>
      </xsd:sequence>
    </xsd:complexType>
  </xsd:element>
  <xsd:simpleType name="SSIDType">
    <xsd:restriction base="xsd:string" />
  </xsd:simpleType>
  <xsd:simpleType name="SubjType">
    <xsd:restriction base="xsd:string" />
  </xsd:simpleType>
  <xsd:simpleType name="SeverityType">
    <xsd:restriction base="xsd:string">
      <xsd:enumeration value="Critical" />
      <xsd:enumeration value="Error" />
      <xsd:enumeration value="Warning" />
      <xsd:enumeration value="Info" />
      <xsd:enumeration value="Trace" />
      <xsd:enumeration value="UserDefined1" />
      <xsd:enumeration value="UserDefined2" />
      <xsd:enumeration value="UserDefined3" />
      <xsd:enumeration value="UserDefined4" />
    </xsd:restriction>
  </xsd:simpleType>
  <xsd:simpleType name="SisErrCodeType">
    <xsd:restriction base="xsd:int" />
  </xsd:simpleType>
</xsd:schema>
```



```
</xsd:simpleType>
  <xsd:simpleType name="SisErrTextType">
    <xsd:restriction base="xsd:string" />
  </xsd:simpleType>
  <xsd:complexType name="SisErrQualifierType">
    <xsd:sequence>
      <xsd:element name="SisSSID" type="evt:SSIDType" />
      <xsd:element name="SisSubject" type="evt:SubjType" />
      <xsd:element name="SisAction" type="xsd:string" />
      <xsd:element name="SisActionStatus" type="xsd:string" />
      <xsd:element name="SisErrorDetail" type="evt:SisErrDetailType" minOccurs="0"
maxOccurs="5" />
    </xsd:sequence>
  </xsd:complexType>
  <xsd:complexType name="SisErrDetailType">
    <xsd:sequence>
      <xsd:element name="SisErrCode" type="evt:SisErrCodeType" />
      <xsd:element name="SisErrText" type="evt:SisErrTextType" />
    </xsd:sequence>
  </xsd:complexType>

```

Integrating the ACI Event Adapter with the IBM Tivoli Enterprise Console (TEC) or Tivoli Netcool/OMNibus

Use the following procedures to integrate the ACI Event Adapter with the IBM Tivoli Enterprise Console (TEC) or Tivoli Netcool/OMNibus. The first three steps are performed on the IBM System z or a Unix platform and are the same whether integrating with TEC or Tivoli Netcool/OMNibus. The remaining steps are performed on the server on which TEC or Netcool/OMNibus reside.

Event Adapter Procedures

Perform the following procedures on the IBM System z platform.

1. Copy the Tivoli EIFSDK directory to the ACI Event Adapter server directory.

The Tivoli EIFSDK directory must be added to the JAVA classpath in the Event Adapter startup shell script.

The Tivoli EIFSDK directory typically resides in the following IBM Unix System Services (USS) directory on the IBM System z platform:

```
/public/dev/b24/users/IBM/Tivoli/EIFSDK_zos
```

2. Edit the ACI Event Adapter configuration files. The ACI Event Adapter configuration files are as follows:

- EventConfig.xml
- TivoliConfigFile

These configuration files are located in the config subdirectory of the ACI Event Adapter installation directory. In this example installation, the configuration files reside in the following directory:

```
/db01/b24dev/users/event_user/EvtAdapter/config
```

- a. Edit the EventConfig.xml file. The areas to edit are toward the end and are marked in blue in the excerpt provided below.

```

<!-- Appender for the output of events to Tivoli Enterprise Console-->
<appender name="EVENTTIVOLIAPPENDER"
  class="com.aciworldwide.event.appender.EventTivoliAppender">
  <param name="ConfigFile" value="/db01/b24dev/users/event_user/EvtAdapter/config/
TivoliConfigFile"/>
</appender>

<appender name="NULL" class="com.aciworldwide.framework.log.NullAppender">
  <layout class="org.apache.log4j.SimpleLayout"/>
</appender>

<!-- The category used by Error Manager. All event messages are logged under this
category. -->
<!-- ErrorManager automatically prepends this category name to all requests. -->
<!-- This example illustrates sending CRITICAL events to a different destination. In
this case, -->
<!-- it is to a different file; but it could be, for example, an SMTPAppender (to send an
E-mail alert). -->

<category name="event" additivity="false">
  <priority value="info"/>
  <appender-ref ref="EVENT"/>
  <appender-ref ref="SYSOUT"/>
  <appender-ref ref="CRITICAL_EVENTS"/>
  <appender-ref ref="EVENTTEXTAPPENDER"/>
  <appender-ref ref="EVENTDATABASEAPPENDER"/>
  <appender-ref ref="EVENTTIVOLIAPPENDER"/>
</category>

<!-- The required root category. Because the 'event' category has it's-->
<!-- additivity flag set to false, nothing should use this category. -->
<root>
  <priority value="debug"/>
  <appender-ref ref="SYSOUT"/>
</root>
</log4j:configuration>

```

- b. Create the Tivoli Configuration script file (TivoliConfigFile) in the location specified by the ConfigFile param in the EventConfig.xml file (e.g., /db01/b24dev/users/event_user/EvtAdapter/config/TivoliConfigFile).

Note: When integrating the ACI Event Adapter residing on the IBM System z platform with the IBM Tivoli Enterprise Console (TEC), you must use the non-Tivoli Management Environment (TME) configuration file. If integrating the ACI Event Adapter residing on a Unix platform with TEC, you can use a TME or non-TME configuration file. Non-TME adapters establish connections using standard interprocess communication mechanisms (e.g., opening an IP socket) rather than using the oserv services provided by Tivoli Management Framework. For more information on TME and non-TME communications, refer to the *IBM Tivoli Enterprise Console Adapters Guide*.

Non-TME Tivoli Configuration Script File. The non-TME Tivoli Configuration script file specifies the server IP address and port as well as a buffer cache location as in the following example.

```
ServerLocation=172.21.33.117
ServerPort=5529
BufEvtPath=/public/dev/b24/es/ibmbct/B682/EvtAdapter/TivoliEventAdapter.cachefile
```

The ServerLocation and ServerPort keywords must be set to the IP address and listening port of the Event Server on the Tivoli Enterprise Console server.

The BufEvtPath keyword defines the cache file used if the Event Server is temporarily unavailable. This keyword should be set to a static location to ensure that the same location is used regardless of the path from which the runevtadapter script is invoked.

TME Tivoli Configuration Script File. Among other keywords, a TME Tivoli Configuration script file specifies TME-specific host, port, user ID, and password keywords. An example TME Tivoli Configuration script file is shown below.

```
#BASE24-eps custom event adapter configuration file.
#
#
#Cache Configuration
BufferEvents=NO
BufEvtPath=/public/dev/b24/es/ibmbct/B682/EvtAdapter/TivoliEventAdapter.cachefile
MaxPacketSize=130
BufferFlushRate=96
ConnectionMode=CO
#Transport Configuration
TransportList=tl_
tl_Type=TME
tl_Channels=c1_
c1_ServerLocation=@EventServer
c1_TMEHost=oma2tivoli01
c1_TMEUserID=your_userid_here
c1_TMEPassword=your_password_here
c1_TMEPort=94
#Tracing - set to ALL to enable
TraceLevel=NONE
TraceFileName=./trace_file
```

For detailed descriptions of all keywords you can use in a TME or non-TME Tivoli Configuration script file, refer to the IBM's *Tivoli Event Integration Facility: User's Guide*

3. Edit the runeventadapter startup script in the \$ES_HOME/bin directory (e.g., /db01/b24dev/users/event_user/bin/runeventadapter). The areas to edit are marked in blue below.

```
#!/usr/bin/ksh
#
#
# Example BASE24-es Event Adapter Startup Script
#
# The following variables must be updated for your environment:
#
# MQJMS is location where your MQ Java Jars are "e.g. /usr/mqm/java"
# JAVASF is the directory of your Java Server Framework
# JAVABIN is the directory where "java" executable is
# EA_HOME is the directory where Event Adapter is
# LOGS is the directory where output from Event Adapter is stored
# SISLIB is where JDAL2 shared object is located (ES only)
#
export MQJMS=/usr/lpp/mqm/V6R0M0/java
export JAVASF=/public/dev/b24/es/ibmbct/B682/JavaSF
export JAVABIN=/usr/lpp/java/IBM/J1.4/bin
export EA_HOME=/public/dev/b24/es/ibmbct/B682/EvtAdapter
export LOGS=/public/dev/b24/es/ibmbct/B682/logs
# Add for EIFSDK:
EIFSDK=/public/dev/b24/users/IBM/Tivoli/EIFSDK_zos

case `uname -s` in
    "SunOS")
        export LD_LIBRARY_PATH=/public/dev/b24/es/ibmbct/B682/lib/sislib:$LD_LIBRARY_PATH
        ;;
    "AIX" | "OS/390")
        export LIBPATH=/public/dev/b24/es/ibmbct/B682/lib/sislib:$LIBPATH
        ;;
    "HP-UX")
        export SHLIB_PATH=/public/dev/b24/es/ibmbct/B682/lib/sislib:$SHLIB_PATH
        ;;
    "OS/390")
        export LIBPATH=/public/dev/b24/es/ibmbct/B682/lib/sislib:$LIBPATH
        export STEPLIB=$STEPLIB
        ;;
esac

if [[ -z "$JAVABIN" ]]; then
    echo JAVABIN not set
    exit 1
fi

#
# INSTALL NOTE: Please make sure that the directories listed below match
#               those on your system. Remember that Unix file systems are
#               case sensitive.
#
CP=$MQJMS/lib/fscontext.jar
CP=$CP:$MQJMS/lib/com.ibm.mq.jar
CP=$CP:$MQJMS/lib/com.ibm.mqjms.jar
CP=$CP:$MQJMS/lib/providerutil.jar
CP=$CP:$MQJMS/lib/jndi.jar
CP=$CP:$MQJMS/lib/connector.jar
```

```

CP=$CP:$MQJMS/lib/jms.jar
CP=$CP:$MQJMS/lib/jta.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/bouncycastle.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/commons-lang.jar
# CP=$CP:$JAVASF/webapps/Tomcat/lib/commons-logging.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/concurrent.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/jdom.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/log4j.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/xercesImpl.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/xml-apis.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/xsdl.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/relaxngDatatype.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/jakarta-oro.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/namespace.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/jaxb-libs.jar
# CP=$CP:$JAVASF/webapps/Tomcat/lib/jaxp-api.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/jaxb-xjc.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/jaxb-impl.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/jaxb-api.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/jax-qname.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/common-core.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/common-jms.jar
CP=$CP:$JAVASF/lib/common-websphereMQ.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/commerce.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/framework-core.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/framework-jms.jar
#CP=$CP:$JAVASF/webapps/Tomcat/lib/framework-naming.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/mds.jar
CP=$CP:$JAVASF/lib/jdal2.jar
CP=$CP:$JAVASF/lib/jmsMQappender.jar
CP=$CP:$JAVASF/lib/application.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/uif_sh.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/uif_srv.jar
CP=$CP:$JAVASF/webapps/Tomcat/lib/uexec_sh.jar
CP=$CP:$EA_HOME/lib/eventadapter.jar

# Add for EIFSDK:
CP=$CP:$EIFSDK/jars/evd.jar
CP=$CP:$EIFSDK/jars/ibmjsse.jar
CP=$CP:$EIFSDK/jars/jcf.jar
CP=$CP:$EIFSDK/jars/log.jar
CP=$CP:$EIFSDK/jars/jsafe.zip
CP=$CP:$EIFSDK/jars/zce.jar

echo " "
echo "-----CLASS_PATH-----"
echo $CP
echo "----END_OF_CLASS_PATH-----"
echo " "

export PROCESS_ID=EVENT_ADAPTER

export RUNTIME_OPTS="-Djdal.trace=false -Djdal.trace.method=false -Xmx256M -Xms256M -
Xnoclassgc"
export SOLARIS_OPTS="-XX:CompileThreshold=2000"

$JAVABIN/java -Dprocess.id=$PROCESS_ID -cp $CP $RUNTIME_OPTS -Dconfig.filename=$EA_HOME/
config/config.properties com.aciworldwide.event.EventMain > $LOGS/
$PROCESS_ID.stdout 2> $LOGS/$PROCESS_ID.stderr &

echo $! >> $LOGS/.$(PROCESS_ID).pid
echo $PROCESS_ID started in the background

```

Tivoli Enterprise Console Server Procedures

Perform the following procedures on the Tivoli Enterprise Console Server platform if you are integrating the Event Adapter with TEC. If you are integrating the Event Adapter with Tivoli Netcool/OMNIBus, perform the procedures provided under the [“Netcool/Omnibus Integration Procedures”](#) topic instead.

4. Configure the Enterprise Console Server. This example shows the commands as they are entered on a Windows server.
 - a. Start a Tivoli Shell command window using the Tivoli Shell desktop icon created during the Tivoli Enterprise Console install.
 - b. Make sure the aci_event.baroc file is located in a directory accessible to the Enterprise Console Server. For this example, the aci_event.baroc file is placed in the following location:
`c:\IBM\Tivoli\aci\aci_event.baroc.`
 - c. Enter the bash shell:
`sh`
 - d. Create a new rulebase:
`wrb -crtrb -path C:\IBM\Tivoli\bin\w32-ix86\TME\TEC\aci_rb`
 - e. Copy the default rulebase into the new rulebase:
`wrb -cprb -o -f Default aci_rb`
 - f. Import aci_event.baroc into the new rulebase:
`wrb -imprbclass C:\IBM\Tivoli\aci\ aci_event.baroc aci_rb`
 - g. Compile the rulebase:
`wrb -comprules aci_rb`
 - h. Load the new rulebase in the Enterprise Console Server:
`wrb -loadrb -use aci_rb`
 - i. Stop and restart the Enterprise Console Server:
`wstopesvr && wstartesvr`
 - j. Verify that the Enterprise Console Server is running:
`wstatesvr`
 - k. Verify that the new rulebase is in use by the server:
`wrb -lscurrb`

Netcool/Omnibus Integration Procedures

Perform the following procedures on the Tivoli Netcool/OMNIBus server platform and the TEC server platform if you are integrating with Tivoli Netcool/OMNIBus. If you are integrating with TEC, perform the procedures provided under the [“Tivoli Enterprise Console Server Procedures”](#) topic instead.

Note: These procedures assume that Tivoli Netcool/OMNIBus is installed on a Windows server.

Refer to the appropriate IBM Tivoli Netcool/Omnibus documentation for steps 4–6.

4. Install Netcool Omnibus 7.2.1.
5. Install non_native_probe support 3.0.
6. Install tivoli_eif_probe.
7. Update the Windows server database to support generic TEC event message fields by entering the following ObjectServer commands:

Note: As of Feb-2009, the commands documented in the *Netcool/OMNIBus Probe for Tivoli EIF Reference Guide* were incorrect.

```
%OMNIHOME%\bin\isql -S server_name -U username -P password -i  
%OMNIHOME%\probes\win32\tec_db_update.sql
```

8. Make sure that the path to Java 1.5 is in the %PATH% system environment variable. For example:

```
C:\Program Files\IBM\Java50\jre\bin
```

The IBM Tivoli Netcool/OMNIBus Probe for Tivoli Event Integration Facility (EIF) messages will not run unless Java is in the %PATH% variable.

Note: Steps 10–13 are not required if ACI distributes the isql script that is produced in step 13. Steps 10–12 are performed on the TEC server platform.

9. Enter the following command from the bash shell to create an ACI rulebase and use it to update the Netcool/Omnibus Object Server to recognize the ACI_EVENT class.

```
bash$ wrb -crtrb -path tecrb aci_netcool_rb
```

A response similar to the following should be returned.

```
Parsing C:\data\aci_netcool\tecrb\TEC_CLASSES\root.baroc  
Parsing C:\data\aci_netcool\tecrb\TEC_CLASSES\tec.baroc  
Loading Rule Sets  
Loading Rule Packs  
Loading RB Targets  
Parsing C:\data\aci_netcool\tecrb\TEC_RULES\EventServer  
bash$ pwd  
c:/data/aci_netcool
```

10. Enter the following command from the bash shell to import the ACI event class into the new rulebase.

```
bash$ wrb -imprbclass aci_event.baroc aci_netcool_rb
```


A response similar to the following should be returned.

```
Loading Classes
Parsing C:\data\aci_netcool\tecrb\TEC_CLASSES\root.baroc
Parsing C:\data\aci_netcool\tecrb\TEC_CLASSES\tec.baroc
Parsing C:\data\aci_netcool\tecrb\TEC_CLASSES\aci_event.baroc
```

11. Enter the following command from the bash shell to compile the new rulebase.

```
bash$ wrb -comprules aci_netcool_rb
```

A response similar to the following should be returned.

```
Compiling aci_netcool_rb
Loading Classes
Parsing C:\data\aci_netcool\tecrb\TEC_CLASSES\root.baroc
Parsing C:\data\aci_netcool\tecrb\TEC_CLASSES\tec.baroc
Parsing C:\data\aci_netcool\tecrb\TEC_CLASSES\aci_event.baroc
Loading Rule Sets
Loading Rule Packs
Loading RB Targets
Parsing C:\data\aci_netcool\tecrb\TEC_RULES\EventServer
Translating Rules
Compiling Rules
Building RB target EventServer
Compiling C:\data\aci_netcool\tecrb\
rbtargets\EventServer\TEC_RULES\load_rules.pro
Compilation Complete.
bash$
```

The remaining steps are performed on the Tivoli Netcool/OMNIBus server platform.

12. Enter the following commands to create the sql insert script used to update the Netcool Object Server database:

```
C:\Program Files\IBM\Tivoli\Netcool\omnibus\bin>nco_baroc2sql -baroc
C:\data\aci_netcool\tecrb\TEC_CLASSES\load_classes -sql c:\Data\aci_
netcool\aci_tec_insert.sql
```

A response similar to the following should be returned.

```
Info - Processing file 'C:\data\aci_netcool\tecrb\TEC_CLASSES\root.baroc'
Info - Processing file 'C:\data\aci_netcool\tecrb\TEC_CLASSES\tec.baroc'
Info - Processing file 'C:\data\aci_netcool\tecrb\TEC_CLASSES\aci_event.baroc'
Info - Processed 24 BAROC classes.
```

```
C:\Program Files\IBM\Tivoli\Netcool\omnibus\bin>
```

An example aci_tec_insert.sql file is shown below.

```
insert into master.class_membership (Class, ClassName, Parent) values ( 76013, 'DB_Cleanup_event', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76014, 'TEC_Generic', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76006, 'TEC_Stop', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76010, 'TEC_Maintenance', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76021, 'TEC_GWR_Notice', 76017);
insert into master.class_membership (Class, ClassName, Parent) values ( 76012, 'Escalate_event', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76011, 'TEC_Cleanup_event', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76000, 'EVENT', -1 );
insert into master.class_membership (Class, ClassName, Parent) values ( 76008, 'TEC_Heartbeat', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76023, 'ACI_EVENT', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76015, 'TEC_Tick', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76009, 'TEC_Heartbeat_missed', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76001, 'TASK', -1 );
insert into master.class_membership (Class, ClassName, Parent) values ( 76017, 'TEC_GWR_Event', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76003, 'TEC_Error', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76005, 'TEC_Start', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76004, 'TEC_Notice', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76018, 'TEC_GWR_Start', 76017);
insert into master.class_membership (Class, ClassName, Parent) values ( 76019, 'TEC_GWR_Stop', 76017);
insert into master.class_membership (Class, ClassName, Parent) values ( 76022, 'TEC_GWR_Error', 76017);
insert into master.class_membership (Class, ClassName, Parent) values ( 76002, 'TASK_COMPLETE', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76007, 'TEC_DB', 76000);
insert into master.class_membership (Class, ClassName, Parent) values ( 76020, 'TEC_GWR_ReStart', 76017);
insert into master.class_membership (Class, ClassName, Parent) values ( 76016, 'TEC_LOGGING_BASE', 76000);
go
insert into alerts.conversions values ( 'Class76013', 'Class', 76013, 'DB_Cleanup_event');
insert into alerts.conversions values ( 'Class76014', 'Class', 76014, 'TEC_Generic');
insert into alerts.conversions values ( 'Class76006', 'Class', 76006, 'TEC_Stop');
insert into alerts.conversions values ( 'Class76010', 'Class', 76010, 'TEC_Maintenance');
insert into alerts.conversions values ( 'Class76021', 'Class', 76021, 'TEC_GWR_Notice');
insert into alerts.conversions values ( 'Class76012', 'Class', 76012, 'Escalate_event');
insert into alerts.conversions values ( 'Class76011', 'Class', 76011, 'TEC_Cleanup_event');
insert into alerts.conversions values ( 'Class76000', 'Class', 76000, 'EVENT');
insert into alerts.conversions values ( 'Class76008', 'Class', 76008, 'TEC_Heartbeat');
insert into alerts.conversions values ( 'Class76023', 'Class', 76023, 'ACI_EVENT');
insert into alerts.conversions values ( 'Class76015', 'Class', 76015, 'TEC_Tick');
insert into alerts.conversions values ( 'Class76009', 'Class', 76009, 'TEC_Heartbeat_missed');
insert into alerts.conversions values ( 'Class76001', 'Class', 76001, 'TASK');
insert into alerts.conversions values ( 'Class76017', 'Class', 76017, 'TEC_GWR_Event');
insert into alerts.conversions values ( 'Class76003', 'Class', 76003, 'TEC_Error');
insert into alerts.conversions values ( 'Class76005', 'Class', 76005, 'TEC_Start');
insert into alerts.conversions values ( 'Class76004', 'Class', 76004, 'TEC_Notice');
insert into alerts.conversions values ( 'Class76018', 'Class', 76018, 'TEC_GWR_Start');
insert into alerts.conversions values ( 'Class76019', 'Class', 76019, 'TEC_GWR_Stop');
insert into alerts.conversions values ( 'Class76022', 'Class', 76022, 'TEC_GWR_Error');
insert into alerts.conversions values ( 'Class76002', 'Class', 76002, 'TASK_COMPLETE');
insert into alerts.conversions values ( 'Class76007', 'Class', 76007, 'TEC_DB');
insert into alerts.conversions values ( 'Class76020', 'Class', 76020, 'TEC_GWR_ReStart');
insert into alerts.conversions values ( 'Class76016', 'Class', 76016, 'TEC_LOGGING_BASE');
go
```

13. Update the Netcool Object Server's database using isql. In the following command syntax, the Netcool Object Server is named NCO_ACI. You must be in the directory where the isql.bat file is located when entering this command, or the isql.bat file must be added to your path.

```
C:\Program Files\IBM\Tivoli\Netcool\omnibus\bin>isql -U root -S
NCO_ACI -P "<your-password>" -i c:\data\aci_netcool\aci_tec_
insert.sql
```

Upon successful completion, response text similar to the following is displayed:

(24 rows affected)

(24 rows affected)

14. Replace the default `tivoli_eif.rules` file with the ACI-specific rules file. Note that the file is named the same as the default file.

```
C:\Program Files\IBM\Tivoli\Netcool\omnibus\probes\win32\tivoli_eif.rules
```

15. Verify that the Event Adapter is connected to the Tivoli EIF probe.

This step should be performed on the Tivoli Netcool/OMNIBus server platform. You can use the TEC bash shell or whatever facilities are available to retrieve information.

The previously installed Netcool/Omnibus Tivoli EIF probe must be running and the Event Adapter must have sent at least one message before you perform this step. The Event Adapter does not connect until a message is sent. If the probe is running, you will see a LISTENING entry in the response. If the Event Adapter has successfully connected, you will also see an ESTABLISHED entry in the response. The `grep` command shown here uses port 41868. You will need to enter whatever port your Netcool/Omnibus server is listening on, which was configured when you set up Netcool according to the Netcool installation documentation from IBM. If you do not have the `grep` command, you will have to scan through the output from `netstat` to see if you have a listener and an established connection.

Note: The `grep` command is not available if you are just running a vanilla Netcool install on a Windows platform without TEC present.

```
bash$ netstat -a | grep 41868
```

If a successful connection is established, a response similar to the following should be returned.

TCP	oma2tivoli01:41868	oma2tivoli01.am.tsacorp.com:0	LISTENING
TCP	oma2tivoli01:41868	oma3s031.am.tsacorp.com:33937	ESTABLISHED

ACI Worldwide, Inc.

8: ACI Java Presentation Framework (AJPF)

This section discusses configuration items specifically related to AJPF applications such as web.xml entries, how to configure Faces (for navigation, beans, and components), AJI bootstrap and runtime touch points, and configurable aspects for the presentation, look and feel, and text displayed for AJPF applications.

Web.xml Configuration

Web applications, such as the BASE24-eps Voice Authorization web pages, are configured in a web.xml file. Servlet containers, such as Apache Tomcat or IBM WebSphere Application Server, use the web.xml file to start a web application. One web.xml file exists for each application Web ARchive (WAR) file. The web.xml file configures such items as the following:

- The name in the title bar of the browser
- Faces configuration files
- The default file extension for Facelets (i.e., xhtml)
- Filters, some of which are related to SSO (Single Sign On)

Faces Configuration Files

Although by default, there is only one Faces configuration file—the `face-config.xml` file, the AJPF uses four Faces configuration files. One each for configuration, beans, navigation, and components.

Faces-config.xml File

The AJPF's `faces-config.xml` file is used to hold general configuration that does not fall into the other beans, navigation, or component configuration categories. The `faces-config.xml` file contains fully qualified Java classes for the application's resource bundle, view handler, variable resolver, and phase listener.

Faces-config-beans.xml File

The AJPF `faces-config-beans.xml` file declares the Java classes (beans) that Faces manages and the scope in which to place each managed bean. Beans can be managed in the following scopes.

Scope	Description
none	A managed bean with this scope is not stored anywhere, it is created <i>on demand</i> whenever it is needed.
request	A bean with a request scope is stored only for the duration of a single request. Upon subsequent requests on the same object, a new version is instantiated.
session	A bean with session scope is stored on to the session meaning that the bean's properties remain alive over multiple requests.
application	Objects created with application scope exist during the entire lifetime of the container.
conversation	A bean with conversation scope spans multiple requests, but is not kept in memory as long as session-scoped beans. This scope is used for

Faces-config-navigation.xml File

The AJPF faces-config-navigation.xml file contains the declarations of which page to navigate to next based on responses from the Controller bean.

Custom Tags, Components, and Functions

The AJPF faces-config-component.xml file contains definitions for custom Faces tags or components. This file is only present if the AJPF application uses custom tags or components; most AJPF applications just use framework components.

The faces-config-component.xml tells faces which class to put into memory when build the component tree. It also specifies any renderers we need for our custom components.

The custom.taglib.xml file declares custom tags, composite components, and functions used by the AJPF application, while the custom.tld file defines any custom tags.

AJI Runtime and Database Configuration

AJPF and AJSF applications share the same runtime system and database configuration properties. For AJSF applications, the process ID and configuration file name properties must be defined in the config.properties file or specified at runtime. For some AJPF applications, the process ID may be defined in the web.xml file instead.

Both AJPF and AJSF applications require that the minimum database connection configuration information be defined in the following properties in the config.properties file:

- db.type
- db.user.id
- db.user.password
- db.datasource
- db.driver
- db.max.connections

Logging and Tracing

AJPF applications use the same log4J configuration files for events and tracing that AJSF applications use.

Presentation

The presentation provided by AJPF applications is HTML generated from Java Server Faces and ICE Faces components. The components are placed in XHTML files (XML and HTML markup) and these files are preprocessed and compiled into Java that outputs HTML. The .xhtml files for an AJPF application typically reside in the \WebContent\page directory.

Faces Templates

AJPF applications use Facelets templates. A template is an .xhtml file that contains reusable markup. The goal is to place common markup of a page in a single file rather than in each .xhtml file in the application.

Cascading Style Sheets

AJPF applications use cascading style sheets (e.g., app.css, app_blue.css) to define the look and feel of the web page presentation. Users can select different style sheets to apply from the User Preferences web page, which maps to the Web User Preferences data source (WEBUSROPTS). Cascading style sheets must reside in the installed \$ES_HOME\apf\styles directory for the web application.

Resource Bundles and Localizing Text

The AJPF uses Java resource bundles to hold any text that is displayed on application web pages. This is done so that text can be customized or translated into other languages. The text is defined in the messages.properties file in the installed \$ES_HOME\apf\resources directory for the web application (e.g., \ESWeb\ESConfig\apf\resources).

The messages.properties file defines navigation breadcrumb and menu title text, screen section headings, field labels, and button labels. Basically, it defines any text that can be displayed on a web page view that is not retrieved from or written to a data source (e.g, user-entered field text).

The keys for values in the message.properties are defined in the common code values (CCV) properties file (ccv.properties). The location of the ccv.properties file is configured in the Java property in the APPLCNFG (e.g., va.ccv.properties for the BASE24-eps Voice Authorization application). Thus, changes to the messages.properties file should be done in concert with the ccv.properties.

Localization

Text in the messages.properties file can be localized by appending the two-character ISO 639-1 international-standard language-code locale identifier, and optional two-character ISO 3166-1 country code, to the file name (e.g., messages_en.properties, messages_en_us.properties, messages_en_uk.properties, messages_sp.properties, etc.) and placing the file in the appropriate application installation directory (e.g., \$ES_HOME\ESWeb\ESConfig\apf).

Users can switch locales by selecting the locale on the User Preferences web page (stored in the Web User Preferences data source (WEBUSROPTS)). In the following example, users could switch between U.S. English, U.K. English, French, German, Spanish, and Chinese.

```
$ES_HOME\apf\messages_en_us.properties
      messages_en_uk.properties
      messages_fr.properties
      messages_de.properties
      messages_es.properties
      messages_zh.properties
```

If needed, you can also change or localize the text in messages generated by *stock* Java Faces or ICEFaces components used by the AJPF application. These are located in the messages.properties files in the jsf-impl.jar and icefaces-comps.jar, respectively. The icefaces-comps.jar only contains text for the inputfile and selectinputdate components.

After you make changes to any messages.properties file, you must stop and restart the AJPF application before the changes take effect.

App-def.xml File

The app-def.xml file defines the navigation attributes of a AJPF application.

The main elements in the app-def.xml describe the menu. The following example depicts how tags in the app-def.xml file relate to navigational items on the web application menu. The numbers in the XML code are shown to their corresponding values on the example web page. The actual text displayed for each value is retrieved from the messages.properties file.



```

<menubar>
  1 <crumb>crumb.home</crumb>
    <viewid>/page/apf/home.xhtml</viewid>
    <title>text.user.welcome</title>

  2 <menu property="menu.maintenance">

    <!-- Language root -->
    3 <menu property="menu.maintenance.language">
      <viewid>/page/admin/maintenance/language_search.xhtml</viewid>
      4 <title>text.admin.language</title>
      5 <crumb>crumb.language</crumb>
      <scopeid>LanguageScope</scopeid>
      <task name="LanguageAdmin"/>

      <!-- Language: Create -->
      6 <menu property="text.admin.language.create"
        display="false" isSecondary="true">
        <viewid>/page/admin/maintenance/language_create.xhtml</viewid>
        <title>text.admin.language.create</title>
        <action>#{LanguageController.doCreateLink}</action>
        <crumb>crumb.create</crumb>
        <task name="LanguageAdmin">
          <function name="Save"/>
        </task>
      </menu>
    </menu>
  </menubar>

```

The web application menu and breadcrumbs is created from the app-def.xml file. The menu is a hierarchy of menu nodes (menu group #2, primary navigation #3 and title#4, and secondary navigation #6). Each node describes a level in the application hierarchy. Breadcrumbs display the current hierarchy of nodes accessed (home#1 and language#5).

AJPF-Specific Data Sources

Besides the data sources provided by the AJSF, AJPF applications use three data sources specific to AJPF applications—WEBAPPS, WEBAPPOPTS, and WEBUSROPTS.

Web Applications Data Source (WEBAPPS)

WEBAPPS is used to create a *jump* page (Other Applications web page) for single sign-on (SSO) access to multiple web applications. By default, the Other Applications web page is not displayed. To display it, change the display value of the text.other.applications menu property in the app-def.xml file from false to true.

```
<!-- Disable the "Other Applications" and "Application
Preferences" options -->
    <menu property="text.other.applications" display="false"/>
    <menu property="text.application.preferences"
display="false"/>
```

The layout of WEBAPPS is provided below.

Column	DB Type	Description	Key	Null
ApplID	varchar(16)	Application ID	*	No
ApplNamTx	varchar(100)	Name used on screen as default		Yes
ApplBundleNamTx	varchar(32)	Bundle key to look up name		Yes
ProductID	varchar(16)	Product ID		Yes
WebAppURL	varchar(256)	URL to get to this app		Yes
HomePage	Varchar(64)	Home page for the application		Yes
LastUpdtTm	Datetime	Last update time		Yes

Web Application Options Data Source (WEBAPPOPTS)

WEBAPPOPTS stores the default user preferences options for the web application defined on the Options > Application Preferences page. These options can be overridden by individual users in the Web Application User Options data source (WEBUSROPTS). By default, the Applications Preferences page is not displayed. To display it, change display value of the text.application.preferences menu property in the app-def.xml file from false to true.

```
<!-- Disable the "Other Applications" and "Application
Preferences" options -->
    <menu property="text.other.applications" display="false"/>
    <menu property="text.application.preferences"
display="false"/>
```

The layout of WEBAPPOPTS is provided below.

Column	DB Type	Description	Key	Null
AppIID	varchar(16)	Application ID	*	No
OptionKey	varchar(100)	Key	*	No
OptionValue	varchar(100)	Value for key		No
OptionDesc	varchar(255)	Description		Yes
LastUpdtTm	Datetime	Last update time		No

Web Application User Options Data Source (WEBUSROPTS)

WEBUSROPTS stores user preferences selected by the web application's users on the User Preferences page.

A sample web application User Preferences page is shown below.

User Preferences

[Change Password](#) | [Session Management](#)

Cascading Style Sheet (CSS):

Page Layout:

Locale:

» Favorites Menu Configuration

Available Menu Items (select to assign or unassign)	Assigned Menu Items (drag to change order)
Acquirer Transaction	
Acquirer Override	
Issuer Transaction	
Issuer Override	
Card Status Search	
Code 10 Issuer	
Code 10 Acquirer	
About	
Sitemap	

[Cancel](#)

The layout of WEBUSROPTS is provided below.

column	DB Type	Description	Key	Null
ApplID	varchar(16)	Application ID	*	No
UserID	varchar(64)	Username	*	No
OptionKey	varchar(100)	Key	*	No
OptionValue	varchar(100)	Value for key		No
OptionDesc	varchar(255)	Description		Yes
LastUpdtTm	Datetime	Last update time		No

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9: Initialization Maps

The AJI environment provides the following XML initialization map files that AJI applications can use to control static database content.

- AJI Initialization Map
- AJI Event Log Map
- AJI Authentication Security Map
- AJI Authorization Security Map
- AJI Application Configuration Map

These XML map files are deployed by an AJI application and are treated as a resource by the application's class loader. They are defined and used to initially *seed* a database upon server startup during the initialization phase. The maps are versioned and controlled so that they are not loaded and executed upon every startup, but only when a new version is introduced. Typically, this is only when an AJI application is first started. You can also configure XML initialization map files so that they only pertain to a specific AJI application.

In addition, you can provide your own custom XML initialization map files to load application-specific database tables.

Note: To use XML initialization map files, you cannot configure initialization. function<text-string> properties in the APPLCNFG because they are mutually exclusive. To be used, XML initialization map files must reside in the \config directory of an AJI application.

Map Versioning

Map versioning is supported using the `<Implementation>` element with the required “version” attribute and optional “appId” attributes. The `<Implementation>` element must be provided in each XML initialization map file, except for the AJI Initialization Map (`AJIInitializationMap.xml`), which does not use versions. The version and appId attributes are used to control whether or not a map should be applied (*seeded*).

The version attribute contains a version value in the form of “*major.minor.revision*” numbers (e.g., 1.0.10). These numbers should be incremented each time the XML initialization map file is modified so that the AJI application can determine whether the map needs to be reloaded. The appId attribute indicates the AJI application that *owns* the map. If the appId attribute is not defined, it defaults to “AJI”.

The version and application information is used to manage the version information from one map to the next. It also prevents the reloading of a map that has already been loaded so that initialization does not *reseed* data over and over again.

Examples:

```
<Implementation version="1.0.5" appId="XYZ"/>
<Implementation version="1.0.10" appId="AJI"/>
<Implementation version="1.0.0" appId="AJI"/>
<Implementation version="1.0.4" appId="AJI"/>
<Implementation version="1.0.26" appId="AJI" />
```

If the Versions data source (VERSIONS) is present in your system, the Initialization Manager writes a record to the data source each time a new implementation version is loaded. Before loading an XML initialization map file, it checks the version information in the file against VERSIONS to determine whether or not to load the map.

Note: If the Versions data source (VERSIONS) is not present in your system, you can use autoload properties in the APPLCNFG to determine whether an XML initialization map file is loaded. In this case, you must configure the autoload.versions.table property in the config.properties with a value of false and set the “autoload” property for each XML initialization map file as desired in the APPLCNFG.

Using versioned XML initialization maps, developers can check out the XML map, make changes and increment the version, and check the file back in. A subsequent build and deployment of the server code causes the application to read the map and reload the modified information if the version is more current than the version currently loaded. This causes the new information to be loaded into the

database automatically without reseeding data. Once the load completes, the version is updated in the VERSIONS table or the given XML resource map. The next time the server starts, the map will not be loaded (as long as the version from the XML map matches the version saved in the VERSIONS table record containing the XML map resource name.

Autoload Properties

The following autoload properties determine whether map versioning is checked before loading an XML initialization map files or whether specific XML initialization map files are loaded.

autoload.versions.table — A flag indicating whether the Initialization Manager uses the Versions data source (VERSIONS) to check version information to determine whether seed data should be loaded. Valid values are as follows:

true = Yes, use the Versions data source (VERSIONS). Default.
false No, do not use the Versions data source (VERSIONS). Use the APPLCNFG instead.

Required: No
Example Entry: autoload.versions.table=false
Usage: Initialization Manager

FirstInstanceInitProcess — The process ID of the first instance initializer server. The first process (AJI application server) to insert the “FirstInstance” row in the APPLCNFG becomes the first instance process.

Required: No
Example Entry: FirstInstanceInitProcess=P92^ESWEB
Usage: FirstInstanceInitializer

autoload.applcnfg — A flag indicating whether to load the AJI Application Configuration XML initialization map file (AJIAppConfigMap.xml) upon initialization. Valid values are as follows:

true = Yes, load the AJI Application configuration map file (AJIAppConfigMap.xml) upon initialization.
false No, do not load the AJI Application configuration map file (AJIAppConfigMap.xml) upon initialization. Default.

Required: No
Example Entry: `autoload.applcnfg=true`
Usage: Initialization Manager

autoload.events — A flag indicating whether to load the AJI Event log map XML initialization file (e.g., `AJIEventLogMap_en_US.xml`) upon initialization. Valid values are as follows:

`true` = Yes, load the AJI Event log map XML initialization file upon initialization. Default.
`false` No, do not load the AJI Event log map XML initialization file upon initialization.

Required: No
Example Entry: `autoload.events=false`
Usage: Initialization Manager

autoload.security — A flag indicating whether to load the AJI Authentication security map XML initialization file (`AJIAuthenticationSecurityMap.xml`) and AJI Authorization security map XML initialization file (`AJIAuthorizationSecurityMap.xml`) upon initialization. Valid values are as follows:

`true` = Yes, load the AJI Authentication and Authorization security map XML initialization files upon initialization. Default.
`false` No, do not load the AJI Authentication and Authorization security map XML initialization files upon initialization.

Required: No
Example Entry: `autoload.security=false`
Usage: Initialization Manager

AJI Initialization Map

The AJI Initialization XML map file contains a list of initializer objects that must implement the `InitializationInterface` from the `com.aciworldwide.framework` package. These initializer objects are initialized and loaded using the standard initialization interface by the `AJI InitializationManager`. This map file is not versioned.

When the AJI base server initializes, performs a resource lookup for this xml map, but only if the initializers are not configured in the `APPLCNFG` in initialization. function<text-string> properties. Initializers are either loaded from the XML map or through specific configuration properties in the `APPLCNFG` table. They are mutually exclusive.

Resource Naming Conventions

AJI applications can leverage the XML map approach to define their own initialization XML map file as long as it uses the following resource naming convention:

```
config/InitializationMap.xml
```

You can refer to the example contents of the `AJIInitializationMap.xml` file for descriptions and usage of how an AJI application-specific XML initialization map file needs to be configured.

For each *initializer* entry, you must define a class attribute with the fully qualified class name and an optional description. If a class is only present in one server, then define the type to be “FirstInstance”. There is always only one first instance server and it is determined by the first server that starts up. The server registers its process ID in the `FirstInstanceInitProcess` property in the `APPLCNFG`.

Example Contents

The contents of the `AJIInitializationMap.xml` file are shown below as an example for defining an AJI application-specific `InitializationMap.xml`.

```
<InitializationMapContent>  
  <Init class="com.aciworldwide.common.ApplConfigMapInitializer"
```

```
        description="This initializer loads (seeds) application configuration
properties."/>

    <Init class="com.aciworldwide.framework.TransactionMapManager"
        description="The fully qualified class name to be created upon
initialization."/>

    <Init class="com.aciworldwide.application.security.SecurityMapInitializer"
        description="The fully qualified class name to be created upon
initialization."/>

    <Init class="com.aciworldwide.application.security.USECManager"
        description="The fully qualified class name to be created upon
initialization."/>

    <Init class="com.aciworldwide.application.pci.masking.formatter.jaxb.
JAXBContextManager"
        description="The fully qualified class name to be created upon
initialization."/>

    <Init class="com.aciworldwide.common.MessageRouterService"
        description="The fully qualified class name to be created upon
initialization."/>

</InitializationMapContent>
```

AJI Event log Map

The AJI Event Log Map XML initialization file (e.g., `AJIEventLogMap<locale>.xml`) contains descriptions of all events that can be generated by an AJI application. The locale is derived by from the web server's locale and is appended to the name of the map. The default locale is “_en_US”.

The event loader class is an initialization class that is statically loaded and is not required to exist in the AJI Initialization XML map file (see [“AJI Initialization Map”](#)).

The `autoload.events` property in the `APPLCNFG` indicates whether the AJI Event Log Map XML initialization file is loaded during initialization. This property defaults to “true”, but can be set to “false” if you do not want to autoload events during initialization.

Resource Naming Conventions

AJI applications can leverage the XML map approach to define their own event initialization XML map file as long as it uses the following resource naming convention:

```
config/EventLogMap[locale].xml
```

where: `[locale]` is the locale of the AJI application's web server.

You can refer to the example contents of the `AJIEventLogMap_en_US.xml` file for descriptions and usage of how an AJI application-specific event XML initialization map file needs to be configured.

The AJI event log initializer resolves the resource, applies the locale, reads the map, and determines whether a load is required after the AJI-based event map is loaded. An application-specific Event Log Map XML initialization file is version-checked and loaded as applicable.

Example Contents

Example contents of the `AJIEventLogMap_en_US.xml` file are shown below as an example for defining an AJI application-specific Event Log Map XML initialization file.

```

<EventLogContent>
  <!-- This version should be bumped after each modification so the server
        knows to redeploy the content. An application id can be optionally
specified to indicate
        who owns the map when determining if a map should processed or not. -->
  <Implementation version="1.0.0" appId="AJI" />

  <SubSystem id="3001" name="AJI" class="com.aciworldwide.application.User">
    <Event number="1" severity="Warn" suppressType="AlwaysLog"
suppressCntLimit="0"
        suppressTimeLimit="0">
      <Message>valueUnbound exception: {0} event.getSession failed.</
Message>
      <ActionTaken>This is a Tomcat bug.</ActionTaken>
      <ProbableCause>Running Tomcat.</ProbableCause>
      <Remedy>Upgrade to a newer Tomcat version.</Remedy>
    </Event>
  </SubSystem>
  <SubSystem id="3002" name="AJI"
        class="com.aciworldwide.commerce.common.formatter.
BitMapMessage">
    <Event number="1" severity="Warn" suppressType="AlwaysLog"
suppressCntLimit="0"
        suppressTimeLimit="0">
      <Message>Trying to set a null element at position ({0}).</Message>
      <ActionTaken>Messages are not being constructed correctly.</
ActionTaken>
      <ProbableCause>Programming error.</ProbableCause>
      <Remedy>Notify ACI.</Remedy>
    </Event>
    <Event number="2" severity="Error" suppressType="AlwaysLog"
suppressCntLimit="0"
        suppressTimeLimit="0">
      <Message>{0} is a variable length field without a length header.
Data: {1}.</Message>
      <ActionTaken>Messages are not being constructed correctly.</
ActionTaken>
      <ProbableCause>Programming error.</ProbableCause>
      <Remedy>Notify ACI.</Remedy>
    </Event>
    <Event number="3" severity="Warn" suppressType="AlwaysLog"
suppressCntLimit="0"
        suppressTimeLimit="0">
      <Message>{0} is not numeric. Data: {1}.</Message>
      <ActionTaken>Messages are not being constructed correctly.</
ActionTaken>
      <ProbableCause>Programming error.</ProbableCause>
      <Remedy>Notify ACI.</Remedy>
    </Event>

```



```
<Event number="4" severity="Warn" suppressType="AlwaysLog"
suppressCntLimit="0"
    suppressTimeLimit="0">
    <Message>The length of {0} is too long. Data: {1}.</Message>
    <ActionTaken>Messages are not being constructed correctly.</
ActionTaken>
    <ProbableCause>Programming error.</ProbableCause>
    <Remedy>Notify ACI.</Remedy>
</Event>
<Event number="5" severity="Warn" suppressType="AlwaysLog"
suppressCntLimit="0"
    suppressTimeLimit="0">
    <Message>{0} contains an invalid character at position ({1}). Data:
{2}.</Message>
    <ActionTaken>Messages are not being constructed correctly.</
ActionTaken>
    <ProbableCause>Programming error.</ProbableCause>
    <Remedy>Notify ACI.</Remedy>
</Event>
</SubSystem>
</EventLogMap>
```

AJI Authentication Security Map

The AJI Authentication Security XML initialization map file contains a list of all predefined users (i.e., SysAdmin, SecAdmin, AudAdmin, AJMFAdmin) as well as a default security group and password group for administrators.

The authentication loader class is an initialization class that is executed during initialization if specified in an initialization.function<text-string> property or resolved from the initialization map (see [“AJI Initialization Map”](#)).

The autoload.security property in the APPLCNFG indicates whether the AJI Authentication Security Map XML initialization file is loaded during initialization. This property defaults to “true”, but can be set to “false” if you do not want to autoload authentication security configuration information during initialization.

Resource Naming Conventions

AJI applications can leverage the XML map approach to define their own authentication security initialization XML map file as long as it uses the following resource naming convention:

```
config/AuthenticationSecurityMap.xml
```

You can refer to the example contents of the AJIAuthenticationSecurityMap.xml file for descriptions and usage of how an AJI application-specific authentication security XML initialization map file needs to be configured.

The AJI authentication security initializer resolves the resource, reads the map, and determines whether a load is required after the AJI-based authentication security map is loaded. An application-specific Authentication Security Map XML initialization file is version- checked and loaded as applicable.

Example Contents

Example contents of the AJIAuthenticationSecurityMap.xml file are shown below as an example for defining an AJI application-specific Authentication Security Map XML initialization file.

Note: In this example, the default passwords represent a SHA-1 hash of “newuser01” that is set to expire on the first login.

```

<SecurityAuthenticationContent>
  <!-- This version should be bumped after each modification so the server
        knows to redeploy the content. An application id can be optionally
specified to indicate
        who owns the map when determining if a map should processed or not. -->
  <Implementation version="1.0.0" appId="AJI" />

  <SecurityGroupList>
    <!--Note: You can also configure a PasswordGroup element in a
SecurityGroup element'
  <SecurityGroup
    name="SysAdminSecurityGrp"
    description="Default security constraints for administrators."
    startTime="0000"
    endTime="0000"
    daysOfWeek="YYYYYYY"
    maxBadLoginAttempts="4"
    badLoginAttemptsReset="300"
    maxSessions="99"
    sessionTimeout="7200"
    uiTimeout="1200"
    inActiveIDSuspend="365"
    sysAdminRoleProfile="*"
    creationID="*"
    passwordGroup="SysAdminPswdGrp"/>

    <!--OR '

  <SecurityGroup
    name="SysAdminSecurityGrp"
    description="Default security constraints for administrators."
    startTime="0000"
    endTime="0000"
    daysOfWeek="YYYYYYY"
    maxBadLoginAttempts="4"
    badLoginAttemptsReset="300"
    maxSessions="99"
    sessionTimeout="7200"
    uiTimeout="1200"
    inActiveIDSuspend="365"
    sysAdminRoleProfile="*"
    creationID="*">
  <PasswordGroup
    <!--Name attribute is not needed as it matches securitygroup name-->
    <!-- name="SysAdminPswdGrp" '
    expireWarnDays="0"
    lifetime="0"
    minLength="6"
    maxLength="16"
    minNumericChars="1"
    minAlphaChars="1"

```

```

        maxHistory="4"
        maxChangesPerDay="3"/>
    </SecurityGroup>

</SecurityGroupList>

<!--Not needed if PasswordGroup elements are a 1 to 1 with a security
group. In this case just add
    the PasswordGroup element under the SecurityGroup element'
<PasswordGroupList>
    <PasswordGroup
        name="SysAdminPswdGrp"
        description="Default security constraints for administrators."
        expireWarnDays="0"
        lifetime="0"
        minLength="6"
        maxLength="16"
        minNumericChars="1"
        minAlphaChars="1"
        maxHistory="4"
        maxChangesPerDay="3"/>
</PasswordGroupList>

<UserList>
    <User
        name="SecAdmin"
        password="jJ57hwD+gMN4RD8Cijmv6xPSYAA="
        statusCode="A"
        firstName="Security"
        lastName="Administrator"
        securityGroup="SysAdminSecurityGrp"/>
    <User
        name="SysAdmin"
        password="jJ57hwD+gMN4RD8Cijmv6xPSYAA="
        statusCode="A"
        firstName="System"
        lastName="Administrator"
        securityGroup="SysAdminSecurityGrp"/>
    <User
        name="AudAdmin"
        password="jJ57hwD+gMN4RD8Cijmv6xPSYAA="
        statusCode="A"
        firstName="Audit"
        lastName="Administrator"
        securityGroup="SysAdminSecurityGrp"/>
</UserList>
</SecurityAuthenticationContent>

```

AJI Authorization Security Map

The AJI Authorization Security Map XML initialization file contains a list of all predefined tasks, functions, permissions, permission groups, and roles for predefined users.

The authorization loader class is an initialization class that is executed during initialization if specified in an `initialization.function<text-string>` property or resolved from the initialization map (see [“AJI Initialization Map”](#)).

The `autoload.security` property in the `APPLCNFG` indicates whether the AJI Authorization Security Map XML initialization file is loaded during initialization. This property defaults to “true”, but can be set to “false” if you do not want to autoload authorization security configuration information during initialization.

Resource Naming Conventions

AJI applications can leverage the XML map approach to define their own authorization security initialization XML map file as long as it uses the following resource naming convention:

```
config/AuthorizationSecurityMap.xml
```

You can refer to the example contents of the `AJIAuthorizationSecurityMap.xml` file for descriptions and usage of how an AJI application-specific authorization security XML initialization map file needs to be configured.

The AJI authorization security initializer resolves the resource, reads the map, and determines whether a load is required after the AJI-based authorization security map is loaded. An application-specific Authorization Security Map XML initialization file is version- checked and loaded as applicable.

Filtering

AJI applications that support filtering may need to seed a wildcard filter value for each filter type they use. Wildcard filter values are defined as an asterisk “*”.

For example, if an AJI application supports a filter type of “Institution” and wants to create an “AllInstitution” filter, you would define a filter value of “*” only as shown in the following `<FilterTypeList>` and `<FilterList>` tags.

```
<FilterTypeList>
  <FilterType name="Institution" description="product filter type
institution">
</FilterTypeList>
<FilterList>
  <Filter name="AllInstitutions" description="Access to all institutions">
    <FilterType name="Institution"/>
    <FilterValue value="*" description="Wildcard access"/>
  </Filter>
</FilterList>
```

Example Contents

Example contents of the AJIAuthorizationSecurityMap.xml file are shown below as an example for defining an AJI application-specific Authorization Security Map XML initialization file.

```
<SecurityAuthorizationContent>

<!-- This version should be bumped after each modification so the server
      knows to redeploy the content. An application id can be optionally
specified to indicate
      who owns the map when determining if a map should processed or not. -->
<Implementation version="1.0.0" appId="AJI" />

<!-- Tasks -->
  <TaskList>
    <Task name="AllowPostLogonDebug"      description="Utilize client debug
once logged in."/>
    <Task name="AllowPostLogonTrace"      description="Utilize client
channel trace once logged in."/>
    <Task name="FilterValueAccess"        description="Server Access of
Filters task."/>
    <Task name="UserAuditReport"          description="Audit reporting
query/extract task."/>
    <Task name="UserSecurityControl"      description="User Security
Control task."/>
    <Task name="UserSecurityGroupConfig"  description="User Security
Group Configuration task."/>
    <Task name="UserSecurityProfile"      description="User Security
Profile task."/>
```

1.

```

        <Task name="UserSecurityRoleConfig"      description="User Security
Role Configuration task."/>
    </TaskList>

<!-- Functions -->
    <FunctionList>
        <Function name="ChangePassword" description="Allows admin. to change a
user password."/>
        <Function name="Delete" description="Delete function."/>
        <Function name="ExpireAll"              description="Expire all
passwords function."/>
        <Function name="ExpirePassword"        description="Expire password
function."/>
        <Function name="ExpireSelected"        description="Expire selected
passwords function."/>
        <Function name="Filters"               description="UI Filters Page
Access."/>
        <Function name="LogoutAll"             description="Logout all
function."/>
        <Function name="LogoutSelected"        description="Logout selected
function."/>
        <Function name="ModifyUserStatus"      description="Allows admin. to
change a user status."/>
        <Function name="Permissions"          description="UI Permissions
Page Access."/>
        <Function name="PermissionGroups"      description="UI Permission
Groups Page Access."/>
        <Function name="Save"                 description="Save function -
inserts and updates."/>
        <Function name="View"                 description="View function."/>
    </FunctionList>

<!-- Filter types or Fields example
    <FilterTypeList>
        <FilterType name="Institution"         description="Used assigning values
associated with institutions."/>
        <FilterType name="BusinessUnit"        description="Used assigning values
associated with institutions."/>
    </FilterTypeList>
-->

<!-- Permissions can be defined or omitted if task/functions are assigned
directly to role.-->
    <PermissionList>
        <Permission name="AllowPostLogonDebugPerm" description="Utilize
client debug once logged in.">
            <Task name="AllowPostLogonDebug">
                <Function name="View"/>
            </Task>
        </Permission>

```

```

        <Permission name="AllowPostLogonTracePerm"    description="Utilize
client channel trace.">
        <Task name="AllowPostLogonTrace">
        <Function name="View"/>
        </Task>
        </Permission>
        <Permission name="AuditReportingPermission"    description="User
audit reporting permission.">
        <Task name="UserAuditReport">
        <Function name="View"/>
        </Task>
        </Permission>
        <Permission name="FullFilterValueAccessPerm"    description="Full
filter value access for Filter Maint.">
        <Task name="FilterValueAccess">
        <Function name="View"/>
        <Function name="Delete"/>
        <Function name="Save"/>
        </Task>
        </Permission>
        <Permission name="FullUSCtlPermission"    description="Full User
Security Control permission set.">
        <Task name="UserSecurityControl">
        <Function name="ExpireAll"/>
        <Function name="ExpireSelected"/>
        <Function name="LogoutAll"/>
        <Function name="LogoutSelected"/>
        </Task>
        </Permission>
        <Permission name="FullUSGCfgPermission"    description="Full User Sec
Group Config permission set.">
        <Task name="UserSecurityGroupConfig">
        <Function name="View"/>
        <Function name="Delete"/>
        <Function name="Save"/>
        </Task>
        </Permission>
        <Permission name="FullUSPPPermission"    description="Full User
Security Profile permission set.">
        <Task name="UserSecurityProfile">
        <Function name="LogoutSelected"/>
        <Function name="LogoutAll"/>
        <Function name="ExpirePassword"/>
        <Function name="View"/>
        <Function name="Delete"/>
        <Function name="Save"/>
        <Function name="ModifyUserStatus"/>
        </Task>
        </Permission>
        <Permission name="ViewUSPPPermission"    description="View User
Security Profile permission set.">
        <Task name="UserSecurityProfile">

```



```

        <Function name="View"/>
    </Task>
    </Permission>
    <Permission name="ChangePasswordPerm" description="Change password
perm from User Profile.">
        <Task name="UserSecurityProfile">
            <Function name="View"/>
            <Function name="ExpirePassword"/>
            <Function name="ChangePassword"/>
        </Task>
    </Permission>
    <Permission name="ModifyUserStatusPerm" description="Modify user
status permis from User Profile.">
        <Task name="UserSecurityProfile">
            <Function name="View"/>
            <Function name="ModifyUserStatus"/>
        </Task>
    </Permission>
    <Permission name="FullUSRCfgPermission" description="Full User
Security Role permission group set.">
        <Task name="UserSecurityRoleConfig">
            <Function name="Filters"/>
            <Function name="PermissionGroups"/>
            <Function name="Permissions"/>
            <Function name="View"/>
            <Function name="Delete"/>
            <Function name="Save"/>
        </Task>
    </Permission>
</PermissionList>

<!-- Permission Groups can be defined or omitted if task/functions are assigned
directly to a role.-->
    <PermGroupList>
        <PermGroup name="SecAdminPermGroup" description="System
administrator group.">
            <Permission name="FullFilterValueAccessPerm"/>
            <Permission name="FullUSCtlPermission"/>
            <Permission name="FullUSGCfgPermission"/>
            <Permission name="FullUSPPPermission"/>
            <Permission name="FullUSRCfgPermission"/>
        </PermGroup>
        <PermGroup name="AllowPostLogonDebugPermGrp" description="Utilize
client debug once
logged in.">
            <Permission name="AllowPostLogonDebugPerm"/>
        </PermGroup>
        <PermGroup name="AllowPostLogonTracePermGrp" description="Utilize
client channel trace once
logged in.">
            <Permission name="AllowPostLogonTracePerm"/>

```

```

        </PermGroup>
        <PermGroup name="AuditReportingPermGroup" description="Permission
group for audit
                                reporting/
extract.">
        <Permission name="AuditReportingPermission"/>
        </PermGroup>
        <PermGroup name="ViewUserProfileGroup" description="View user
profiles.">
        <Permission name="ViewUSPPPermission"/>
        </PermGroup>
        <PermGroup name="ChangePasswordGroup" description="Change password
from user profile.">
        <Permission name="ChangePasswordPerm"/>
        </PermGroup>
        <PermGroup name="ModifyUserStatusGroup" description="Modify user
status from user profile.">
        <Permission name="ModifyUserStatusPerm"/>
        </PermGroup>
    </PermGroupList>

<!-- Roles can be defined and have permission groups assigned and/or task/
functions assigned directly.-->
    <RoleList>
        <Role name="AllowPostLogonDebugRole" description="Utilize client
debug once logged in.">
        <PermGroup name="AllowPostLogonDebugPermGrp"/>
        </Role>
        <Role name="AllowPostLogonTraceRole" description="Utilize client
channel trace once logged
in.">
        <PermGroup name="AllowPostLogonTracePermGrp"/>
        </Role>
        <Role name="AuditRole" description="Audit
Reporting/Extract role.">
        <PermGroup name="AuditReportingPermGroup"/>
        </Role>
        <Role name="SecAdminRole" description="Security
Administrator role.">
        <PermGroup name="SecAdminPermGroup"/>
        </Role>
        <Role name="SysAdminRole" description="System
Administrator role.">
        <PermGroup name="SecAdminPermGroup"/>
        </Role>
        <Role name="ViewUserProfileRole" description="View user
profiles role.">
        <PermGroup name="ViewUserProfileGroup"/>
        </Role>

```

```

        <Role name="ChangePasswordRole"          description="Change password
from user profile.">
        <PermGroup name="ChangePasswordGroup"/>
    </Role>
    <Role name="ModifyUserStatusRole"          description="Modify user
status from user profile.">
        <PermGroup name="ModifyUserStatusGroup"/>
    </Role>
    </Role>

    <!--Example assigning a task and some functions assoc. to a task and
bypassing perms/permgrps.-->
    <Role name="UserProfileAssignGroupsRole"      description="Assign
roles/filters to users.">
        <Task name="UserProfileConfig"/>
        <Function name="AssignGroupsOnly"/>
        <Function name="View"/>
    </Task>
</Role>

</RoleList>

<!--Example of defining seeded filter data
<FilterList>
    <Filter name="MyInstFilter" description="my filter description">
        <FilterType name="BusinessUnit"/>
        <FilterValue value="12345" description="Bank XYZ"/>
        <FilterValue value="98765" description="Bank ABC"/>
    </Filter>
    <Filter name="AllInstitutions" description="wildcarded all institution
access">
        <FilterType name="Institution"/>
        <FilterValue value="*" description="wildcard"/>
    </Filter>
</FilterList>
-->

<!-- Users w/Roles for authorization relationships -->
    <UserList>
        <User name="AudAdmin">
            <Role name="AuditRole"/>
        </User>

        <User name="SecAdmin">
            <Role name="AllowPostLogonDebugRole"/>
            <Role name="AllowPostLogonTraceRole"/>
            <Role name="SecAdminRole"/>
        </User>

    <!-- Example of a filter to user assignment
    <Filter name="MyInstFilter"/>
    -->

```

```
    </User>

    <User name="SysAdmin">
    <Role name="AllowPostLogonDebugRole"/>
    <Role name="AllowPostLogonTraceRole"/>
    <Role name="AuditRole"/>
    <Role name="SysAdminRole"/>
    </User>
  </UserList>
</SecurityAuthorizationContent>
```

AJI Application Configuration Map

The AJI Application Configuration XML initialization map file contains a list of all Java properties for an AJI application, other than properties required to *bootstrap* the application (bootstrap properties are defined in the config.properties file). These properties are stored in the Application Configuration data source (APPLCNFG).

The application configuration loader class is an initialization class that is executed during initialization if specified in an initialization.function<text-string> property or resolved from the initialization map (see [“AJI Initialization Map”](#)).

The autoload.applcnfg property in the APPLCNFG indicates whether the AJI Application Configuration XML initialization file is loaded during initialization. This property defaults to “false”, but can be set to “true” if you want to autoload application configuration information during initialization.

Resource Naming Conventions

AJI applications can leverage the XML map approach to define their own application configuration initialization XML map file as long as it uses the following resource naming convention:

```
config/AplConfigMap.xml
```

You can refer to the example contents of the AJIAplConfigMap.xml file for descriptions and usage of how an AJI application-specific application configuration XML initialization map file needs to be configured.

The AJI application configuration initializer resolves the resource, reads the map, and determines whether a load is required after the AJI-based application configuration map is loaded. An application-specific Application Configuration Map XML initialization file is version- checked and loaded as applicable.

Example Contents

Example contents of the AJIAplConfigMap.xml file are shown below as an example for defining an AJI application-specific Application Configuration Map XML initialization file.

```
<ApplicationConfigurationContent>
```

```
<!-- This version should be bumped after each modification so the server
      knows to redeploy the content. An application id can be optionally
specified to indicate
      who owns the map when determining if a map should processed or not. -->
<Implementation version="1.0.0" appId="AJI" />

<!-- Process cleanup properties -->
<Property name="cleanup.userauditlog.cleanup.starttime"value="23:59"
      description="The time of day the User Audit Log cleanup starts
removing old rows from
      the User Audit Log table (HH:MM format)."/>
<Property name="cleanup.userauditlog.retention.days"value="30"
      description="The number of days to leave audit rows in the User Audit
Log table."/>
<Property name="cleanup.userauditlog.timeperiod.hours"value="24"
      description="The number of hours between User Audit Log cleanup
activity."/>

<!-- Alternate Datasource propert(ies)-->
<Property name="db.alt.datasource.name.usec" value="USEC"
      description="USEC datasource name"/>

<!-- Database auditing service name properties -->
<Property name="db.audited" value="true"
      description="Identifies if UI database actions are to be audited."/>
<Property name="db.audited.actvsesn" value="full"
      description="Audit setting for the Active Sessions table."/>
<Property name="db.audited.applcnfgservice" value="full"
      description="Audit setting for the APPLCNFG table."/>
<Property name="db.audited.eventdocservice" value="full"
      description="Audit setting for the EVDOD table."/>

....
....
....

<!-- Encoding flag property -->
<Property name="encoding.use.default.flag" value="UTF-8" description="Default
encoding flage."/>
</ApplicationConfigurationContent>
```

XML Map Lists

You can use XML map lists to group multiple maps of the same kind (e.g., events, security authorization, etc.). Each of the AJI XML load maps, except for the AJI Initialization Map, also supports a corresponding map *list*.

Map lists provide a list of more than one map to be processed under a given load map initializer. In some cases, a map list can be used to load several maps of the same kind (e.g., events, security authorization, etc.).

For example, you may want to break up the authorization seed data by the type of services use them (e.g., interfaces, device handlers, user interface windows, etc.).

The following four map lists are supported:

- config/AuthenticationSecurityMapList.xml
- config/AuthorizationSecurityMapList.xml
- config/EventLogMapList.xml
- config/ApplConfigMapList.xml

The AJI load map initializers always check first for a map list. If a map list is found, the initializer parses the map list and returns an array of resources. It then checks for each resource and loads each found resource into the appropriate data sources

XML Scheme

The XML scheme for a map list contains one or more <Resource> tags within a <ResourceListContent> tag as shown below.

```
<?xml version="1.0" encoding="UTF-8"?>
<ResourceListContent>
<!--A list can contain 1 or more Resource elements contains a value of a resource
such as:
    config/AuthorizationSecurityInfsMap.xml -->
<Resource>name of resource</Resource>
<Resource>name of resource2</Resource>
<ResourceListContent>
```

Example Contents

The following is an example of a map list for Authorization Security Map XML initialization files.

```
<?xml version="1.0" encoding="UTF-8"?>
<ResourceListContent>
<!--Authorization security load maps: -->
<Resource>esconfig/ESAuthorizationSecBaseMap.xml</Resource>
<Resource>esconfig/ESAuthorizationSecVisaMap.xml</Resource>
<Resource>esconfig/ESAuthorizationSecMCMap.xml</Resource>
</ResourceListContent>
```


Supported AJI Map Loaders

The AJI AJSF supports the following map loaders, which are actually initializers that follow the `InitializationInterface` but extend the `TransactionCommandBase`. These map loaders must be configured within the initialization list derived from the initialization map or from the `initialization.function<text-string>` properties from the APPLCNFG.

- `com.aciworldwide.application.security.SecurityMapInitializer`
This initializer/loader includes both the Authentication and Authorization initializers.
- `com.aciworldwide.common.ApplConfigMapInitializer`
This initializer/loader must be explicitly set to true with the `autoload.applconfig=true` property because the default value is false.

AJI also supports the following static initializers that are always invoked by the `InitializationManager` and do not need to be specified in an initialization list.

- `com.aciworldwide.common.EventLogMapInitializer`
- `com.aciworldwide.common.FirstInstanceInitializer`

Custom Map Loaders/Initializers

You can use custom initializers to load custom maps (as they pertain to any AJI-based application). You can write your own initializer to interpret a custom XML map initialization file, parse the map file's contents, and apply the contents to an AJI application's database.

You create a customer initializer by performing the following steps:

1. Create an initializer class that extends `com.aciworldwide.common.MapLoadInitializerBase`.

This class is an *abstract* class, which means the custom initializer class must implement the following methods:

- `buildMap(...)`
This method is called to parse the custom XML map and store contents into a hashmap used later by the “loadMap” method. Alternatively, you can just store the XML root element in the hashmap and deal with it later during load processing.
- `getMapDescription()`
This method is called to obtain a description that corresponds to the map you are initializing for.
- `getResourceNames()`
This method is called to obtain a list of resources that need to be processed (one call to `buildMap` each if found).
- `isEnabled()`
This method is called to allow your initializer to tell the base class whether or not processing is enabled. If false is returned, no other abstract methods are called.
- `loadMap(...)`
This method is called (using an inherent transactional command pattern). A hashmap is provided based on content parsed from the invocations of each call to the `loadMap(...)` method. (see above).

The base class “`MapLoadInitializerBase`” implements the `InitializationInterface`. This means that in order to be invoked during the initialization phase of a server, the custom initializer that implements this base class must be configured as an initializer either through initialization map logic or explicitly in the `APPLCNFG` as an `initialization.function<text-string>` property. This base class:

- ❑ Handles all versioning for the custom map.
 - ❑ Makes callbacks to the abstract methods.
 - ❑ Provides transaction protection during loadMap method invocation because it also extends the TransactionCommandBase of the AJSF.
2. Add the abstract methods to your initializer class. Use the buildMap and loadMap methods to parse and load your database tables, accordingly.
 - It is important to perform the database access within the loadMap method.
 - You can complete the parsing of the custom XML load file within the buildMap method.
 3. Code the custom loader/initializer as shown in the following com.aciworldwide.application.MyMapLoader class example.
-

```
package com.aciworldwide.application;

import java.util.HashMap;
import org.jdom.Element;
import com.aciworldwide.common.DataObject;
import com.aciworldwide.common.MapLoadInitializerBase;

public class MyMapLoader extends MapLoadInitializerBase {

    /**
     * This method is called by the base class to allow this loader to build
     * it's buildMap (HashMap) for the given root jdom element obtained from
     * a document of an xml map resource.
     * This method may be called 0 or more times depending upon the number
     * of resources located and parsed into a jdom document.
     *
     * The rootElmt can be accessed to obtain data to be seeded. This
     * seed data can be stored in the buildMap (HashMap) that will be used
     * later at "build" time. Or one can just store the root element now in the
     * hashmap "buildMap" and continue.
     *
     * @param rootElmt The root element to be parsed.
     * @param buildMap The hash map to contain data obtained from the root
     * element which is
     * used later at load callback time.
     * @return true if the parsing and map building is successful.
     */
    protected boolean buildMap(Element rootElmt, HashMap buildMap) {
        //
        // TODO - Add custom parsing of elements and store content
        // in buildMap here. Or store off the elements as needed.
        //
    }
}
```

```
        // Parse each map's rootElmt and obtain content accordingly.

        return true;
    }

    /**
     * This method should return a description of your loader.
     */
    protected String getMapDescription() {
        return "MyMapLoader";
    }

    /**
     * This method should return 1 or more resource names that will
     * be resolved to a URL if the resource name(s) are found under the
     * applications deployment (via the classloader). Each resource
     * is read into a jdom document (and should correspond to xml resources).
     * (i.e. - "config/mydata/myloadmap.xml")
     *
     * This method is called prior to looking up resources and building
     * document(s) and calling the "buildMap" method.
     *
     * Note: To use map lists, you may want to reference the
     * base method(s). Map list is an xml resource that contains a list
     * of resources.
     * return getResourceNamesFromMapList("config/mylist/myloadmaplist.xml");
     *
     * @see #getResourceNamesFromMapList(String)
     * @see #getResourceNamesFromMapList(String, String[])
     */
    protected String[] getResourceNames() {
        //
        //TODO - return your own resources
        //
        String[] rsrcs = {"config/mymaps/appXMap1.xml" +
            "config/mymaps/appXMap2.xml"};

        // Example of using a map list approach instead of above.
        // return getResourceNamesFromMapList("config/mymaps/mylist.xml");

        return rsrcs;
    }

    /**
     * This method gives the custom loader an opportunity to check
     * configuration to see if it should be enabled for load.
     */
    protected boolean isEnabled() {
        return true;
    }
}
```

```
/**
 * This method is the last callback method to be called. It
 * will provide the customer loader with transaction protection,
 * inside a transaction command pattern. The MapLoadInitializerBase extends
 * the TransactionCommandBase class and this method is invoked within
 * the "execute". So the unit of work done within this method
 * is protected.
 *
 * @param dataObj This is the dataObject that can be used as this is done
within the
 * confines of a transaction command base.
 * @param buildMap contains the hash map which contains all the contents to
be loaded based
 * on the xml maps that were parsed and built.
 * <p>
 * @return true if the load was successful.
 *
 * Note: One can use the super.commitBatch method if needed to commit
 * data in chunks.
 *
 * @see #commitBatch(DataObject)
 */
protected boolean loadMap(DataObject dataObj, HashMap buildMap) {
    //
    // TODO - Perform custom loading (whatever that means to this loader)
using
    // contents stored during the buildMap phase in the HashMap
provided.
    //

    // Do your load logic here.

    return true;
}
```

Dynamic Initializers

The `com.aciworldwide.framework.InitializationManager` class supports the feature of *dynamic initializers*. AJI applications can write their own initializers that implement the “`com.aciworldwide.framework.InitializationInterface`” interface. Instead of adding the initializer class to the AJI application’s initialization map or `initialization.function<text-string>` properties list, you can *register* the class for initialization upon demand (initial access). The `InitializationManager` class offers the following methods for this purpose:

- `boolean isRegistered(Class cls)`
This method checks to see if the given class is an instance of `InitializationInterface`, and if so, whether the class is already registered and initialized.
- `boolean register(Class cls)`
This method can be called by an initialization class static method like a “`getInstance()`” method.

Example

The following is an example of dynamic initialization.

```
package com.aciworldwide.application;

import com.aciworldwide.framework.InitializationInterface;
import com.aciworldwide.framework.InitializationManager;

public class MyDynamicInitializer implements InitializationInterface {

    private static boolean regInd = false;

    public void destroy() {
        // TODO Auto-generated method stub
    }

    public String getRefreshKey() {
        // TODO Auto-generated method stub
        return null;
    }

    public void init() {
```

```
        // TODO Auto-generated method stub

    }

    public void refresh() {
        // TODO Auto-generated method stub

    }

    private synchronized static void register () {
        if (! InitializationManager.isRegistered(MyDynamicInitializer.class)) {
            InitializationManager.register(MyDynamicInitializer.class);
        }
        regInd = true;
    }

    /**
     * This is a custom method accessible to all from an initializer class.
     * Each method would check the "regInd" flag and if false, then register
     * on demand.
     */
    public static void myMethod() {
        if (! regInd) {
            register();
        }

        // Do my work here.
    }
}
```

ACI Worldwide, Inc.

A: BASE24-eps User Interface (ESWEB)

This appendix describes ACI Java Server Framework (AJSF) properties specific to the BASE24-eps Java User Interface servlets in the ESWEB server process. The ESWEB server process supports the ACI desktop user interface. This appendix also provides an example config.properties file for the ESWEB server process.

System Properties

The following system properties are specific to the BASE24-eps Java User Interface servlets. These properties are typically specified in a `-D` option JVM runtime argument in the startup script for the ESWEB server (e.g., `startesweb`).

root.config — Identifies the root location of configuration information for User Interface Framework (UIF) support in the BASE24-eps Java User Interface servlets instance.

Required: Yes

Example Entry: `-Droot.config=C:/ESWeb/ESConfig`

SCH — A flag indicating whether the BASE24-eps Java User Interface servlets run in channel header support mode. Valid values are as follows:

`true` = Yes, the BASE24-eps Java User Interface servlets run in Channel header support mode.

`false` = No, the BASE24-eps Java User Interface servlets do not run in channel header support mode.

When this property is set to `true`, inbound request messages and responses to those requests will be processed with or without channel header support from the client. Channel header support is a feature of the User Interface Framework (UIF). If this property is set to `false`, channel header support is turned off.

Note: Support of this feature from the ACI desktop user interface client requires that the configuration file of `../ES/desktop/DesktopLauncher.lax` includes the system property setting of `-DSCH=true`.

Required: Yes

Example Entry: `-DSCH=true`

Service System Properties

The following optional system properties name the auditing, security, and framework services used by the ACI desktop user interface application.

AuditService — The name of the service to be used for auditing. If this system property is not configured, it defaults to “com.aciworldwide.application.audit.UAUDManager”. This property is not required when the user.security.version property is set to a value of usec1.1 or greater.

Required: No

Example Entry: -DAuditService=com.aciworldwide.application.audit.
UAUDManager

SecurityServerService — The name of the service to be used for authentication, authorization, and maintenance. If this system property is not configured, it defaults to “com.aciworldwide.application.security.USECManager”.

Required: No

Example Entry: -DSecurityServerService=com.aciworldwide.application.
security.USECManager

FrameworkService — The name of the provider used to derive a desktop client’s framework service. If this system property is not configured, it defaults to “com.aciworldwide.application.security.USECManager”.

Required: No

Example Entry: -DFrameworkService=com.aciworldwide.application.security.
USECManager

Application Properties

The following application properties configured in the `config.properties` file or the `APPLCNFG` that are specific to the BASE24-eps Java User Interface servlets in the ESWEB server process.

config.home — Specifies the home location of the configuration files used by the Java User Interface servlets. The value of this property is used by components and services to resolve relative URIs to absolute URIs or file specifications. The value of this property must be in one of the following formats and must end in a forward slash (/):

```
config.home=file:///D:/config/
```

Required: Yes

Example Entries:

```
config.home=/user/B24es_054/ESWeb/ESConfig/
```

```
config.home=file:///C:/user/B24es054/ESWeb/ESConfig/
```

datatype.jar.aci.ESXMLDataTypes — Identifies the relative Universal Resource Identifier (URI) location of the BASE24-eps data types JAR file. The value of the `config.home` property is used to resolve the relative URI to an absolute location.

Required: Yes

Example Entries: `lib/es-datatypes.jar`

dataValidation.schema.ESXMLDataTypes — Identifies the relative Universal Resource Identifier (URI) location of the data type schema definition file. The value of the `config.home` property is used to resolve the relative URI to an absolute location.

Required: Yes

Example Entry: `schemas/lib/ESXMLDataTypes.xsd`

db.audited.*table-name* — Indicates the type of user auditing performed for a particular database table, represented by *table-name*, when the db.audited property is set to true. Valid values are as follows:

- none = No auditing is performed for this database table.
- medium = Medium auditing is performed for this database table. For this value, only header information is logged in auditing records.
- full = Full auditing is performed for this database table. For this value, header information as well as detail information about the before and after images of the affected record are logged.

If this property is not defined for a database table, it defaults to a value of full.

Required: No
Example Entry: db.audited.applcnfg=full
Usage: Database Component

Note: Because property names and values are case-sensitive in the APPLCNFG, database table names are case-sensitive as well.

The following table lists all Java database table names specific to BASE24-eps as well as the data sources and ACI desktop user interface windows or browser web pages associated with each database table. Refer to the [db.audited.*table-name*](#) property description in section 5 for a listing of all Java database table names generic to the AJSF.

If a BASE24-eps database table for a data source is not listed in this table, the data source is maintained by a C++ server. For these servers, auditing is controlled by the UI_AUDIT_LEVEL environment attribute rather than a property. All audit records are written to the User Audit Log data source, regardless of whether the data source is maintained by a Java or C++ server. For more information on the UI_AUDIT_LEVEL attribute, refer to the *BASE24-eps Environment User Guide*.

Note: All non-database activity on the ACI desktop user interface (e.g., user login, logoff selected, logoff, password verification, password change, password expiration, session expired, session timeout, and audit report generation) is always audited.

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
AbiKey	ABI Key Configuration data source (ABI_Key)	ABI (Italian Banking Association) MAC Key Configuration window
AccountRouting	Account Routing data source (Account_Routing)	Account Routing Configuration window
AcctStatusMappingInb	Account Status External to Internal data source (Acct_Status_Extrn_to_Intrn)	Inbound Mapping tab of the Account Status Mapping Profile Configuration window
AcctStatusMappingOut	Account Status Internal to External data source (Acct_Status_Intrn_to_Extrn)	Outbound Mapping tab of the Account Status Mapping Profile Configuration window
AcqInstRelation	Acquirer Instrument Type Relationship data source (Acquirer_Instrument_Type_Rel)	Acquirer/Instrument Type Relationship Profile Configuration window
acqissrelationship	Acquirer Issuer Relationship data source (Acquirer_Issuer_Relation)	Routing Configuration - Destination/Source Relationship window
acqrteprfl	Acquirer Route Profile data source (Acquirer_Route_Profile)	Source Routing Profile Configuration window
acquirertransallowed	Acquirer Transactions Allowed data source (Acquirer_Txn_Allowed)	Acquirer Transactions Allowed Profile Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
actioncodeext	Action Code External to Internal data source (Action_Code_Extrn_to_Intrn)	Inbound Mapping tab of the Action Code Profile Configuration window
actioncodeint	Action Code Internal to External data source (Action_Code_Intrn_to_Extrn)	Outbound Mapping tab of the Action Code Profile Configuration window
activescript	Active Script Statistics data source (Active_Script_Statistics)	Active Script Statistics and Management
admincard	Administrative Card data source (Admin_Card)	Administrative Card Configuration window
apacsconfiguration	APACS Configuration data source (APACS_Configuration)	APACS tab of the Merchant Channel Profile window
AS2805Configuration	AS2805 Configuration data source (AS2805_CONFIGURATION)	Message Profile > AS2805 tab of the Merchant Channel Profile Configuration window
atdd1	ATM Terminal Data Dynamic data source—general data (ESATDD1)	Hopper and Terminal Activity tabs of the ATM (Java) > ATM Channel Configuration window and the ATM (Java) > ATM Channel Administration window
atdd2	ATM Terminal Data Dynamic data source—scratch pad (ESATDD2)	Not applicable
atds1	ATM Terminal Data Static data source—general data (ESATDS1)	Location and Processing tabs of the ATM (Java) > ATM Channel Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
atmconfig	ATM Configuration Load data source (ATMCNFG)	ATM (Java) > ATM Load Profile window
atmdhchannelprofile	ATM Channel Profile Configuration data source (ATM_PRFL)	ATM Channel Profile Configuration window
atmdhchannel	ATM Terminal Data Static data source—general data (ATM_DS1)	ATM Channel Configuration window
ATMDHChnlDialupMgmt	ATM Dial-Up Command Data Source (ATM_DIAL_UP_CMD)	ATM Dial-Up Channel Management window
atmdhconsumermedia	ATM Consumer Media data source (ATM_CONSUMER_MEDIA)	ATM Consumer Media Configuration window
atmdhdepbinprofile	ATM Depository Bin Profile Configuration data source (ATM_SSB_BIN)	ATM Depository Bin Profile Configuration window
ATMDHDkmProfile	ATM Dynamic Key Management Profile data source (ATM_DKM)	ATM DKM Profile window
atmdhemvprofile	ATM EMV Profile Configuration data source (ATM_EMV)	ATM EMV Profile Configuration window
atmdhfastcashprofile	ATM Fast Cash Profile Configuration data source (ATM_FAST_CASH)	ATM Fast Cash Profile Configuration window
atmdhflow	ATM Flow Control Configuration data source (ATM_FLOW_CNTL)	ATM Flow Configuration window
atmdhhopperprofile	ATM Hopper Profile Configuration data source (ATM_HOPR)	ATM Hopper Profile Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
atmdhinactivityprofile	ATM Inactivity Alert Profile Configuration data source (ATM_INACT)	ATM Inactivity Alert Profile Configuration window
atmdhloadprofile	ATM DH Load Configuration data source (ATM_CONF)	ATM Load Profile window
atmdd1	ATM Data Dynamic data source—general data (ATM_DD1)	Previous day totals tabs on the ATM Totals window
atmdd2	ATM Data Dynamic data source—scratch pad (ATM_DD2)	Not applicable
atmdhmacprofile	ATM MAC Profile Configuration data source (ATM_MAC)	ATM MAC Profile Configuration window
atmdhopcode	ATM Operation Code data source (ATM_OP_CDE)	ATM Operation Code Configuration
atmdhoperid	Operator ID data source (ATM_OPER)	ATM Diebold Channel Operator Configuration window
atmdhpeakloadprofile	ATM Peak Load data source (ATM_PEAK_LOAD)	ATM Peak Load Profile Configuration window
atmdhreceiptformat	ATM Receipt Template data source (ATM_RECEIPT)	ATM Receipt Format Profile Configuration window
atmdhreceiptreqtype	ATM Receipt Required data source (ATM_REQ_RCPT)	ATM Receipt Required Profile Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
atmdhstatescreen	ATM State and Screen data source (ATM_STATE_SCRN)	ATM State and Screen Configuration window
atmdhstmntprintprofile	ATM Statement Print Profile Configuration data source (ATM_STMNT_PRNT)	ATM Statement Print Profile Configuration window
atmdhsurchargeprofile	ATM Surcharge Profile Configuration data source (ATM_SUR)	ATM Surcharge Profile Configuration window
atmifx	ATM IFX Data source (ATM_IFXDATA)	For future use
atmprv	ATM Previous Day Totals data source (ATM_PREV_TTL)	ATM Totals window
auditsafcfg	Audit Store and Forward Configuration data source (Audit_Store_and_Forward_Config)	Audit Store and Forward (SAF) Configuration window
authscript	Authorization Script Configuration data source (Authorization_Script)	Authorization Script Configuration window
b24atmtranmappinginb	ATM Transaction Code to Processing Code data source (ATM_TC_Proc_Code)	Inbound Messaging tab of the BASE24-atm Transaction Mapping Profile Configuration window
b24atmtranmappingout	Processing Code to ATM Transaction Code data source (Proc_Code_ATM_TC)	Outbound Messaging tab of the BASE24-atm Transaction Mapping Profile Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
b24postranmappinginb	POS Transaction Code to Processing Code data source (POS_TC_Proc_Code)	Inbound Messaging tab of the BASE24-pos Transaction Mapping Profile Configuration window
b24postranmappingout	Processing Code to POS Transaction Code data source (Proc_Code_POS_TC)	Outbound Messaging tab of the BASE24-pos Transaction Mapping Profile Configuration window
b53institution	B53 Institution data source (B53_Institution)	Armed Forces Financial Network (AFFN) Institution Configuration window
		Credit Union 24 (CU24) Institution Configuration window
		JEANIE Institution Configuration window
banknetconfiguration	Banknet Configuration data source (Banknet_Configuration)	Banknet Values and Message Management tabs of the Banknet ISO Interface Configuration window
banknetinstitutionid	Banknet Institution Identification data source (Banknet_Institution_Id)	Banknet Acquirers tab of the Banknet ISO Interface Configuration window
banknetmerchantprogram	Banknet Merchant Program data source (Banknet_Merchant_Program)	Merchant Programs tab of the Banknet ISO Interface Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
BatchAuth	Batch Authorization Configuration data source (Batch_Auth_Config)	Batch Authorization Configuration window
bntngheader	BNTng Header data source (BNTng_Header)	BNTng Key Header Configuration window
cardaccount	Card Account data source (Card_Account)	Accounts tab of the Card Management window
card	Card data source (Card)	General, Status, and Limits tabs of the Card Management window
		Voice Authorization Card Status Search and Update web page
		Voice Authorization Code 10 Issuer web page
		Voice Authorization Code 10 Acquirer web page
		Voice Authorization Issuer Override web page
		Voice Authorization Acquirer Override web page
channelinstrumenttypereletion	Channel_Instrm_Type_Relation	Instrument Types tab of the Merchant Channel Configuration and Statistics window
channelprofileinstrmtypereletion	Chan_Profile_Instrm_Typ_Relation	Instrument Types tab of the Merchant Channel Profile window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
channelpubkeyperusal	Channel Public Key Security data source (CHAN_PKEY_SEC)	Channel Public Key Perusal window
channelsec	Channel Security data source (CHANNEL_SECURITY)	Channel Security Configuration window
Check	Check Status data source (Check_Status)	Check Management window
CheckType	Check Type data source (Check_Type)	Check Type Configuration window
chipauthprofile	Chip Authentication data source (CHIP_AUTHENTICATION)	Chip Auth tab of the Authentication Profile Configuration window
CircuitInterface	Circuit Interface data source (Circuit_Interface)	Circuit tab of the Circuit Interface Configuration window
code10	Code 10 data source (COD10)	Voice Authorization Code 10 Acquirer web page
		Voice Authorization Code 10 Issuer web page
compatibilityprofile	ATM BASE24 Compatibility Profile data source (CMPTPRFL)	ATM (Java) > ATM BASE24 Compatibility Profile Configuration window
configprofile	ATM Device Handler Configuration data source (ATDCNFG)	ATM (Java) > ATM Channel Profile Configuration window
contextcfg	Context Configuration data source (Context_Configuration)	Context Profile Configuration

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
contingencyinterface	Contingency Interface data source (Contingency_Interface)	Contingency tab of the Contingency Interface Configuration window
currencyconvrate	Currency Conversion Rate data source (Currency_Conversion_Rate)	Currency Conversion Rate Configuration window
derivationkey ¹	Derivation Key data source (Derivation_Key)	DUKPT Derivation Key Profile Configuration window
DiscoverInstitution	Discover Institution data source (Discover_Institution)	Discover Acquirers Configuration window
DiscoverInterface	Discover_Interface	Discover tab of the Discover Interface Configuration window
dsoltprelation ¹	Data Source OLTP Relationship data source (DS_OLTP_Relation)	Not applicable
efundsintf	Efunds Interface Configuration data source (EFunds_Interface)	Visa DPS tab of the Visa DPS Interface Configuration window
ElanInstitution	Elan Institution data source (ÉLAN_INSTITUTION)	Elan Acquirers Configuration window
emvcardvrfnresultsstatus	EMV Card Verification Results (CVR) Status data source (EMV_CRD_VRF_RSLT)	EMV Card Verification Results (CVR) Status Profile Configuration window
emvreasononlinecodedescr	Reason Online Code Description data source (RSN_ONL_CDE_DSC)	EMV Reason Online Code (ROC) Description Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
emvreasononlinecode	EMV Reason Online Code Status data source (EMV_RSN_ONL_CDE)	EMV Reason Online Code (ROC) Status Profile Configuration window
emvtermvrfrnresultsstatus	EMV Terminal Verification Results Status data source (EMV_TRM_VRF_RSLT)	EMV Terminal Verification Results (TVR) Status Profile Configuration window
emvprofile	ATM EMV Profile data source (EMVPRFL)	ATM (Java) > ATM EMV Profile Configuration window
environment ¹	Environment data source (Environment)	Environment Configuration window
eppserialnumber	EPP Serial Number data source (EPP_Serial_Number)	EPP Serial Number Configuration window
epsnetinstitution	EPSNET_Inst	EPS-Net Institution Configuration window
epsnetinterface	EPSNET_Interface	EPS-Net ISO Interface Configuration window
EquensInstrumentTypeMappingInb	Equens Instrument Type Internal data source (EQUENS_IT_INT)	Inbound Mapping tab of the Equens BIM Instrument Type Mapping Profile Configuration window
EquensInstrumentTypeMappingOut	Equens Instrument Type External data source (EQUENS_IT_EXT)	Outbound Mapping tab of the Equens BIM Instrument Type Mapping Profile Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
ErsbConfiguration	E-RSB Interface data source (ERSB_Interface)	E-RSB tab of the E-RSB ISO Interface Configuration window
ESATDL	ATM Terminal Load data source (ESATDL)	Not applicable
eurooptedcurrency	Euro Opted Currency data source (Euro_Opted_Currency)	Euro Opted Currency Configuration window
eventlog ¹	AJSF Event Log data source (EVENTLOG)	Not applicable
eventsubsystem ¹	Event Document data source (Event_Document)	Event Message Documentation tab of the Event Configuration and Management window
exportedoperators	Exported Operator Documentation data source (Exported_Operator_Documentation)	Exported Operators Configuration window
fastcashprofile	ATM Fast Cash data source (FCSHPRFL)	ATM (Java) > ATM Fast Cash Profile Configuration window
filepartctlg ¹	File Partition Catalog data source (file_part_catalog)	Catalog tab of the File Partition Configuration window
filepartfields ¹	File Partition Fields data source (file_part_fields)	Fields tab of the File Partition Configuration window
filepartlist ¹	File Partition List data source (file_part_list)	Catalog tab of the File Partition Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
fileupdaterouter	File Update Router data source (File_Update_Route)	File Update Router Configuration window
hiso87	ISO 8583-1987 Host Interface data source (HISO87_Interface)	BASE24 ISO8583 Host tab of the BASE24 ISO8583 Host Interface Configuration window
hiso93	ISO 8583-1993 Host Interface configuration data source (HISO93_Interface)	ISO 93 Host tab of the ISO 93 Host Interface Configuration window
hopperprofile	ATM Hopper Profile data source (HPRPRFL)	ATM (Java) > ATM Hopper Profile Configuration window
hostpubkeyperusal	Host Public Key Security data source (HOST_PUBLIC_KEY)	Host Public Key Perusal window
hpdhconfig	(HPDH_CONFIG)	Message Profile > Hypercom tab of the Merchant Channel Profile Configuration window
hypercomnac	NAC Terminal Station data source (HPDH_NAC_TERM)	Hypercom NAC Profile Configuration window
hypercommns	NMS Terminal Station data source (HPDH_NMS_TERM)	Hypercom NMS Profile Configuration window
IfAbi	ABI Interface Configuration data source (ABI_Interface)	ABI (Italian Banking Association) ISO Interface Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
ifaccel	ACCEL Interface data source (ACCEL_INTERFACE)	Accel/Exchange tab of the Accel/Exchange Interface Configuration window
IfAmexCAPN	American Express CAPN Interface data source (AMEX_CAPN_Interface)	Amex CAPN tab of the American Express CAPN Interface Configuration window
ifamexgns	AEGN Interface data source (AEGN_INTERFACE)	Amex GNS tab of the American Express Global Network Services Interface Configuration window
IfAs28	AS2805 ISO Interface data source (AS2805_Interface)	AS2805 ISO tab of the AS2805 ISO Interface Configuration window
IfB53Iso	Fifth Third ISO Interface data source (Fifth_Third_Intf)	Fifth Third ISO Interface Configuration window
IfBenefit	Benefit Interface data source (BNFT_Interface)	Benefit tab of the Benefit ISO Interface Configuration window
IfBicIso	BIC Interface data source (BIC_INTERFACE)	BIC ISO (87) tab of the BIC ISO (87) Interface Configuration window
IfBicSettlKeyXchnng	BIC Settlement Key Exchange data source (BIC_SETTLEMENT_KEY_XCHNG)	BIC ISO (87) Status tab of the BIC ISO (87) Interface Configuration window
IfCup	CUP Interface data source (CUP_Interface)	CUP tab of the China Union Pay (CUP) Interface Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
IfDias	DIAS Interface data source (DIAS_Interface)	DIAS tab of the DIAS Interface Configuration window
IfEquens	Equens Interface data source (Equens_Interface)	Equens BIM tab of the Equens BIM Interface Configuration window
IfInterac	Interac Interface data source (Interac_Interface)	Interac tab of the Interac Interface Configuration window
IfMnetIso	MNET Interface data source (MNET_Interface)	MNET tab of the MNET ISO Interface Configuration window
IfPresto	Presto Interface data source (PRESTO_INTERFACE)	Presto tab of the Presto Interface Configuration window
IfPrice	PRICE Interface data source (PRICE_Interface)	Price tab of the PRICE Interface Configuration window
IfSpan2Iso	SPAN2 Interface data source (SPAN2_INTERFACE)	SPAN2 tab of the SAMA Saudi Payments Network (SPAN2) ISO Interface Configuration window
ifxdata	IFX data source (IFXDATA)	Not applicable
ifxhostactioncode	IFX Host Interface Action Code Internal to External data source (IFX_Host_Action_Code_Int_Ext)	IFX Host Action Code Profile Configuration window
ifxhostinterface	IFX Host Interface data source (IFX_Host_Interface)	IFX Host tab of the IFX Host Interface window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
imtatmtranmappings	IMT Processing Code to ATM Transaction Code data source (IMT_Proc_Code_ATM_Trان_Code)	Outbound Messaging tab of the Internal Message Translation (IMT) Transaction Mapping - ATM window
imtatmtranstoproc	IMT ATM Transaction Code to Processing Code data source (IMT_ATM_Trان_Code_Proc_Code)	Inbound Messaging tab of the Internal Message Translation (IMT) Transaction Mapping - ATM window
imtpostranmappings	IMT Processing Code to POS Transaction Code data source (IMT_Proc_Code_POS_Trان_Code)	Outbound Messaging tab of the Internal Message Translation (IMT) Transaction Mapping - POS window
imtpostranstoproc	IMT POS Transaction Code to Processing Code data source (IMT_POS_Trان_Code_Proc_Code)	Inbound Messaging tab of the Internal Message Translation (IMT) Transaction Mapping - POS window
imt	IMT Interface data source (IMT_Interface)	IMT Values tab of the IMT Interface Configuration window
inactivitytimerprofile	ATM Inactivity data source (ITMRPRFL)	ATM (Java) > ATM Inactivity Alert Profile Configuration window
institutionconfig	Institution data source (Institution)	Institution Configuration window
InstrumentStatusOut	Instrument Status Internal to External data source (Instrm_Status_Intrn_to_Extrn)	Outbound Mapping tab of the Instrument Status Mapping Profile Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
instrumentstatus	Instrument Status External to Internal data source (Instrm_Status_ Extrn_to_Intrn)	Instrument Status Mapping Profile Configuration window
InstrumentTypeMappingInb	Instrument Type External to Internal data source (Instrm_Typ_Extrn_to_ Intrn)	Inbound Mapping tab of the Instrument Type Mapping Profile Configuration window
InstrumentTypeMappingOut	Instrument Type Internal to External data source (Instrm_Typ_Intrn_to_ Extrn)	Outbound Mapping tab of the Instrument Type Mapping Profile Configuration window
instrumenttypereletion	Instrument Type to Interchange Relation data source (Instrum_Typ_ Interchange_Rel)	Instrument Type/ Interchange Relationship Configuration window
instrumenttype	Instrument Type data source (Instrument_Type)	Instrument Type Configuration window
interchangeprefix	Interface-specific Interchange Prefix data sources (e.g., IPRFX_ MDS, IPRFX_VISA, etc.)	Interchange Prefix Configuration window
interfacemanager	Interface Manager data source (Interface_ Manager)	Various tabs of all Interface Configuration windows
interfacesaf	Interface SAF data source (Interface_SAF)	Store-and-Forward (SAF) tab of all Interface Configuration windows
interface	Interface data source (Interface)	Various tabs of all Interface Configuration windows

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
interfacestation	Interface Station data source (Interface_Station)	Station(s) tab of all Interface Configuration windows
interfaceterminal	Interface Terminal data source (Interface_Terminal)	Interface Terminal Configuration window
interpay	Interpay Interface data source (Ipay_Interface)	Interpay tab of the Interpay Interface Configuration window
isofield	ISO Field data source (ISO_FIELD)	BCD/MAC Information tab of the ISO Message Profile Configuration window
isomessage	ISO Message data source (ISO_MESSAGE)	Message Type Bit Maps tab of the ISO Message Profile Configuration window
isotranmappinginb	External ISO Processing Code to Internal Processing Code Mapping data source (ISO_PROC_CODE)	Inbound Messaging tab of the ISO Transaction Mapping Profile Configuration window
isotranmappingout	Internal Processing Code to External ISO Processing Code Mapping data source (PROC_CODE_ISO)	Outbound Messaging tab of the ISO Transaction Mapping Profile Configuration window
issrteprfl	Issuer Route Profile data source (ISS_ROUTE_PRFL)	Destination Routing Profile Configuration window
issuerprofilemap	Issuer Profile Map data source (ISS_PROFILE_MAP)	Journal Issuer Profile Relationship Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
issuertransallowed	Issuer Transactions Allowed data source (ISS_TXN_ALLOWED)	Issuer Transactions Allowed Profile Configuration window
JcbAcquirers	JCB Institution data source (JCB_Institution)	JCB Acquirers Configuration window
jcbinterface	JCB Interface data source (JCB_INTERFACE)	JCB tab of the JCB Interface Configuration window
journalprofilegroupassign	Journal Profile Group Assign data source (JRNL_PF_GRP_ASGN)	Data Source Assigns section of the Journal Profile Configuration window
journalprofilegroup	Journal Profile Group data source (JRNL_PF_GRP)	Journal Profile table section of the Journal Profile Configuration window
journaltdesuppression	Journal Transaction Data Element Suppression data source (Journal_Data_Suppression)	Journal TDE Suppression Configuration window
limit	Limits data source (Limits)	Limit Profile Management window
linkedcurrency	Linked Currency data source (Linked_Currency)	Linked Currency Configuration window
lis5interface	LIS5 (LINK) Interface Configuration data source (LIS5_Interface)	LINK tab of the LINK ISO Interface (LIS5) Configuration window
loadtimerprofile	ATM Channel Load Timeout data source (LTMRPRFL)	ATM (Java) > ATM Load Timeout Profile Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
macprofile	ATM MAC Profile data source (MACPRFL)	ATM (Java) > ATM MAC Profile Configuration window
mdsbrandrel	MasterCard Brand/Instrument Type Relationship data source (MDS_BRAND_INSTRM_REL)	MasterCard Brand/Instrument Type Relationship Configuration window
mdsinterface	MasterCard Debit Switch Interface data source (MDS_Interface)	MDS tab of the MasterCard Debit Switch (MDS) Interface Configuration window
merchantchannelprofile	Merchant Delivery Channel Profile data source (Merchant_Delivery_Channel_Profil)	Merchant Channel Profile Configuration window
merchantchannel	Merchant Delivery Channel data source (Merchant_Delivery_Channel)	Merchant Channel Configuration window
merchantinstrmtype	Merchant Instrument Type Relation data source (Merch_Instr_Typ_Relation)	Instrument Types tab of the Merchant Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
merchant	Merchant data source (Merchant)	Merchant Configuration window
		Voice Authorization Acquirer Transaction web page
		Voice Authorization Acquirer Override web page
		Voice Authorization Code 10 Acquirer web page
netwkbrandreln	Network Brand Instrument Relationship data source (NET_BRND_IM_RELN)	Interface-specific Network ID/Instrument Type Relationship Configuration window
nyceacquirer	NYCE Institution data source (NYCE_INSTITUTION)	NYCE Acquirers Configuration window
nyceinterface	NYCE Interface data source (NYCE_INTERFACE)	NYCE tab of the NYCE ISO Interface Configuration window
oltpwarmboot ¹	OLTP Warmboot data source (OLTP_Warmboot)	OLTP Warmboot Management window
operid	Operator ID data source (OPERID)	ATM (Java) > ATM Diebold Channel Operator ID Configuration window
optionalreceiptprofile	Optional Receipts Profile data source (OPTRCPT)	ATM (Java) > ATM Optional Receipt Profile window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
positivebalance	Positive Balance data source (Positive_Balance)	Positive Balance Management window
		Voice Authorization Acquirer Override web page
preauthhold	Preauthorization Hold data source (Pre_Auth_Hold)	Preauthorization Hold Management window
prefixonus	Prefix data source (Prefix)	On-Us Prefix Configuration window
prminterface	PRM Interface data source (PRM_Interface)	PRM Credit/Debit, Enterprise Risk, or Merchant tabs of the Proactive Risk Management (PRM) Credit/Debit, Enterprise Risk, or Merchant Near-Real-Time (NRT) or Real-Time (RT) Interface Configuration window
pulseinstitution	Pulse Institution data source (PULSE_INSTITUTION)	Pulse Institution Configuration window
pulseinterface	Pulse Interface data source (PULSE_INTERFACE)	Pulse tab of the Pulse Interface Configuration window
realtimefeedinterface	Real Time Feed Configuration data source (Real_Time_Feed_Config)	Real Time Feed tab of the Real Time Feed Interface Configuration window
receipt	Receipt data source (RECEIPT)	ATM (Java) > ATM Receipt Format Profile Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
receiptrequired ¹	Receipt Required data source (RCPTRQRD)	ATM (Java) > ATM Receipt Required Type Configuration window
referral	Referral data source (REFERRAL)	Voice Authorization Issuer Override web page
		Voice Authorization Acquirer Override web page
route	Route data source (ROUTE)	Routing Configuration - Destination window
secdevcfg ¹	Security Device Configuration data source (Security_Device_Config)	Security Device Configuration window
spdhacd	SPDH Action Code Description data source (SPDH_act_cde_descr)	Alternate Descriptions tab of the SPDH Device Profile Configuration window
spdhconfiguration	SPDH Configuration data source (SPDH_CONFIG)	Message Profile > SPDH tab of the Merchant Channel Profile Configuration window
spdhmessage	SPDH Message data source (SPDH_Message)	Message Configuration tab of the SPDH Device Profile Configuration window
spdhoptresp	SPDH Optional Response data source (SPDH_opt_resp_data)	Optional Response Data tab of the SPDH Device Profile Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
spdhtcd	SPDH Transaction Code Description data source (SPDH_txn_cde_descr)	Alternate Descriptions tab of the SPDH Device Profile Configuration window
starinstitution	STAR Institution data source (Star_Institution)	STAR Institution Configuration window
starinterface	STAR Interface data source (Star_Interface)	Star Processing tab of the STAR ISO Interface Configuration window
starnetworkrel	STAR NE Network Identifier/Instrument Type Relation data source (STAR_Network_Rel)	STAR NE Network Identifier/Instrument Type Relationship Configuration window
starnetworkrel	STAR Network Identifier/Instrument Type Relation data source (STAR_Network_Rel)	STAR Network Identifier/Instrument Type Relationship Configuration window
statementprofile	ATM Statement Print Profile data source (STMPPRFL)	ATM (Java) > ATM Statement Print Profile Configuration window
StopPayment	Stop Payment data source (Stop_Payment)	Stop Payment Management window
surchargeprofile	ATM Surcharge Profile data source (SRCHPRFL)	ATM (Java) > ATM Surcharge Profile Configuration window
Surcharge	Surcharge data source (Surcharge)	Surcharge Profile Configuration window
surcountrycode	Surcharge Country Code data source (Surcharge_Country_Code)	Surcharge Country Code Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
systemalgorithm	System Algorithm Configuration data source (System_Algorithm_Config)	System Algorithm Configuration window
systemprefix	System Prefix data source (System_Prefix)	System Prefix Configuration window
terminallnfiidmapping	Terminal Logical Network FIID Map data source (TERM_LN_FIID_MAP)	Terminal Logical Network/FIID Mapping window
terminalpreviousdaytotals	Previous Day ATM Terminal Data Dynamic data source—general data (ESAPDT)	Previous Days Totals tab of the ATM (Java) > ATM Channel Configuration window
tokenid	BASE24 Token ID List data source (Token_Id_List)	Not Applicable
tokenprofile	Token Configuration data source (Token_Config)	Token Profile Configuration window
totalsconfiguration	Totals Configuration data source (TOTALS_CONFIGURATION)	Totals tab of the interface-specific Configuration window (e.g., BIC ISO (87) Interface, SAMA SPAN 2 Interface)
totals	Totals data source (TOTALS)	Interface Totals window
totalstagdocumentation	Totals Tag Documentation data source (TOTALS_TAG_DOCUMENTATION)	Totals Tag Documentation Configuration window
tracecfg ¹	Trace Configuration data source (Trace_Config)	Trace Configuration window

Database Table Name	Data Source	ACI Desktop Window or Browser Web Page
tracefilter ¹	Trace Filter data source (Trace_Filter)	Trace Filter Profile Configuration window
transportkey	Transport Key data source (TRANSPORT_KEY)	PIN Transport Key Profile Configuration window
usage	Usages data source (Usage)	Usages Management window
Usage	Usage data source (Usage)	Usages Management window
userperusalprofilerelationship	User Perusal Profile Relationship data source (USER_PRSL_PF_RLN)	User Profile Relationship window
visabrandrel	Visa Brand/Instrument Type Relationship data source (VISA_BRAND_INSTRM_REL)	Visa Brand/Instrument Type Relationship Configuration window
visadpsinstitution	Visa DPS Institution data source (Visa_DPS_Institution)	Visa DPS Acquirers Configuration window
visadpsinterface	Efunds Interface data source (EFunds_Interface)	Visa DPS tab of the Visa DPS Interface Configuration window
visainstitution	Visa Institution data source (Visa_Institution)	Visa Acquirers Configuration window
visainterface	Visa Interface data source (Visa_Interface)	Visa Processing tab of the VisaNet ISO Interface Configuration window

¹ Available for BASE24 customers using the Transaction Security Services process.

db.cached.*table-name* — Indicates whether the data source indicated by the physical database table name, represented by *table-name*, is accessed from cache memory or from disk. Valid values are as follows:

true = Yes, perform caching for this database table.
false = No, do not perform caching for this database table. Default.

Refer to the [db.audited.*table-name*](#) property description in this appendix and the [db.audited.*table-name*](#) property description in section 5 for a complete list of database tables for the Java User Interface servlets and browser pages and the data sources they access.

ACI recommends that the following db.cached.*table-name* properties be set as indicated in the APPLCNFG to support caching and auditing of the USEC data sources tables as a result of in-process (onboard) USEC processing.

```
db.cached.applcnfg=true
db.cached.event_document=true
db.cached.event=true
db.cached.extcnct=false
db.cached.fltxrole=true
db.cached.fields=true
db.cached.fldvals=true
db.cached.flxfldvl=true
db.cached.filters=true
db.cached.functs=true
db.cached.language=false
db.cached.listener=false
db.cached.permdetl=true
db.cached.permgrp=true
db.cached.pmgpxprm=true
db.cached.perms=true
db.cached.process=false
db.cached.pswdgrp=false
db.cached.pswdhist=false
db.cached.rolxpmgp=true
db.cached.roles=true
db.cached.secgrp=false
db.cached.taskflds=false
db.cached.taskfncs=true
db.cached.tasks=true
db.cached.usrxfilt=true
db.cached.usrxrole=true
```

```
db.cached.users=false  
db.cached.usrstat=false  
db.cached.usraulog=false
```

If this property is not defined for a database table, it defaults to a value of false.

Note: The caching mechanism of the AJSF is completely independent from, and should not be confused with, external memory tables used in online transaction processing (OLTP).

Required: No
Example Entry: db.cached.applcnfg=false
Usage: Cache Manager

db.char.set.ebcdic — Indicates the EBCDIC encoding scheme (in a host code page) to use when BASE24-eps is installed on an IBM System z platform. This attribute is used in conjunction with the CCSID_VARIANTS Environment attribute to define the IBM Coded Character Set Identifier (CCSID) variants supported by ATM Device Handler components. For more information on the CCSID_VARIANT attribute, refer to the ***BASE24-eps Environment Management User Guide***. For all other platforms, this property should be set to None. Valid values are as follows:

None = No, do not use EBCDIC encoding.
CP1047 = Yes, use EBCDIC encoding specified in host code page 1047.

Required: Yes
Example Entry: db.char.set.ebcdic=CP1047

db.upshiftdata — Indicates whether the Java application supports the automatic upshifting of string data. Upshifting of string data is handled at the application level when you are using Enscribe for VSAM data sources. In this case, you should set this property to true. If you are using an SQL database, the database may handle upshifting requirements, in which case, this property can be set to false. Valid values are as follows:

true = Yes, the Java application automatically upshifts string data.
false = No, the Java application does not automatically upshift string data.

Required: Yes
Example Entry: db.upshiftdata=false

db.sqlmp — A flag indicating whether the database used in the current JVM executing environment is NonStop SQL/MP. Valid values are as follows:

true = Yes, the database is NonStop SQL/MP.
false = No, the database is not NonStop SQL/MP.

Required: No
Example Entry: db.sqlmp=true

elmtmap.jar.aci.es-elmtmaps — The name of the JAR file containing BASE24-eps element to data validation map properties files.

Required: Yes
Example Entry: jars/es-elmtmaps.jar
Usage: Data Validation Manager

initialization.function<text-string> — Specifies the fully-qualified Java package and class name to be started at initialization by the Initialization Manager, where <text-string> is a unique text string for this initialization.function property name. These properties should be defined in the APPLCNFG. Beginning with AJSF 07.4, initialization function properties are only loaded from the APPLCNFG and not from the config.properties file.

Required: Yes
Example Entries: initialization.function.0.classname=es.server.framework.
TransactionMapManager
initialization.function.1.classname=es.server.common.
DataValidatorManager
initialization.function.2.classname=es.server.formatter.xml.
MessageSchemaManager
initialization.function.3.classname=com.aciworldwide.
application.security.USECManger
initialization.function.4.classname=com.aciworldwide.
framework.pools.SAX2ParserManager
initialization.function.5.classname=com.aciworldwide.
application.formatter.jaxb.JAXBContextManager
initialization.function.6.classname=com.aciworldwide.
common.crypto.hsm.HSMService
Usage: InitializationManager

jsf.jaxb.formatter.formatted.output — A flag indicating whether the Java User Interface Servlets use the JAXB XML parser to format (i.e., clean up) XML response output for user audit reporting. Although there are no programmatic requirements for this property to be set to true, you may want to set this property to true to make user audit reports more readable. If this property is not configured or invalid, it defaults to false. Valid values are as follows:

true = Yes, parse the XML output for user audit reporting.
false = No, do not parse the XML output for user audit reporting. Default.

Required: Yes, for user audit reporting.
Example Entry: jsf.jaxb.formatter.formatted.output=true
Usage: User Audit Reporting

jsf.pci.masking.jaxb.context — the Java API for XML binding (JAXB) context path for PCI field masking.

Required: Yes, for PCI field masking.
Example Entry: jsf.pci.masking.jaxb.context=com.aciworldwide.application.
pci.masking.formatter.jaxb
Usage: PCI masking

LogChannelTimeouts — A flag indicating whether the UAUD application generates error event messages when an audit record it sends to the Java User Interface Servlets for logging to the User Audit Log data source (USRAULOG) is not responded to within the allotted time limit. Valid values are as follows:

true = Yes, generate error event messages for late responses to UAUD messages. Default
false = No, do not generate error event messages for late responses to UAUD messages.

Required: No
Example Entries: LogChannelTimeouts=false
Usage: UAUD application

ptc.connectionID — Identifies the unformatted connection ID to use for straight pass-through processing of XML request and response messages. The value of this property must match the name of an external connection ID in the EXTRCNCT that uses a null formatter class.

The Java User Interface servlets support a pass-through component that allows external entities to have work combined within workflow guidelines of other BASE24-eps-based components. Additionally, for customization, the pass-through component class can be extended to provide enhanced data handling requirements.

Required: No

Example Entry: `ptc.connectionID=BASE24-eps/C++XMLSERVER`

ptcf.connectionID — Identifies the formatted connection ID to use for pass-through processing of XML request and response messages. The value of this property must match the name of an external connection ID in the EXTRCNCT that uses a valid formatter class.

The Java User Interface servlets support a pass-through component that allows external entities to have work combined within workflow guidelines of other BASE24-eps-based components. The pass-through component allows request messages to be formatted and response messages to be parsed to and from external entities when the original XML request is not in the required format for those entities. Additionally, for customization, the pass-thru component class can be extended to provide enhanced data handling requirements.

Required: No

Example Entry: `ptc.connectionID=BASE24-eps/C++XMLSERVER
FORMATTED`

timertask.function<text-string> — The name of a class to be executed by the UI Timer process, where <text-string> is a unique text string for this timertask. function property name. Each class designates cleanup tasks or the automatic reporting tasks to be performed for the AJSF application. Additional properties must be configured in conjunction with most timed tasks to be performed. These properties are identified in parentheses following each example timer task class entry.

Required: Yes

Example Entries: `timertask.function.0.classname=com.aciworldwide.common.timertask.UserAuditLogCleanup` (see also the following property descriptions: `cleanup.userauditlog.retention.days`, `cleanup.userauditlog.cleanup.starttime`, `cleanup.userauditlog.timeperiod.hours`, `cleanup.userauditlog.batch.size`)

`timertask.function.1.classname=com.aciworldwide.common.timertask.UserAuditReport` (see also the following property descriptions: `report.userauditlog.offset.days`, `report.userauditlog.report.starttime`, `report.userauditlog.loc`, `report.userauditlog.name`, and `xml.parser.validate.document`)

`timertask.function.2.classname=com.aciworldwide.application.security.timertask.UserPermissionsReport` (see also the following property descriptions: `report.userpermissions.loc`, `report.userpermissions.name`, `report.userpermissions.starttime`, `report.userpermissions.timeperiod.hours`)

`timertask.function.3.classname=com.aciworldwide.application.security.timertask.UserDetailReport` (see also the following property descriptions: `report.userdetail.name`, `report.userdetail.timeperiod.hours`, `report.userdetail.starttime`, `report.userdetail.loc`)

Usage: Timer Task Manager

timertask.manager.processid — The process ID of the BASE24-eps Java User Interface servlets (Web Server) process designated to run timed tasks only. This process can be referred to as the UI Timer process, which performs the functions specified in the `timertask.function<text-string>` properties.

Required: Yes

Example Entry: `timertask.manager.processid=JUITIMR1`

Usage: Timer Task Manager

trace.dataobject.enabled — A flag indicating whether trace messages are enabled for data object output from the Java server process. Tracing of messages is configured within the `TraceConfig.xml` file that is specified by the `trace.log4j.config.url` property. When this property is set to true, the `MessageSchema` class dumps traces of the contents of the primary data object during request message processing and response message building.

Note: The Trace facility was designed as a diagnostic tool for development and test environments and not for general use in a production environment. ACI strongly recommends that it not be used in a production environment due to the overhead and impacts on performance.

Required: No
Example Entry: `trace.dataobject.enabled=false`
Usage: Primary data object

tranmap.jar.aci.tranmap-name — The relative URI location of a Transaction Map XML JAR file, where *tranmap-name* is the name of the XML JAR file without the .jar file extension. A separate property is required for every Transaction Map XML JAR file used. To meet performance requirements, the TransactionMapManager preloads all information contained in this configuration JAR file into individual TransactionMapField objects, with each object representing an <entry> from the configuration file. The value of the config.home property is used to resolve the relative URI location to an absolute URI or file specification.

Required: Yes
Example Entries
`tranmap.jar.aci.atm-transmaps=jars/atm-transmaps.jar`
`tranmap.jar.aci.es-transmaps=jars/es-transmaps.jar`
Usage: Transaction Map Manager

va.bin.n — The first variable number of digits of a primary account number (PAN) used to identify a MasterCard bank identification number (BIN) or prefix, where *n* is a unique number used to differentiate between multiple MasterCard BIN properties. The Voice Authorization Card Status function uses the value of this property to determine the MasterCard-assigned issuer customer ID number or ICA (specified in a *va.bin.n.id* property) for a given BIN (or prefix) as follows. Starting with 1, the function attempts to find property *va.bin.1*. If it exists and its value is not empty, then the function checks whether the card number (PAN) starts with this value. If it does, then the function finds *va.bin.1.id* property; if this property exists and its value is not empty then the function uses this value as the MasterCard customer ID. The function continues looking for *va.bin.n* properties in increasing order (2,3...), until it finds the MasterCard customer ID or until a *va.bin.n* property does not exist or has an empty value. Note that the MasterCard customer ID is mandatory. Without it, the Banknet MCC103 file update will fail.

Required: Yes

Example Entries: va.bin.1=54
va.bin.1.id=123456
va.bin.2=52
va.bin.2.id=234567
va.bin.3=53
va.bin3.id=345678
va.bin.4=54
va.bin.4.id=456789
va.bin.5=55
va.bin.5.id=567891

Usage: Voice Authorization

va.bin.n.id — The six-digit bank MasterCard-assigned issuer customer identification number (ICA) associated with a particular MasterCard BIN (specified in a va.bin.n), where *n* is a unique number used to differentiate between multiple ICA properties. The Voice Authorization Card Status function uses the value of this property to determine the MasterCard-assigned issuer customer ID number or ICA (specified in a va.bin.n.id property) for a given BIN (or prefix) as described for the va.bin.n property above.

Required: Yes

Example Entries: va.bin.1=54
va.bin.1.id=123456
va.bin.2=52
va.bin.2.id=234567
va.bin.3=53
va.bin3.id=345678
va.bin.4=54
va.bin.4.id=456789
va.bin.5=55
va.bin.5.id=567891

Usage: Voice Authorization

va.ccv.properties — The location of the Common Code Values (CCV) properties file (ccv.properties) for voice authorization. This properties file defines common code values such as currency codes, card acceptor business codes, country codes, action codes, instrument types, and result codes used by the Voice Authorization application.

Required: Yes
Example Entry: \${config.home}/VAConfig
Usage: Voice Authorization

va.connection.hiso — The name of the external connection ID used to connect to the BASE24-eps ISO '93 Host Interface component in the Integrated Server process. The value of this property must match an external connection ID record on the External Connection Configuration window.

Required: Yes
Example Entry: VOICE-AUTH-INTF
Usage: Voice Authorization

va.connection.esweb — The name of the external connection ID used to connect to the BASE24-eps User Interface (ESWEB) server. The value of this property must match an external connection ID record on the External Connection Configuration window.

Required: Yes
Example Entry: ESWEB
Usage: Voice Authorization

va.default.acquirer.inst.id.code — The default acquirer institution ID code for the Voice Authorization application. This code identifies the acquiring institution for the transaction, or its agent.

Required: Yes
Usage: Voice Authorization

va.default.card.acceptor.terminal.id — The default card acceptor terminal ID code for the Voice Authorization application. This is a unique code identifying the terminal at the card acceptor location.

Required: Yes
Usage: Voice Authorization

va.default.currency.code — The default currency code for the Voice Authorization application. This code defines the default currency code used in currency code selectable list fields on Voice Authorization web pages.

Required: Yes
 Example Entry: 840
 Usage: Voice Authorization

va.default.forwarding.inst.id.code — The default forwarding institution ID code for the Voice Authorization application. This code identifies the forwarding institution (or service provider).

Required: Yes
 Usage: Voice Authorization

va.default.query.detail.name — The default journal query name to be used to look up the referred transaction in the Journal.

Required: Yes
 Example Entry: VA_QUERY_DETAIL
 Usage: Voice Authorization.

va.min.year — The minimum year used for validating year entries on the Voice Authorization web pages.

Required: Yes
 Example Entry: 1900
 Usage: Voice Authorization

va.max.year — The maximum year used for validating year entries on the Voice Authorization web pages.

Required: Yes
 Example Entry: 2100
 Usage: Voice Authorization

website.helpURL — The URL for Voice Authorization web page help.

Required: Yes
 Example Entry: http://{host}:{port}/{server}/help/help.html
 Usage: Voice Authorization

xml.msgschema.jar.aci.schema-name — The relative URI location of an XML Schema Definition (XSD) JAR file, where *schema-name* is the name of the JAR file without the .jar file extension. A separate property is required for every XSD JAR file used. The MessageSchemaManager uses the specified JAR file at initialization to locate and preload request and response message schema definitions. The value of the config.home property is used to resolve the relative URI location to an absolute URI or file specification.

Required: Yes

Example Entries:

xml.msgschema.jar.aci.atm-schemas=jars/atm-schemas.jar

xml.msgschema.jar.aci.es-schemas=jars/es-schemas.jar

Usage: Message Schema Manager

xml.parser.validate.document — A flag indicating whether the BASE24-eps Java User Interface servlets validate XML documents while parsing them.

true = Yes, validate XML documents while parsing them.

false = No, do not validate XML documents while parsing them. Default.

Required: No

Example Entry: xml.parser.validate.document=false

Usage: User Audit Reporting

Sample Config.properties File

The following is an example of the config.properties file used for the ESWEB server process. This file may need to be edited during installation for site-specific information.

```
#
# Database properties
#
db.max.connections=10

#
# Database {user ID} and {password}. Usage is determined by the JDBC database
# driver.
#
# {user id} = user ID for application access to the database.
# {password} = corresponding password for the user ID. If a db.user.password
# value is specified, it must be encrypted.
#
#
db.user.id=pserve
db.user.password=c67834a7b2e28c4e

#
# {driver specification} - should be set according to JDBC driver requirements
# and
# may include information such as:
#
# - host or IP address where database resides
# - port to connect to for database
# - name of database on the host
# - other JDBC driver settings
#
db.datasource=jdbc:JDAL://172.19.96.18:20726

#
# db.static.connections defines the number of connections to acquire to the
# database
# when the jvm starts. These connections will stay active through the life of
# the
# executing JVM.
#
db.static.connections=2

#
# db.connection.wait.time defines the amount of time, in milliseconds, to wait
# for
```

```
# another thread to free a Database connection in the Database connection
manager.
# This is used when the maximum number of connections have been acquired as
specified
# in the db.max.connections property.
#
db.connection.wait.time=30000

#
# db.useisolationlevel indicates that the database services will automatically
manipulate
# connection transaction isolation levels. This should be set to true for
Microsoft
# SQL Server and IBM's DB2 databases. It should be set to false for Oracle and
Tandem SML/MP
# databases. When set to true database access will automatically be set to
reduce
# database contention on queries where reading through locks is valid.
#
db.useisolationlevel=true

#
# db.driver specifies the class name of the JDBC driver to use for all database
connections.
# The whole package name as it is found in the jar file containing the driver
should be specified.
#
# {driver package} - should be set to the Java package specification of the
JDBC driver being used
#
db.driver=com.aciworldwide.jdbcdriver.jdal.JDALDriver
db.sqlmp=false
db.type=JDAL

#
# Error Manager properties. If the filename contains the characters
# "YYYYMMDD" in its name this will cause the ErrorManager to roll log
# files on a daily basis. If these characters are not found in the name
# the same log file will remain open.
#
# LogManager properties.
# The Log4J config file location for trace and event
#
# DOS specifier - change {root} to drive: or drive:/{other directory} to
# be prefixed to value below (Example - C: or C:/ESWeb)
#
# UNIX specifier - change {root} to root location to be prefixed to value
# below (Example - db01, db01/ESWeb)
#
# {root} setting should be same as that for TraceConfig.xml below.
#
```

```
event.config=/user/esbcoss/bct_092_1/ESWeb/ESConfig/BC91/P91^NODE/EventConfig.xml
```

```
#
# Trace processing properties.
#
# DOS specifier - change {root} to drive: or drive:/{other directory} to
#                 be prefixed to value below (Examples - C:, C:/ESWeb)
#
# UNIX specifier - change {root} to root location to be prefixed to value
#                 below (Examples - db01, db01/ESWeb)
#
# {root} setting should be same as that for EventConfig.xml above.
#
trace.config=/user/esbcoss/bct_092_1/ESWeb/ESConfig/BC91/P91^NODE/TraceConfig.xml
#trace.enabled=false
```

```
#
# The process id of the one process that is to run
# the cleanup tasks.
#
timertask.manager.processid=BASE24-ESUI
#timertask.function.3.classname=es.server.timertask.ContingencySAFProcess
```

```
#
# Messaging properties.
#
#
# Valid values for message.environment are:
#   DIRECT
#   JMS
#
#message.environment=JMS
#
#   OPENJMS
#
#message.jms.jndi.initialcontextfactory=com.sun.jndi.fscontext.
RefFSContextFactory
#message.jms.jndi.providerurl={jms_url}
#message.jms.QueueConnectionFactory=JmsQueueConnectionFactory
#message.jms.TopicConnectionFactory=JmsTopicConnectionFactory
#
#   XPNET with embedded JNDI
#
#message.jms.jndi.initialcontextfactory=com.aci.jms.jndi.embedded.
EmbeddedXpnetInitialContextFactory
#message.jms.jndi.providerurl=jms://dummy
#message.jms.QueueConnectionFactory=JmsQueueConnectionFactory
#message.jms.TopicConnectionFactory=JmsTopicConnectionFactory
#
```

```
# The following MUST be set to true if running on the HP NSK platform
# due to how it handles threading.
#
#message.jms.use.nsk.threads=false
SSL.client.truststore=/user/esbcoss/bct_092_1/ESWeb/ESConfig/eswebclient.
truststore.key
SSL.client.truststore.password=f41aa10b62c1bb4401d2e7728a898538
context.type=DISK-WITH-CACHE
```

ACI Worldwide, Inc.

B: BASE24-eps ATM (Java) Device Handler and IFX ATM Manager Processes

This appendix describes the third-party JARs and ACI Java Server Framework (AJSF) properties specific to BASE24-eps ATM (Java) Device Handler or IFX ATM Manager processes.

Third-Party JARs

This section describes the third-party JARs used exclusively by BASE24-eps ATM (Java) Device Handler and IFX ATM Manager processes.

Jakarta-ORO

The IFX ATM Manager process uses the Jakarta-ORO Java classes, which are a set of text-processing Java classes that provide the following:

- Perl5 compatible regular expressions
- Aho, Weinberger, and Kernighan (AWK) text-processing program language-like regular expressions
- Glob expressions
- Utility classes for performing substitutions, splits, filtering filenames, etc.

These Jakarta-ORO Java classes are provided in a JAR named `jakarta-oro-vernum.jar`, where *vernum* is the version number.

Glob expressions are a type of string that can be compared and matched against several other strings using special characters such as `*` and `?`. An `*` can match zero or more characters of any kind, while `?` can match one, and only one, character of any kind. All other characters match themselves.

Properties

The following application properties configured in the config.properties file or the APPLCNFG are specific to the BASE24-eps ATM (Java) Device Handler or IFX ATM Manager processes. Some of the properties defined depend on the platform on which BASE24-eps is installed.

In some property names, the following values indicate whether a property is specific to a device type or contains a default value.

01 = NCR NDC+
02 = Diebold 911/912
03 = IFX
Dflt = Default

amptc.connectionID — Identifies the external HTTP connection to which requests for the Application Management Agent are sent. The value of this property must match an external ID configured in the External Connections data source (EXTRCNCT).

Required: Yes, if using the Application Management Agent
Example Entry: APPLICATION-MGMT
Usage: BASE24-eps Java User Interface servlets

ATMCHNLADMNPTC.connectionID — Identifies the queue on which channel administrative requests are to be placed. The value of this property must match an external ID configured in the External Connections data source (EXTRCNCT).

Required: Yes
Example Entry: ATM_CHANNEL_ADMIN
Usage: BASE24-eps Java User Interface servlets

ATMCHANNEL.connectionID.01 — Identifies the name of the connection ID to be used from the External Connections data source (EXTRCNCT) for ATM terminals using the NCR NDC+ message format. Messages other than administrative messages from the BASE24-eps ATM (Java) Device Handler process are sent to this connection.

Required: Yes, if using NCR NDC+ terminals.
Example Entry: NCR-JMS
Usage: BASE24-eps Java User Interface servlets

ATMCHANNEL.connectionID.02 — Identifies the name of the connection ID to be used from the External Connections data source (EXTRCNCT) for ATM terminals that use the Diebold 911 or 912 message format. Messages other than administrative messages from the BASE24-eps ATM (Java) Device Handler process are sent to this connection.

Required: Yes, if using Diebold 911/912 terminals.
Example Entry: DIEBOLD-JMS
Usage: BASE24-eps Java User Interface servlets

ATMCHANNEL.connectionID.03 — Identifies the name of the connection ID to be used from the External Connections data source (EXTRCNCT) for IFX terminals. Messages other than administrative messages from the BASE24-eps ATM IFX ATM Manager process are sent to this connection.

Required: Yes, if using IFX terminals.
Example Entry: IFX-JMS
Usage: BASE24-eps Java User Interface servlets

ATMRELOAD.connectionID — Identifies the name of the connection ID to be used from the External Connections data source (EXTRCNCT) for requests to reload XML configuration documents into memory.

Required: Yes
Example Entry: ATM-RELOAD
Usage: BASE24-eps Java User Interface servlets

ca.confirmation.required.for.verification — A flag indicating whether confirmation messages to the certificate authority are required for new signature verification key certificates generated during certificate rollover. Valid values are as follows:

true = Yes, confirmation messages to the certificate authority are required for new signature verification key certificates generated during certificate rollover.

false = No, confirmation messages to the certificate authority are not required for new signature verification key certificates generated during certificate rollover. Default.

Required: No

Usage: IFX ATM Manager process

ca.confirmation.saf.max.rowcount — The maximum number of rows to retrieve from the Certificate Authority Confirmation SAF (CACSAF) in one fetch operation. This property defaults to 10.

Required: No

Usage: IFX

ca.confirmation.saf.msg.retries — The number of times the IFX ATM Manager process attempts to send a confirmation message to the certificate authority before the message is written to the Exception Log data source (Exception_Log). This property defaults to 10 retries.

Required: No

Usage: IFX

ca.confirmation.saf.run.delay.minutes — The number of minutes to delay between runs of the Certificate Authority Confirmation SAF Timer Task. This property defaults to 10 minutes.

Required: No

Usage: IFX

ca.confirmation.saf.send.interval — The number of seconds the Certificate Authority Confirmation SAF Timer Task delays between sending confirmation messages to the certificate authority. This property defaults to 0.25 seconds.

Required: No
Usage: IFX

ca.confirmation.saf.start.delay.minutes — The number of minutes to delay the start of Certificate Authority Confirmation SAF processing after the Certificate Authority Confirmation SAF process is started. This property defaults to two minutes.

Required: No
Usage: IFX

ca.message.queue.externalconnectionid — The ID of the external connection that the IFX ATM Manager process uses to send request commands to the certificate authority. This property defaults to CARQST.

Required: No
Usage: IFX

datatype.jar.aci.data-types — Identifies the relative URI location of a data types JAR file, where *data-types* is ATMDHXMLDataTypes, ATMXMLDataTypes, or ESXMLDataTypes. More than one data types JAR file may exist.

Required: Yes
Example Entries: lib/atmdh-datatypes.jar
lib/atm-datatypes.jar
lib/es-datatypes.jar

dataValidation.schema.ATMXMLDataTypes — Identifies the relative URI location of BASE24-eps ATM (Java) Device Handler data type schema definition files. More than one data type schema definition file may exist.

Required: Yes
Example Entry: schemas/lib/ATMXMLDataTypes.xsd

db.cached.*table-name* — Indicates whether the data source indicated by the database table name, represented by *table-name*, is accessed from cache memory or from disk. Valid values are as follows:

true = Yes, this data source is accessed from cache memory.

false = No, this data source is not accessed from cache memory.

table-name values used by the BASE24-eps ATM (Java) Device Handler or IFX ATM Manager processes are shown in the following table.

ACI recommends that all of these data sources be cached.

Database Table Name	Data Source	ACI Desktop Window
acqtxnal	Acquirer Transactions Allowed data source (Acquirer_Txn_Allowed)	Acquirer Transactions Allowed Configuration window
atds1	BASE24-eps ATM Terminal Data Static data source—general data (ESATDS1)	Location and Processing tabs of the ATM Channel Configuration window
atdd1	ATM Terminal Data Dynamic data source—general data (ESATDD1)	Hopper and Terminal Activity tabs of the ATM Channel Configuration window, ATM Channel Management windows, and the ATM Channel Administration window
configprofile	ATM Device Handler Configuration data source (ATDCNFG)	ATM Channel Profile Configuration window
externalconnection	External Connections data source (EXTRCNCT)	External Connection Configuration window
hopperprofile	Hopper Profile data source (HPRPRFL)	ATM Hopper Profile Configuration window
inst	Institution data source (Institution)	Institution Configuration window

Database Table Name	Data Source	ACI Desktop Window
listener	Listener data source (LISTENER)	Listener Configuration window
operid	Operator ID data source (OPERID)	ATM Diebold Channel Operator ID Configuration window
optionalreceiptprofile	Optional Receipts data source (OPTRCPT)	ATM Optional Receipt Profile window
receipt	Receipt data source (RECEIPT)	ATM Receipt Format Profile Configuration window
receiptrequired	Receipt Required data source (RCPTRQRD)	ATM Receipt Required Type Configuration

dh.atmlog.cleanup.timeperiod.hours — The number of hours to delay between runs of the ATM Native Message Log Cleanup task to delete old ATM native messages from the ATM Native Message Log data source (ATMLOG). If this property is not configured, it defaults to 24 hours.

Required: No
Example Entry: 48
Usage: ATM Native Message Log Cleanup Timer Task

dh.atmlog.cleanup.starttime — The time of day, in hh:mm format, the ATM Native Message Log Cleanup timer task is to run. If this property is not configured, it defaults to a value of 23:59. This time is in the time of the executing server where the application is running.

Required: No
Example Entry: 01:30
Usage: ATM Native Message Log Cleanup Timer Task

dh.atmlog.cleanup.retention.days — The number of days statement data rows are to be retained in the ATM Native Message Log data source (ATMLOG). Rows older than this retention period are deleted when the ATM Native Message Log Cleanup timer task is run. If this property is not configured, it defaults to seven days.

Required: No
Example Entry: 2
Usage: ATM Native Message Log Cleanup Timer Task

dh.allow.balance.enquiry.reversals — A flag indicating whether balance inquiry transactions are reversed. Valid values are as follows:

true = Yes, balance inquiry transactions are reversed.
false = No, balance inquiry transactions are not reversed. Default.

Required: No
Example Entry: true
Usage: All ATM device types supporting balance inquiry reversal commands

device.outofservice.mode — The default IFX Effective Urgency (<EffUrgency>) value to send in commands to close the ATM. Valid values are as follows:

Immediate = Close the ATM immediately.
CurrentTransactionEnd = Close the ATM after the current transaction is completed.
SessionEnd = Close the ATM after the current session is completed.

Required: No
Usage: IFX ATM Manager process

dh.clear.terminal.totals.on.cutover — A flag indicating whether the ATM Cutover process clears ATM hopper totals and transaction amounts and counts from the BASE24-eps ATM Terminal Data Dynamic data source—general data source (ESATDD1) during terminal cutover processing. Valid values are as follows:

true = Yes, the ATM Cutover process clears totals from the ESATDD1 during terminal cutover processing.
false = No, the ATM Cutover process does not clear totals from the ESATDD1 during terminal cutover processing. Default.

Required: No
Example Entry: false
Usage: ATM Cutover process

dh.cutover.saf.institution.id — The ID of the institution for records that the Cutover SAF Timer Task is to process from the Cutover SAF. An asterisk is a wildcard that indicates to process records for all institutions.

Required: No
Example Entry: *
Usage: Cutover SAF Timer Task

dh.cutover.saf.intra.batch.delay — The number of seconds that the Cutover SAF Timer Task waits between batches. This delay allows control of the thread to be passed to another message.

Required: No
Example Entry: 1
Usage: Cutover SAF Timer Task

dh.cutover.saf.max.rowcount — The maximum number of rows the Cutover SAF process can retrieve from the BASE24-eps ATM Cutover SAF data source (CTVRSF) in a single fetch operation. If this property is not configured, it defaults to 10.

Required: No
Example Entry: 5
Usage: Cutover SAF Timer Task

dh.cutover.saf.msg.retries — The maximum number of times the Cutover SAF process attempts to send a cutover message to the authorizer for logging. If the cutover message cannot be successfully sent after the maximum number of retries is reached, it is written to the Exception Log data source (Exception_Log). If this property is not configured, it defaults to a value of 2.

Required: No
Example Entry: 3
Usage: Cutover SAF Timer Task

dh.cutover.saf.send.interval — The number of seconds the Cutover SAF process waits between sending cutover record messages to the authorizer for logging. A value of 0 indicates that there is no delay between the sending of messages. If this property is not configured, it defaults to a value of .25 seconds.

Required: No
Example Entry: 1
Usage: Cutover SAF Timer Task

dh.cutover.saf.run.delay.minutes — The number of minutes the Cutover SAF process waits between searching the BASE24-eps ATM Cutover SAF data source (CTVRSaf) for cutover rows to be sent to the authorizer for logging. If this property is not configured, it defaults to a value of 15 minutes.

Required: No
Example Entry: 20
Usage: Cutover SAF Timer Task

dh.cutover.saf.start.delay.minutes — The number of minutes to delay the start of cutover SAF processing after the Cutover SAF process is started. If this property is not configured, it defaults to a value of 15 minutes.

Required: No
Example Entry: 1
Usage: Cutover SAF Timer Task

dh.default.action.code — The action code value that is used when the BASE24-eps ATM (Java) Device Handler process cannot retrieve the StateMap entry for the transaction action code.

Required: No
Example Entry: 100
Usage: Diebold 911/912, NCR NDC+

dh.ifx.device.suppress.na — A flag indicating whether the IFX ATM Manager process suppresses the reporting of DevAdvise.DevDetail elements in event messages that are not sent in the message. N/A is reported for these elements if they are not suppressed. Nothing is reported for these elements if they are suppressed. Valid values are as follows:

true = Yes, suppress the reporting of N/A for DevAdvise.DevDetail elements.
Default.
false = No, do not suppress the reporting of N/A for DevAdvise.DevDetail elements.

Required: No
Usage: IFX

dh.ifx.duplicate.checking.mode — A code indicating whether the IFX ATM Manager process checks for duplicate transactions in the RQUID (LONGTERM), ESATDD1 (SHORTTERM), or not at all (NONE). The RQUID can maintain the Request Unique Identifier (RqUID) from the service wrapper of an IFX document for multiple transactions while the ESATDD1 holds the RqUID for the last transaction processed. If this property is not configured, it defaults to a value of LONGTERM. Valid values are as follows:

LONGTERM = Check for duplicates using the RQUID. Default.
SHORTTERM = Check for duplicates using the ESATDD1.
NONE = Do not check for duplicates.

Required: No
Usage: IFX

dh.ifx.open.close.supported — A flag indicating whether the IFX ATM Manager process supports open and close messages. Valid values are as follows:

true = Yes, the IFX ATM Manager process supports open and close messages.
false = No, the IFX ATM Manager process does not support open and close messages. Default.

Required: No
Usage: IFX

dh.dkm.certificate.authority.exists — A flag indicating whether a certificate authority is configured to process automated certificate rollovers. The IFX ATM Manager process does not attempt to send certificate rollover messages unless this property is set to true. Certificate rollover messages are also not sent for IFX devices that do not support certificate rollover. Valid values are as follows:

true = Yes, a certificate authority is configured to process automated certificate rollovers.
false = No, a certificate authority is not configured to process automated certificate rollovers. Default.

Required: No
Usage: IFX

dh.dkm.certificate.rollover.retry.count — The number of retries to be attempted if the automatic rollover of a public key certificate fails. This property defaults to five retries.

Required: No
Usage: IFX

dh.dkm.certificate.rollover.retry.delay — The number of minutes to delay between retries if the automatic rollover of a public key certificate fails. This property defaults to two minutes.

Required: No
Usage: IFX

dh.dkm.key.change.interval — The maximum time allowed (in hours) between PIN, MAC, and message data key changes for an ATM terminal. The default value is 0, which indicates that a key change interval is not used. The difference between the current time and the DKM key change timestamp for a terminal can not exceed the value specified by this property without a key change being performed. When a key change is performed, due to a dynamic key change limit or this interval being exceeded, the DKM key change timestamp for the terminal is updated. Valid values for this property are 0–32767. The default value is 0, which indicates that a key change interval is not used.

Required: No
Example Entry: 144
Usage: All ATM device types

dh.dkm.key.change.retry.count — The number of times to retry the Load Working Key Change command in case it fails. This property defaults to five retries.

Required: No
Usage: IFX

dh.dkm.mac.error.limit — The maximum number of MAC errors for an ATM, before both the MAC key and PIN key must be changed in the ATM. If this property is not configured, it defaults to a value of 3. A value of 0 for this property indicates that a MAC error limit is not used. When a dynamic key management limit or interval is exceeded, the BASE24-eps ATM (Java) Device Handler process downloads new MAC and PIN keys to the terminal, resets all dynamic key management counters for the terminal to 0, and updates the key change timestamp for the terminal.

Required: No
Example Entry: 4
Usage: All ATM device types

dh.dkm.mac.consecutive.error.limit — The maximum number of consecutive MAC errors for an ATM, before both the MAC key and PIN key must be changed in the ATM. If this property is not configured, it defaults to a value of 3. A value of 0 for this property indicates that a consecutive MAC error limit is not used. When a dynamic key management limit or interval is exceeded, the BASE24-eps ATM (Java) Device Handler process downloads new MAC and PIN keys to the terminal, resets all dynamic key management counters for the terminal to 0, and updates the key change timestamp for the terminal.

Required: No
Example Entry: 4
Usage: All ATM device types

dh.dkm.mac.transaction.limit — The maximum number of transactions for an ATM for which the MAC is authenticated or generated, before both the MAC key and PIN key must be changed in the ATM. If this property is not configured, it defaults to 100 transactions. A value of 0 for this property indicates that a MAC transaction limit is not used. When a dynamic key management limit or interval is exceeded, the BASE24-eps ATM (Java) Device Handler process downloads new MAC and PIN keys to the terminal, resets all dynamic key management counters for the terminal to 0, and updates the key change timestamp for the terminal.

Required: No
Example Entry: 200
Usage: All ATM device types

dh.dkm.msg.error.limit — The maximum number of message data verification errors for an ATM, before the data encryption communications key must be changed in the ATM. If this property is not configured, it defaults to a value of 3. A value of 0 for this property indicates that a maximum number of message data verification errors limit is not used. When a dynamic key management limit or interval is exceeded, the IFX ATM Manager process downloads a new data encryption communications key to the terminal, resets both data encryption communications key dynamic key management counters for the terminal to 0, and updates the key change timestamp for the terminal.

Required: No
Example Entry: 4
Usage: IFX

dh.dkm.msg.transaction.limit — The maximum number of transactions for an ATM for which message data is verified or encrypted, before both the data encryption communications key must be changed in the ATM. If this property is not configured, it defaults to 100 transactions. A value of 0 for this property indicates that a message data transaction limit is not used. When a dynamic key management limit or interval is exceeded, the IFX ATM Manager process downloads a new data encryption communications key to the terminal, resets both data encryption communications key dynamic key management counters for the terminal to 0, and updates the key change timestamp for the terminal.

Required: No
Example Entry: 200
Usage: IFX

dh.dkm.pin.error.limit — The maximum number of PIN errors for an ATM, before both the MAC key and PIN key must be changed in the ATM. If this property is not configured, it defaults to a value of 3. A value of 0 for this property indicates that a PIN error limit is not used. When a dynamic key management limit or interval is exceeded, the BASE24-eps ATM (Java) Device Handler process downloads new MAC and PIN keys to the terminal, resets all dynamic key management counters for the terminal to 0, and updates the key change timestamp for the terminal.

Required: No
Example Entry: 4
Usage: All ATM device types

dh.dkm.pin.transaction.limit — The maximum number of transactions for an ATM for which the PIN is verified, before both the MAC key and PIN key must be changed in the ATM. If this property is not configured, it defaults to a value of 1000 transactions. A value of 0 for this property indicates that a PIN transaction limit is not used. When a dynamic key management limit or interval is exceeded, the BASE24-eps ATM (Java) Device Handler process downloads new MAC and PIN keys to the terminal, resets all dynamic key management counters for the terminal to 0, and updates the key change timestamp for the terminal.

Required: No
Example Entry: 2000
Usage: All ATM device types

dh.dkm.premature.certificate.rollover — The number of days before a certificate's end of validity date that the automatic rollover of public key certificate will be started. The IFX ATM Manager process subtracts the value of this property from the certificate's end of validity date for the automated certificate rollover timer. By starting the automated certificate rollover process well in advance of the certificate's end of validity date, it allows time for manual intervention in case the automatic rollover fails. This property defaults to seven days.

Required: No
Usage: IFX

dh.dkm.tmk.change.retry.count — The number of times to retry the Load Master Key command in case it fails. This property defaults to five retries.

Required: No
Usage: IFX

dh.generate.reversal.event — A flag indicating whether the BASE24-eps ATM (Java) Device Handler process outputs an event message when it generates a reversal. Valid values are as follows:

true = Yes, the BASE24-eps ATM (Java) Device Handler or IFX ATM Manager process outputs an event message when it generates a reversal.
false = No, the BASE24-eps ATM (Java) Device Handler or IFX ATM Manager process does not output an event message when it generates a reversal. Default.

Required: No
Usage: All device types

dh.inactivity.intra.batch.delay — The number of seconds that the Transaction Inactivity Timer Task waits between batches. This delay allows control of the thread to be passed to another message.

Required: No
Example Entry: 1
Usage: Transaction Inactivity Timer Task

dh.inactivity.max.rowcount — The maximum number of rows the Transaction Inactivity Timer task can retrieve from the BASE24-eps ATM Terminal Data Dynamic data source—general data (ESATDD1) in a single fetch operation. If this property is not configured, it defaults to 10.

Required: No
Usage: Transaction Inactivity Timer Task

dh.inactivity.run.delay.minutes — The number of minutes the Transaction Inactivity Timer task waits between searching the BASE24-eps ATM Terminal Data Dynamic data source—general data (ESATDD1) for ATM terminals that have exceeded their transaction inactivity timer limit. The BASE24-eps ATM (Java) Device Handler process generates event messages and resets the transaction inactivity timer when a transaction has not been initiated from the terminal within the specified time interval. If this property is not configured, it defaults to five minutes.

Required: No
Example Entry: 10
Usage: Transaction Inactivity Timer Task

dh.inactivity.start.delay.minutes — The number of minutes to delay the start of transaction inactivity timer processing after the BASE24-eps ATM (Java) Device Handler process is started. If this property is not configured, it defaults to five minutes.

Required: No
Example Entry: 1
Usage: Transaction Inactivity Timer Task

dh.inactivity.timeout.period — The number of milliseconds used to determine the device inactivity duration if it cannot be calculated using the ATM's configured inactivity periods.

Required: No
Example Entry: 86400000
Usage: Inactivity Timer Tasks

dh.init.tmkchange.after.rollcert — A flag indicating whether the terminal's master key and all working keys are automatically changed and replaced after a public key certificate rollover. The mutual authentication and exchange of public keys is always executed automatically during a rollover. Valid values are as follows:

true = Yes, automatically load a new master key and working keys after a public key certificate rollover. Default.
false = No, do not automatically load a new master key and working keys after a public key certificate rollover.

dh.load.inactivity.intra.batch.delay — The number of seconds that the Load Inactivity Timer Task waits between batches. This delay allows control of the thread to be passed to another message.

Required: No
Example Entry: 3
Usage: Load Inactivity Timer Task

dh.load.inactivity.max.rowcount — The maximum number of rows the Load Inactivity Timer task can retrieve from the BASE24-eps ATM Terminal Data Dynamic data source—general data (ESATDD1) in a single fetch operation. If this property is not configured, it defaults to 10.

Required: No
Usage: Load Inactivity Timer Task

dh.load.inactivity.run.delay.minutes — The number of minutes the Load Inactivity Timer task waits between searching the BASE24-eps ATM Terminal Data Dynamic data source—general data (ESATDD1) for ATM terminals that have exceeded their transaction load inactivity timer limit. The BASE24-eps ATM (Java) Device Handler process generates event messages and resets the load inactivity timer when a transaction has not been initiated from the terminal within the specified time interval. If this property is not configured, it defaults to five minutes.

Required: No
Example Entry: 10
Usage: Load Inactivity Timer Task

dh.load.inactivity.start.delay.minutes — The number of minutes to delay the start of load inactivity timer processing after the BASE24-eps ATM (Java) Device Handler process is started.

Required: No
Example Entry: 1
Usage: Load Inactivity Timer Task

dh.load.inactivity.timeout.period — The number of milliseconds used to determine the device load inactivity duration if it cannot be calculated using the ATM's configured load inactivity periods.

Required: No
Example Entry: 240000
Usage: Load Inactivity Timer Tasks

dh.log.terminal.state.messages — A flag indicating whether the BASE24-eps ATM (Java) Device Handler process logs terminal state messages as events to the BASE24-eps system log. If this property is set to true, the native messages for terminal states (fitness, hardware configuration, supply, and sensor) are output on the event log. Valid values are as follows:

true = Yes, log native terminal state messages as events.
false = No, do not log native terminal state messages as events. Default.

Required: No
Usage: Diebold and NCR

dh.maximum.statement.data.chunk.size — Specifies how many bytes of data the BASE24-eps ATM (Java) Device Handler process can store in a single row in the Statement Print data source (STMTDATA). The maximum size is 4000 bytes. The actual size is determined by the configured size of the StmtData element in meta data. The configured size may be less than 4000 bytes.

Required: No
Usage: All ATM device types supporting statement print transactions

dh.message.queue.externalconnectionid.terminal — The ID of the outbound terminal queue in a generic queue environment.

Required: No
Usage: All ATM device types

dh.message.staletimer.window — The number of milliseconds the IFX ATM Manager process waits for a <SvcProfInqRq> message from the ATM before the message is considered *stale* and dropped. If this property is set to zero, then no messages are considered stale.

Required: No
Usage: IFX

dh.native.message.logging — A flag indicating whether the BASE24-eps ATM (Java) Device Handler process logs all native messages sent to and received from an ATM terminal to the ATM Native Message Log data source (ATMLOG). Valid values are as follows:

true = Yes, native messages are logged to the ATMLOG.
false = No, native messages are not logged to the ATMLOG. Default.

Required: No
Usage: Diebold and NCR

dh.phonecard.amount.value.currency-code.n — An amount to display on the phone card purchase transaction selection screen at the ATM. *Currency-code* specifies the currency in which the amount is displayed and *n* is a number from 1 to 4 that indicates whether the amount is the first, second, third, or fourth choice

displayed. You must define `dh.phonecard.amount.value.currency-code.n` properties for each currency in which amounts can be displayed on the phone card purchase transaction selection screen at the ATM.

Required: Yes, for phone card purchase transactions

Example Entries: `dh.phonecard.amount.value.826.1=165`
`dh.phonecard.amount.value.826.2=440`
`dh.phonecard.amount.value.826.3=825`
`dh.phonecard.amount.value.826.4=1375`
`dh.phonecard.amount.value.840.1=300`
`dh.phonecard.amount.value.840.2=800`
`dh.phonecard.amount.value.840.3=1500`
`dh.phonecard.amount.value.840.4=2500`

Usage: All ATM device types

dh.phonecard.minutes.value.n — The number of minutes to be purchased in a phone card purchase transaction, where *n* indicates the choice (first, second, third, or fourth), this value is associated with on the phone card purchase transaction selection screen at the ATM.

Required: Yes, for phone card purchase transactions

Example Entries: `dh.phonecard.minutes.value.1=10`
`dh.phonecard.minutes.value.2=30`
`dh.phonecard.minutes.value.3=50`
`dh.phonecard.minutes.value.4=120`

Usage: All ATM device types

dh.pin.change.use.ndc.buffer.v — A flag indicating whether the BASE24-eps ATM (Java) Device Handler process uses Buffer V with the ATMulator product. This property enables the BASE24-eps ATM (Java) Device Handler process to work around an encryption error in the ATMulator product with Buffer V and should be set to true for actual NCR ATMs or when the error is fixed. Valid values are as follows:

true = Yes, use Buffer V when present. Default.

false = No, do not use Buffer V when present.

Required: No

Usage: NCR

dh.reversal.saf.intra.batch.delay — The number of seconds that the Reversal SAF Timer Task waits between batches. This delay allows control of the thread to be passed to another message.

Required: No
Example Entry: 1
Usage: Reversal SAF Timer Task

dh.reversal.saf.max.rowcount — The maximum number of rows the Reversal SAF process can retrieve from the BASE24-eps ATM Reversal SAF data source (RVSLSAF) in a single fetch operation. If this property is not configured, it defaults to 10.

Required: No
Example Entry: 5
Usage: Reversal SAF Timer Task

dh.reversal.saf.run.delay.minutes — The number of minutes the Reversal SAF process waits between searching the BASE24-eps ATM Reversal SAF data source (RVSLSAF) for reversal rows to be sent to the authorizer for processing. If this property is not configured, it defaults to 15 minutes.

Required: No
Example Entry: 20
Usage: Reversal SAF Timer Task

dh.reversal.saf.start.delay.minutes — The number of minutes to delay the start of reversal SAF processing after the Reversal SAF process is started. If this property is not configured, it defaults to 15 minutes.

Required: No
Example Entry: 1
Usage: Reversal SAF Timer Task

dh.reversal.saf.msg.retries — The maximum number of times the Cutover SAF process attempts to send a reversal message to the authorizer for processing. If the reversal message cannot be successfully sent after the maximum number of retries is reached, it is written to the Exception Log data source (Exception_Log). If this property is not configured, it defaults to a value of 2.

Required: No
 Example Entry: 3
 Usage: Reversal SAF Timer Task

dh.reversal.saf.send.interval — The number of seconds the Reversal SAF process waits between sending reversal messages to the authorizer for processing. A value of 0 indicates that there is no delay between the sending of messages. If this property is not configured, it defaults to a value of .25 seconds.

Required: No
 Example Entry: 1
 Usage: Reversal SAF Timer Task

dh.save.previous.day.totals — A flag indicating whether the previous day's totals for an ATM terminal are written to the Previous Day ATM Terminal Data Dynamic data source—general data (ESAPDT) when you enter a balancing transaction for the ATM or only during forced terminal cutover processing by the End-of-Period Processing (EOPP) process.

When this property is set to a value of false, the Previous Day Activity tab is disabled on the ATM Channel Configuration window. When this property is set to a value of true, this tab is enabled. Valid values are as follows:

true = Yes, previous day totals are written to the ESAPDT before terminal cutover processing.
 false = No, previous day totals are not written to the ESAPDT. Default.

Required: No
 Usage: ATM Cutover process

dh.atm.screen.error.notification.enabled — A flag indicating whether error notification messages are to be displayed using overlay screens to customers at ATM terminals. Valid values are as follows:

true = Yes, ATM screen error notifications are displayed to customers at ATM terminals using overlay screens.
 false = No, ATM screen error notifications are not displayed to customers at ATM terminals using overlay screens. Default.

Required: No
 Example Entries: dh.atm.screen.error.notification.enabled=true
 Usage: Diebold and NCR device types

dh.statement.interleaving — A flag indicating whether statement print messages are interleaved with other messages between the BASE24-eps ATM (Java) Device Handler process and the ATM and between the BASE24-eps ATM (Java) Device Handler process and the ATM ISO Host Interface component. Valid values are as follows:

true = Yes, statement print messages are interleaved.
false = No, statement print messages are not interleaved. Default.

Required: No
Usage: Diebold 911/912 and NCR NDC+

dh.statementdata.cleanup.starttime — The time of day, in hh:mm format, the Statement Data Cleanup timer task is to run. If this property is not configured, it defaults to a value of 23:59. This time is in the time of the executing server where the application is running.

Required: No
Example Entry: 01:30
Usage: Statement Data Cleanup Timer Task

dh.statementdata.cleanup.timeperiod.hours — The number of hours to delay between runs of the Statement Data Cleanup task to delete old statement data from the Statement Print data source (STMTDATA). If this property is not configured, it defaults to 24 hours.

Required: No
Example Entry: 48
Usage: Statement Data Cleanup Timer Task

dh.statementdata.cleanup.retention.days — The number of days statement data rows are to be retained in the Statement Print data source (STMTDATA). Rows older than this retention period are deleted when the Statement Data Cleanup timer task is run. If this property is not configured, it defaults to 1 day.

Required: No
Example Entry: 2
Usage: Statement Data Cleanup Timer Task

dh.tss.epp.serial.number.verify — A flag indicating whether the BASE24-eps ATM (Java) Device Handler or IFX ATM Manager process optionally verifies EPP serial numbers received from an ATM terminal when it authenticates and exchanges public keys against valid EPP serial numbers entered in the EPP Serial Number data source (EPP_Serial_Number). For Diebold and IFX devices, only the serial number in the EPP signature verification public key certificate is verified. Currently, the serial number in the EPP encipherment key certificate is not verified for Diebold and IFX devices. Valid values are as follows:

true = Yes, verify EPP serial numbers.
false = No, do not verify EPP serial numbers. Default.

Required: No
Usage: All ATM device types

dh.tss.external.connection — The ID of the external connection to use for sending requests to the Transaction Security Services process. The value of this property must match an external connection ID in the External Connection Configuration data source (EXTRCNCT).

Required: Yes
Example Entry: ATM_TSS
Usage: All ATM device types

dh.tss.key.signer.n — The name of the BASE24-eps system to be used as the host name for the Rivest, Shamir, and Adleman (RSA) key pair generated for the BASE24-eps system in the KEYGEN process control command, where *n* is 1, 2, or 3 representing one of the following encryption types:

- 1 = NCR signature
- 2 = Diebold certificates
- 3 = Shared Wincor Nixdorf signature

This property can contain from 1–16 alphanumeric characters and is used in conjunction with the Automated Key Distribution System. If this property is not configured, the host name for the BASE24-eps system RSA key pair defaults to a value of HOST.

Required: No
Example Entry: ATM_TSS
Usage: All ATM device types

diebold.deposit.bin.threshold1 — The amount of an authorized check transaction that determines which bin the check is deposited into. If the transaction amount is less than the amount of this property, the check is deposited into bin 1. If the transaction amount is equal to or greater than the amount of this property, the check is deposited into bin 2. If this property is not configured, the BASE24-eps ATM (Java) Device Handler process checks the value of the diebold.deposit.bin.cashback property.

Required: No
Example Entries: diebold.deposit.bin.threshold=100
diebold.deposit.bin.threshold=1000
Usage: Diebold 911/912

diebold.deposit.bin.cashcheck1 — A flag indicating which bin the check is deposited into for a cash check transaction when the diebold.deposit.bin.threshold property is not configured. If this property is set to Y and the transaction is a cash check transaction, the check is deposited into bin 1. Otherwise, the check is deposited into bin 2. If this parameter is not configured or contains a value other than Y, then the check is deposited into bin 0.

Required: No
Example Entry: diebold.deposit.bin.cashcheck=Y
Usage: Diebold 911/912

diebold.override.operator.id.access — A flag indicating whether the BASE24-eps Diebold 911/912 ATM (Java) Device Handler process checks the Diebold Operator ID data source (OPERID) to determine access to the ATM chest door. This property must be set to true for the host-controlled solenoid-activated lock feature. Valid values are as follows:

true = Yes, use the OPERID to determine access to the ATM chest door.
false = No, do not use the OPERID to determine access to the ATM chest door.
Default.

Required: Yes, if using the host-controlled solenoid-activated lock feature for the BASE24-eps Diebold 911/912 ATM (Java) Device Handler process
Usage: Diebold 911/912 terminals

elmtmap.jar.aci.atm-elmtmaps — The name of the JAR file containing ATM Framework element to data validation map properties files.

Required: Yes
Example Entry: jars/atm-elmtmaps.jar

elmtmap.jar.aci.atmdh-elmtmaps — The name of the JAR file containing BASE24-eps ATM (Java) Device Handler element to data validation map properties files.

Required: Yes
Example Entry: jars/atmdh-elmtmaps.jar

ifx.formatter.formatted.output — A flag indicating whether output IFX documents are to be formatted for easier human interpretation.

true = Yes, format output IFX documents.
false = No, do not format output IFX documents. Default.

Required: No
Usage: IFX

ifx.formatter.ifx.ns.is.default.ns — A flag indicating whether the IFX namespace is the default namespace of XML output documents. XML namespaces provide a simple method for qualifying element and attribute names used in Extensible Markup Language documents by associating them with namespaces identified by URI references. Valid values are as follows:

true = Yes, the IFX namespace is the default namespace of XML output documents. Default.
false = No, the IFX namespace is not the default namespace of XML output documents.

Required: Yes, for the BASE24-eps IFX ATM Manager process
Example Entry: true
Usage: IFX terminals

ifx.message.support.filename.n — Specifies the name of an IFX message support map XML file, where n is a sequential number starting at 0 and increased by 1 for each file to be initialized. IFX message support map XML files identify the IFX service wrappers supported by the IFX ATM Manager process and the messages within each wrapper that are supported. The IFX ATM Manager process uses this file in two ways—to validate IFX documents received and to list the messages supported in a Service Profile Inquiry response. This property is defined

in the Application Configuration data source (APPLCNFG). The Message Support Map Manager reads these files into memory at initialization. The data is held in memory for as long as the application is running.

Required: Yes, for IFX ATM Manager process.
Example Entries: ifx.message.support.filename.0=/user/ESDH041/config/IFX/
IFXFrameworkMessageSupportMap.xml
ifx.message.support.filename.1=/user/ESDH041/config/IFX/
IFXATMMessageSupportMap.xml
Usage: Message Support Map Manager

ifx.schema.version — The schema version that the IFX ATM Manager process is using. This property defaults to a value of 1.7.

Required: No
Usage: IFX

ifx.status.code.map.*n* — Specifies the name of an IFX status code map XML file, where *n* is a sequential number starting at 0 and increased by 1 for each file to be initialized. IFX status code map XML files map IFX status codes to a severity (Info, Warn, or Error) and an optional description. Status codes are provided in the response to an IFX device. The StatusCode and Severity are required elements (as they are required to be sent by the IFX schema), while the StatusDesc element is optional. These files should be configured in the Application Configuration data source (APPLCNFG). The IFX Status Code Map Manager reads them at initialization to preload status code mapping details into memory. The data is held in memory for as long as the application is running.

Required: Yes, for IFX ATM Manager process.
Example Entries: ifx.status.code.map.0=/user/ESDH041/config/IFX/
IFXStatusCodeMap.xml
Usage: IFX Status Code Map Manager

initialization.function<text-string> — Specifies the fully-qualified Java package and class name to be started at initialization by the Initialization Manager, where <text-string> is a unique text string for this initialization.function property name. These properties should be defined in the APPLCNFG. Beginning with AJSF 07.4, initialization function properties are only loaded from the APPLCNFG and not from the config.properties file.

Required: Yes

Example Entries:

Diebold 911/912 and NCR NDC+ ATM (Java) Device Handler processes:

initialization.function.0.classname=com.aciworldwide.
framework.TransactionMapManager
initialization.function.1.classname=com.aciworldwide.
common.TransMetaMapManager
initialization.function.2.classname=com.aciworldwide.base24.
es.dh.common.config.OpCodeMapManager
initialization.function.3.classname=com.aciworldwide.base24.
es.dh.common.config.CurrencySymbolMapManager
initialization.function.4.classname=com.aciworldwide.base24.
es.dh.common.config.ReceiptTemplateService
initialization.function.5.classname=com.aciworldwide.base24.
es.dh.common.config.ReceiptRequiredFlagsService
initialization.function.6.classname=com.aciworldwide.base24.
es.dh.common.config.StateMapManager
initialization.function.7.classname=com.aciworldwide.base24.
es.dh.common.config.ReceiptMapManager
initialization.function.8.classname=com.aciworldwide.base24.
es.dh.common.config.LocaleMapManager
initialization.function.9.classname=com.aciworldwide.base24.
es.dh.common.config.DispenseAlgorithmMapManager
initialization.function.10.classname=com.aciworldwide.
base24.es.common.config.ISO93MessageMapManager
initialization.function.11.classname=com.aciworldwide.
base24.es.dh.common.config.PrintTagConversionMap
Manager
initialization.function.12.classname=com.aciworldwide.
base24.es.dh.common.config.PrivateUseDataMapManager
initialization.function.13.classname=com.aciworldwide.
base24.es.dh.common.config.AuthMapManager
initialization.function.14.classname=com.aciworldwide.
base24.es.common.config.MessageClassifierMapManager
initialization.function.15.classname=com.aciworldwide.
common.MessageRouterService

IFX ATM Manager process:

initialization.function.0.classname=com.aciworldwide.
framework.TransactionMapManager
initialization.function.1.classname=com.aciworldwide.
common.TransMetaMapManager

```

initialization.function.2.classname=com.aciworldwide.base24.
es.ifx.framework.config.ActionCodeStatusCodeMapManager
initialization.function.3.classname=com.aciworldwide.base24.
es.ifx.framework.config.ReversalReasonCodeMapManager
initialization.function.4.classname=com.aciworldwide.base24.
es.ifx.framework.config.MessageSupportMapManager
initialization.function.5.classname=com.aciworldwide.base24.
es.dh.common.config.TokenMapManager*
initialization.function.6.classname=com.aciworldwide.base24.
es.dh.common.config.TokenRequiredMapManager*
initialization.function.7.classname=com.aciworldwide.base24.
es.dh.common.config.AuthMapManager
initialization.function.8.classname=com.aciworldwide.base24.
es.common.config.MessageClassifierMapManager
initialization.function.9.classname=com.aciworldwide.base24.
ifx.jaxb.JAXBContextManager
initialization.function.10.classname=com.aciworldwide.
base24.es.dh.common.config.classic.LCONFManager*
initialization.function.11.classname=com.aciworldwide.
base24.es.ifx.framework.config.IFXStatusCodeMapManager
initialization.function.12.classname=com.aciworldwide.
common.MessageRouterService

```

* These properties are only used in a BASE24 environment.

Usage: InitializationManager

jaxb.context.*n* — The Java API for XML binding (JAXB) context path for IFX messages, where *n* is a number starting at 0 and increased by 1 for each class to be executed.

Required: Yes, for the BASE24-eps IFX ATM Manager process
 Example Entry: jaxb.context.0=com.aciworldwide.base24.ifx.schema:com.aciworldwide.base24.ifx.schema.ACImods
 Usage: IFX terminals

message.jms.TopicConnectionFactory — The name of the topic connection factory used to automatically create topic connection objects with a publish/subscribe JMS provider.

Required: No
Example Entry: message.jms.TopicConnectionFactory=JmsTopicConnectionFactory
Usage: JMS Connection Manager

ncr.mac.field.select.data.classname — The pathname of the class used to select fields to be included in a message authentication code (MAC) for the BASE24-eps NCR NDC+ ATM (Java) Device Handler process. This property is required only if you are performing message authentication between BASE24-eps and NCR NDC+ terminals.

Note: Message authentication is not currently supported.

Required: Yes, if performing message authentication between BASE24-eps and NCR NDC+ terminals
Example Entry: com.aciworldwide.base24.es.dh.ncr.valueobjects.DefaultMacFieldSelectionData
Usage: NCR NDC+ terminals

orig.process.name — A symbolic name use to construct a ClassicData value object.

Required: No
Example Entry: P1A^JDH
Usage: All ATM device types

terminal.logical.network — The logical network for an ATM terminal.

Required: No
Example Entry: PRO1
Usage: All ATM device types

timertask.function<text-string> — The name of a class to be executed by the ATM Timer process, where <text-string> is a unique text string for this timertask. function property name. Each class designates cleanup tasks to be performed for the AJSF application. Each class designates cleanup tasks or the automatic reporting tasks to be performed for the AJSF application. Additional properties must be configured in conjunction with most timed tasks to be performed. These properties are identified in parentheses following each example timer task class entry.

Required:	Yes
Example Entries:	timertask.function.0.classname=com.aciworldwide.base24.es.dh.common.timer.timertask.ReversalSAFTimerTask (see also the following property descriptions: dh.reversal.saf.intra.batch.delay, dh.reversal.saf.max.rowcount, dh.reversal.saf.run.delay.minutes, dh.reversal.saf.start.delay.minutes, dh.reversal.saf.msg.retries, dh.reversal.saf.send.interval) timertask.function.1.classname=com.aciworldwide.base24.es.dh.common.timer.timertask.CutoverSAFTimerTask (see also the following property descriptions: dh.cutover.saf.institution.id, dh.cutover.saf.intra.batch.delay, dh.cutover.saf.max.rowcount, dh.cutover.saf.msg.retries, dh.cutover.saf.send.interval, dh.cutover.saf.run.delay.minutes, dh.cutover.saf.start.delay.minutes) timertask.function.2.classname=com.aciworldwide.base24.es.dh.common.timer.timertask.InactivityTimerTask (see also the following property descriptions: dh.inactivity.intra.batch.delay, dh.inactivity.max.rowcount, dh.inactivity.run.delay.minutes, dh.inactivity.start.delay.minutes, dh.inactivity.timeout.period) timertask.function.3.classname=com.aciworldwide.base24.es.dh.common.timer.timertask.LoadInactivityTimerTask (see also the following property descriptions: dh.load.inactivity.intra.batch.delay, dh.load.inactivity.max.rowcount, dh.load.inactivity.run.delay.minutes, dh.load.inactivity.start.delay.minutes, dh.load.inactivity.timeout.period)
Usage:	Timer Task Manager

tss.external.message.encoding — The type of encoding that is used for messages exchanged between the BASE24-eps ATM (Java) Device Handler or IFX ATM Manager process and Transaction Security Services process. If this property is not configured, it defaults to the native encoding format of the BASE24-eps ATM (Java) Device Handler or IFX ATM Manager process. Valid values are as follows:

- | | |
|--------|--|
| ASCII | = Use ASCII encoding for messages exchanged between the BASE24-eps ATM (Java) Device Handler process and Transaction Security Services process. |
| EBCDIC | = Use EBCDIC encoding for messages exchanged between the BASE24-eps ATM (Java) Device Handler process and Transaction Security Services process. |

Required: No
Usage: All ATM device types

xml.action.code.status.code.map.filename. — The name of the Action Code to Status Code Map XML file that maps ISO and internal action codes to IFX status codes. This property should be defined in the Application Configuration data source (APPLCNFG). The Action Code to Status Code Map Manager reads this file into memory at initialization. The data is held in memory for as long as the application is running.

Required: Yes, for an IFX ATM Manager process.
Example Entries: `xml.action.code.status.code.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/IFX/ActionCodeStatusCodeMap.xml`
Usage: Action Code to Status Code Map Manager

xml.auth.map.filename — The name of the Authorization Destination Map XML file that indicates the authorization destination to which the BASE24-eps ATM (Java) Device Handler processes send transactions for specified transaction types and institution IDs. The transaction type and institution ID can be wildcarded. Currently, the BASE24-eps ATM (Java) Device Handler process supports authorization destination values of AUTH_ES and AUTH_CLASSIC, which correspond to conditions in the appropriate Transaction Meta Map file. Conditions for AUTH_ES are defined in the Base DH Transaction Meta Map file while conditions for AUTH_CLASSIC are defined in the Classic DH Transaction Meta Map file. This property should be defined in the Application Configuration data source (APPLCNFG). The Authorization Map Manager reads it at initialization to preload message routing details into memory. The data is held in memory for as long as the application is running.

Required: Yes, for all types of BASE24-eps ATM (Java) Device Handler processes.
Example Entry: `xml.auth.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Common/AuthMap.xml`
Usage: Authorization Map Manager

xml.currency.symbol.map.filename — The name of the Currency Symbol Map XML file that defines currency symbols for ATM statements and receipts by device type. This allows you to configure multiple currency symbols. This property should be defined in the Application Configuration data source

(APPLCNFG). The Currency Symbol Map Manager reads it at initialization to preload currency symbol details into memory. The data is held in memory for as long as the application is running.

Required: Yes, for BASE24-eps Diebold 911/912 and NCR NDC+ ATM (Java) Device Handler processes. This property is not used by the IFX ATM Manager process.

Example Entry: `xml.currency.symbol.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/CurrencySymbolMap.xml`

Usage: Currency Symbol Map Manager

xml.dispense.algorithm.map.filename — The name of the Dispense Algorithm Map XML file that defines a mapping between a dispense algorithm code and the class that implements the dispense algorithm. Currently, only the least bills method is supported. This property should be defined in the Application Configuration data source (APPLCNFG). The Dispense Algorithm Map Manager reads it at initialization to preload dispense algorithm handling details into memory. The data is held in memory for as long as the application is running. Refer to the ***BASE24-eps ATM Acquirer Management Guide*** for a description of the lease bills method dispense algorithm.

Required: Yes, for BASE24-eps Diebold 911/912 and NCR NDC+ ATM (Java) Device Handler processes. This property is not used by the IFX ATM Manager process.

Example Entry: `xml.dispense.algorithm.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DispenseAlgorithmMap.xml`

Usage: Dispense Algorithm Map Manager

xml.iso.message.map.filename — The name of the ISO Message Map XML file that defines a list of classes to format individual bits (data elements) in the ISO message sent to the Integrated Server process for custom software modifications (CSMs). If needed for CSMs, this property should be configured in the Application Configuration data source (APPLCNFG). The ISO 93 Message Map Manager reads it at initialization to preload ISO message handling details into memory. The data is held in memory for as long as the application is running.

Required: No, unless needed for CSMs for any BASE24-eps ATM (Java) Device Handler process.

Example Entry: `xml.iso.message.map.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/ISOMessageMap.xml`

Usage: ISO 93 Message Map Manager

xml.locale.map.filename — The name of the Locale Map XML file that maps language codes (received in the last byte of the eight-byte operations code buffer) to locale strings for specified terminal types and load profiles. The ATM terminal provides the language code in the eighth byte of the operations code buffer. Locale strings indicate the language in which the host should format text in a response (e.g., for a statement print transaction) and are used to access the appropriate ATM Bundler properties file. This property should be configured in the Application Configuration data source (APPLCNFG). The Locale Map Manager reads it at initialization to preload language handling details into memory. The data is held in memory for as long as the application is running.

Required: Yes, for BASE24-eps Diebold 911/912 and NCR NDC+ ATM (Java) Device Handler processes. This property is not used by the IFX ATM Manager process.

Example Entry: `xml.locale.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/LocaleMap.xml`

Usage: Locale Map Manager

xml.message.classifier.map.filename — The name of the Message Classifier Map XML file that specifies the mapping between the first digit of the message type of a response message to the formatter controller class for that type of message. This property should be defined in the Application Configuration data source (APPLCNFG). The Message Classifier Map Manager reads this file into memory at initialization. The data is held in memory for as long as the application is running.

Required: Yes, for all types of BASE24-eps ATM (Java) Device Handler processes.

Example Entry: `xml.message.classifier.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/MessageClassifierMap.xml`

Usage: Message Classifier Map Manager

xml.opcode.map.filename.*n* — The name of an Operation Code Map XML file, where *n* is a sequential number starting at 0 and increased by 1 for each filename specified. Operation Code Map XML files translate the first five bytes of the eight-character ATM operations code buffer received from the ATM into a BASE24-eps processing code and pre-processing steps. This property should be defined in the Application Configuration data source (APPLCNFG). The Operation Code Map Manager reads it at initialization to preload operations code translation details into memory. The data is held in memory for as long as the application is running.

Required: Yes, for BASE24-eps Diebold 911/912 and NCR NDC+ ATM (Java) Device Handler processes. This property is not used by the IFX ATM Manager process.

Example Entries: `xml.opcode.map.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DieboldOpCodeMap.xml`
`xml.opcode.map.filename.1=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/NCROpCodeMap.xml`

Usage: Operation Code Map Manager

xml.parser.validate.document — A flag indicating whether the BASE24-eps ATM (Java) Device Handler or IFX ATM Manager process validates XML documents while parsing them.

true = Yes, validate XML documents while parsing them.

false = No, do not validate XML documents while parsing them. Default.

Required: No

Usage: All ATM device types

xml.print.tag.conversion.map.filename.*n* — The name of a Print Tag Conversion Map XML file, where *n* is a sequential number starting at 0 and increased by 1 for each filename specified. Print Tag Conversion Map XML files define print tags for printer control sequences such as line breaks, form feeds, etc., as well as space and text compression. This property should be defined in the Application Configuration data source (APPLCNFG). The Print Tag Conversion Map Manager reads it at initialization to preload printer control handling details into memory. The data is held in memory for as long as the application is running.

Required: Yes, for BASE24-eps Diebold 911/912 and NCR NDC+ ATM (Java) Device Handler processes. This property is not used by the IFX ATM Manager process.

Example Entries: `xml.print.tag.conversion.map.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DieboldPrintTagConversionMap.xml`
`xml.print.tag.conversion.map.filename.1=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/NCRPrintTagConversionMap.xml`

Usage: Print Tag Conversion Map Manager

xml.private.use.data.map.filename — The name of the Private Use Field Formatter Map XML file that defines tags for private use fields (e.g., data elements P-62 and P-63) in the ISO message sent between the BASE24-eps ATM (Java)

Device Handler process and the Integrated Server process. Any tags defined in a private use data element for CSMs (e.g., S-127) must be defined in this file. This property should be defined in the Application Configuration data source (APPLCNFG). The Private Use Data Map Manager reads it at initialization to preload ISO message handling details into memory. The data is held in memory for as long as the application is running.

Required: Yes, for all BASE24-eps ATM (Java) Device Handler processes.

Example Entry: `xml.private.use.data.map.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/PrivateUseFieldFormatterMap.xml`

Usage: Private Use Data Map Manager

xml.reason.code.map.filename — The name of the Reason Code Map XML file that maps IFX reversal reason codes to ISO reason codes. This property should be configured in the Application Configuration data source (APPLCNFG). The Reversal Reason Code Map Manager reads it at initialization to preload reason code mapping details into memory. The data is held in memory for as long as the application is running.

Required: Yes, for IFX ATM Manager process.

Example Entries: `xml.reason.code.map.filename=/user/ESDH041/config/IFX/ReversalReasonCodeMap.xml`

Usage: Reversal Reason Code Map Manager

xml.receipt.map.filename.*n* — The name of a Receipt Map XML file, where *n* is a sequential number starting at 0 and increased by 1 for each filename specified. Receipt Map XML files define field codes specific to a device type for formatting receipts. Field codes common to all device types are defined in the Receipt Formatter Map XML file. Receipt formatting codes are tokens that represent fields to be printed on a receipt. This property should be configured in the Application Configuration data source (APPLCNFG). The Receipt Map Manager reads it at initialization to preload receipt handling details into memory. The data is held in memory for as long as the application is running.

Required: Yes, for BASE24-eps Diebold 911/912 and NCR NDC+ ATM (Java) Device Handler processes. This property is not used by the IFX ATM Manager process.

Example Entries: `xml.receipt.map.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/ReceiptFormatterMap.xml`
`xml.receipt.map.filename.1=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DieboldReceiptMap.xml`
`xml.receipt.map.filename.2=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/NCRReceiptMap.xml`

Usage: Receipt Map Manager

xml.state.map.filename.*n* — The name of a State Map XML file, where *n* is a sequential number starting at 0 and increased by 1 for each filename specified. State Map XML files determine the next state and screen to be displayed by the ATM based on the ISO response code returned and specified conditions. This property should be defined in the Application Configuration data source (APPLCNFG). The State Map Manager reads it at initialization to preload state handling details into memory. The data is held in memory for as long as the application is running.

Required: Yes, for BASE24-eps Diebold 911/912 and NCR NDC+ ATM (Java) Device Handler processes. This property is not used by the IFX ATM Manager process.

Example Entries: `xml.state.map.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DieboldStateMap.xml`
`xml.state.map.filename.1=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/NCRStateMap.xml`

Usage: State Map Manager

xml.token.map.filename — The name of the Token Map XML file that specifies the mapping between token identifiers and the formatters used to parse and build those tokens when sending transactions to BASE24-atm for authorization. If routing messages to BASE24-atm for authorization, this property should be defined in the Application Configuration data source (APPLCNFG). The Token Map Manager reads it at initialization to preload token handling details into memory. The data is held in memory for as long as the application is running.

Required: Yes, for any BASE24-eps ATM (Java) Device Handler processes if routing to BASE24-atm for authorization.

Example Entry: `xml.token.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Common/TokenMap.xml`

Usage: Token Map Manager

xml.token.required.map.filename — The name of the Token Required Map XML file that specifies which tokens are required to be added to STM request messages when sending transactions to BASE24-atm for authorization. This file is not used to validate tokens received in messages from BASE24-atm. If routing messages to BASE24-atm for authorization, this property should be defined in the Application Configuration data source (APPLCNFG). The Token Required Map Manager reads it at initialization to preload token handling details into memory. The data is held in memory for as long as the application is running.

Required:	Yes, for any BASE24-eps ATM (Java) Device Handler process if routing to BASE24-atm for authorization.
Example Entry:	xml.token.required.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Common/TokenRequiredMap.xml
Usage:	Token Required Map Manager

Sample Config.properties File

Different sample config.properties files are provided for the BASE24-eps ATM (Java) Device Handler processes, IFX ATM Manager process, and ATM Timer process as listed below. These sample config.properties files are located in the DH-Common\Properties folder.

Config.properties Files	Usage
combined.properties	Properties for a combined BASE24-eps ATM (Java) Device Handler process for NCR NDC+, Diebold 911/912, and IFX devices running in a BASE24-eps system
ifx.base24.config.properties	IFX ATM Manager process running in a BASE24-atm system
ifx.config.properties	BASE24-eps IFX ATM Manager process (IFX terminals)
ndc.config.properties	BASE24-eps ATM (Java) Device Handler process (NCR NDC+ terminals)
tcs.config.properties	BASE24-eps ATM (Java) Device Handler process (Diebold 911/912 terminals)
tcsndc.combined.properties	Properties for a combined BASE24-eps ATM (Java) Device Handler process for NCR NDC+ and Diebold 911/912 terminals running in a BASE24-eps system
timer.config.properties	ATM Timer process

The following is an example of a jdih.config.properties file used for the BASE24-eps ATM (Java) Device Handler process and NCR NDC+ or Diebold 911/912 terminals. Any of the sample config.properties files you use may need to be edited during installation for site-specific information.

```
db.user.id=pserve
db.user.password=c67834a7b2e28c4e

db.datasource=jdbc:JDAL:-MDBV=CSV:-AI=ES:-STRT=/G/CONFIG/SH2RDATA

#
# db.max.connections defines the maximum number of connections allowed to
# connect to
# the database.
#
db.max.connections=50

#
# db.static.connections defines the number of connections to acquire to the
# database
# when the jvm starts. These connections will stay active through the life of
# the
# executing JVM.
#
db.static.connections=1
#
# db.connection.wait.time defines the amount of time, in milliseconds, to wait
# for
# another thread to free a Database connection in the Database connection
# manager.
# This is used when the maximum number of connections have been acquired as
# specified
# in the db.max.connections property.
#
db.connection.wait.time=10
#
# db.driver specifies the class name of the JDBC driver to use for all database
#
connections.
# The whole package name as it is found in the jar file containing the driver
# should be
#
specified.
#
db.driver=com.aciworldwide.jdbcdriver.jdal2.JDAL2Driver
#
#
# db.type specifies the type of database the application is installed on.
# Valid values are:
# DB2
# ORACLE
# MSSQL
# SQLMX
#
db.type=JDAL2
```

```
#
# Error Manager properties.  If the filename contains the characters
# "YYYYMMDD" in its name this will cause the ErrorManager to roll log
# files on a daily basis. If these characters are not found in the name
# the same log file will remain open.
#
#
# LogManager properties.
# The Log4J config file location for trace and event
#
event.log4j.config.url=file:///user/B24es_054/ESDH/uaf2/P2R^GATE/config/
EventConfig.xml
#
# Timer Task Manager properties.
#
timertask.manager.processid=P2R^JTIMER
timertask.function.0.classname=com.aciworldwide.base24.es.dh.common.timer.
timertask.ReversalSAFTimerTask
timertask.function.1.classname=com.aciworldwide.base24.es.dh.common.timer.
timertask.CutoverSAFTimerTask
timertask.function.2.classname=com.aciworldwide.base24.es.dh.common.timer.
timertask.StatementDataCleanupTimerTask
timertask.function.3.classname=com.aciworldwide.base24.es.dh.common.timer.
timertask.LoadInactivityTimerTask
timertask.function.4.classname=com.aciworldwide.base24.es.dh.common.timer.
timertask.InactivityTimerTask
timertask.function.5.classname=com.aciworldwide.base24.es.dh.common.timer.
timertask.ATMLogCleanupTimerTask

#
# Messaging properties.
#
#
# Valid values for message.environment are:
# DIRECT
# JMS
# EJB (Not yet fully implemented)
#
message.environment=JMS
#
# OPENJMS
#
message.jms.jndi.initialcontextfactory=com.aci.jms.jndi.embedded.
EmbeddedXpnetInitialContext

Factory
message.jms.jndi.providerurl=jms://dummy
message.jms.QueueConnectionFactory=JmsQueueConnectionFactory
message.jms.TopicConnectionFactory=JmsTopicConnectionFactory
#
# MQSeries
```



```
#
#message.jms.jndi.initialcontextfactory=com.sun.jndi.fscontext.
#RefFSContextFactory
#message.jms.jndi.providerurl=file:C:/Program files/ibm/MQSeries/Java/JNDI-
#Directory

#
# The following MUST be set to true if running on the HP NSK platform
# due to how it handles threading.
#
message.jms.use.nsk.threads=true

#
# Trace processing properties.
#
trace.enabled=true
trace.log4j.config.url=file:///user/B24es_054/ESDH/uaf2/P2R^GATE/config/
TraceConfig.xml

#####
# XML Message Builder properties.
#
# The Use Indent property may be used to force the building of XML
# messages with indentation and new line characters. This formatted
# XML message is easier to read and may be useful in development.
# IT IS NOT ADVISABLE TO USE INDENTATION IN A PRODUCTION SYSTEM.
#####
xml.builder.use.indent=false

#
# XML configuration files
#

#
# Tran Map files
#
xmltranmap.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Common/
DHCommonTranMap.xml
xmltranmap.filename.1=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Diebold/
DieboldDHTranMap.xml
xmltranmap.filename.2=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-NCR/
NCRDHTranMap.xml
xmltranmap.filename.3=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Surcharge/
BaseDHSurchargeTranMap.xml

#
# Trans Metamap files
#
xmltransmetamap.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Common/
DHCommonTransMetaMap.xml
xmltransmetamap.filename.1=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Diebold/
DieboldDHTransMetaMap.xml
```

```
xmltransmetamap.filename.2=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-NCR/
NCRDHTransMetaMap.xml
xmltransmetamap.filename.3=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-
Surcharge/BaseDHSurchargeTransMetaMap.xml
xmltransmetamap.filename.4=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Diebold/
DieboldDHSurchargeTransMetaMap.xml
xmltransmetamap.filename.5=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-NCR/
NCRDHSurchargeTransMetaMap.xml

#
#      DH State Map files
#
xml.state.map.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Diebold/
DieboldStateMap.xml
xml.state.map.filename.1=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-NCR/
NCRStateMap.xml

#
#      DH Op Code Map files
#
xmlopcodemap.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Diebold/
DieboldOpCodeMap.xml
xmlopcodemap.filename.1=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-NCR/
NCROpCodeMap.xml

#
#      Currency Symbol Map file
#
#      Specified the escape sequences that are used for each device to
#      display currency symbols on the screen and on printers. Used by
#      the NCR and Diebold (Java) Device Handlers.
#
xml.currency.symbol.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-
Common/CurrencySymbolMap.xml

#
#      Receipt Map files
#
xml.receipt.map.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Common/
ReceiptFormatterMap.xml
xml.receipt.map.filename.1=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Diebold/
DieboldReceiptMap.xml
xml.receipt.map.filename.2=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-NCR/
NCRReceiptMap.xml

#
#      Locale Map
#
xml.locale.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Common/
LocaleMap.xml

#
```

```
# ISO93MessageMap
#
xml.iso.message.map.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/BASE24-
Common/ISOMessageMap.xml

#
# Dispense Algorithm Map
#
# Used to allow configuration of the dispense algorithms used by
# the NCR and Diebold (Java) Device Handlers.
#
xml.dispense.algorithm.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/
DH-Common/DispenseAlgorithmMap.xml

#
# PrintTag Conversion Map
#
# Used to allow configuration of the control characters that are inserted into
# statement responses for each of the NCR and Diebold (Java) Device Handlers.
#
xml.print.tag.conversion.map.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/
config/DH-Diebold/DieboldPrintTagConversionMap.xml
xml.print.tag.conversion.map.filename.1=/user/B24es_054/ESDH/uaf2/P2R^GATE/
config/DH-NCR/NCRPrintTagConversionMap.xml

#
# Private Use Field Formatter
#
# Allows configuration of the Private Use Data fields P-62 and P-63 in the
# ISO messages to the IS. This file specifies the Tags that are supported
# for each message type and which class is used for retrieve the data.
#
xml.private.use.data.map.filename.0=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/
DH-Common/PrivateUseFieldFormatterMap.xml

#
# Classic Token Support files
#
xml.token.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Common/
TokenMap.xml
xml.token.required.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-
Common/TokenRequiredMap.xml

#
# Routing
#
xml.auth.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/DH-Common/
AuthMap.xml

#
# Response message classification
#
```

```
xml.message.classifier.map.filename=/user/B24es_054/ESDH/uaf2/P2R^GATE/config/  
BASE24-Common/MessageClassifierMap.xml
```

C: ACI Application Management

The ACI Application Management agent is an application based on the ACI Java Server Framework (AJSF) that manages system resources.

As an AJSF-based application, the ACI Application Management agent is configured in the following AJSF data sources and in the User Security (USEC) and user auditing subsystems of the AJSF:

- Application Configuration data source (APPLCNFG)
- Config.properties
- EventConfig.xml
- TraceConfig.xml

The ACI Application Management agent is also configured in the following data sources specific to the agent:

- Adaptor definition files (adaptor.xml)
- Connector definition files (connector.xml)

The AJSF configuration requirements for the ACI Application Management agent are provided in the *ACI Application Management User Guide*.

ACI Worldwide, Inc.

D: Remote Database Server

The Remote Database Server is an application based on the ACI Java Server Framework (AJSF) that is designed to provide a portable, high performance mechanism to insert records received from the Real Time Feed Interface into an application database, such as the ACI Payments Manager Transaction Warehouse database.

As an AJSF-based application, the Remote Database Server is configured in the following AJSF data sources:

- Alternate Data Source data source (DATASRC)
- Application Configuration data source (APPLCNFG)
- Config.properties
- EventConfig.xml
- External Connections data source (EXTRCNCT)
- Listener data source (LISTENER)
- Process data source (PROCESS)
- TraceConfig.xml

The AJSF configuration requirements for the Remote Database Server are provided in the ***BASE24-eps Extract and Reporting User Guide***.

ACI Worldwide, Inc.

Glossary

The following entries describe technical and industry terms used in ACI's Java Server Framework.

caching — A technique whereby records initially retrieved from a data source on disk are maintained in cache memory for all subsequent data access requests until the cache memory is cleared. This technique increases the overall performance of data access.

clustered system — A server with a group of independent processors managed as a single system for higher availability, manageability, and scalability. Each processor has its own operating system and memory subsystem. Applications run in single-threaded mode on clustered systems such as the HP NonStop platform.

coarse-grained object — An object that maps to more than one data source. A fine-grained object maps to a single data source.

daemon — A process that runs in the background and performs a specified operation at predefined times or in response to certain events. The term daemon is a UNIX term, though many other operating systems provide support for daemons, though they're sometimes called other names. Windows, for example, refers to daemons as System Agents and services.

Enterprise JavaBeans (EJBs) — An architecture for setting up program components, written in the Java programming language, that run in the server parts of a computer network that uses the client/server model. Enterprise JavaBeans is built on the JavaBeans technology for distributing program components (which are called Beans, using the coffee metaphor) to clients in a network. Enterprise JavaBeans offers enterprises the advantage of being able to control change at the server rather than having to update each individual computer with a client whenever a new program component is changed or added. EJB components have the advantage of being reusable in multiple applications. To deploy an EJB Bean or component, it must be part of a specific application, which is called a container.

Extensible Markup Language (XML) — Describes a class of data objects called XML documents and partially describes the behavior of computer programs which process them. XML is an application profile or restricted form of the Standard Generalized Markup Language SGML. By construction, XML documents are conforming SGML documents.

Java Message Service (JMS) — A messaging standard that allows application components based on the Java 2 Platform, Enterprise Edition (J2EE) to create, send, receive, and read messages. It enables distributed communication that is loosely coupled, reliable, and asynchronous.

Java Server Page (JSP) — A technology for controlling the content or appearance of web pages through the use of servlets. Sun Microsystems also refers to JSP technology as the Servlet application program interface (API). JSP is comparable to Microsoft's Active Server Page (ASP) technology. Whereas a Java Server Page calls a Java program that is executed by a web server, an Active Server Page contains a script that is interpreted by a script interpreter (such as VBScript or JScript) before the page is sent to the user.

An HTML page that contains a link to a Java servlet is sometimes given the file name suffix of .JSP.

Java Virtual Machine (JVM) — The Java interpreter and runtime environment for Java applications. Each runtime instance of the JVM runs a single Java application and implements all the Java APIs for the platform on which it is deployed. A JVM runtime instance is created when you start a Java application. When you stop a Java application, the runtime instance is destroyed. A JVM instance starts running the single Java application by invoking the main () method of an initial class.

Java DataBase Connectivity (JDBC) — An industry standard database access method for database-independent connectivity between the Java programming language and a wide range of databases. The JDBC API provides a call-level API for SQL-based database access. JDBC technology allows you to use the Java programming language to exploit "Write Once, Run Anywhere" capabilities for applications that require access to enterprise data.

multithreaded — The ability of an application to process multiple transactions simultaneously without having to have multiple copies of the application running in the computer. Each transaction is kept track of as a thread with a separate identity. As applications work on behalf of the initial request for that thread and are

interrupted by other requests, the status of work on behalf of that thread is kept track of until the work is completed. AJSF applications can run multithreaded on SMP systems.

Open DataBase Connectivity (ODBC) — A standard database access method developed by Microsoft Corporation. The goal of ODBC is to make it possible to access any data from any application, regardless of which database management system (DBMS) is handling the data. ODBC manages this by inserting a middle layer, called a database driver between an application and the DBMS. The purpose of this layer is to translate the application's data queries into commands that the DBMS understands. For this to work, both the application and the DBMS must be ODBC-compliant—that is, the application must be capable of issuing ODBC commands and the DBMS must be capable of responding to them.

optimistic locking — A technique by which the data to be updated is not locked in advance (i.e., when the data is read). This technique proceeds on the assumption that the data being updated has not changed since the read and does not provide concurrency control. Concurrency control is the mechanism that ensures that the data being written back to the database is consistent with what was read from the database in the first place (i.e., that no other transaction has updated the data after it was read).

pessimistic locking — A technique by which the data to be updated is locked in advance. Once the data to be updated has been locked, the application can make the required changes, and then commit or rollback the update, during which time the lock is automatically dropped. If the application attempts to acquire a lock of the same data during this process, it must wait until the first transaction has completed. This approach is called pessimistic because it assumes that another transaction might change the data between the read and the update. In order to prevent that change from happening, and the data inconsistency that would result, the read statement locks the data to prevent any other transaction from changing it. Pessimistic locking provides concurrency control. Concurrency control is the mechanism that ensures that the data being written back to the database is consistent with what was read from the database in the first place (i.e., that no other transaction has updated the data after it was read).

reflection — Enables Java code to discover information about the fields, methods and constructors of loaded classes, and to use reflected fields, methods, and constructors to operate on their underlying counterparts on objects, within security restrictions. The API accommodates applications that need access to either the public members of a target object (based on its runtime class) or the members declared by a given class.

Secure Sockets Layer (SSL) — A protocol developed by Netscape for transmitting private documents via the Internet. SSL works by using a private key to encrypt data transferred over the SSL connection from a client to a server.

serialized data object — A whole object written to and read from a raw byte stream. Object serialization allows Java objects and primitives to be encoded into a byte stream suitable for streaming to some type of network or to a file-system, or more generally, to a transmission medium or storage facility.

servlet — A small program specified in a web page and run on a web server to modify the web page before it is sent to the user who requested it.

single-threaded — The processing of transactions one-at-a-time by an application. The application must complete processing of a transaction before it starts processing another transaction. AJSF applications run single-threaded on clustered systems, such as the HP NonStop platform.

socket — A connection. A secure socket is a connection that uses some type of security, such as Secure Sockets Layer (SSL), to secure the connection.

symmetric multiprocessing (SMP) — The processing of program code by multiple processors that share a common operating system and memory subsystem. In symmetric (or *tightly coupled*) multiprocessing, the processors share memory and the I/O bus. A single copy of the operating system manages all the processors. SMP systems do not usually exceed 16 processors, although some of the latest machines released by several UNIX vendors support up to 64 processors. Applications can run in multithreaded mode on SMP systems, such as Windows NT, UNIX, and IBM System z platforms.

XML document — A document made up of storage units called entities, which contain either parsed or unparsed data. Parsed data is made up of characters, some of which form character data, and some of which form markup. Markup encodes a description of the document's storage layout and logical structure. XML provides a mechanism to impose constraints on the storage layout and logical structure.

XML processor — A software module used to read XML documents and provide access to their content and structure.

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